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Marc Nieper-Wißkirchen

Elementary Galois Theory

A Constructive Approach

Springer

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Marc Nieper-Wißkirchen

Elementary Galois Theory

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To Konrad, Karla, and Irmgard

Preface

Galois theory, which can certainly be counted among the pearls of pure mathematics with some justification, occupies a central position in algebra courses for bachelor students or teacher trainees in mathematics.

The starting point is the question of whether solutions to polynomial equations can be described using radical expressions. To this end, symmetries between the solutions of such an equation are investigated. It is shown that essential information about the polynomial can be obtained from these symmetries and that these in particular provide answers to the original question of solvability by radicals.

The importance of Galois' theory is also due to the fact that the three most important structures of algebra have developed from it: The symmetries of the solutions of an equation together form a *group*. The coefficients of the polynomial equations take on values in a *field*, such as the rational or algebraic numbers. Finally, the polynomials as a whole form a *ring*, which in many respects has similarities with the ring of integers.

The aim of this book is to provide an elementary, classical introduction to Galois theory without using higher concepts and, at the same time, to reap the fruits of the theory in the form of applications to millennia-old problems.

The material in this volume is designed in such a way that it can be covered in a one-semester algebra course directly at the beginning of studying. As it deals with the fundamentals of theory, all the material covered is relevant for examinations in algebra.

The appendix contains important results for algebra from linear algebra and analysis. This material can be particularly useful if this book forms the basis of a first or second semester lecture or is worked through in self-study.

The book will be continued by a second volume in which the more modern, abstract view of Galois theory will be emphasized and the abstract structures group, ring, and field will be motivated, defined, and then studied in detail.

As far as the present volume is concerned, emphasis has been placed on the fact that even the study of this book alone will make a significant contribution to the general mathematical education of an interested layman or a student who does not wish to delve further into the field of algebra.

Many introductory textbooks have been written on algebra and especially on Galois theory. The author of this book, for example, has learned a lot from the book by Serge

Lang [5], and there are also many well-established textbooks in German, such as the one by Siegfried Bosch [2]. I would also like to point out Jörg Bewersdorff's excellent book [1] on elementary Galois theory with a historical perspective.

So what are the special features of this book, the contents of which also form the basis of lectures by the author?

A number of principles have been followed in the creation of the "Galois theory":

Abstract theories and definitions are derived from concrete problems and solutions and, conversely, the concrete is not regarded as a special case of the abstract. Even if the abstract theory is ultimately what represents the actual further development of mathematics, it can only be sufficiently appreciated and understood with knowledge of the original concrete framework.

Furthermore, with any advanced mathematical theory there is a danger that the great theorems may ultimately follow elegantly from a chain of abstractions and lemmas, but the actual reason for their truth is no longer seen. It is therefore also the declared aim of the book to ensure that the common thread in the reader's mind is not broken and that the reader is at all times in a position to be able to explain at least roughly what the contents of the theory are to someone who does not know the theory and to be able to provide approximate justifications for its statements. In this respect, it is also important to the author that the applications of the theory are mentioned early and not only at the end, after a huge theoretical edifice has been erected in a seemingly unmotivated manner.

This book contains three elements that are intended to help achieve this self-imposed goal:

Firstly, the continuous text is regularly interrupted by boxes with a gray background. These boxes contain important facts from the main text in the form of catch phrases. On the one hand, they are intended to encourage the first reader to pause and reflect on whether the key points of the actual content have been understood; on the other hand, they allow the reader to focus on the most important points when repeating the text (in exams, for example).

Secondly, at the end of each chapter there are summaries in the form of bulleted lists with a gray background. In these summaries, the essential mathematical arguments of the respective chapter are retraced step by step.

Thirdly, these summaries are followed by numerous exercises. It is the author's firm opinion that mathematics, like swimming or playing an instrument, cannot be learned just by passively studying a book. The level of the exercises deliberately starts with very simple questions. These make it easy to check learning success. The level of the exercises rises in each case and goes as far as difficult brain teasers, which should also encourage the particularly gifted among the readers.

One thing that distinguishes this book from most, but not all—we are thinking of Harold Edwards' [4]—algebra textbooks is that it consistently takes the viewpoint of a constructive mathematician:

Constructive mathematics means the following: In order to prove the existence of a mathematical object, a construction rule for it must be given. An abstract proof of existence without specifying such a construction rule is rejected.

This means that the law of the excluded third and the axiom of choice, both of which are not constructive, must be dispensed with (i.e., classical logic must be replaced by intuitionistic logic), but this not only allows a sharper view of things but also prevents us from deriving statements that are logical consequences of the axioms but have no practical relevance in themselves:

For example, the classical fundamental theorem of algebra states that every non-trivial polynomial equation over the complex numbers has a zero. However, there is no general method for calculating such a zero, which is essentially due to the fact that it is undecidable for a general complex number whether it vanishes or not.

Many of today's mathematicians do not come into contact with constructive mathematics in the course of their training and may perceive constructivism or intuitionistic logic as esoteric mathematics. However, this must be vehemently contradicted: On the one hand, every constructively proven statement is also classical (i.e., non-constructively) true; on the other hand, every classical proof of a statement can be translated into a constructive proof of a correspondingly reformulated statement. Moreover, intuitionistic logic also appears in classical mathematics, for example in the study of sheaves over a topological space (Fig. 1).



Fig. 1 Évariste Galois, 25.10.1811–31.05.1832 [3]

However, unlike other books on constructive algebra, such as [6], this book does not emphasize the constructive point of view. Someone unfamiliar with constructivism who reads this book will essentially notice that some statements are formulated somewhat more cautiously than is classically usual—for example, we will only prove the fundamental theorem of algebra for equations whose coefficients are algebraic numbers (in this case there is an effective method of finding zeros!)—and perhaps that some proofs are more elaborate, but have greater clarity.

Enough preamble, now it's time for the algebra. I hope you enjoy browsing and working through the chapters.

Augsburg, Germany Marc Nieper-Wißkirchen
March 2024

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Symbols

A_n	Alternating group
$\mathbf{C}$	Ring of complex numbers
C_n	Cyclic group
D_n	Dihedral group
$e_k(X_1, \ldots, X_n)$	Elementary symmetric function
G_σ	Centralizer of the group element σ
$[G : 1]$	Order of the group
$[G : G_\sigma]$	Number of group elements conjugate to σ
$[G : H]$	Index of a subgroup
$\mathrm{Gal}_K(x_1, \ldots, x_n)$	Galois group
$\mathrm{Gal}_{\mathbf{Q}}(x_1, \ldots, x_n)$	Galois group over the rational numbers
gcd	Greatest common divisor
$H \cdot N$	Product of a subgroup with a normal subgroup
$K[X]$	Ring of polynomials in the indeterminate X
$K[X_1, \ldots, X_n]$	Ring of polynomials in the indeterminates $X_1, \ldots, X_n$
$K(x_1, \ldots, x_n)^{\sigma_1, \ldots, \sigma_m}$	Invariants under the Galois action
$\Phi_n(X)$	Cyclotomic polynomial
$\mathbf{Q}$	Field of rational numbers
$\overline{\mathbf{Q}}$	Field of algebraic numbers
$\mathbf{Q}_+$	Set of positive rational numbers
$\mathbf{Q}(t_1, \ldots, t_n)$	Set of numbers rational in $t_1, \ldots, t_n$
$\mathbf{Q}[X]$	Ring of polynomials with rational coefficients in the indeterminate X
$\mathbf{Q}[X_1, \ldots, X_n]$	Ring of polynomials with rational coefficients in the indeterminates $X_1, \ldots, X_n$
$\mathbf{R}$	Ring of real numbers
S_n	Permutation group
sgn σ	Sign of a permutation
$\mathbf{Z}$	Field of integers
ζ_n	Primitive root of unity
V_4	Klein four-group

Introduction

1

Ask Jacobi or Gauss publicly for their opinion, not regarding the correctness, but regarding the meaning of these sentences. Later on, I hope, some people will find it worth the effort to decipher this whole scribble.
Letter from É. Galois to A. Chevalier ([1, S. 25–32])

Among the most important tasks in the applications of mathematics, but also in mathematics itself, is certainly the solving of equations. The simplest equations are perhaps the linear ones, i.e., equations of the form

$$aX + b = 0,$$

or several such equations combined into linear systems of equations, for example

$$\begin{aligned} a_{11}X_1 + \cdots + a_{1m}X_m + b_1 &= 0, \\ \vdots \qquad\qquad\qquad\quad &\vdots \\ a_{n1}X_1 + \cdots + a_{nm}X_m + b_n &= 0, \end{aligned}$$

where the a_{ij} and b_i are, for example, rational numbers. A satisfactory solution theory for such linear systems of equations is provided by linear algebra.

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A whole lot more complicated are polynomial equations, i.e., equations of the form

$$a_n X^n + a_{n-1} X^{n-1} + \cdots + a_1 X + a_0 = 0. \tag{1.1}$$

The solution theory of such equations is the subject of *algebra* and thus a central topic of this book.

The next more complicated step would then lead to polynomial systems of equations in several variables. These are dealt with in algebraic geometry, which builds on algebra and whose beginnings we will go into in a later volume.

If we can divide by a_n, Eq. (1.1) can be transformed into the equivalent equation

$$X^n + b_{n-1} X^{n-1} + \cdots + b_1 X + b_0 = 0 \tag{1.2}$$

with $b_i = a_n^{-1} a_i$ for $i \in \{0, \ldots, n-1\}$. We then call $n \in \mathbf{N}_0$ the *degree of the polynomial equation*. In the case of $n = 1$ we get the linear equation $X + b_0 = 0$, which has exactly one solution, namely $X = -b_0$.

Things get interesting starting with the *quadratic equations*, i.e., the case $n = 2$. With $p = b_1$ and $q = b_0$, the general quadratic equation is given by

$$X^2 + pX + q = 0. \tag{1.3}$$

The solution theory of such equations is taught in school and was also well known to the Babylonians four thousand years ago [2]: If the *discriminant*

$$\Delta = p^2 - 4q$$

is a square, the solutions of (1.3) are given by

$$x_{1,2} = \frac{1}{2}\left(-p \pm \sqrt{\Delta}\right);$$

otherwise, the equation is unsolvable.

This already highlights a significant difference to the linear equation in one variable: there can also be several, in this case up to two solutions. Looking more closely at the two possible solutions, we recognize an interesting thing: If we define a root $\sqrt{\Delta}$ of Δ simply as a number whose square is Δ, this number is generally only defined up to a sign. And it is exactly this choice of sign that distinguishes the two solutions x_1 and x_2. We can consider the change of this sign, that is, the swapping of x_1 and x_2, as a symmetry operation on the set of solutions of the equation.

It is also worth mentioning that we have to divide by 2 to obtain the solution. We will learn in the volume following this book that this is not possible in every domain, i.e., there are domains over which the solution theory of the quadratic equation must look different.

Next, let's move on to polynomial equations of higher degree. It turns out that for Eq. (1.2) of degree n, at most n different solutions occur, in the case of the equation

$$X^3 + aX^2 + bX + c = 0 \tag{1.4}$$

of the third degree, therefore, up to three solutions. In contrast to the quadratic equation, it took until the sixteenth century for the general solution of the cubic equation to be found, first by Scipione del Ferro[1] and Nicolo Tartaglia.[2] The solution is as follows: We set

$$p := b - \frac{a^2}{3}, \qquad q := \frac{2a^3 - 9ab + 27c}{27}.$$

Then the *depressed equation*

$$Y^3 + pY + q = 0 \tag{1.5}$$

with $Y = X + \frac{a}{3}$ is equivalent to (1.4) (as can be quickly checked by substitution and term comparison), that is, if y is a solution of (1.5), then $x = y - \frac{a}{3}$ is a solution of the original Eq. (1.4) (and vice versa).

Therefore, it is obviously sufficient to solve the depressed cubic equation. Let

$$u = \sqrt[3]{-\frac{q}{2} + \frac{1}{2}\sqrt{-\frac{\Delta}{27}}},$$

where

$$\Delta := -4p^3 - 27q^2$$

is the *discriminant of the depressed cubic equation*. If we set

$$v = -\frac{p}{3u},$$

then

$$y = u + v = u - \frac{p}{3u}$$

is a solution of (1.5). (It follows that $x = u + v - \frac{a}{3}$ is a solution of the original Eq. (1.4)).

What about the other solutions? To this end, we note that (at least in the *ring of complex numbers*) the extraction of the third root is not a unique operation, that is, if a number z is given, there are usually three solutions u for $u^3 = z$. Thus, we also get three possible values for y.

Similar to the solution formula for the quadratic equation, it stands out that the cubic equation is again solved by roots, in this case, the solutions can be expressed by square and cube roots. The ambiguity of the (in this case third) root again provides the various solutions of the equation. If we look closely, we notice that another root appears, namely the square root $\sqrt{\Delta}$ under the third root of u. What happens if we choose the other root here, i.e., change the sign in front of the root? As can be confirmed by a quick calculation, it is

$$v^3 = -\frac{q}{2} - \frac{1}{2}\sqrt{-\frac{\Delta}{27}},$$

so

$$v = \sqrt[3]{-\frac{q}{2} - \frac{1}{2}\sqrt{-\frac{\Delta}{27}}},$$

that is, u and v swap by changing the sign of a square root, something we considered a symmetry operation in the case of the quadratic equation. And here too, symmetries are recognizable. The solution $y = u + v$ is, for example, *invariant* under the swapping of u and v, so it is symmetrical.

The solution formula for the cubic equation was written down by Gerolamo Cardano[3] in 1545 in his work *Ars magna*. He was informed about this formula by Tartaglia, but only under the oath not to spread it further. Cardano no longer felt bound by this oath after he learned that del Ferro had discovered the equations some time before Tartaglia.

Cardano's student Ludovico Ferrari[4] under the guidance of his teacher, found a solution formula for polynomial equations of the fourth degree, which was also published in the *Ars magna* (for this reason, the solution formulas for the equations of the third and fourth degrees are sometimes also called *Cardanic formulas*).

The formula for the *quartic equation*, i.e., the equation of the fourth degree, we do not want to specify here due to their complexity—but we will derive this formula in the main part of the book—,however, it should be said that the solutions can again be expressed by square and cube roots. (Anyone who absolutely expects a fourth root should be reminded that a fourth root is nothing more than a square root of a square root.)

Historically, this naturally raised the question of corresponding solution formulas for polynomial equations of the fifth and higher degrees, that is, the answer to the following problem: Given the coefficients of a polynomial equation, determine its solution using basic arithmetic and forming radicals from the coefficients.

After Cardano had written down the formula for the equations of the third and fourth degrees in 1545, there was no real progress for centuries. The general equation of the fifth degree resisted all attempts at solution. In the years 1770/1771, Joseph-Louis de Lagrange[5] once again examined the calculation tricks that ultimately led to the solutions for the equations of the third and fourth degrees. He recognized a pattern and applied it to

the equation of the fifth degree. But instead of a simplification to an equation of a lower degree, he got an equation of degree 24, which in any case does not simplify the original equation. For equations higher than the fifth degree, the situation was even more hopeless.

Slowly, mathematicians began to believe that there might not be a solution formula for equations of the fifth or higher degrees. In 1799, Paolo Ruffini[6] published a paper in which he claimed that the general equation of the fifth degree could not be solved by roots. However, there was a small gap in the proof he provided, which was not closed until 1824 by Niels Henrik Abel.[7] Since then, it has been clear that the *general* equation of the fifth and higher degrees cannot be solved by roots, thus any search for solution formulas was in vain. This theorem is now named in honor of its discoverers, the Abel–Ruffini theorem.

What the theorem does not exclude, however, is the existence of a radical expression in the coefficients for the solutions of each individual equation. Because for two different equations, these root expressions could be so different that they cannot be combined into a general formula, the non-existence of which was proven by Abel and Ruffini.

But this assumption was also wrong: There are special equations of the fifth (and higher) degree that cannot be solved by roots, such as

$$X^5 - X + 1 = 0, \tag{1.6}$$

that is, for no radical expression of any kind is its fifth power the same minus one. This was recognized around 1830 by Évariste Galois,[8] who examined the symmetries between the solutions of a polynomial equation and proved that equations with too many symmetries for the solution to still be expressible by roots exist.

Galois himself never found recognition for his groundbreaking achievement during his lifetime—compared with his non-existence theorem, the theorem of Abel and Ruffini is much easier. This was partly due to the fact that he tragically died in 1832 at the age of only 20 from injuries sustained in a pistol duel, the presumed cause of which was an unfortunate love affair. On the other hand, he seemed to have been far ahead of his mathematical contemporaries with his ideas; it was not until 1843 that Joseph Liouville[9] recognized the significance of the *Galois theory* and published it in his journal.

We will trace this theory here (and its further developments into the twenty-first century in the next volume), so that by the end of this book we can also answer the question of why equations of the third and fourth degree can be solved by roots, but the Eq. (1.6), for example, cannot.

In addition to the question of solution formulas for polynomial equations, there were other, millennia-old unsolved problems, this time from plane geometry. The ancient Greeks already asked whether it was possible to trisect any angle using only a compass and straightedge. (The fact that an angle can be bisected with a compass and straightedge has been known since antiquity and is regularly taught in schools.)

Another question was about doubling the cube: Can the side length of a cube with double the volume be constructed from the side length of a given cube using only a

compass and straightedge? In other words, the question of whether there is a construction of a line segment of length $\sqrt[3]{2}$ from a line segment of length 1.

A similar question was about squaring the circle: Given the radius of a circle, can the side length of a square with the same area as the circle be constructed using only a compass and straightedge? A positive answer to this question would provide a construction method for a line of length π from a line with the length 1.

We will see that all three questions are to be answered with no. In the history of mathematics, however, it took a very long time after the Greeks until these answers were found: In 1837, Pierre Laurent Wantzel[10] also building on the ideas of Galois, showed that neither the trisection of a general angle nor the doubling of a cube is possible with compass and straightedge alone. The problem of squaring the circle was more stubborn and required methods of analysis in addition to algebra. Only in 1882 was the impossibility of squaring the circle shown by Carl Louis Ferdinand von Lindemann.[11]

Closely related to the trisection of the angle is also the question of the constructability of regular polygons. With compass and straightedge, the Greeks were able to construct the regular triangle, square, pentagon, and pentadecagon. Due to the possibility of angle halving, this immediately results in construction instructions for the n-gon with $n = 2^m k$, where $k = 3, 5, 15$.

What remained open, for example, was the question of the constructability of the regular heptagon or nonagon. (Since for the nonagon the angle of the equilateral triangle must be trisected, non-constructability of the regular nonagon would generally imply the impossibility of trisecting any arbitrary angle.)

There was no progress about this question until 1796: At the age of nineteen, Carl Friedrich Gauss[12] provided a construction for the regular 17-gon. Five years later, he was able to provide constructions with compass and straightedge for the regular n-gon whenever n is a product of a power of two and pairwise different *Fermat*[13] *primes*. A Fermat prime is a prime number of the form $F_n = 2^{2^n} + 1$. (So far, only five Fermat primes are known, namely $F_0 = 3$, $F_1 = 5$, $F_2 = 17$, $F_3 = 257$, $F_4 = 65537$.)

Gauss suspected that no further regular n-gons are constructible with compass and straightedge—which would particularly include the non-constructability of the regular heptagon and nonagon. A proof of this conjecture was also given in the work of 1837 by Pierre Wantzel.

Endnotes

[1] Scipione del Ferro, 1465–1526, Italian mathematician.
[2] Nicolo Tartaglia, 1499/1500–1557, Italian mathematician.
[3] Gerolamo Cardano, 1501–1576, Italian doctor, philosopher, and mathematician.
[4] Ludovico Ferrari, 1522–1565, Italian mathematician.
[5] Joseph-Louis de Lagrange, 1738–1813, Italian mathematician.
[6] Paolo Ruffini, 1765–1822, Italian mathematician, physician, and philosopher.
[7] Niels Henrik Abel, 1802–1829, Norwegian mathematician.

[8]Évariste Galois, 1811–1832, French mathematician (portrait in Fig. 1).
[9]Joseph Liouville, 1809–1882, French mathematician.
[10]Pierre Laurent Wantzel, 1814–1848, French mathematician.
[11]Carl Louis Ferdinand von Lindemann, 1852–1939, German mathematician.
[12]Carl Friedrich Gauss, 1777–1855, German mathematician, astronomer, geodesist, and physicist.
[13]Pierre de Fermat, 1607–1665, French mathematician and jurist.

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Check for updates

The Fundamental Theorem of Algebra

2

Despite their name, the complex numbers **C** *are in some ways simpler than the rational or real numbers, because polynomial equations can be solved without restriction over* **C***.*

Abstract

The polynomial equation $X^2 + 1 = 0$ has no solution in the ring of real numbers, but it does in the ring of the so-called complex numbers. In this chapter, we introduce the complex numbers **C** as the smallest possible extension of the ring of real numbers in which the aforementioned polynomial equation has a solution.

Just as we can identify the entirety of real numbers with the number line, we can identify the entirety of complex numbers with the *number plane*. Complex numbers are thus points in a plane. If we mark the numbers 0 and 1 in the plane, we call a complex number *constructible* if the corresponding point can be constructed with a compass and straightedge from the two marked points. Questions about the feasibility of certain construction problems—such as squaring the circle—can thus be reduced to algebraic statements about complex numbers.

It is an astonishing fact that in **C** not only $X^2 + 1 = 0$ has a solution, but automatically every polynomial equation $X^n + a_{n-1}X^{n-1} + \cdots + a_1X + a_0 = 0$ with rational coefficients $a_0, \ldots, a_n$, $n \geq 1$. This is the statement of the so-called *fundamental theorem of algebra*. At the end of this chapter, we provide a constructive proof for this theorem, which goes back to Martin Kneser. A complex number that is a solution of such a polynomial equation with rational coefficients is called *algebraic*. In this chapter, we show, among other things, that constructible numbers are always algebraic, which will be a key for the proof of the impossibility of squaring the circle.

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2.1 About Polynomial Equations

As described in the introduction, one of the goals of this book is to describe the solution theory of *polynomial equations*, i.e., equations of the form

$$a_n X^n + a_{n-1} X^{n-1} + \cdots + a_1 X + a_0 = 0, \tag{2.1}$$

where the *coefficients* $a_0, \ldots, a_n$ are numbers and we call X the *indeterminate*. A *solution* of the Eq. (2.1) is a number x such that the substitution $X = x$ in (2.1) satisfies the equation, i.e., that

$$a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 = 0. \tag{2.2}$$

The first question that arises is whether (2.1) has a solution at all. Before we can answer this question, we should first clarify what we mean by a *number* in the definition of the polynomial equation.

For (2.2) to make sense, we must at least be able to add and multiply in the domain of numbers, and the relevant laws, namely the associative laws, the commutative laws, and the distributive law should apply. Furthermore, 0 should be included in the domain—after all, it appears on the right side of the equation—and also 1 should be allowed as a number—because with it we can also bring an equation like $X^2 - 1 = 0$ into the form (2.1), namely as $1 \cdot X^2 - 1 = 0$.[1] (In the future, we will usually omit to write the coefficient 1 explicitly, just as we will not write terms with coefficient 0.)

From the domains known from school with these properties, we have the natural numbers $\mathbf{N}_0$, the integers $\mathbf{Z}$, the rational numbers $\mathbf{Q}$ and the real numbers $\mathbf{R}$. The domains are nested, that is, we can consider each natural number as a integer, each integer as a rational number, etc. When setting up polynomial equations, we can thus allow the sought solutions to be drawn from a larger domains than the coefficients.

Let's start with the consideration of the coefficient range: An equation of the form

$$X - a = 0, \tag{2.3}$$

we can bring into the form (2.1) by writing $X + (-a) = 0$. Obviously, the negation $-a$ of a must exist in the coefficient range, i.e., a number $-a$ with the property $a + (-a) = 0$. Since we also want to consider equations of the form (2.3) as polynomial equations, we need to exclude the range of natural numbers from the allowed coefficient ranges. The remaining domains $\mathbf{Z}, \mathbf{Q}, \mathbf{R}$ possess negations. (Before mathematicians learned to calculate with negative numbers, for example, the quadratic equations $X^2 + pX + q = 0$ and $X^2 - pX + q = 0$ were equations of a different type, which led to a whole series of then necessary case distinctions.) Such domains of numbers are called *rings*.

But also the integers as a coefficient ring have serious limitations: We want to call a polynomial equation of the form

$$X^n + b_{n-1}X^{n-1} + \cdots + b_1 X + b_0 = 0$$

a *normalized polynomial equation of degree n*. For the degrees $n = 1, 2, 3, 4, 5$ we also the adjectives *linear*, *quadratic*, *cubic*, *quartic* or *quintic* or the nouns *linear equation*, *quadratic equation*, *cubic equation*, *quartic equation* or *quintic equation*.

If we have $a_n \neq 0$ in (2.1), we can transform the Eq. (2.1) into the equivalent normalized equation

$$X^n + \frac{a_{n-1}}{a_n}X^{n-1} + \cdots + \frac{a_1}{a_0}X + \frac{a_0}{a_n} = 0 \tag{2.4}$$

of degree n over the rational (or real) numbers, which is not possible over the integers (unless $a_n = \pm 1$). Here, *equivalent* means that x is a solution of (2.1) if and only if it is a solution of (2.4). Over the rational numbers, we can even transform every non-trivial equation of the form (2.1) into a normalized polynomial equation: If $a_n = a_{n-1} = \cdots = a_{m+1} = 0$ and $a_m \neq 0$ in (2.1), we can obviously omit the leading terms, so that the equation is equivalent to $a_m X^m + a_{m-1}X^{m-1} + \cdots + a_1 X + a_0 = 0$, which due to $a_m \neq 0$ is again equivalent to a normalized equation.

Over the rational numbers, every non-trivial polynomial equation is equivalent to a normalized polynomial equation.

Therefore, we agree that when studying polynomial equations we choose (at least) the rational numbers as the ring of coefficients. We can thus limit ourselves to normalized polynomial equations of n-th degree.

Next, let's consider a suitable ring for the solutions. The larger this ring of numbers, the more likely we can expect that the Eq. (2.4) has a solution. Is there a ring of numbers in which the normalized polynomial Eq. (2.4) for $n \geq 1$ always has a solution? (The case $n = 0$ is the equation $1 = 0$, which we obviously have to exclude.)

For solutions to exist at all, the ring of numbers for the solutions must certainly be at least as large as the ring of coefficients, because the equation $X - a = 0$ has no solution in the natural numbers or integers if a is a non-integer rational number. But even in the rational numbers, polynomial equations generally have no solution, as already recognized by Hippasus of Metapontum.[2] A polynomial equation (with rational coefficients) that does not have a solution in the rational numbers, for example, is

$$X^2 - 2 = 0. \tag{2.5}$$

According to the Pythagorean theorem,[3] the solution is given by the length of the diagonal of a square with side length 1.

We want to briefly recall why no rational number x can be a solution of (2.5). Suppose $x^2 = 2$, where $x = \frac{p}{q}$ and p, q are two integers with $q \neq 0$. Then $p^2 = 2q^2$ would follow. Since in the prime factor decomposition of a square of a integer each prime factor appears in even number, the exponent of 2 of the prime factorization of the left side is an even number, but on the right side it is an odd number. Contradiction! Since we can construct a square over a line segment and its diagonal just with compass and straightedge as known from school, we have particularly shown that we can construct from a line segment of length 1 a line segment with a length that is not a rational number.

For the Pythagoreans of antiquity, this was an astonishing discovery, namely that lines segments can be constructed from a line segment that are *incommensurable* with it, that is, they are not in a rational proportion with it.

As is well known, (2.5) can be solved if we allow solutions from the larger ring of real numbers: One solution is

$$\sqrt{2} = 1.4142135623\ldots,$$

the other $-\sqrt{2}$. Often in school, the transition from the rational numbers to the real numbers is motivated by the search for a ring of numbers in which even equations like (2.5) have a solution. However, this can be misleading, because in fact, as we will see, in addition to the solutions of such polynomial equations, many more numbers are added when transitioning from the rational numbers to the real numbers.

On the other hand, there are still polynomial equations (as we agreed with rational coefficients) that also have no solution in the real numbers. The simplest example is the equation

$$X^2 + 1 = 0. \tag{2.6}$$

A solution would be a number whose square is -1. But the square of any real number is known to be nonnegative. However, we can extend the solution space once more to obtain a solution to this equation, namely to the complex numbers, which will be discussed in Sect. 2.2.

A polynomial equation generally has no solution in $\mathbf{Q}$, nor in $\mathbf{R}$.

2.2 The Complex Numbers

Perhaps the complex numbers are already known from school (if the reader is lucky) or from some beginner's lectures; nevertheless, we want to introduce them again at this point and especially motivate their definition, because we will encounter similar constructions several times later.

First, let's imagine that there is a ring of numbers in which (2.6) has a solution, which contains the real numbers and in which addition and multiplication are defined, which extends addition and multiplication of real numbers and for which the relevant laws of arithmetic (associativity laws, commutativity laws, distributive law and laws about zero and one) also apply. Let C be such a minimal ring of numbers.

Let's denote the solution of the Eq. (2.6) in this larger ring of numbers with i, a number that we call *imaginary unit*. For this number, obviously

$$\mathrm{i}^2 = -1. \tag{2.7}$$

Since multiplication has to be defined on our (still hypothetical) ring of numbers C, numbers of the form $b\,\mathrm{i}$ must also be included, where b is any real number. In addition, we have to be able to add in the ring of numbers. Thus, from the existence of numbers of the form $b\,\mathrm{i}$, the existence of numbers of the form $a + b\,\mathrm{i}$ follows, where a and b are both real numbers. In particular, the real numbers themselves are of this form, because $a = a + 0\,\mathrm{i}$.

We claim that due to the assumed minimality, all numbers in C are of this form. To show this, we need to show that neither addition nor multiplication lead out of the ring of numbers of the form $a + b\,\mathrm{i}$: From the relevant calculation rules and (2.7), it follows that

$$(a + b\,\mathrm{i}) + (a' + b'\,\mathrm{i}) = (a + a') + (b + b')\,\mathrm{i}$$

and that

$$(a + b\,\mathrm{i}) \cdot (a' + b'\,\mathrm{i}) = aa' + (ab' + ba')\mathrm{i} + bb'\,\mathrm{i}^2 = (aa' - bb') + (ab' + ba')\,\mathrm{i}$$

for real numbers a, a', b, b'. These equations also tell us how to calculate in C.

This analysis of this hypothetical ring of numbers C also shows us how to construct such a ring of numbers: We define a *complex number* as an expression of the form $a + b\,\mathrm{i}$, where a and b are real numbers, and for which the relevant calculation rules and also (2.7) apply. We denote the ring of numbers of complex numbers with $\mathbf{C}$.

The ring of complex numbers $\mathbf{C}$ is a minimal extension of the ring of real numbers $\mathbf{R}$ in which $X^2 + 1 = 0$ has a solution.

When are two complex numbers equal? When is $a_1 + b_1\,\mathrm{i} = a_2 + b_2\,\mathrm{i}$? Obviously, this is exactly the case when $(a_1 - a_2) + (b_1 - b_2)\,\mathrm{i} = 0$. Therefore, it is sufficient to investigate when $a + b\,\mathrm{i} = 0$ for a complex number. If we multiply this equation by $a - b\,\mathrm{i}$, we get

$$0 = (a + b\,\mathrm{i})\,(a - b\,\mathrm{i}) = a^2 - b^2\,\mathrm{i}^2 = a^2 + b^2,$$

that is, the square sum of the two real numbers a and b vanishes. Since squares of real numbers are always nonnegative, a^2 and b^2 must vanish, and therefore a and b. Consequently, we have

$$a + b\,\mathrm{i} = 0 \iff (a = 0 \wedge b = 0)$$

and thus

$$a_1 + b_1\,\mathrm{i} = a_2 + b_2\,\mathrm{i} \iff (a_1 = a_2 \wedge b_1 = b_2).$$

It follows that we can identify a complex number $z = a + b\,\mathrm{i}$ with the pair (a, b) of real numbers. We call a the *real part* of z and b the *imaginary part* of z. Two complex numbers are therefore the same if and only if their real and imaginary parts match.

If z is a complex number with real part a and imaginary part b, we call the complex number

$$\overline{z} = a - b\,\mathrm{i}$$

the *conjugate* of z. Note that $\overline{\mathrm{i}} = -\mathrm{i}$ is a (in fact the) other solution of (2.6), so complex conjugation swaps the two solutions of this equation. From the rules of calculation for complex numbers, it follows that

$$\overline{z_1 + z_2} = \overline{z}_1 + \overline{z}_2,$$

and

$$\overline{z_1 z_2} = \overline{z}_1 \overline{z}_2$$

hold for complex numbers z_1 and z_2 and that $\overline{z} = z$ is true if and only if z is a real number.

With

$$|z| \coloneqq \sqrt{z\overline{z}} = \sqrt{(a + b\,\mathrm{i})(a - b\,\mathrm{i})} = \sqrt{a^2 + b^2}$$

we denote the *magnitude* of z, which is a real number. The complex magnitude behaves multiplicatively, that is

$$|1| = 1$$

and

$$|z \cdot z'| = |z| \cdot |z'|$$

for two complex numbers z and z', as can be immediately derived from the definitions.

With our above considerations about the vanishing of a complex number, it also follows that

$$|z| = 0 \iff z = 0.$$

Let z_1, z_2 be two complex numbers. We write $z_1 \neq z_2$, if $|z_1 - z_2| > 0$. With this we have[4]

$$\neg(z_1 \neq z_2) \iff z_1 = z_2.$$

For two real numbers a and b, the root $\sqrt{a^2 + b^2}$ is close to zero if and only if both $|a|$ and $|b|$ are close to zero.[5]

Applied to a complex number $z = a + b\,\mathrm{i}$ with real part a and imaginary part b, this means because of $|z| = \sqrt{a^2 + b^2}$ that the distance from z to 0 is close to zero (we then simply say, z is close to zero), when both its real part a and its imaginary part b are close to zero.

We use this for the following considerations: Let for each natural number n a construction of a complex number z_n be given, and let these constructions fulfill the property that $|z_n - z_{n'}|$ becomes arbitrarily small for sufficiently large n and n'. (Here, choices may be involved in the construction of z_n, that is, for given n the construction does not have to produce the same z_n every time. However, $|z_n - z_{n'}|$ must become arbitrarily small regardless of the choices.) If we write $z_n = a_n + b_n\,\mathrm{i}$ with real a_n and b_n, it follows that $|a_n - a_{n'}|$ and $|b_n - b_{n'}|$ become arbitrarily small for sufficiently large n and n'. Due to the completeness of the real numbers, there exist exactly two real numbers a and b, such that $|a_n - a|$ and $|b_n - b|$ become arbitrarily small for sufficiently large n. If we set $z = a + b\,\mathrm{i}$, it follows that $|z - z_n|$ becomes arbitrarily small for sufficiently large n. Conversely, for a complex number z with real part a and imaginary part b and the property that $|z_n - z|$ for large n becomes arbitrarily small, we have that $|a_n - a|$ and $|b_n - b|$ become arbitrarily small. But this uniquely determines a and b. So z depends only on the construction prescription for the z_n.

(In particular, if (z_n) is a *Cauchy sequence*[6] of complex numbers,[7] then (z_n) converges to exactly one complex number, that is, there exists exactly one complex number z so that $|z - z_n|$ becomes arbitrarily small for sufficiently large n.)

What about the invertibility of complex numbers z, that is, when is there a (then unique) *inverse* z^{-1} with $zz^{-1} = 1$? Let's first consider the case that there is an inverse z^{-1} of z. Then $\overline{z}$ is also invertible, and we have

$$\overline{z}^{-1} = \overline{z^{-1}}.$$

(We can prove this by multiplying the right side by $\overline{z}$ and using the equalities $\overline{zz'} = \overline{z}\overline{z'}$ and $\overline{1} = 1$.) From this follows

$$(z^{-1}\overline{z^{-1}}) \cdot (z\overline{z}) = (z^{-1}z) \cdot (\overline{z^{-1}}\overline{z}) = 1 \cdot (\overline{z}^{-1}\overline{z}) = 1.$$

Taking the square root, we get

$$|z|^{-1} = |z^{-1}|.$$

In particular, we see that the magnitude of an invertible complex number is itself invertible.

Conversely, let the magnitude of a complex number z be invertible. We make the ansatz

$$z^{-1} = \frac{\overline{z}}{z\overline{z}} = \frac{\overline{z}}{|z|^2}. \tag{2.8}$$

The right side is well-defined due to the invertibility of the magnitude. And indeed, the right side multiplied with z gives the complex number 1, that is, the complex number z is invertible, and its inverse is given by (2.8). With this, we have shown that a complex number is exactly invertible when its magnitude is invertible.

Since we can consider a complex number z with real part a and imaginary part b as a pair (a, b), it makes sense as in Fig. 2.1 to identify complex numbers with points in a Cartesian plane, one axis consisting of the real numbers and the other axis consisting of the *imaginary numbers*, that is, of the numbers of the form $(0, b)$. We call this plane the *Gaussian (or Argand[8]) number plane*. Its origin is the complex number 0.

The complex numbers form the points of a number plane.

In this picture, we can geometrically interpret the operations on the complex numbers: The complex conjugation is the reflection along the real axis, the magnitude (according to the theorem of Pythagoras) is the distance to the origin.

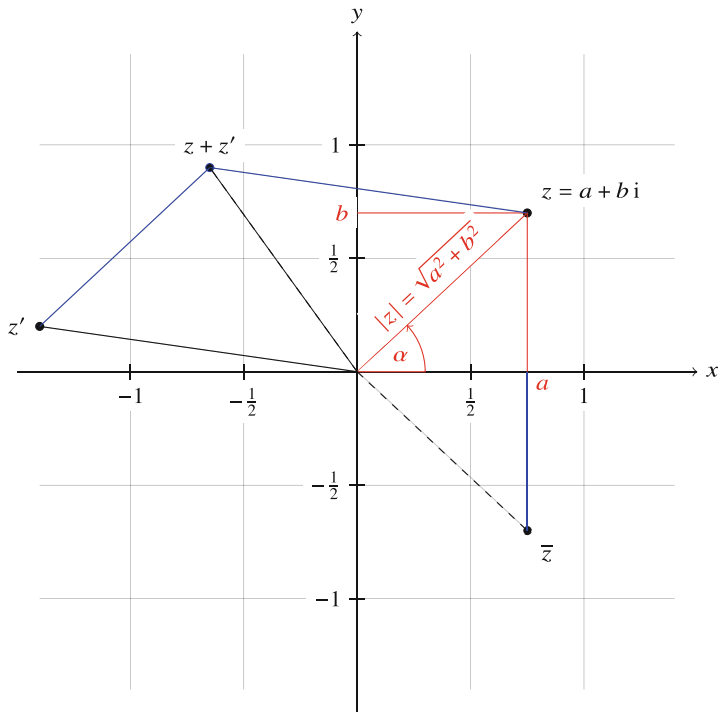


Fig. 2.1 The Gaussian number plane

If z_1 and z_2 are two complex numbers considered as points in the Gaussian number plane, then their complex sum $z_1 + z_2$ is the point that lies on the parallelogram spanned by z_1, z_2 and 0 opposite the corner 0. Since a diagonal of a parallelogram is at most as long as the sum of the two side lengths, the triangle inequality follows

$$|z_1 + z_2| \leq |z_1| + |z_2|$$

for the complex magnitude.

The complex conjugation is geometrically the reflection along the real axis. The complex magnitude is the distance to the origin. The addition with a complex number is geometrically a parallel shift.

The interpretation of multiplication of two complex numbers $z_1 = (a_1, b_1)$ and $z_2 = (a_2, b_2)$ is a bit more involved. In the case of $z_1 = 0$ or $z_2 = 0$, we have $z_1 z_2 = 0$. Let's consider the case $z_1 \neq 0$ and $z_2 \neq 0$. Then the real axis together with the connecting line

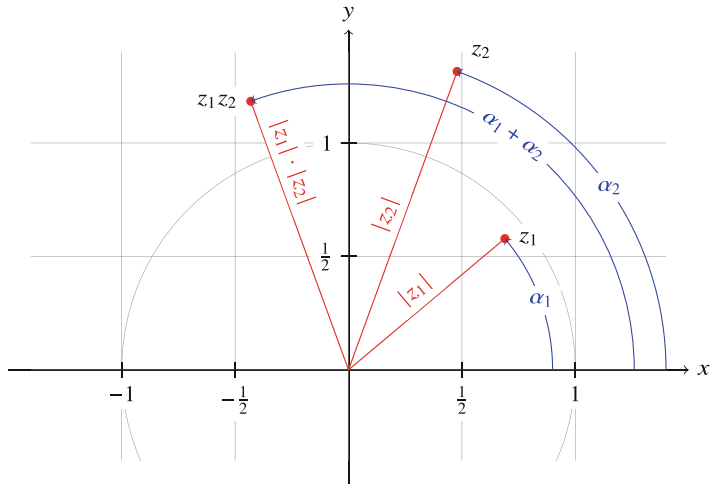


Fig. 2.2 Multiplication of complex numbers

between the origin of the Gaussian number plane and z_i forms an angle α_i as in Fig. 2.2. By definition of cosine and sine, we have

$$b_i \cos \alpha_i = a_i \sin \alpha_i$$

for $i \in \{1, 2\}$.

We call α_1 an[9] *argument* of z_1 and α_2 an argument of z_2. According to the addition theorems for the sine and cosine, it follows that

$$\begin{aligned}(a_1 b_2 + b_1 a_2)\cos(\alpha_1 + \alpha_2) &= (a_1 b_2 + b_1 a_2)(\cos\alpha_1 \cos\alpha_2 - \sin\alpha_1 \sin\alpha_2) \\ &= (a_1 a_2 - b_1 b_2)(\cos\alpha_1 \sin\alpha_2 + \sin\alpha_1 \cos\alpha_2) \\ &= (a_1 a_2 - b_1 b_2)\sin(\alpha_1 + \alpha_2).\end{aligned}$$

Thus, the complex number $z_1 z_2 = (a_1 a_2 - b_1 b_2, a_1 b_2 + b_1 a_2)$ lies on the half-line with angle $\alpha_1 + \alpha_2$ to the real axis. To fully construct it, it remains to determine its distance to the origin. This is given by the product of the distances from z_1 and z_2 to 0 due to the multiplicativity of the modulus.

Multiplication with a complex number is geometrically a rotation around and dilation from the origin.

2.3 Algebraic Numbers

The fundamental property of the complex number i is that it satisfies the equation $X^2 + 1 = 0$. What about the other complex numbers? Are they also solutions to a polynomial equation with rational coefficients? Let's consider just those complex numbers which are solutions to such a polynomial equation, and define:

Definition 2.1 A complex number z is called *algebraic*, if there is a normalized polynomial equation

$$X^n + a_{n-1}X^{n-1} + \cdots + a_1 X + a_0 = 0$$

with rational coefficients that has z as a solution.

To express that z is an algebraic number, we also write

$$z \in \overline{\mathbf{Q}}.$$

We will be able to explain this notation further below in Sect. 2.6.

The imaginary unit i is an example of an algebraic number. The real number $\sqrt{2}$ is another example of an algebraic number, after all it satisfies the equation $X^2 - 2 = 0$, which obviously has rational coefficients. Furthermore, every rational number a is trivially an algebraic number; indeed, it the solution of the equation $X - a = 0$. This and the fact that every algebraic number is by definition a complex number, can be expressed by

$$\mathbf{Q} \subseteq \overline{\mathbf{Q}} \subseteq \mathbf{C}.$$

Since we can multiply every polynomial equation with rational coefficients by the main denominator of all coefficients to obtain an equivalent polynomial equation with integer coefficients, a complex number z is obviously algebraic if and only if there is an equality of the form

$$b_n z^n + b_{n-1} z^{n-1} + \ldots + b_1 z + b_0 = 0$$

with integers b_i where not all b_i are simultaneously zero.

This equality should remind the reader of the concept of linear dependence from linear algebra: A complex number z is algebraic if and only if there exists a natural number n such that the tuple $(1, z, z^2, \ldots, z^n)$ of complex numbers is linearly dependent over the integers (or, equivalently, over the rational numbers).

This should sufficiently motivate the following term:

Definition 2.2 A complex number z is called *transcendent* if an equation of the form

$$b_n z^n + b_{n-1} z^{n-1} + \ldots + b_1 z + b_0 = 0$$

with integers b_i implies that $b_0 = b_1 = \cdots = b_n = 0$.

A complex number z is therefore transcendent if and only if for all natural numbers n the tuple $(1, z, z^2, \ldots, z^n)$ of complex numbers is linearly independent over the integers (or, equivalently, over the rational numbers). A transcendent number is therefore the same as a non-algebraic complex number.

So far, we have seen examples of algebraic numbers, but not a single example of a transcendent number. However, once we have an example of a transcendent number, we have also shown that not every complex number is algebraic, which means that there are complex numbers which are not solutions of a normalized polynomial equation with rational coefficients. An example of a transcendent number will be given in Sect. 3.4, where we will show that π is a transcendent number.[10]

Let's return to the algebraic numbers. The following proposition allows us to produce new algebraic numbers from known ones:

Proposition 2.1 *If x and y are two algebraic numbers, then $x + y$ and $x \cdot y$ are also algebraic numbers.*

For example, from the algebraicity of the rational numbers and the imaginary unit. the algebraicity of all complex numbers of the form $a + b\,\mathrm{i}$, where a and b are any rational numbers, follows. We prove this using a lemma, which will also underlie another proposition later.

Lemma 2.1 *Let $c_1, c_2, \ldots, c_n$ be complex numbers, where we assume $c_1 = 1$. Assume further that the complex number x has the property that the numbers $xc_1, \ldots, xc_n$ are rational linear combinations of $c_1, \ldots, c_n$; that is, there exists a matrix*

$$A = \begin{pmatrix} a_{11} & \ldots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \ldots & a_{nn} \end{pmatrix}$$

of rational numbers a_{ij}, such that

$$\begin{aligned} xc_1 &= a_{11}c_1 + \ldots + a_{1n}c_n, \\ &\vdots \\ xc_n &= a_{n1}c_1 + \ldots + a_{nn}c_n. \end{aligned} \tag{2.9}$$

Then x is an algebraic number.

Proof of Lemma 2.1 Let $B = x \cdot I - A$, where I is the identity matrix with n rows, that is

$$B = \begin{pmatrix} x - a_{11} & -a_{12} & \dots & \dots & -a_{1n} \\ -a_{21} & x - a_{22} & \ddots & & \vdots \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ \vdots & & \ddots & x - a_{n-1,n-1} & -a_{n-1,n} \\ -a_{n1} & \dots & \dots & -a_{n,n-1} & x - a_{nn} \end{pmatrix}.$$

Then we can rewrite (2.9) in the form

$$\begin{aligned} 0 &= b_{11}c_1 + \ldots + b_{1n}c_n, \\ &\vdots \\ 0 &= b_{n1}c_1 + \ldots + b_{nn}c_n \end{aligned}$$

Since $c_1 = 1$, we have

$$\begin{aligned} b_{11} &= -b_{12}c_2 - \ldots - b_{1n}c_n, \\ &\vdots \\ b_{n1} &= -b_{n2}c_2 - \ldots - b_{nn}c_n. \end{aligned}$$

In other words, the first column of the matrix B is a linear combination of the remaining columns. Consequently, $\det B = 0$. According to the Leibniz formula for the determinant, we have

$$0 = \sum_{\sigma \in \mathrm{S}_n} \operatorname{sgn}\sigma \cdot \prod_{i=1}^{n} b_{i\sigma(i)} = b_{11}b_{22}b_{33}\cdots b_{nn} - b_{12}b_{21}b_{33}\cdots b_{nn} + \cdots, \tag{2.10}$$

where the sum runs over all n-letter permutations and $\operatorname{sgn}\sigma \in \{1, -1\}$ is the so-called signum of the permutation. Looking again at the definition of the matrix entries of B, we recognize that Eq. (2.10) is a (normalized) polynomial equation for x which only has rational numbers as coefficients. Thus, x is the solution of a polynomial equation with rational coefficients, hence an algebraic number. □

Proof of Proposition 2.1 Since x is an algebraic number, there exist a natural number n and rational numbers $a_0, \ldots, a_{n-1}$ such that

$$x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 = 0,$$

thus

$$x^n = -a_{n-1}x^{n-1} - \cdots - a_1 x - a_0. \tag{2.11}$$

Since y is also an algebraic number by assumption, there exist similarly a natural number m and rational numbers $b_0, \ldots, b_{m-1}$ such that

$$y^m = -b_{m-1}y^{m-1} - \cdots - b_1 y - b_0. \tag{2.12}$$

Let the complex numbers $c_1, \ldots, c_{nm}$ be the $n \cdot m$ products $x^i y^j$ with $i \in \{0, \ldots, n-1\}$ and $j \in \{0, \ldots, m-1\}$; in particular, $c_1 = 1$.

Let $z = x + y$. Each product $zx^i y^j$ can be written, using (2.11) and (2.12), as a linear combination of $(c_1, \ldots, c_{nm})$ with rational coefficients. According to Lemma 2.1, $z = x + y$ is therefore an algebraic number.

We can use the same reasoning for $z = x \cdot y$ and conclude that the product of two algebraic numbers is also an algebraic number. □

The sum and the product of two algebraic numbers are again algebraic.

Thus, forming sums and products (and therefore also forming differences, which can be reduced to multiplying by -1 and adding) does not lead out of the domain of algebraic numbers. What about taking roots? That is, the following question arises: Assume $x^n = a$, the number x is therefore an n-th root of a number a. Does the algebraicity of a also imply the algebraicity of x? The question obviously allows the following generalization: Let x be the solution of a normalized polynomial equation with algebraic coefficients. Is then x also algebraic? The following proposition gives the positive answer:

Proposition 2.2 *Let a complex number z be the solution of a normalized polynomial equation*

$$X^n + a_{n-1}X^{n-1} + \cdots + a_1 X + a_0 = 0 \tag{2.13}$$

with algebraic coefficients. Then z is an algebraic number.

In particular, roots of algebraic numbers are again algebraic.

What does this proposition mean for our original question about polynomial equations? We had initially agreed that the coefficients of such an equation should be rational numbers and that we should look for the solutions at least in the complex numbers. By definition, all complex solutions are already algebraic, that is, we can look for solutions of

polynomial equations with rational coefficients just as well in the algebraic numbers. But what if we dare to generalize and consider polynomial equations with algebraic numbers as coefficients? According to Proposition 2.2, every such solution is a solution of some polynomial equation with rational coefficients, which we could, in principle, write down explicitly. So the transition to algebraic numbers as the coefficient domain is a conservative step.

In the following, however, we will see that we can solve many problems about algebraic numbers more easily when not only considering polynomial equations with rational coefficients that an algebraic number fulfills, but more generally equations with algebraic coefficients which the algebraic number satisfies.

Proof Since z satisfies Eq. (2.13), it follows that

$$z^n = -a_{n-1}z^{n-1} - \cdots - a_1 z - a_0. \tag{2.14}$$

Since the a_i are algebraic, i.e., solutions of a normalized polynomial equation with rational coefficients, there exist natural numbers m_i and rational numbers b_{ij} such that

$$a_i^{m_i} = -b_{i,m_i-1}a_i^{m_i-1} - \ldots - b_{i,1}a_i - b_{i,0}. \tag{2.15}$$

Let $c_1, \ldots, c_N$ be all products of the form

$$z^i \cdot a_0^{j_0} \cdots a_{n-1}^{j_{n-1}}$$

with $i \in \{0, \ldots, n-1\}$, $j_0 \in \{0, \ldots, m_0 - 1\}$, $\ldots$, $j_{n-1} \in \{0, \ldots, m_{n-1}\}$. We can sort the products so that $c_1 = 1$. Because of (2.14) and (2.15), the product of z with one of the complex numbers $c_1, \ldots, c_N$ is a linear combination of $(c_1, \ldots, c_N)$ with rational coefficients. According to Lemma 2.1, z is therefore an algebraic number. □

Solutions of polynomial equations with algebraic coefficients are again algebraic.

At this point, we want to introduce another term: We call an algebraic number x an *algebraic integer* (not to be confused with the term of an algebraic number that is an integer, which is the same as an integer) if it satisfies a normalized (!) polynomial equation

$$X^n + a_{n-1}X^{n-1} + \cdots + a_1 X + a_0 = 0$$

whose coefficients $a_0, \ldots, a_n$ are all integers.

For example, the imaginary unit i and the root of $\sqrt{2}$ are not only algebraic, but even algebraic integers. Also, every integer a, being a solution of $X - a$, is an algebraic integer. Let's go through the above-noted statements we have obtained for algebraic numbers again: If we require in the assumption of Lemma 2.1 that the numbers xc_1, ..., xc_n are linear combinations of c_1, ..., c_n with integer coefficients, then a review of the proof of Lemma 2.1 shows that then the polynomial equation constructed there that has x as a root is the eigenvalue equation of a matrix with integer entries and thus a normalized polynomial equation with integer entries. Consequently, under the extra integral assumption in Lemma 2.1 the complex number x is an algebraic integer. Lemma 2.1 goes into the proofs of Propositions 2.1 and 2.2. If we go through these again, we see that the statement of Propositions 2.1 and 2.2 can be analogously strengthened: If x and y are two algebraic integers, then their sum $x + y$ and their product $x \cdot y$ are algebraic integers. If x is a root of a normalized polynomial equation, whose coefficients are all algebraic integers, then x is also an algebraic integer.

What is still missing is an answer to the question to what extent the division or formation of the inverse is possible within the domain of algebraic numbers:

Proposition 2.3 *Let z be an algebraic number. Then either $z = 0$, or there exists a (then unique) algebraic number z^{-1} with $zz^{-1} = 1$.*

In particular, we can therefore determine whether two algebraic numbers x and y are equal.[11] In the first case, $x - y = 0$. In the second case $x - y$ has a multiplicative inverse.

Proof Since z is an algebraic number, there exists a normalized polynomial equation

$$z^n + a_{n-1}z^{n-1} + \cdots + a_1 z + a_0 = 0 \tag{2.16}$$

with rational coefficients $a_0, \ldots, a_{n-1}$. Let $m \in \{0, \ldots, n\}$ so that $a_m \neq 0$ but $a_0 = a_1 = \cdots = a_{m-1} = 0$. (Here we set $a_n := 1$.) With $k := n - m$ and $b_j := a_{i+m}$, Eq. (2.16) becomes

$$(b_k z^k + b_{k-1}z^{k-1} + \ldots + b_1 z + b_0)\, z^m = 0, \tag{2.17}$$

where $b_k = 1$ and $b_0 \neq 0$. Let ε be a positive real number with

$$\varepsilon < \sqrt[j]{\frac{|b_0|}{k|b_j|}} \tag{2.18}$$

for all $j \in \{1, \ldots, n\}$. (Here we set $\frac{1}{0} := \infty$.) Because $\varepsilon > 0$ we have

$$|z| > 0 \quad \vee \quad |z| < \varepsilon.$$

We examine both cases individually.

In the first case, i.e. $|z| > 0$, z is invertible in the complex numbers. Thus, we can multiply Eq. (2.16) by z^{-m} and after subtracting b_0 we get the equation

$$b_k z^k + b_{k-1} z^{k-1} + \cdots + b_1 z = -b_0.$$

Dividing by $-b_0$ (recall that b_0 is a non-vanishing rational number) and subsequent factoring out of z yields

$$z\left(-\frac{b_k}{b_0} z^{k-1} - \frac{b_{k-1}}{b_0} z^{k-2} - \cdots - \frac{b_1}{b_0}\right) = 1.$$

Thus,

$$z^{-1} = -\frac{b_k}{b_0} z^{k-1} - \frac{b_{k-1}}{b_0} z^{k-2} - \cdots - \frac{b_1}{b_0}, \tag{2.19}$$

and since z and the sum and product of algebraic numbers are again algebraic, the algebraicity of z^{-1} follows. Thus, z is in particular invertible in the algebraic numbers.

Now let's consider the second case, $|z| < \varepsilon$. Under this assumption we have

$$\begin{aligned} & |b_k z^k + b_{k-1} z^{k-1} + \cdots + b_1 z + b_0| \\ > & |b_0| - |b_k| \cdot |z|^k - |b_{k-1}| \cdot |z|^{k-1} - \cdots - |b_1| \cdot |z| \\ > & |b_0| - |b_k| \varepsilon^k - |b_{k-1}| \varepsilon^{k-1} - \cdots - |b_1| \varepsilon \\ > & 0 \end{aligned}$$

due to the triangle inequality and (2.18). Thus the bracketed expression on the left side of (2.17) is invertible in the complex numbers. If we multiply Eq. (2.17) with its inverse, we get $z^m = 0$. This implies $|z|^m = 0$, thus $|z| = 0$, which in turn implies $z = 0$. □

An algebraic number is either zero or it is (as an algebraic number) invertible.

2.4 Complex Roots of Unity

The real exponential function is usually defined by its power series, i.e.,

$$\mathrm{e}^x = \sum_{\ell=0}^{\infty} \frac{x^\ell}{\ell!}.$$

For a complex number z, we have

$$\left| \sum_{\ell=n_1}^{n_2} \frac{z^\ell}{\ell!} \right| \leq \sum_{\ell=n_1}^{n_2} \frac{|z|^\ell}{\ell!}$$

due to the triangle inequality of the complex modulus. Since the real exponential series converges, it follows that the partial sums of the series

$$\mathrm{e}^z := \sum_{\ell=0}^{\infty} \frac{z^\ell}{\ell!} \tag{2.20}$$

form a Cauchy sequence. Thus, this series converges in the complex numbers. Through (2.20) we have defined the complex exponential function for all complex numbers. The usual calculation rules for the exponential function, that is,

$$\mathrm{e}^{z_1+z_2} = \mathrm{e}^{z_1}\mathrm{e}^{z_2}$$

for two complex numbers z_1 and z_2 and

$$\mathrm{e}^{nz} = (\mathrm{e}^z)^n \tag{2.21}$$

for a complex number z and a natural number n, can be proven just like the corresponding rules for the exponential function restricted to the real numbers.

We can compute the real and imaginary part of e^z as follows: First, let $z = x + y\,i$ with real x and y. Then the equalities

$$\begin{aligned}
\mathrm{e}^z = \mathrm{e}^{x+y\mathrm{i}} = \mathrm{e}^x\mathrm{e}^{y\mathrm{i}} &= \mathrm{e}^x \sum_{\ell=0}^{\infty} \frac{(y\,\mathrm{i})^n}{n!} \\
&= \mathrm{e}^x \cdot \left(\sum_{n=0}^{\infty} (-1)^n \frac{y^{2n}}{(2n)!} + \mathrm{i} \sum_{n=0}^{\infty} (-1)^n \frac{y^{2n+1}}{(2n+1)!} \right) \\
&= \mathrm{e}^x \cdot (\cos y + \mathrm{i} \sin y)
\end{aligned}$$

hold; that is, the real part of e^z is given by $\mathrm{e}^x \cos y$ and the imaginary part by $\mathrm{e}^x \sin y$. Because $\mathrm{e}^0 = 1$, we get by setting $x = 0$ the particular equality

$$\mathrm{e}^{y\mathrm{i}} = \cos y + \mathrm{i} \sin y,$$

the so-called *Euler's*[12] *formula*.

The value of e^{yi} always lies on the unit circle in the Gaussian plane, and conversely, every point on the unit circle is of the form e^{yi}. Specifically, we have (for $y = \pi$)

$$e^{\pi i} = -1,$$

so $e^{\pi i} + 1 = 0$, a remarkable formula that connects the numbers e, π, i, 0, and 1. The addition theorems for the cosine and sine can be deduced from the equality $e^{y_1 i} \cdot e^{y_2 i} = e^{(y_1+y_2)i}$.

In the geometric interpretation of the multiplication of two complex numbers, we have represented a complex number $z \neq 0$ (i.e., $|z| > 0$) by specifying its argument y, that is, an angle between the real axis and the line connecting the origin and the complex number z, and its distance r to the origin:

$$z = r \cdot (\cos y + i \sin y). \tag{2.22}$$

This representation is called the *polar representation* of the complex number z.

The real number y is defined by z up to multiples of the whole angle 2π; that is, two different possible choices of y differ exactly by an integer multiple of 2π, and conversely, the addition of an integer multiple of 2π to a choice results in another possible choice. Since $r > 0$, there exists exactly one real number x (namely the natural logarithm of r) with $r = e^x$. Thus, we can write (2.22) compactly as

$$z = r\, e^{yi} = e^{x+yi}. \tag{2.23}$$

We can specify a complex number either by its real and imaginary part or by its magnitude and an argument.

In the case that $z \neq 0$ is an algebraic number, we can choose the argument y in (2.23) so that $0 \leq y < 2\pi$, that is, for each algebraic number z not equal to zero there exists exactly one positive real (algebraic) number r and exactly one real number y with $0 \leq y < 2\pi$ (the number y is then called *the* argument of z):

First, we write $z = r\, e^{\alpha i}$, where r is a uniquely determined positive real number (namely $|z|$) and α is a real number determined up to a multiple of 2π. Let ε be a small, positive real number. By possibly adding integer multiples of 2π to α, we can assume that $-\varepsilon < \alpha < 2\pi$. In the case of $\alpha > \frac{\varepsilon}{2}$ we can simply set $y = \alpha$. Otherwise, in any case $-\varepsilon < \alpha < \varepsilon$.

If ε is small enough, we can read off the sign of α from the sign of the imaginary part of z. And this sign is determinable, because the imaginary part is an algebraic number, and for an algebraic number we can determine whether it is equal to zero or not equal to zero;

in the case of a real algebraic number in particular, whether it is zero, positive or negative. If $\alpha \geq 0$, we set $y = \alpha$. If $\alpha < 0$, we set $y = \alpha + 2\pi \geq 0$.

Let's use these insights to continue with our original question about a suitable ring for solutions of polynomial equations with rational coefficients. Our original goal was to find a ring in which as many as possible polynomial equations (with rational or algebraic) coefficients have a solution. To this end, we first extended the real numbers by formally adding a root of -1[13] to the complex numbers, so that in particular the Eq. (2.6) has a solution in the complex numbers.

What about other polynomial equations? Consider the equation

$$X^4 + 1 = 0, \tag{2.24}$$

whose solution is a fourth root of -1, i.e., a square root of i. Do we need to extend our ring of numbers further to be able to solve this equation as well; in other words, do we need to add a solution of $X^2 = \mathrm{i}$, let's call it j, to our ring?

It is an astonishing fact that this is not the case. It was enough to add a square root of -1 to the real numbers so that a fourth root z of -1 exists. To see this, we recall the geometric interpretation of the multiplication of complex numbers:

The quadruple of an argument of z must be an argument of -1. Thus, we can choose as arguments of z for example $\frac{1}{4}\pi$, $\frac{3}{4}\pi$, $\frac{5}{4}\pi$, $\frac{7}{4}\pi$. On the other hand, the fourth power of the distance from z to the origin 0 is given by the distance from 1 to 0, thus by 1. It follows that the distance from z to 0 is also 1. Thus, there are four solutions of (2.24) in the complex numbers by

$$\begin{aligned} x_1 &= \frac{1}{2}\sqrt{2} + \frac{\mathrm{i}}{2}\sqrt{2}, & x_2 &= -\frac{1}{2}\sqrt{2} + \frac{\mathrm{i}}{2}\sqrt{2}, \\ x_3 &= -\frac{1}{2}\sqrt{2} - \frac{\mathrm{i}}{2}\sqrt{2}, & x_4 &= \frac{1}{2}\sqrt{2} - \frac{\mathrm{i}}{2}\sqrt{2} \end{aligned} \tag{2.25}$$

as shown in Fig. 2.3. With the help of the complex exponential function, these solutions can also be easily found and stated: If z is a solution of (2.24), i.e., $z^4 = -1$, then z has the representation

$$z = \mathrm{e}^{\varphi \mathrm{i}}$$

for a real number φ that is determined up to multiples of 2π. If we write out $z^4 = -1$ in this representation, we get

$$\mathrm{e}^{\pi \mathrm{i}} = -1 = z^4 = (\mathrm{e}^{\varphi \mathrm{i}})^4 = \mathrm{e}^{4\varphi \mathrm{i}}.$$

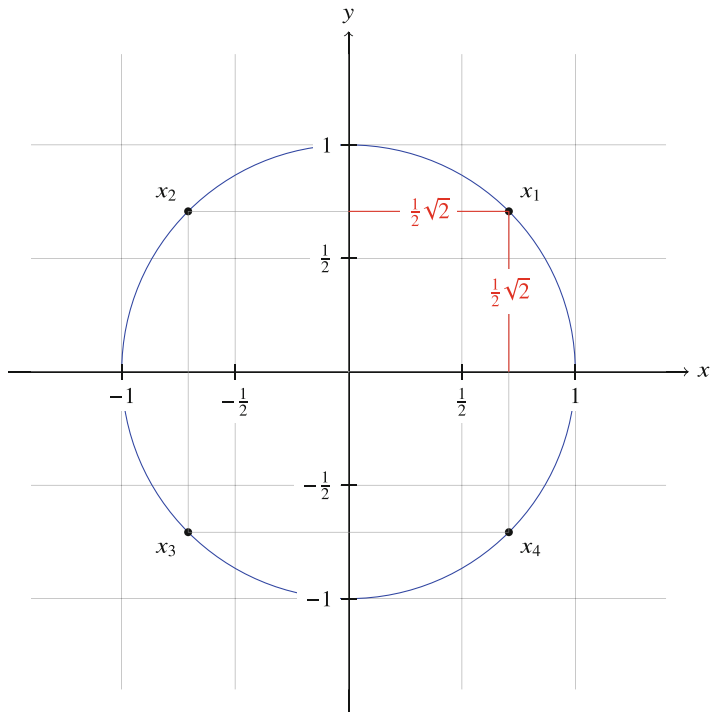


Fig. 2.3 The fourth roots of -1

Since the argument of a complex number not equal to zero is determined up to multiples of 2π, it follows

$$4\varphi = \pi + 2k\pi,$$

where k is an integer, so $\varphi = \frac{2k+1}{4}\pi$. Thus,

$$z = \mathrm{e}^{\varphi \mathrm{i}} = \mathrm{e}^{\frac{2k+1}{4}\pi \mathrm{i}},$$

so one of the following four different expressions

$$x_1 = \mathrm{e}^{\frac{1}{4}\pi \mathrm{i}}, \qquad x_2 = \mathrm{e}^{\frac{3}{4}\pi \mathrm{i}}, \qquad x_3 = \mathrm{e}^{\frac{5}{4}\pi \mathrm{i}}, \qquad x_4 = \mathrm{e}^{\frac{7}{4}\pi \mathrm{i}},$$

which are all solutions of (2.24) and coincide with the solutions (2.25). (There are no further solutions. For example, $\mathrm{e}^{\frac{9}{4}\pi \mathrm{i}} = \mathrm{e}^{\frac{1}{4}\pi \mathrm{i} + 2\pi \mathrm{i}} = \mathrm{e}^{\frac{1}{4}\pi \mathrm{i}}$.)

This procedure for extracting fourth roots from -1 can be generalized: We can extract arbitrary roots from non-vanishing complex numbers:

Proposition 2.4 *Let a be a complex number. Let n be a positive natural number. If $a = 0$, then the polynomial equation*

$$X^n - a = 0 \tag{2.26}$$

has exactly one solution x in the complex numbers, namely $x = 0$. If $a \neq 0$, i.e. $a = r\,\mathrm{e}^{\varphi \mathrm{i}}$ for a positive real number r and a real number φ, then (2.26) has exactly n different solutions in the complex numbers, namely

$$x_k = \sqrt[n]{r}\,\mathrm{e}^{\frac{2k\pi\mathrm{i}+\varphi\mathrm{i}}{n}} \tag{2.27}$$

for $k \in \{0, \ldots, n-1\}$.

Let's set

$$\zeta_n := \mathrm{e}^{\frac{2\pi\mathrm{i}}{n}}, \tag{2.28}$$

so we can rewrite (2.27) according to the rules of computation for the exponential function to

$$x_k = \zeta_n^k \sqrt[n]{r}\,\mathrm{e}^{\frac{\mathrm{i}\varphi}{n}}.$$

Proof If $a = 0$, then $x = 0$ is obviously a solution of (2.26). Conversely, in the case $a = 0$ for any solution x of (2.26), it holds that

$$|x|^n = |x^n| = |0| = 0,$$

in particular $|x| = 0$. But this is only possible for $x = 0$.

This leaves us to consider the case $a = r\,\mathrm{e}^{\varphi \mathrm{i}}$ for a positive real number r and a real number φ. Let the complex number x be a solution of the Eq. (2.26), i.e. $x^n = r\,\mathrm{e}^{\varphi \mathrm{i}}$. In this case, we have

$$|x|^n = |r\,\mathrm{e}^{\varphi \mathrm{i}}| = r,$$

so $|x| = \sqrt[n]{r} > 0$. Thus, the solution is of the form

$$x = \sqrt[n]{r}\,\mathrm{e}^{\alpha \mathrm{i}}.$$

Substituting into the equation $x^n = r\,\mathrm{e}^{\varphi \mathrm{i}}$ we get

$$\mathrm{e}^{n\alpha \mathrm{i}} = \mathrm{e}^{\varphi \mathrm{i}}$$

after dividing by r and using (2.21). As similarly seen in the preliminary considerations, we have $n\alpha = \varphi + 2k\,\pi$, so

$$\alpha = \frac{2k\,\pi + \varphi}{n}$$

for an integer k. It follows that the solution x must be one of the n different values

$$x_k = \sqrt[n]{r}\,\mathrm{e}^{\frac{2k\pi\mathrm{i}+\varphi\mathrm{i}}{n}}$$

with $k \in \{0, \ldots, n-1\}$. Direct substitution into (2.26) shows that all these values x_k are indeed solutions. □

We can extract n-th roots from any complex number. A complex number not equal to zero has n different n-th roots.

If a is an algebraic number, we can even construct a canonical solution of (2.26), which we simply want to denote by

$$\sqrt[n]{a}\,:$$

In the case of $a = 0$, we set $\sqrt[n]{a} = 0$. In the case of $a \neq 0$, we first write $a = r\,\mathrm{e}^{\alpha\mathrm{i}}$, where r is a uniquely determined positive real number and α is a uniquely determined real number with $0 \leq \alpha < 2\pi$, i.e. the argument of a. Then we set

$$\sqrt[n]{a} = \sqrt[n]{r}\,\mathrm{e}^{\frac{\alpha\mathrm{i}}{n}},$$

which means, $\sqrt[n]{a}$ is the n-th root of a that has a minimal argument value.

Remark 2.1 Since sums, products, and quotients of algebraic numbers are again algebraic numbers, the complex numbers

$$1 + \sqrt{2},\ \sqrt[3]{1+\sqrt{2}},\ \sqrt[17]{1-\sqrt[3]{1+\sqrt{2}}}$$

are all algebraic numbers. Mathematicians have long wondered whether all algebraic numbers, i.e., all solutions to polynomial equations (with rational coefficients), are of this form, i.e., can be obtained from rational numbers by iteratively applying basic arithmetic operations and taking roots. At the end of this book, we will be able to answer this question,

and indeed, as much can be revealed now, the answer is negative: There are polynomial equations whose solutions cannot be represented by (iterated) radical expressions.

The complex number ζ_n introduced above in (2.28) is a so-called *n-th root of unity*. This is because ζ_n satisfies the equation

$$X^n - 1 = 0, \tag{2.29}$$

or $X^n = 1$. (The number one is referred to as *unity* in this context.)

According to Proposition 2.4, all solutions of (2.29) are given by

$$\zeta_n^k = \mathrm{e}^{\frac{2k\pi\mathrm{i}}{n}},$$

where again $k \in \{0, \ldots, n-1\}$. The complex number ζ_n^k lies on the unit circle, and the angle between the real axis and the connecting line between the origin and ζ_n^k is $\frac{2k\pi}{n}$, that is, the n-th roots of unity form the corners of a regular n-gon on the unit circle of the Gaussian number plane, with one of its corners given by $1 = \zeta_n^0$, as shown in Fig. 2.4.

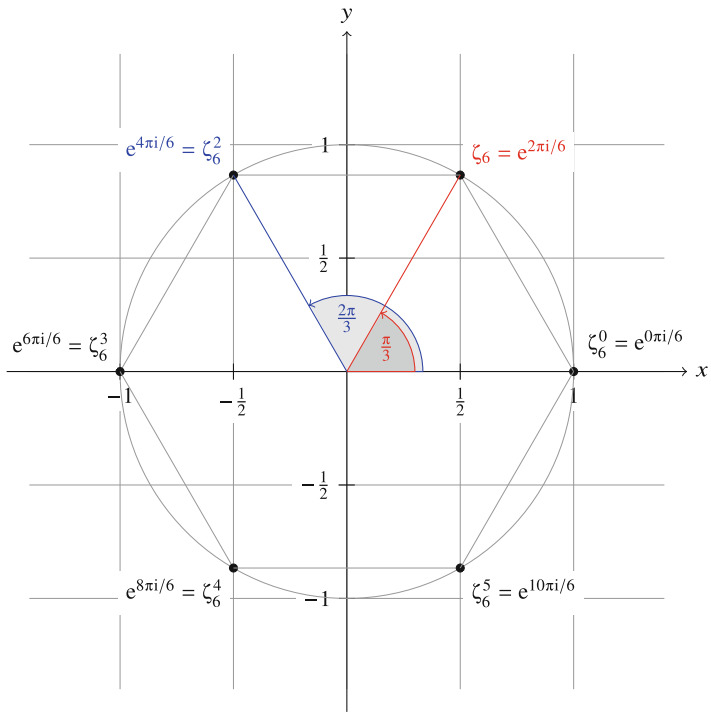


Fig. 2.4 The sixth roots of unity

The solutions of (2.29) thus have a particularly beautiful geometric interpretation. For this geometric reason, (2.29) is also called *n-th cyclotomic equation*.

An n-th root of unity is one of the n different n-th roots of 1.

2.5 Constructions with Compass and Straightedge

We stay for a moment with plane geometry and consider the drawing plane, on which n points $x_1, \ldots, x_n$ are given. What does it mean that a certain point x in the plane can be constructed only with compass and straightedge (without markings) from $x_1, \ldots, x_n$ (we say briefly that x is *constructible* from the given points $x_1, \ldots, x_n$)?

The concept of constructibility can be mathematically formulated as follows:

1. Each of the points $x_1, \ldots, x_n$ is constructible from $x_1, \ldots, x_n$.
2. If x, x', y and y' are points constructible from $x_1, \ldots, x_n$, with $x \neq x'$ and $y \neq y'$, such that the line L_1 through x and x' is not parallel to the line L_2 through y and y', and if z is the intersection of L_1 and L_2, then z can be constructed from $x_1, \ldots, x_n$.
3. Let x, x', y and y' be points that can be constructed from $x_1, \ldots, x_n$ with $x \neq x'$ and $y \neq y'$. If z is an intersection of the line through x and x' and the circle through y' with center y, then z can be constructed from $x_1, \ldots, x_n$.
4. If x, x', y and y' are points that can be constructed from $x_1, \ldots, x_n$ with $x \neq x'$ and $y \neq y'$ such that the circle K_1 through x' with center x is different from the circle K_2 through y' with center y, and if z is an intersection of K_1 and K_2, then z can be constructed from $x_1, \ldots, x_n$.
5. There are no other constructible points.

The definition is obviously inductive. Every constructible point must be constructible in finitely many steps by intersecting lines and circles from the given points $x_1, \ldots, x_n$.

Let's first consider the edge case $n = 0$, that is, no point is given in the plane. Then we lack the induction start, so in this case we cannot construct any other point. In the case $n = 1$, exactly one point x_1 is given. This point is also the only one that can be constructed from x_1, because a single point does not allow, according to the above definition, to draw lines and circles, so that no intersections can be added as further constructible points.

Therefore, we restrict ourselves in the following to the cases in which at least two points x_1 and x_2 with $x_1 \neq x_2$ are given on the drawing plane (particularly $n \geq 2$). In these cases, there is a unique distance- and orientation-preserving identification of the drawing plane with the Gaussian number plane that identifies x_1 with the complex number zero and x_2 with the complex number 1. The line through x_1 and x_2 corresponds under this identification to the real axis, the circle through x_2 with center x_1 to the unit circle of

the Gaussian number plane. After we have made this identification, the points that can be constructed from 0 and 1 and possibly further x_i are complex numbers.

Since we don't need more for later applications, we restrict ourselves in the following to the case $n = 2$, that is, two points, namely 0 and 1 are given. (Essentially, everything that follows remains correct for $n > 2$, that is, if we give more than two points.) A complex number that can be constructed from the numbers 0 and 1 we simply call *constructible*.

The complex number plane is the drawing plane, in which the points 0 and 1 are marked. Each point in the drawing plane corresponds to a complex number and vice versa.

Trivially, 0 and 1 are constructible. Since we can draw the line through 0 and 1 and successively carry forward the distance from 0 to 1 repeatedly in both directions with the compass, all integers are also constructible.

From school mathematics, we remember that perpendiculars through points can be constructed with compass and straightedge. We can construct perpendiculars to the real axis through all integers, these are then parallel to the imaginary axis. On these perpendiculars, the distance between two perpendiculars (i.e., the unit distance) can be successively carried forward in both directions with the compass. It follows that all points of the form $a + b\,\mathrm{i}$, where a and b are integers, can be constructed.

Complex numbers of the form $a + b\,\mathrm{i}$, where a and b are integers, are called *Gaussian integers*.

The integers can be constructed with compass and straightedge from 0 and 1.

Even the entirety of all Gaussian integers $\mathbf{Z}+\mathbf{Z}\mathrm{i}$ can be constructed with compass and straightedge from 0 and 1.

In the following lemmas, we will derive some strong statements about the connection of the arithmetic operations of complex numbers with the constructibility of complex numbers. We start with the addition:

Lemma 2.2 *If z and w are two constructible numbers, then their sum $z + w$ and the negation $-z$ are also constructible.*

Proof We first show the constructibility of the sum. To this end, we first consider as in Fig. 2.5 the case, that one of the two numbers, say z, is different from zero. From school, we know that we can construct the perpendicular from the point w to the line through 0

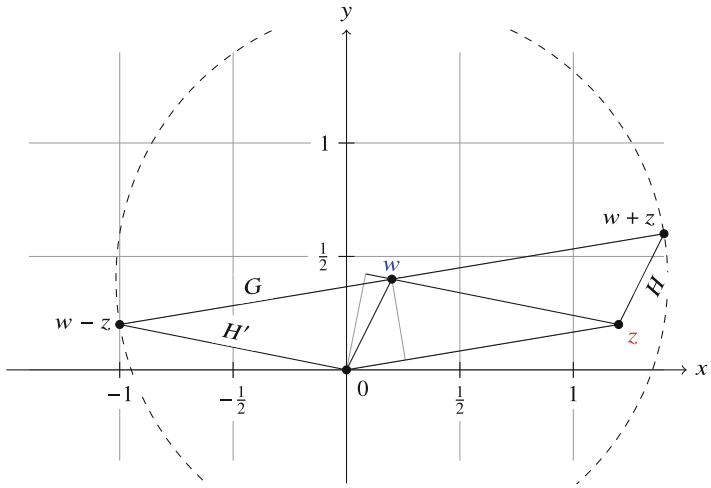


Fig. 2.5 Construction of the sum of two complex numbers

and z with compass and straightedge. Then the construction of a line perpendicular to the perpendicular through w is possible. This line G is parallel to the line through 0 and z. If $w \neq 0$, the construction of a parallel H to the line through 0 and w that passes through the point z is also possible. This line intersects the line G at the point $w + z$. If $w \neq z$ (because $z \neq 0$ at least one of the cases $w \neq 0$ or $w \neq z$ must hold), the construction of a parallel H' to the line through w and z that passes through the point 0 is possible. This line intersects the line G at the point $w - z$. The point to be constructed $w + z$ is then the second intersection point of a circle through $w - z$ with center w with the line G.

Now consider the general case that $z \neq 0$ does not necessarily hold. Then we choose a non-vanishing integer k such that $z + k \neq 0$. According to the already treated special case, we can construct $z + k$ and $w - k$ and also $z + w = z + k + (w - k)$.

It remains to construct the negation of a constructible number z. If $z \neq 0$, we can draw the line through 0 and z, draw a circle through z with center 0 and construct the second intersection point of the circle with the line through 0 and z. This is exactly $-z$.

If $z \neq 0$ is not necessarily the case, there is an integer k with $z + k \neq 0$. We can then construct $-(z + k) = -z - k$, and also $-z = (-z - k) + k$. □

The proof of the first part of Lemma 2.2 thus consists essentially in specifying a procedure to construct a fourth point d from three points a, b and c such that a, b, d and c are the vertices of a parallelogram in this order.

The constructible numbers are closed under complex addition and negation.

Lemma 2.3 *If z and w are two constructible numbers, their product $z \cdot w$ is also constructible.*

Proof We first consider the case where z and w are both different from zero. In this case, we have $z = r\,\mathrm{e}^{\alpha \mathrm{i}}$ and $w = s\,\mathrm{e}^{\beta \mathrm{i}}$ for positive real numbers r and s and for suitable angles α and β. We then have to construct $zw = rs\,\mathrm{e}^{(\alpha+\beta)\mathrm{i}}$.

To do this, it is obviously sufficient to construct the number rs on the real axis and then a ray starting at 0, such that the real axis together with this ray encloses an angle of $\alpha + \beta$.

We start with the construction of rs according to Fig. 2.6. To do this, we draw a circle with center 0 through the point z. This intersects the real axis at the point r, in particular, the real number r is constructible. The same applies to the constructibility of the real number s. On the imaginary axis, we construct the number r i, by drawing a circle through the point r with center 0. We then draw the line through 1 and r i. We have already seen that we can then construct a line parallel to this line through s. Due to the intercept theorem, this line intersects the imaginary axis at the point rs i. The real number rs can then be transferred to the real axis by drawing a circle through rs i with center 0.

Next, we show how to construct the angle $\alpha + \beta$, that is: Given two rays G and H starting at 0, such that the angle between the real axis and the rays is α and β respectively,

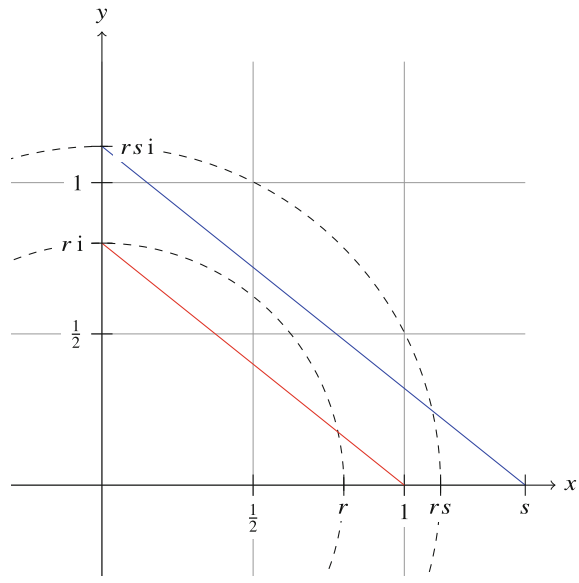


Fig. 2.6 Construction of the product of two real numbers

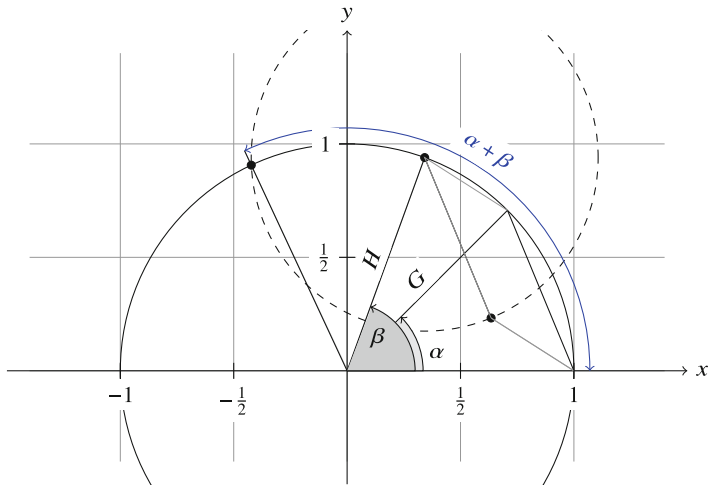


Fig. 2.7 Construction of the sum of two angles

construct a ray such that the angle from the real axis to this ray is exactly $\alpha + \beta$. To do this, as in Fig. 2.7, we first draw the unit circle around the origin, i.e., the circle with radius 1 around 0. Then we move the line segment between 1 and the intersection of the unit circle and G parallel to the intersection of the unit circle and H, so that we can draw a circle around this point with the radius of the line segment, which then intersects the unit circle at two points. The ray starting at 0 through one of the two points is the sought-after one. (The ray starting at 0 through the other point is the one corresponding to the angle $\beta - \alpha$.)

This concludes the case $z \neq 0$ and $w \neq 0$. In the general case, where we cannot assume that $z \neq 0$ and $w \neq 0$, there exist in any case integers k and ℓ such that $z + k \neq 0$ and $w + \ell \neq 0$. The already proven special case provides the constructibility of $(z + k) \cdot (w + \ell) = zw + kw + \ell z + k\ell$. Because of Lemma 2.2, however, also $zw = (z + k)(w + \ell) - kw - \ell z - k\ell$ is constructible. □

The constructible numbers are closed under complex multiplication.

Lemma 2.4 *If z is a constructible number with $z \neq 0$, then z^{-1} is also a constructible number.*

Proof By assumption, $z = r\,\mathrm{e}^{\alpha \mathrm{i}}$ for a positive real number and an angle α. We need to construct $z^{-1} = r^{-1}\,\mathrm{e}^{-\alpha \mathrm{i}}$. We can argue as in the proof of Lemma 2.3, so it suffices to construct the real number r^{-1} and the angle $-\alpha$. In the proof of Lemma 2.3, the construction of $-\alpha = 0 - \alpha$ has already been shown, so it remains to construct the

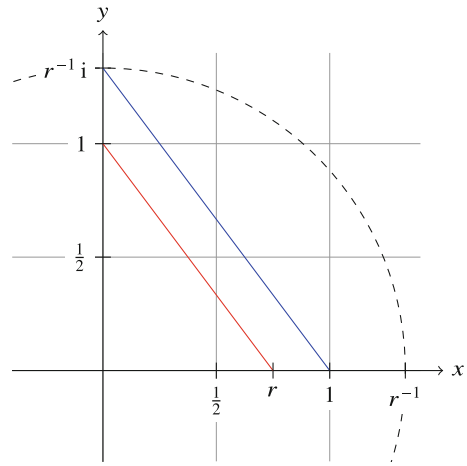


Fig. 2.8 Construction of the inverse of a real number

real number r^{-1} from a real number r. For this, we again use the intercept theorem as in Fig. 2.8: We draw the line through r and i. We shift this parallel to a line through 1. This parallel intersects the imaginary axis at the point r^{-1} i. We can then construct the number r^{-1} by drawing a circle through r^{-1} i and with the center 0 on the real axis. □

Constructible numbers are closed under inversion.

Lemma 2.5 *If $z \neq 0$ is a constructible number and w is a square root of z, that is, a complex number with $w^2 = z$, then w is also constructible.*

In the case of $z = 0$, we can certainly also construct a square root. In fact, zero is the (only) square root.

Proof Since $z \neq 0$, there exist a positive real number r and an angle α with $z = r\,\mathrm{e}^{\alpha\mathrm{i}}$. We have to construct the two possible square roots of z, that is, the numbers $\sqrt{r}\,\mathrm{e}^{\frac{1}{2}\alpha\mathrm{i}}$ and $-\sqrt{r}\,\mathrm{e}^{\frac{1}{2}\alpha\mathrm{i}}$.

We can argue as in the proof of Lemma 2.3, so it suffices to construct the real number $\sqrt{r}$ and the angle $\frac{1}{2}\alpha$. The method of angle bisecting should be known from school: Let G be the ray such that the angle α is enclosed by the real axis and G. Then we construct the midpoint of the line through 1 and the intersection of G with the unit circle. The angle between the real axis and the ray through this midpoint is then $\frac{1}{2}\alpha$.

The construction of the real square root remains and is illustrated in Fig. 2.9: After the previous, the real number $\frac{1}{2}(-r + 1)$, i.e., the midpoint of the line from $-r$ to 1, is constructible. Around this midpoint, we draw a circle that goes through $-r$ and thus also

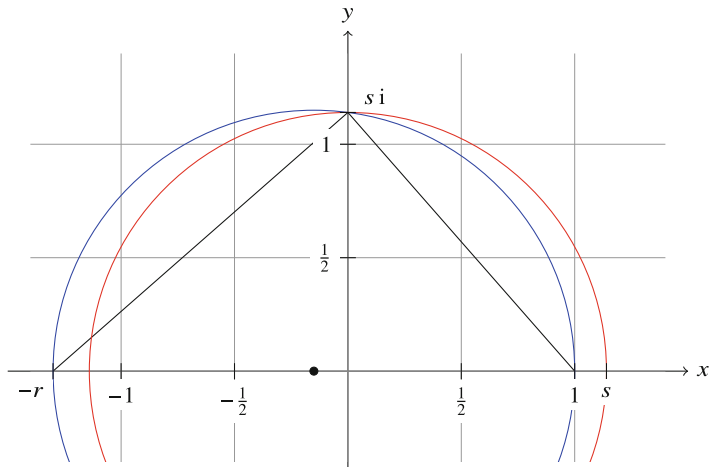


Fig. 2.9 Construction of the real root

through 1. Let an intersection of this circle with the imaginary axis be given by $s\,\mathrm{i}$, where s is a positive real number. From the constructibility of $s\,\mathrm{i}$, the constructibility of the real number s also follows. And s is the height of the triangle with the vertices 1, $s\,\mathrm{i}$ and $-r$. Due to the theorem of Thales,[14] this triangle is right-angled, the square s^2 of its height is determined by the altitude theorem of Euclid[15] as the product of the two hypotenuse sections, i.e., $s^2 = r \cdot 1 = r$. Consequently, $s = \sqrt{r}$, so we have constructed the positive real square root of r. □

Constructible numbers are closed under square root extraction.

Lemma 2.6 *If z is a constructible number, then its conjugate $\overline{z}$ is also a constructible number.*

Proof The number $\overline{z}$ is the reflection of z on the real axis. We construct this reflection as follows: We choose two different constructible points on the real axis (for example two integers), both of which are different from z. Then we can draw two circles through z, the centers of which are each one of the two chosen points on the real axis. Both circles intersect in z by construction. The further intersection point of the two circles is $\overline{z}$. □

Constructible numbers are closed under complex conjugation.

The set of constructible numbers is thus closed under addition, subtraction, multiplication, division by non-zero numbers, square root extraction from non-zero numbers and complex conjugation. In particular, all rational numbers on the real axis can be constructed.

The rational numbers can be constructed with compass and straightedge from 0 and 1.

On the other hand, we will soon show that every constructible number can be obtained by successively applying these arithmetic operations from the given points $x_1, \ldots, x_n$.

To this end, we first agree on the following terminology: Let $t_1, \ldots, t_m$ be complex numbers. A complex number z is then *rational* in $t_1, \ldots, t_m$ if z can be expressed in the rational numbers and the complex numbers $t_1, \ldots, t_m$ only using addition, subtraction, multiplication and division. Thus, a complex number z is rational in $t_1, \ldots, t_m$ if and only if z is of the form

$$z = \frac{\sum\limits_{i_1,\ldots,i_m \geq 0} a_{i_1 \cdots i_m} t_1^{i_1} \cdots t_m^{i_m}}{\sum\limits_{i_1,\ldots,i_m \geq 0} b_{i_1 \cdots i_m} t_1^{i_1} \cdots t_m^{i_m}}, \tag{2.30}$$

where the coefficients $a_{i_1 \cdots i_m}$ and $b_{i_1 \cdots i_m}$ are all rational numbers and where almost all coefficients vanish, so that the sums in the numerator and denominator are effectively finite:

The right side of (2.30) is a rational expression in the t_i, the rational numbers and the t_i are themselves of the form (2.30), and the allowed arithmetic operations addition, subtraction, multiplication and division transform terms in the form of the right side of (2.30) back into such a one.

(By multiplying with a suitable common denominator, we can even achieve that all coefficients $a_{i_1 \cdots i_m}$ and $b_{i_1 \cdots i_m}$ are integers.)

If z is a complex number rational in $t_1, \ldots, t_m$, we write

$$z \in \mathbf{Q}(t_1, \ldots, t_m).$$

Every rational number is in particular also a number rational in $t_1, \ldots, t_m$.

Theorem 2.1 *A complex number z is constructible if there is a sequence of complex numbers $z_1, \ldots, z_{n-1}, z_n = z$ such that for all $i \in \{1, \ldots, n\}$ the complex number z_i is rational in $z_1, \ldots, z_{i-1}$ or a square root of a complex number that is rational in $z_1, \ldots, z_{i-1}$.*

The points constructible from 0 and 1 can be characterized algebraically.

The theorem thus states that the constructible numbers are exactly those that can be obtained from 0 and 1 by successively applying addition, subtraction, multiplication, division, and extracting square roots.

Proof Let $z_1, \ldots, z_{n-1}, z_n = z$ be a sequence of complex numbers ending at z such that z_i is rational in $z_1, \ldots, z_{i-1}$ or a square root of such a number. We have to show that z is constructible. Since sums, differences, products, quotients, and square roots of algebraic numbers are again algebraic, it follows by induction that all z_i are algebraic.

We now show by induction that all z_i are also constructible, that is, that z_i is constructible if $z_1, \ldots, z_{i-1}$ are constructible: Every rational expression x in $z_1, \ldots, z_{i-1}$ is constructible, since sums, differences, products, and quotients of constructible numbers are again constructible. It remains to show that the square roots of x are also constructible. Since x as a rational expression in algebraic numbers is algebraic, we can decide with Proposition 2.3 whether $x = 0$ or $x \neq 0$. In the first case, $x = 0$ itself is the square root, in the second case, the constructibility of the square root follows from Lemma 2.5.

It remains to prove the reverse direction of the theorem's statement: Let z be a constructible number. It is to be shown that z can be obtained from the rational numbers successively through the arithmetic operations of addition, subtraction, multiplication, division, and square rooting. For this, we have to show that the allowed construction methods do not lead out of the set of these numbers.

For the following, we write $x \in K$ if x is a complex number that can be obtained from the rational numbers only through addition, subtraction, multiplication, division, and extracting square roots. First, we note the following: If $x \in K$, then also $\overline{x} \in K$ because the complex conjugation commutes with addition, subtraction, multiplication, and division, that is, the complex conjugate of a rational expression in complex numbers $t_1, \ldots, t_m$ is a rational expression in $\overline{t}_1, \ldots, \overline{t}_m$, and the complex conjugate of a square root of a complex number y is a square root of $\overline{y}$.

Since the real part of a complex number x is given by $\frac{1}{2}(x + \overline{x})$ and the imaginary part by $\frac{1}{2i}(x - \overline{x})$, we get also that the forming of real and imaginary parts does not lead out of K.

We finally go through the allowed construction methods one after the other:

Let $x, x', y, y' \in K$ be points with $x \neq x'$ and $y \neq y'$. Let the line L_1 through x and x' not be parallel to the line L_2 through y and y'. Let w be the intersection of the lines L_1 and L_2. Since there is exactly one intersection point, there exist unique real numbers s and t with

$$w = sx + (1 - s)x' = ty + (1 - t)y'. \tag{2.31}$$

The equality on the right side of (2.31) is an equality between complex numbers, so it corresponds to two real equations, namely one for the real part and one for the imaginary part. The solutions of the resulting linear system of equations in s and t over the real numbers can be expressed, as is well known, according to Cramer's rule[16] as rational expressions in the coefficients, here in the real and imaginary parts of x, x', y and y'. Thus, $s, t \in K$, in particular also $w \in K$.

Let $x, x', y, y' \in K$ be points with $x \neq x'$ and $y \neq y'$. Let w be an intersection of the line through x and x' with the circle through y' and with center y. Then there exists a real number t such that $w = tx + (1-t)x'$ lies on the circle, so

$$|tx + (1-t)x' - y|^2 = |y' - y|^2.$$

This is a quadratic equation in t, whose coefficients are rational expressions in the real and imaginary parts of x, x', y and y'. According to the well-known solution formula for quadratic equations, in which besides rational expressions only a square root occurs, it follows that $t \in K$, in particular also $w \in K$.

It remains to consider the case of an intersection of two circles. For this, let points x, x', $y, y' \in K$ with $x \neq x'$ and $y \neq y'$ be given. Let the circle K_1 through x' and with center x be different from the circle K_2 through y' and with center y. Let w be an intersection of the circles K_1 and K_2. Then the two equations

$$|w - x|^2 = |x' - x|^2, \quad |w - y|^2 = |y' - y|^2 \tag{2.32}$$

hold for w. If we write $w = u + v\,\mathrm{i}$, where u and v are real numbers, we can transform (2.32) into the form

$$(u-a)^2 + (v-b)^2 = r^2, \quad (u-c)^2 + (v-d)^2 = s^2 \tag{2.33}$$

Here, a, b, c, d are the real and imaginary parts of x and y and r and s are the radii of the circles K_1 and K_2, so $a, b, c, d, r, s \in K$. (The circle radii can be written as square roots of rational expressions in the real and imaginary parts of x, x', y and y'.) If we subtract the Eq. (2.33) from each other, we obtain a linear equation of the form

$$(a-c)u + (b-d)v = e \tag{2.34}$$

in u and v, where $e \in K$. Since K_1 and K_2 do not coincide, $a - c \neq 0$ or $b - d \neq 0$ holds.

Without loss of generality, we consider the first case. Then we can solve (2.34) for u and find that u is a rational expression in a, b, c, d, e, and v. If we substitute this expression into one of the Eq. (2.33), we obtain a nontrivial quadratic equation in v, all of whose coefficients are from K. Thus, $v \in K$. Consequently, $u \in K$ and thus $w = u + v\,\mathrm{i} \in K$.

□

From the proof of Theorem 2.1, we particularly note that all constructible numbers are algebraic. However, we will see later that there are algebraic numbers, i.e., roots of polynomial equations with rational coefficients, which are not constructible.

Every constructible number is algebraic. A non-algebraic number is therefore not constructible.

2.6 The Fundamental Theorem of Algebra

After this excursion into the world of constructible complex numbers, we return to our problem: Is our set of possible solutions, given by the complex numbers, large enough so that every (normalized, nontrivial) polynomial equation, whose coefficients are in the rational numbers, has a solution in the complex numbers? The pleasing answer is provided by the so-called *fundamental theorem of algebra* (for algebraic numbers), which states that we have reached the end of our search for a sufficiently large ring of numbers:

Theorem 2.2 *Every normalized polynomial equation*

$$X^n + a_{n-1}X^{n-1} + \cdots + a_1 X + a_0 = 0$$

of positive degree n whose coefficients $a_0, \ldots, a_{n-1}$ are all algebraic numbers has a solution z in the complex numbers, that is, for each tuple of n algebraic numbers $a_0, \ldots, a_{n-1}$, there exists a complex number z with

$$z^n + a_{n-1}z^{n-1} + \cdots + a_1 z + a_0 = 0.$$

Since the solution z, whose existence is claimed in Theorem 2.2, is a root of a polynomial with algebraic coefficients, z is particularly itself algebraic, that is, the set of algebraic numbers can be said to be closed under addition, subtraction, multiplication, division, and the formation of solutions of polynomial equations.

The idea of the proof can be summarized as follows. Let n be a positive natural number. For each sequence $b_0, \ldots, b_{n-1}$ of algebraic numbers, we will construct an algebraic number $z = z_{b_0 \cdots b_{n-1}}$ so that we will be able to prove the following statement about it:

Lemma 2.7 *Let $q = q_n := 1 - \frac{1}{2} \cdot 3^{1-n^2} < 1$. Then the (yet to be defined) algebraic number $z = z_{b_0 \cdots b_n}$ satisfies the inequalities:*

$$|z| \leq \sqrt[n]{|b_0|}, \tag{2.35}$$

$$|z^n + b_{n-1}z^{n-1} + \cdots + b_1 z + b_0| \leq q|b_0|. \tag{2.36}$$

We can interpret this statement as follows: If we consider 0 as an approximate solution of

$$X^n + b_{n-1}X^{n-1} + \cdots + b_1X + b_0 = 0, \tag{2.37}$$

then $z = z_{b_0 \cdots b_{n-1}}$ is a better approximation because if we substitute $X = 0$ in (2.37) on the left side, the absolute difference between the left and right side is given by $|b_0|$. However, if we substitute $X = z$ on the left side of (2.37), the absolute difference is at most $q|b_0| = q_n|b_0|$, thus decreasing the error by at least the fixed factor $q < 1$.

How do we proceed from here? The idea of the proof we are presenting is to consider the polynomial equation

$$(X + z)^n + b_{n-1}(X + z)^{n-1} + \cdots + b_1(X + z) + b_0 = 0 \tag{2.38}$$

in the next step instead of (2.37). A solution x of (2.37) corresponds to a solution $x - z$ of (2.38) and vice versa. In particular, the found approximate solution z of (2.37) corresponds to an approximate solution 0 of the Eq. (2.38), which is at least correct up to $q|b_0|$ in absolute terms. From this, we can obtain a solution, which is correct at least up to $q^2|b_0|$ in absolute terms, etc., namely with another application of Lemma 2.7, this time applied to (2.38).

We implement this idea:

Proof of Theorem 2.2 [17] We define $q = q_n < 1$ as in Lemma 2.7. Let

$$f(x) := x^n + a_{n-1}x^{n-1} + \cdots + a_1x + a_0, \qquad x \in \overline{\mathbf{Q}}. \tag{2.39}$$

We are then looking for a complex number z with $f(z) = 0$, i.e., a root of the function f.

To do this, we will inductively construct an algebraic sequence (z_m) as in Fig. 2.10 with the following properties:

$$|z_{m+1} - z_m| \leq \sqrt[n]{q^m|a_0|}, \tag{2.40}$$

$$|f(z_m)| \leq q^m|a_0|. \tag{2.41}$$

From the first inequality (2.40), it follows that (z_m) is a Cauchy sequence: Let $m' \geq m$. Then, according to the triangle inequality for the complex modulus:

$$|z_{m'} - z_m| \leq \sum_{k=0}^{\infty} |z_{m+k+1} - z_{m+k}| \leq \sqrt[n]{q^m|a_0|} \sum_{k=0}^{\infty} q^{\frac{k}{n}} = \sqrt[n]{q^m|a_0|} \cdot \frac{1}{1 - q^{\frac{1}{n}}}. \tag{2.42}$$

Here, we used the well-known summation formula for the geometric series for the last equality. Since $q < 1$ also implies $q^{\frac{1}{n}} < 1$ and thus $\lim_{m \to \infty} q^{\frac{m}{n}} = 0$, it follows from (2.42) that $|z_{m'} - z_m|$ becomes arbitrarily small for sufficiently large m and any m'. By definition,

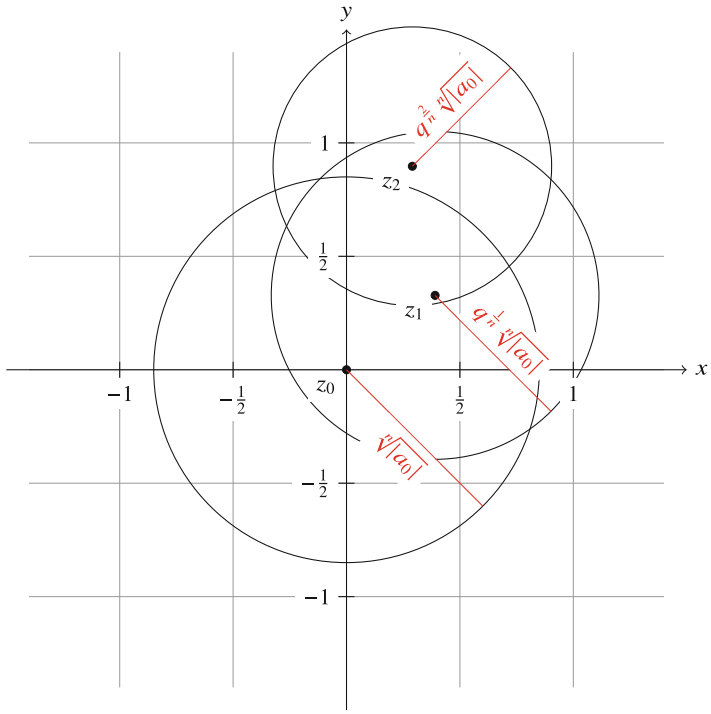


Fig. 2.10 Approximation of a root in the fundamental theorem of algebra

(z_m) is a Cauchy sequence, and the complex number $z = \lim\limits_{m\to\infty} z_m$ exists. The function defined by (2.39) is continuous with respect to the complex modulus (that is, for every algebraic number x, $|f(x') - f(x)|$ is small, provided x' is chosen close enough to x with respect to the complex modulus; this is proven like the continuity of real polynomial functions). This implies

$$|f(z)| = \left|f\left(\lim_{m\to\infty} z_m\right)\right| = \left|\lim_{m\to\infty} f(z_m)\right| = \lim_{m\to\infty} |f(z_m)| \le \lim_{m\to\infty} q^m |a_0| = 0$$

according to (2.41), where the continuity of the complex modulus has also been used at one point.

It remains to construct the sequence (z_m). We start with $z_0 := 0$ and consider this as the first approximation of a root (the value of f at the point z_0 is, however, off by the amount $|a_0|$ from the desired value 0). The further points $z_1, z_2, z_3, \ldots$ are then given as follows: Let the point z_m already be defined. We then set

$$g(x) := f(z_m + x) \tag{2.43}$$

for all algebraic numbers x. Substituting, expanding and collecting on the right side of (2.43) shows that $g(x)$ can again be written as a normalized polynomial in x of degree n, that is, it is of the form

$$g(x) = f(z_m + x) = x^n + b_{n-1}x^{n-1} + \ldots + b_1 x + b_0$$

with algebraic coefficients $b_0, \ldots, b_{n-1}$. In particular, $b_0 = g(0) = f(z_m)$, so by induction hypothesis $|b_0| \le q^m |a_0|$. We can then set

$$z_{m+1} = z_m + z_{b_0 \cdots b_{n-1}}$$

because, due to Lemma 2.7, by induction hypothesis

$$|z_{m+1} - z_m| = |z_{b_0 \cdots b_{n-1}}| \le \sqrt[n]{|b_0|} = \sqrt[n]{f(z_m)} \le \sqrt[n]{q^m |a_0|}$$

and

$$|f(z_{m+1})| = |g(z_{b_0 \cdots b_{n-1}})| \le q|b_0| = qf(z_m) \le q^{m+1}|a_0|.$$

□

What remains is the construction of $z = z_{b_0 \cdots b_{n-1}}$ and the proof of Lemma 2.7. If $|b_0| = 0$, then we set $z := 0$. Otherwise, we proceed as follows: Let $b_n := 1$. For $i \in \{1, \ldots, n\}$ and a positive real algebraic number s, we set

$$f_i(s) := |b_i| s^i.$$

Let

$$J := \{i \in \{1, \ldots, n\} \mid |b_i| > 0\}.$$

If i and $j \in J$ with $i < j$, then the graphs of the functions $f_i(s)$, $f_j(s)$ intersect at exactly one point $\sigma = \sigma_{i,j} > 0$, namely $\sigma = \sqrt[j-i]{|b_i|/|b_j|}$. We also note that

$$f_i(s) \ge f_j(s) \quad \text{for} \quad s \le \sigma,$$
$$f_i(s) \le f_j(s) \quad \text{for} \quad s \ge \sigma.$$

Using the $f_i(s)$ we define

$$m(s) := \max\{f_i(s) \mid i \in J\}$$

as in Fig. 2.11.

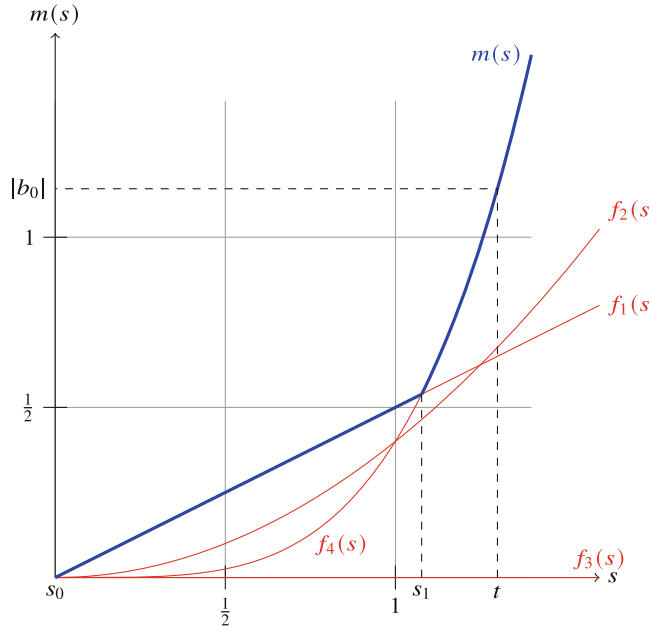


Fig. 2.11 The function $m(s)$

Since $b_n = 1$ we have $m(s) > 0$ for all $s > 0$. There are uniquely determined $0 = s_0 \leq s_1 \leq s_2 \leq \cdots$ such that for $i \in J$ and for $s_{i-1} < s \leq s_i$ it holds

$$m(s) = f_i(s).$$

For each $i \in J$ there is exactly one $t_i > 0$ with $f_i(t_i) = |b_0|$, namely $t_i := \sqrt[i]{|b_0|/|b_i|}$. Then

$$t := \min\{t_i \mid i \in J\}$$

is the smallest positive real algebraic number with $m(t) = |b_0|$.

For all $i \in \{-1, 0, 1, 2, \ldots\}$, let $k_i \in J$ be the number with

$$s_{k_i - 1} < 3^{-i} t \leq s_{k_i}.$$

In particular, $f_{k_i}(3^{-i}t) = m(3^{-i}t)$.

From $3t > t > 3^{-1}t > 3^{-2}t > \cdots$ and $s_0 \leq s_1 \leq s_2 \leq \cdots$ it follows

$$n \geq k_{-1} \geq k_0 \geq k_1 \geq k_2 \geq \cdots \geq 1,$$

that is, we can define j as the smallest natural number with

$$k_{j-1} = k_j = k_{j+1}$$

We then set $k := k_j \in J$ and $r := 3^{-j}t$ and finally

$$z = z_{b_0 \cdots b_{n-1}} := r \sqrt[k]{-\frac{b_0/b_k}{|b_0/b_k|}}.$$

The intuitive meaning is the following: In the Gaussian number plane, the distance from z to the origin is the distance r, and b_0 and $b_k z^k$ lie together with 0 on opposite sides on a straight line, that is, b_0 and $-b_k z^k$ have the same argument.

One proof still has to be done:

Proof of Lemma 2.7 The statement about the magnitude of $z = z_{b_0 \cdots b_{n-1}}$ follows quite quickly: Since $m(s)$ is monotonic in s and $r \le t$, it is

$$r^n \le m(r) \le m(t) = |b_0|,$$

so $|z| = r \le \sqrt[n]{|b_0|}$, which proves (2.35). By definition of z, one has

$$|b_0 + b_k z^k| = \left| b_0 - \frac{|b_k|}{|b_0|} b_0 r^k \right| = |b_0| \cdot \left| 1 - \frac{f_k(r)}{|b_0|} \right|. \tag{2.44}$$

Since $f_k(r) \le f_k(t) \le m(t) = |b_0|$, $1 - \frac{f_k(r)}{|b_0|} \ge 0$, Eq. (2.44) is equivalent to

$$|b_0 + b_k z^k| \le |b_0| \cdot \left(1 - \frac{f_k(r)}{|b_0|} \right) = |b_0| - f_k(r). \tag{2.45}$$

To prove (2.36), we first need another estimate of $m(r)$, this time from the other side: Let $i \ge 0$. By definition of k_{i-1}, $m(3^{-i+1}t) = f_{k_{i-1}}(3^{-i+1}t)$, thus

$$\frac{m(3^{-i}t)}{m(3^{-i+1}t)} = \frac{m(3^{-i}t)}{f_{k_{i-1}}(3^{-i+1}t)} \ge \frac{f_{k_{i-1}}(3^{-i}t)}{f_{k_{i-1}}(3^{-i+1}t)} = 3^{-k_{i-1}},$$

thus $m(3^{-i}t) \ge 3^{-k_{i-1}}\, m(3^{-i+1}t)$. It follows

$$\begin{aligned} m(r) = m(3^{-j}t) &\ge 3^{-k_{j-1}}\, m(3^{-j+1}t) \\ &\ge 3^{-k_{j-1}-k_{j-2}}\, m(3^{-j+2}t) \\ &\ge \cdots \ge 3^{-k_{j-1}-k_{j-2}-\cdots-k_0}\, m(t). \end{aligned} \tag{2.46}$$

We want to find an estimate for the exponent $h = k_0 + k_1 + \cdots + k_{j-1}$. Since j is chosen minimally with $k_{j-1} = k_j = k_{j+1}$, in the inequality chain of numbers from $\{1, \ldots, n\}$

$$k_{-1} \geq k_0 \geq k_1 \geq k_2 \geq \cdots \geq k_j,$$

equality can only hold at most at every second place, that is, for $2i - 1 \leq j$, it is $k_{2i-1} \leq n - i$. Thus,

$$\begin{aligned} h &\leq n + (n-1) + (n-1) + (n-2) + (n-2) + \cdots + (k+1) + (k+1) + k \\ &= n + k + 2 \sum_{\ell=k+1}^{n-1} \ell = n + k + (n-1)n - k(k+1) = n^2 - k^2. \end{aligned}$$

Substituting this into (2.46), we get

$$m(r) \geq 3^{k^2-n^2} m(t) = 3^{k^2-n^2} |b_0|. \tag{2.47}$$

Next, we compare the $f_i(r) = |b_i| r^i$ for $i \neq 0, k$ to $f_k(r) = |b_k| r^k$. Since $k = k_{j+1}$ and thus $s_{k-1} < \frac{r}{3} \leq s_k$ it follows for $i \in \{1, 2, \ldots, k-1\}$, that

$$f_i(r) = 3^i f_i\left(\frac{r}{3}\right) <\leq 3^i m\left(\frac{r}{3}\right) = 3^i f_k\left(\frac{r}{3}\right) = 3^{i-k} f_k(r).$$

Summing up, we get

$$\begin{aligned} \sum_{i=1}^{k-1} f_i(r) &\leq \sum_{i=1}^{k-1} 3^{i-k} f_k(r) = 3^{1-k} f_k(r) \sum_{\ell=0}^{k-2} 3^\ell = 3^{1-k} \frac{3^{k-1} - 1}{3 - 1} f_k(r) \\ &= \frac{1 - 3^{1-k}}{2} f_k(r) \end{aligned} \tag{2.48}$$

according to the known summation formula for the geometric sum. Since also $k = k_{j-1}$ and thus $s_{k-1} < 3r < s_k$, we have for $i \in \{k+1, k+2, \ldots, n\}$, that

$$f_i(r) = 3^{-i} f_i(3r) \leq 3^{-i} m(3r) = 3^{-i} f_k(3r) = 3^{k-i} f_k(r).$$

Summing up, this is

$$\begin{aligned} \sum_{i=k+1}^{n} f_i(r) &\leq \sum_{i=k+1}^{n} 3^{k-i} f_k(r) = 3^{-1} \sum_{\ell=0}^{n-k-1} 3^{-\ell} f_k(r) = 3^{-1} \frac{3^{-(n-k)} - 1}{3^{-1} - 1} f_k(r) \\ &= \frac{1 - 3^{k-n}}{2} f_k(r). \end{aligned} \tag{2.49}$$

Together with (2.48), (2.49) results in

$$\sum_{i\neq 0,k} f_i(r) \leq \left(\frac{1-3^{1-k}}{2} + \frac{1-3^{k-n}}{2}\right) f_k(r) \leq f_k(r) - \frac{3^{1-k}}{2} f_k(r).$$

From this, from (2.45), and from (2.47), it finally follows that

$$\begin{aligned}
|z^n + b_{n-1}z^{n-1} + \cdots + b_1 z + b_0| &\leq |b_0 + b_k z^k| + \sum_{i\neq 0,k} |b_i z^i| \\
&\leq |b_0| - f_k(r) + \sum_{i\neq 0,k} f_i(r) \\
&\leq |b_0| - f_k(r) + f_k(r) - \frac{3^{1-k}}{2} f_k(r) \\
&\leq |b_0| - \frac{3^{1-k}}{2} m(r) \\
&\leq |b_0| - \frac{3^{1-k+k^2-n^2}}{2} |b_0| \leq \left(1 - \frac{3^{1-n^2}}{2}\right) |b_0|,
\end{aligned}$$

which proves the Eq. (2.36) due to $q = 1 - \frac{1}{2}3^{1-n^2}$.

Every normalized polynomial with algebraic coefficients has a root in the complex numbers. This root is also algebraic.

At this point, we can also explain the notation $\overline{\mathbf{Q}}$ for the algebraic numbers: According to Theorem 2.2, $\overline{\mathbf{Q}}$ is the smallest subset of all complex numbers, which contains the rational numbers and which is closed in the following sense: If a normalized polynomial equation of positive degree with coefficients from $\overline{\mathbf{Q}}$ is given, it again has a solution in $\overline{\mathbf{Q}}$.

Summary

- The integers arise from the natural numbers by adjoining numbers that are solutions to equations of the form $X + a = b$. The rational numbers arise from the integers by adjoining numbers that are solutions to equations of the form

(continued)

$a \cdot X = b$. Similarly, we have explained the **complex numbers** by adjoining a number to the real numbers that is the solution to the equation $X^2 + 1 = 0$. This solution is i, the **imaginary unit**.

- An **algebraic number** is a complex number which is the solution to a normalized polynomial equation with rational coefficients. Trivially, a rational number is an algebraic number. Thus, the ring of numbers $\overline{\mathbf{Q}}$ of algebraic numbers is between the ring of numbers of rational numbers and the ring of numbers of complex numbers. There are algebraic numbers like $\sqrt{2}$, which are not rational. Later, we will encounter complex numbers that are not algebraic, the so-called **transcendental numbers**. The inclusions

$$\mathbf{Q} \subseteq \overline{\mathbf{Q}} \subseteq \mathbf{C}$$

  are therefore **proper** inclusions.
- The algebraic numbers are **closed under addition, multiplication and formation of inverses**. For example, if x is the solution to a normalized polynomial equation with rational coefficients and y is the solution to another normalized polynomial equation, there is a third normalized polynomial equation that has $x + y$ as a solution.
- Every normalized polynomial equation of algebraic numbers has a solution in the algebraic numbers. This is the statement of the **fundamental theorem of algebra**. In particular, the algebraic numbers are **closed under extraction of any roots**. Of particular importance are the n-th **roots of unity**, which are those complex numbers ζ, such that $\zeta^n = 1$. For every natural number $n \geq 1$ there are exactly n of these n-th roots of unity.
- The complex numbers form the points of a **drawing plane** in which the points 0 and 1 are marked. The points that can be constructed from the given points 0 and 1 with compass and straightedge are called **constructible**. Since each point corresponds to a complex number, we can also speak of constructible numbers.
- The constructible numbers consist of the rational numbers and all complex numbers that can be obtained from them **by repeated application of the basic arithmetic operations and the extraction of square roots**. In particular, every constructible number is also **algebraic**.

Exercises

About Polynomial Equations

2.1 Show that $\sqrt{3}$ is not a rational number.

2.2 Show that $\sqrt{12}$ is not a rational number.

2.3 Show that $\sqrt[3]{25}$ is not a rational number.

2.4 Show that an integer a has an n-th root in the rational numbers if and only if a has an n-th root in the integers.

2.5 Show that every rational solution of a normalized polynomial equation with integer coefficients is an integer.

2.6 Let

$$X^n + a_{n-1}X^{n-1} + \cdots + a_1X + a_0 = 0$$

be a normalized polynomial equation with integer coefficients. Show that every integer solution must be a divisor of a_0.

2.7 Given rational numbers $x_1, \ldots, x_n$, construct a polynomial equation of degree n with rational coefficients that has $x_1, \ldots, x_n$ as roots.

2.8 Show that every normalized polynomial equation of odd degree with rational coefficients has a solution in the real numbers.

2.9 Provide a normalized polynomial equation of the fourth degree with rational coefficients that has no solution in the real numbers.

2.10 Provide a normalized polynomial equation of the fifth degree with rational coefficients that has the number 1 as its only root.

2.11 Provide a polynomial equation with integer coefficients that has $\sqrt[7]{3 + \sqrt[3]{4}}$ as a solution.

2.12 Provide a polynomial equation with integer coefficients that has the real number $\cos 15°$ as a solution.

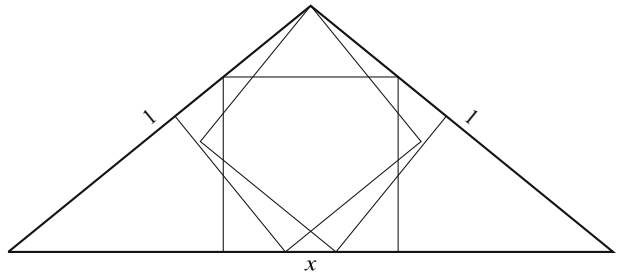


Fig. 2.12 The Calabi triangle

2.13 Eugenio Calabi[18] found a non-equilateral, isosceles triangle in which three equal-sized, largest squares can be inscribed according to Fig. 2.12. Show that the ratio x of the longest to one of the two shorter sides satisfies the equation $2X^3 - 2X^2 - 3X + 2 = 0$.

The Complex Numbers

2.14 Calculate the inverse of the complex number $4 + 3\,\mathrm{i}$.

2.15 Let $z \neq 0$ be a complex number whose real and imaginary parts are rational numbers. Show that z^{-1} also has rational real and imaginary parts.

2.16 Show that the real part of a complex number z is given by $\frac{1}{2}(z + \overline{z})$ and that the imaginary part is given by $\frac{1}{2\mathrm{i}}(z - \overline{z})$.

2.17 Show the following equality, first found by Rafael Bombelli:[19]

$$(2 \pm \sqrt{-1})^3 = 2 \pm \sqrt{-121}$$

Discuss which signs of the square roots should be chosen in each case.

2.18 Let z be an invertible complex number. Deduce the equality

$$\overline{z}^{-1} = \overline{z^{-1}}$$

from the multiplicativity of complex conjugation.

2.19 Show that for two real numbers a and b the root $\sqrt{a^2+b^2}$ is close to zero if and only if both $|a|$ and $|b|$ are close to zero. So show:

$$\forall \varepsilon > 0\, \exists \delta > 0 \colon \sqrt{a^2+b^2} < \delta \implies |a|, |b| < \varepsilon,$$
$$\forall \varepsilon > 0\, \exists \delta > 0 \colon \quad |a|, |b| < \delta \implies \sqrt{a^2+b^2} < \varepsilon,$$

where δ and ε are real numbers.

2.20 Interpret the multiplication with the imaginary unit i as a geometric operation in the Gaussian number plane.

2.21 Explain why $\left(\begin{smallmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{smallmatrix}\right)$ describes a rotation by the angle α around the origin of the Gaussian number plane. Then deduce the addition theorems

$$\cos(\alpha_1+\alpha_2) = \cos\alpha_1\, \cos\alpha_2 - \sin\alpha_1\, \sin\alpha_2,$$
$$\sin(\alpha_1+\alpha_2) = \cos\alpha_1\, \sin\alpha_2 + \sin\alpha_1\, \cos\alpha_2$$

from the known formula for the product of matrices.

2.22 Show that the polynomial equation $X^2 + X + 1 = 0$ has no solution in the real numbers.

2.23 Construct a minimal ring of numbers $\mathbf{R}(\omega)$ which contains the real numbers and a solution ω of the polynomial equation $X^2 + X + 1 = 0$ and in which addition and multiplication are defined, extending addition and multiplication of real numbers and for which also the relevant laws of arithmetic apply. Show that $\omega^3 = 1$ holds in $\mathbf{R}(\omega)$ and that there is a solution to the equation $X^2 + 1 = 0$ in $\mathbf{R}(\omega)$.

Algebraic Numbers

2.24 Is $\cos 10^\circ$ an algebraic number?

2.25 Provide a normalized polynomial equation with rational coefficients that has $\sqrt{2} + \sqrt[3]{7}$ as a solution.

2.26 Provide a non-trivial polynomial equation with integer coefficients that has $\sqrt{3} \cdot \sqrt[3]{7}$ as a solution.

2.27 Let z be a solution of the polynomial equation

$$X^3 - \sqrt{2 - \sqrt[3]{4}}\, X^2 + 3 = 0.$$

Provide a normalized polynomial equation with rational coefficients that has z as a solution.

2.28 Show that the polynomial equation $X^3 - 2X + 5 = 0$ has exactly one real solution x. Further show that x is an invertible algebraic number, and provide a normalized polynomial equation with rational coefficients that has x^{-1} as a solution.

2.29 Let x and y be algebraic numbers that are solutions of normalized polynomial equations with rational coefficients of degrees n and m respectively. Provide an estimate of the degree that a normalized polynomial equation with rational coefficients must have at most, so that it has $x \cdot y$ as a solution.

2.30 Let x be a rational number that is also an algebraic integer. Show that x is already an integer.

2.31 Let $a_0, \ldots, a_{n-1}$ be rational numbers. Let z be a transcendental number. Show that then $z^n + a_{n-1}z^{n-1} + \cdots + a_1 z + a_0$ is also a transcendental number.

Complex Roots of Unity

2.32 Let ζ be a fourth root of unity, and let θ be a sixth root of unity. Show that $\zeta \cdot \theta$ is a twelfth root of unity.

2.33 Let m and n be two positive natural numbers, whose greatest common divisor we denote by (m, n). Let ζ be an m-th and θ be an n-th root of unity. Show that $\zeta \cdot \theta$ is a k-th root of unity, where $k = \frac{mn}{(m,n)}$.

2.34 How many tenth roots of unity ζ, such that all other tenth roots of unity are an integer power of ζ, are there?

2.35 Let n be a natural number. Let $\varphi(n)$ be the number of positive natural numbers less than n that are coprime to n. (The function $\varphi(n)$ is called the *Euler's φ-function*.) Show that the number of n-th roots of unity ζ such that every other n-th root of unity is an integer power of ζ is given by $\varphi(n)$.

2.36 Let an algebraic number z be given in the form $z = r\,\mathrm{e}^{\varphi \mathrm{i}}$ with a positive real number r and a real number φ. Show that r is an algebraic number.

2.37 Give all complex solutions of $X^6 + 1 = 0$ in the form $a + b\,\mathrm{i}$ where a and b are real numbers.

2.38 Give a normalized polynomial equation whose solutions are exactly the corners of a regular heptagon in the complex plane, whose center is the origin of the plane and one of whose corners is given by the complex number $1 + \frac{1}{2}\mathrm{i}$.

2.39 Show that the equation $X^3 + X^2 + X + 1 = 0$ has exactly three complex solutions, namely all fourth roots of unity except 1.

2.40 Show that the equation $X^{n-1} + X^{n-2} + \cdots + X + 1 = 0$ has exactly $n - 1$ complex solutions, namely all n-th roots of unity except the 1.

2.41 Deduce the addition theorems for the sine and cosine function from the identity $\mathrm{e}^{y_1 \mathrm{i}} \cdot \mathrm{e}^{y_2 \mathrm{i}} = \mathrm{e}^{(y_1 + y_2)\mathrm{i}}$.

Constructions with Compass and Straightedge

2.42 Show that all Gaussian integers are algebraic integers.

2.43 Show how to construct a line segment of length $\frac{1}{3}$ from a line segment of length 1.

2.44 Show how to construct a line segment of length $\sqrt[4]{7}$ from a line segment of length 1.

2.45 Construct an angle of 15°.

2.46 Let z be a solution of $X^4 + X^3 + X^2 + X + 1 = 0$. Show that $z + z^{-1}$ is a solution of $X^2 + X - 1 = 0$.

2.47 Show that the four solutions of $X^4 + X^3 + X^2 + X + 1 = 0$ are all constructible complex numbers.

2.48 Provide a construction manual for the regular pentagon.

2.49 Let x, x', y, y' be four complex numbers with $x \neq x'$ and $y \neq y'$. Let K_1 be the circle through x' and with center x, and let K_2 be the circle through y' and with center y. Let K_1 and K_2 be different. Provide explicit formulas for the intersection point or points of the circles K_1 and K_2.

The Fundamental Theorem of Algebra

2.50 Let z_n be a convergent sequence of complex numbers. Why is $\lim_{n\to\infty} |z_n| = |\lim_{n\to\infty} z_n|$?

2.51 Let $f(z)$ be a complex polynomial function, that is, let $f(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0$ for a complex number z and with complex coefficients $a_0, \ldots, a_n$. Show that $f(z)$ is continuous with respect to the complex modulus, that is, show:

$$\begin{aligned} &\forall C > 0\, \forall \varepsilon > 0\, \exists \delta > 0\, \forall z, z' \text{ with } |z|, |z'| \\ &\quad \leq C \colon |z - z'| < \delta \implies |f(z) - f(z')| < \varepsilon, \end{aligned}$$

where C, ε and δ are real numbers and z and z' are two complex numbers.

2.52 Let q be a complex number. Prove the formula $\sum_{k=0}^{n-1} q^k = \frac{1-q^n}{1-q}$.

2.53 Provide an upper estimate of the modulus of all complex solutions of $X^4 - 2X^3 + 5X^2 - 4X + 5 = 0$.

2.54 Let $X^n + a_{n-1}X^{n-1} + \cdots + a_1 X + a_0 = 0$ be a normalized polynomial equation with complex coefficients. Show that every complex solution z lies in the closed disk of radius $1 + \max\{|a_0|, \ldots, |a_{n-1}|\}$.

2.55 In our proof of the fundamental theorem of algebra, the number 3 appears repeatedly. Can it be replaced by another number?

Endnotes

[1] An algebraic structure with such properties is called *commutative semiring*.

[2] Hippasus of Metapontum, sixth/fifth century BC, Greek mathematician, music theorist and philosopher.

[3] Pythagoras of Samos, about 570–495 BC, Greek philosopher.

[4] The equivalence $\neg(z_1 = z_2) \iff z_1 \neq z_2$, more precisely, the implication $\neg(z_1 = z_2) \implies z_1 \neq z_2$, is constructively false: Just because I can show that two complex numbers cannot be equal, I do not necessarily find any number $\varepsilon > 0$, such that $|z_1 - z_2| \geq \varepsilon$.

[5] Formulated in the epsilon-delta language of classical analysis, this means: $\forall \varepsilon > 0\, \exists \delta > 0\, \forall a, b \in \mathbf{R} \colon \left(|a|, |b| < \delta \implies \sqrt{a^2 + b^2} < \varepsilon\right)$. In non-standard analysis (for example, according to Edward Nelson, 1932–2014, American mathematician), this looks like this: $\forall a, b \in \mathbf{R} \colon \left(a, b \approx 0 \implies \sqrt{a^2 + b^2} \approx 0\right)$.

[6]Augustin-Louis Cauchy, 1789–1857, French mathematician.

[7]A *Cauchy sequence* is a sequence (z_n), that is, a *function* that assigns a complex number (z_n) to each natural number n, so that $|z_n - z_{n'}| \to 0$, when $n, n' \to \infty$. A sequence is a special case of a more general construction rule. A mapping rule $n \mapsto z_n$ for a function must be independent of any choices.

[8]Jean-Robert Argand, 1768–1822, French bookseller and mathematician.

[9]We say *an* argument and not *the* argument, since α_1 is obviously not uniquely determined: Besides α_1, for example, $\alpha_1 + 2\pi$ is also possible.

[10]There are complex numbers whose transcendence is much easier to show than the transcendence of π (the so-called Liouville numbers), but the transcendence of π is much more interesting for us.

[11]Thus, for two algebraic numbers x and y, it also holds constructively that $x = y \vee x \neq y$.

[12]Leonhard Euler, 1707–1783, Swiss mathematician.

[13]The mathematician also speaks of an *adjunction* of a root of -1, i.e., a solution of the equation $X^2 + 1 = 0$.

[14]Thales of Miletus, ca. 624–ca. 546 BC, Greek philosopher, mathematician, and astronomer.

[15]Euclid of Alexandria, ca. 360–280 BC, Greek mathematician.

[16]Gabriel Cramer, 1704–1752, Swiss mathematician.

[17]We essentially follow Martin Kneser's (1928–2004, German mathematician) proof in [1].

[18]Eugenio Calabi, 1923–2023, Italian-American mathematician.

[19]Rafael Bombelli, 1526–1572, Italian mathematician.

Reference

1. Kneser, M.: Ergänzung zu einer Arbeit von Hellmuth Kneser über den Fundamentalsatz der Algebra. Math. Z. **177**(2), 285–287 (1981)

On the Impossibility of Squaring the Circle

3

Modern algebra begins with the study of polynomials.

Abstract

A polynomial is a formal expression of the form $a_n X^n + a_{n-1} X^{n-1} + \cdots + a_1 X + a_0$ in an indeterminate X. Polynomials are fundamental objects of algebra, and we can calculate with them as with numbers. In particular, we can add and multiply polynomials.

Just as we can assign the absolute value to each integer as a measure of its size, we can assign the degree to polynomials as a measure. This finds application in division with remainder. If the dividend and divisor are polynomials, we find a quotient so that the remainder has a smaller degree than the divisor.

The divisibility theory of integers is also mirrored by polynomials. To what extent a polynomial can be factorized into polynomials of smaller degrees depends crucially on the chosen coefficient ring. We show that the fundamental theorem of algebra implies that polynomials over the algebraic numbers always decompose into a product of linear terms.

At the end of this chapter, we apply the results obtained about polynomials to show with an elementary proof that π is transcendental, i.e., it is not the root of a polynomial equation with rational coefficients. We infer from this that we cannot construct the lengths π and also $\sqrt{\pi}$ with compass and straightedge. From the latter, it follows that squaring the circle, i.e., the construction of a square with the same area as a given circle with compass and straightedge, is impossible.

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3.1 Polynomials

By now, it is time to formalize the concept of a polynomial, especially since we will need it in Sect. 3.2. For motivation, consider two integers a and b. As is well known, the (first) binomial formula is

$$(a+b)^2 = a^2 + 2ab + b^2. \tag{3.1}$$

We write here "*the* (first) binomial formula", although strictly speaking we do not have one, but infinitely many formulas, namely one for each pair (a, b), let's say, of integers.

If we want to view these infinitely many formulas as one formula, we must understand a and b in (3.1) as placeholders for integers. The left and right sides are then formal expressions in a and b, and *the* binomial formula is then the statement that these two formal expressions are equal. We can substitute concrete integers for a and b into these formal expressions. This is how we regain the original infinitely many concrete formulas.

The mentioned formal expressions in a and b on the left and right side in (3.1) are examples of polynomials and products of polynomials (with integer coefficients) in a and b. The binomial formula (3.1) is therefore an equality between polynomial expressions.

Mathematically precisely, the concept of a polynomial can be defined as follows:

Definition 3.1 A *polynomial* f with rational coefficients in the *indeterminate* X is a formal sum of the form

$$a_n X^n + a_{n-1} X^{n-1} + \cdots + a_1 X + a_0,$$

where the a_i are rational numbers. Let j be a natural number.

The j-th *coefficient* of f is a_j if $j \leq n$ and 0 otherwise.

Two polynomials are equal if and only if they match in all coefficients.

If $a_n = 1$, then f is called a *normalized* polynomial of degree n, written $\deg f = n$. Independent of the value of a_n, we say f is a polynomial of at most *degree* n, written $\deg f \leq n$.[1]

If f is a polynomial with rational coefficients in the indeterminate X, we also write

$$f \in \mathbf{Q}[X].$$

Polynomials can be added and multiplied by applying the associative, distributive, and commutative laws and the rule $X^i \cdot X^j = X^{i+j}$. For example,

$$\begin{aligned}(X^2 + 4X) \cdot (X - 2) &= X^2 \cdot X + (4X) \cdot X + X^2 \cdot (-2) + (4X) \cdot (-2) \\ &= X^3 + 4X^2 - 2X^2 - 8X \\ &= X^3 + 2X^2 - 8X.\end{aligned}$$

We can consider the rational numbers as polynomials, namely as those, whose i-th coefficients for $i > 0$ all vanish. We also call such polynomials *constant polynomials*.

We can subtract two polynomials f and g by defining, in analogy to subtraction of, say, the complex numbers,

$$f - g \coloneqq f + (-1) \cdot g.$$

Since subtraction can be reduced to addition and multiplication in this way, we usually do not need to specifically address it. It turns out that a polynomial with rational coefficients in the indeterminate X is simply an expression that can be obtained in a finite number of steps from the rational numbers and the indeterminate X by addition and multiplication taking into account the associative, distributive, and commutative laws.

The polynomials over a ring (of numbers) form a ring themselves, an extension of the ring of coefficients. We can therefore add, subtract and multiply polynomials, and the associative, distributive, and commutative laws apply accordingly. We find the ring of the coefficients in the form of the constant polynomials.

In Definition 3.1, we defined polynomials with rational coefficients. Analogous definitions for polynomials with algebraic, real, complex or also integer coefficients are possible; that is, as coefficients we allow algebraic, real, complex or integer coefficients. If f is a polynomial with such coefficients, then we write accordingly $f \in \overline{\mathbf{Q}}[X]$, $f \in \mathbf{R}[X]$, $f \in \mathbf{C}[X]$ or $f \in \mathbf{Z}[X]$.

To practice dealing with the concept of a polynomial, we prove a first lemma about the degree concept of a polynomial. Since the ring where we draw the coefficients from does not matter, we do not specify it further.

Lemma 3.1 *Let f and g be polynomials (in the indeterminate X). Let g be normalized. Then for every natural number n it holds that*

$$\deg f \leq n \iff \deg(f \cdot g) \leq n + \deg g.$$

If f and g are both normalized polynomials, it follows that

$$\deg(f \cdot g) = (\deg f) + (\deg g). \tag{3.2}$$

Proof Let

$$f = a_m X^m + a_{m-1} X^{m-1} + \cdots + a_1 X + a_0$$

and

$$g = X^k + b_{k-1}X^{k-1} + \cdots + b_1 X + b_0,$$

in particular, $\deg g = k$. Then

$$f \cdot g = a_m X^{m+k} + \sum_{\ell=0}^{m+k-1} \sum_{i+j=\ell} a_i b_j \, X^\ell.$$

In particular, $\deg(f \cdot g) \leq m+k = m+\deg g$. If $\deg f \leq n$, we can assume $m \leq n$, and thus $\deg(f \cdot g) \leq m + \deg g \leq n + \deg g$, which proves one direction of the equivalence (3.2).

We prove the other direction by induction over m. If $m \leq n$, then $\deg f \leq n$, and there is nothing further to show. Otherwise, we consider the inequality $\deg(f \cdot g) \leq n + \deg g = n + k$. This can only be fulfilled for $a_m = 0$ because $m > n$. Thus, we can write f also in the form

$$f = a_{m-1}X^{m-1} + \cdots + a_1 X + a_0,$$

and can apply the induction hypothesis. □

When we multiply (normalized) polynomials, their degree adds up.

From Lemma 3.1 a cancellation rule for polynomials can be derived directly: For three polynomials f, g and h, it holds that

$$f \cdot g = f \cdot h \implies g = h \tag{3.3}$$

if f is a normalized polynomial. From $f \cdot g = f \cdot h$ it follows that $f \cdot (g - h) = 0$. The estimate in Lemma 3.1 then yields $\deg(g - h) \leq -\deg f$. If $\deg f$ is positive, i.e., $\deg(g - h)$ is negative, then $g - h = 0$, thus $g = h$ must hold. Otherwise, $\deg f = 0$, thus $f = 1$, since it is normalized. In this case, (3.3) is trivial.

In (3.1), polynomials in two indeterminates, namely a and b, appear on both sides, but in Definition 3.1 we have so far only considered polynomials in one indeterminate. The generalization is not difficult:

A *polynomial* (with rational coefficients) in the indeterminates $X_1, \ldots, X_m$ is an expression that can be obtained in a finite number of steps from the rational numbers

and the indeterminates $X_1, \ldots, X_m$ by addition and multiplication, while respecting the associative, distributive and commutative laws. If f is such a polynomial, we write

$$f \in \mathbf{Q}[X_1, \ldots, X_m],$$

where we may again replace $\mathbf{Q}$ with some other (number) ring.

For a polynomial f in m variables, we write

$$\deg f \leq d$$

if f is a sum of terms of the form $a\, X^{k_1} \cdots X^{k_m}$ with a constant a and $k_1 + \cdots + k_m \leq d$.

To avoid unnecessary repetition of words, we will henceforth simply call a polynomial with algebraic coefficients a polynomial and formulate the remaining definitions and statements of this section usually only for these polynomials. Unless otherwise stated, the following remains correct if we choose a different coefficient ring, such as the integers, rational, real, or complex numbers.

The first important operation that we can perform with polynomials is substitution: If

$$f = a_n X^n + a_{n-1} X^{n-1} + \cdots + a_1 X + a_0$$

is a polynomial in X and x is an algebraic number, we call

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 \tag{3.4}$$

the *value* of f at x, and we say we *substitute* x for X in f.

We can substitute numbers into polynomials, which yields numbers. Each polynomial thus defines a function, the corresponding polynomial function.

A polynomial equation is thus an equation of the form

$$f(X) = 0,$$

and a solution to this equation is an algebraic number x with $f(x) = 0$. (We generally call a number x with $f(x) = 0$ a *root* of f.) The advantage of describing polynomial equations through polynomials is that we have the additional structure of addition and multiplication operations on the polynomials.

Solving a polynomial equation is equivalent to finding the roots of a polynomial.

If Y is another indeterminate, we write in formal analogy

$$f(Y) = a_n Y^n + a_{n-1} Y^{n-1} + \cdots + a_1 Y + a_0. \tag{3.5}$$

In this way, we can consider the polynomial f in X as one in Y.

If we substitute X into f, we get $f(X) = f$ again. The Definitions (3.4) and (3.5) can be combined in the following general definition: If $g = g(Y)$ is a polynomial in Y, we set

$$f(g(Y)) = f(g) = a_n g^n + a_{n-1} g^{n-1} + \cdots + a_1 g + a_0.$$

In the case that g is the constant polynomial x, $f(g)$ coincides with $f(x)$; in the case that g is the indeterminate Y, considered as a (normalized) polynomial of degree 1, $f(g)$ coincides with $f(Y)$.

We can substitute polynomials into polynomials, which yields polynomials.

The process of substitution is compatible with the operations defined on polynomials, that is, if $f(X)$ and $g(X)$ are two polynomials, then

$$(f + g)(x) = f(x) + g(x),$$
$$(f \cdot g)(x) = f(x) \cdot g(x).$$

On the other hand, in general, $f(x + y) \neq f(x) + f(y)$ or $f(xy) \neq f(x) \cdot f(y)$. A counterexample for the first equality is, for example, the polynomial $f = X^2$, as follows from the first binomial formula.

If f is a polynomial in several variables $X_1, \ldots, X_n$, and if $x_1, \ldots, x_n$ are numbers (or more generally polynomials), then we write in analogy $f(x_1, \ldots, x_n)$ for the value, which we obtain when we replace each variable X_i in the expression defining f by the given number (or the given polynomial) x_i.

We recall the definition of a rational expression from Sect. 2.5. These are complex numbers that can be obtained from given complex numbers by the basic arithmetic operations addition, subtraction, multiplication and division. If we leave out the division, we come to the concept of a polynomial expression:

If $t_1, \ldots, t_n$ are given complex numbers, then a complex number z is called *polynomial* (or more precisely *polynomial with rational coefficients*) *in* $t_1, \ldots, t_n$ if z can be obtained

from all rational numbers and the n complex numbers $t_1, \ldots, t_n$ only by means of addition and multiplication. This is equivalent to the existence of a polynomial $f(X_1, \ldots, X_n)$ with rational coefficients such that $z = f(t_1, \ldots, t_n)$. If z is a number polynomial in $t_1, \ldots, t_n$, we accordingly write

$$z \in \mathbf{Q}[t_1, \ldots, t_n].$$

Every expression polynomial in $t_1, \ldots, t_n$ is also a rational expression in $t_1, \ldots, t_n$. There is no obvious reason why the reverse should hold, why every rational expression should be convertible into a polynomial one. On the other hand, we can always bring a complex number z rational in $t_1, \ldots, t_n$ into the form $z = p/q$, where p and q are polynomials in $t_1, \ldots, t_n$.

Now let's consider the special case that $t_1, \ldots, t_n$ are algebraic numbers. Then q as a sum of products of algebraic numbers is again algebraic. If we look again at the proof of Proposition 2.3, more precisely at (2.19), we see that the inverse of an invertible algebraic number t is a polynomial (with rational coefficients) in t. In particular, q^{-1} is a polynomial (with rational coefficients) in q, and therefore also in $t_1, \ldots, t_n$. Thus, $z = p/q = p \cdot q^{-1}$ is polynomial in $t_1, \ldots, t_n$. We have thus shown that every rational expression in algebraic numbers can be converted into a polynomial expression in these algebraic numbers.

We can briefly write this as

$$\mathbf{Q}(t_1, \ldots, t_n) = \mathbf{Q}[t_1, \ldots, t_n], \quad t_1, \ldots, t_n \in \overline{\mathbf{Q}}$$

.

Every rational expression in the n algebraic numbers $t_1, \ldots, t_n$ can be written as a polynomial expression in $t_1, \ldots, t_n$.

With this knowledge, we can further tighten Theorem 2.1 about the characterization of constructible complex numbers: Since every constructible number is algebraic, we have:

Corollary 3.1 *A complex number z is constructible if and only if there is a sequence of complex numbers $z_1, \ldots, z_{n-1}, z_n = z$ such that for all $i \in \{1, \ldots, n\}$ the complex number z_i is a rational polynomial in $z_1, \ldots, z_{i-1}$ or a square root of a rational polynomial in $z_1, \ldots, z_{i-1}$.*

The constructible numbers are precisely those numbers which can be obtained from the rational numbers by successively applying addition, subtraction, multiplication, and square root extraction.

In the ring of integers, there is the concept of divisibility: If a and b are two integers, we say, as is well known, *a divides b*, written $a \mid b$, if b is an integer multiple of a, thus if we can write $b = ra$ for another integer r. As special cases, we have that every number is a divisor of zero and that one divides every number.

We could try to extend the concept of divisibility and the concept of prime numbers to the rational numbers, that is, we could say, a rational number a is a divisor of a rational number b, if a rational number r with $b = ra$ exists. Since we can set $r = b/a$ in the case $a \neq 0$, we find that any two arbitrary rational numbers not equal to zero always divide each other. We see that divisibility in the ring of rational numbers is not a particularly exciting relation. The same applies also for the algebraic numbers.

In the domain of rational or algebraic numbers, divisibility is a boring concept; divisibility between polynomials, on the other hand, is of great interest.

However, as soon as we move on to the polynomials (in one variable X), it becomes more interesting again: Let $f(X)$ and $g(X)$ be two polynomials. We say analogously, $f(X)$ *divides* $g(X)$, written $f(X) \mid g(X)$, if some polynomial $q(X)$ with $f(X) = q(X) \cdot g(X)$ exists. For example, $X - 1$ is a divisor of $X^2 - 1$, because we have

$$X^2 - 1 = (X + 1) \cdot (X - 1).$$

On the other hand, for example, $X - 2$ is not a divisor of $X^2 - 1$. A quick way to see this is the following: Assume $(X - 2) \mid (X^2 - 1)$. Then $X^2 - 1 = q(X) \cdot (X - 2)$ would hold for a polynomial $q(X)$. It is $x = 2$ a root of $X - 2$. Because $q(x) \cdot (x - 2) = q(2) \cdot (2 - 2) = q(2) \cdot 0 = 0$, the number $x = 2$ would thus also be a root of $X^2 - 1$. But it holds $x^2 - 1 = 2^2 - 1 = 3 \neq 0$, a contradiction.

If a polynomial $f(X)$ divides another polynomial $g(X)$, then every root of $f(X)$ is a root of $g(X)$. If $f(X)$ has a root x, such that $g(x) \neq 0$, then $f(X)$ cannot be a divisor of $g(X)$.

In the case of integers, there is the concept of division with remainder. The number b is divisible by a if and only if the division of b by a yields a remainder of zero. The same applies also for polynomials in one variable.

Proposition 3.1 *Let $f(X)$ be a polynomial and $g(X)$ a normalized polynomial. Then there exist uniquely determined polynomials $q(X)$ and $r(X)$ with*

$$f(X) = q(X) \cdot g(X) + r(X) \tag{3.6}$$

and $\deg r(X) < \deg g(X)$.

We call $r(X)$ the *remainder* of the division of $f(X)$ by $g(X)$. If we replace the term "polynomial" with "integer", "normalized polynomial" with "positive integer", and the degree $\deg(\cdot)$ with the absolute value $|\cdot|$ in Proposition 3.1, and additionally require that the remainder is nonnegative, we obtain the analogous statement about the division of integers with remainder.

Proof First, we prove the uniqueness of $q(X)$ and $r(X)$ in the case of their existence. To do this, we assume that we have two representations of $f(X)$ as in (3.6), say

$$q(X) \cdot g(X) + r(X) = f(X) = q'(X) \cdot g(X) + r'(X)$$

for polynomials $q(X)$, $r(X)$, $q'(X)$, $r'(X)$ with $\deg r(X)$, $\deg r'(X) < \deg g(X)$. If we subtract the left side from the right side, we get $(q(X)-q'(X)) \cdot g(X)+r(X)-r'(X) = 0$, so

$$(q(X) - q'(X)) \cdot g(X) = r'(X) - r(X). \tag{3.7}$$

Since for the polynomial on the right side, $\deg(r'(X) - r(X)) < \deg g(X)$ holds, we have the corresponding inequality also on the left side, so

$$\deg((q(X) - q'(X)) \cdot g(X)) < \deg g(X).$$

According to Lemma 3.1, $\deg(q(X) - q'(X)) < 0$, so $q(X) - q'(X) = 0$, hence $q(X) = q'(X)$. If we insert the obtained equality into (3.7), it follows $r'(X) - r(X) = 0$, which then proves $r(X) = r'(X)$.

It remains to show the existence of $q(X)$ and $r(X)$. For this, we choose a natural number n with $\deg f(X) \leq n$ and carry out the proof of existence by induction over n. In the case of $n < \deg g(X)$, we can set $q(X) = 0$ and $r(X) = f(X)$, and there is nothing further to show. In the case of $n \geq \deg g$ we proceed as follows: Let $f(X) = a_n X^n + a_{n-1} X^{n-1} + \cdots + a_0$. Set $q'(X) := a_n X^{n-\deg g}$. Since $g(X)$ is normalized, then $\deg(f(X) - q'(X) g(X)) < n$, so we can apply the induction hypothesis to $f(X) - q'(X) \cdot g(X)$, that is, there exist polynomials $q''(X)$ and $r(X)$ with $f(X) - q'(X) \cdot g(X) = q''(X) \cdot g(X) + r(X)$ and $\deg r < \deg g$. We obtain

$$f(X) = (q'(X) + q''(X)) \cdot g(X) + r(X).$$

With $q(X) = q'(X) + q''(X)$ the proposition is thus proven. □

As the first important consequence of division with remainder, we obtain a connection between the concept of divisibility and roots of polynomials, i.e., solutions of polynomial equations:

Proposition 3.2 *Let $f(X)$ be a polynomial. An algebraic number a is a root of $f(X)$ if and only if $f(X)$ is divisible by the polynomial $X - a$.*

Proof Let a be a root of $f(X)$. According to Proposition 3.1, there exist polynomials $q(X)$ and $r(X)$ such that

$$f(X) = q(X) \cdot (X - a) + r(X), \tag{3.8}$$

where $\deg r(X) < \deg(X - a) = 1$. Thus, $r(X)$ is a constant polynomial, so an algebraic number. If we set $X = a$ in (3.8), we get

$$0 = f(a) = q(a) \cdot (a - a) + r = r.$$

Thus, (3.8) is equivalent to $f(X) = q(X) \cdot (X - a)$, and thus $X - a$ is a divisor of $f(X)$.

Conversely, let $X - a$ be a divisor of $f(X)$, that is

$$f(X) = q(X) \cdot (X - a) \tag{3.9}$$

for a polynomial $q(X)$. If we set $X = a$ in (3.9), we get

$$f(a) = q(a) \cdot (a - a) = 0,$$

so a is a root of $f(X)$. □

Finding the roots of a polynomial is equivalent to splitting off linear factors.

3.2 Vieta's Theorem

Every normalized polynomial equation of positive degree with algebraic coefficients has a solution; this is the content of the fundamental theorem of algebra. Usually, there is not a single solution but multiple ones. In the case of the cyclotomic equation $X^n - a = 0$, $a \neq 0$ we have exactly n solutions according to Proposition 2.4.

We will show that every normalized polynomial equation of degree n has at most n different solutions, or even exactly n solutions if we count the number of solutions appropriately.

For this, we consider a normalized polynomial $f(X)$ of degree n with algebraic coefficients and the associated polynomial equation

$$f(X) = 0. \tag{3.10}$$

Let a_1 be an existing solution of (3.10) according to Theorem 2.2. According to Proposition 3.2 we can then write

$$f(X) = (X - a_1) \cdot f_1(X)$$

where $f_1(X)$ is a normalized polynomial of degree $n - 1$ with algebraic coefficients.

If $n \geq 2$, then $f_1(X)$ is a non-constant polynomial. Thus, $f_1(X)$ also has a root in the algebraic numbers, say a_2. According to Proposition 3.2, $f_1(X) = (X - a_2) \cdot f_2(X)$ for a normalized polynomial $f_2(X)$ of degree $n - 2$ with algebraic coefficients and thus

$$f(X) = (X - a_1) \cdot (X - a_2) \cdot f_2(X).$$

If $n \geq 3$, we can continue analogously with $f_2(X)$ and so on. In the end, we obtain that

$$f(X) = (X - a_1) \cdot (X - a_2) \cdots (X - a_n) \tag{3.11}$$

for algebraic numbers $a_1, \ldots, a_n$, that is, we have just shown that every normalized polynomial $f(X)$ with algebraic coefficients can be written as a product of n linear factors of the form $X - a$ (we also say, $f(X)$ *completely decomposes* into linear factors).

If a is a solution of (3.10), i.e., $f(a) = 0$, then from (3.11), it follows that $(a - a_1) \cdot (a - a_2) \cdots (a - a_n) = 0$. Since a product of algebraic numbers is exactly zero when one of the factors is zero, $a = a_i$ for an $i \in \{1, \ldots, n\}$ must hold. Thus, a is one of at most n different values, so we have proven that a normalized polynomial equation of n-th degree (with algebraic coefficients) has at most n different solutions in the algebraic (and thus also in the complex) numbers. (From (3.11) it follows conversely that each a_i is a root of $f(X)$, thus a solution of (3.10).) Since different a_i can coincide, there will generally be fewer than n different solutions of (3.10).

To what extent is the decomposition (3.11) unique? First we can certainly only expect uniqueness of the linear factor decomposition up to order. (Comparable to this question is the one about the uniqueness of the prime factor decomposition of a natural number. It is not obvious why a natural number has essentially, that is up to order, only one prime factor decomposition.)

So the question arises whether the sequence $a_1, a_2, \ldots, a_n$ is unique up to order. We are going to answer this question positively: Let $b_1, \ldots, b_k$ be the different values of the a_i,

that is the pairwise different solutions of (3.10). For $j \in \{1, \ldots, k\}$, let e_j be the frequency with which b_j appears among $a_1, \ldots, a_n$, that is, (3.11) becomes

$$f(X) = (X - b_1)^{e_1} \cdot (X - b_2)^{e_2} \cdots (X - b_k)^{e_k}.$$

We have to show that the $e_1, \ldots, e_k$ are uniquely determined by $f(X)$ under the assumption that the $b_1, \ldots, b_k$ are pairwise different. To do this, we write

$$(X - b_1)^{e_1} \cdot (X - b_2)^{e_2} \cdots (X - b_k)^{e_k} = (X - b_1)^{e'_1} \cdot (X - b_2)^{e'_2} \cdots (X - b_k)^{e'_k} \tag{3.12}$$

and want to show that $e_1 = e'_1, e_2 = e'_2, \ldots, e_k = e'_k$.

Let's start with e_1: Without loss of generality, let $e_1 \leq e'_1$. Since the polynomial $(X - b_1)^{e_1}$ is normalized, we can apply the cancellation rule (3.3) to (3.12) and obtain

$$(X - b_2)^{e_2} \cdots (X - b_k)^{e_k} = (X - b_1)^{e'_1 - e_1} \cdot (X - b_2)^{e_2} \cdots (X - b_k)^{e_k}. \tag{3.13}$$

Since the b_j have been assumed to be pairwise different, the left side of Eq. (3.13) does not have the algebraic number b_1 as a root. But then the right side cannot have b_1 as a root either, so $e'_1 - e_1 = 0$, thus $e'_1 = e_1$ must hold. The equality of e_2 and e'_2 and so on is proven in the same way.

So we have shown:

Theorem 3.1 *Let $f(X)$ be a normalized polynomial with algebraic coefficients. Then there exist algebraic numbers $a_1, \ldots, a_n$ such that*

$$f(X) = (X - a_1) \cdot (X - a_2) \cdots (X - a_n). \tag{3.14}$$

The $a_1, \ldots, a_n$ are uniquely determined by $f(X)$, up to order. □

Over the algebraic numbers, every polynomial (up to order) uniquely decomposes into a product of linear factors.

We say $a_1, \ldots, a_n$ are the *solutions with multiplicities* of (3.10). Thus, a normalized polynomial equation of degree n has exactly n solutions when counting solutions with multiplicities.

We illustrate this using the equation

$$X^n = 0, \tag{3.15}$$

which has 0 as its only solution. Since $X^n = (X-0)^n$, 0 is an n-fold solution of (3.15). Therefore, the equation has n solutions when we count them with multiplicities.

Counting with multiplicities, every (normalized) polynomial equation of degree n over the algebraic numbers has exactly n roots.

If a is a root of the polynomial $f(X)$, we say that a is a k-fold root of $f(X)$ if a is a k-fold solution of $f(X) = 0$; in other words, in the linear factor decomposition (3.14) the term $X - a$ appears exactly k times.

In calculus, there is also the concept of a multiple root. If g is a differentiable function on the real line, a real number a is called a multiple root of g if $g(a) = 0$ and $g'(a) = 0$. In the case of polynomials, we can understand this algebraically:

Definition 3.2 Let

$$g(X) = a_n X^n + a_{n-1} X^{n-1} + \ldots + a_2 X^2 + a_1 X + a_0$$

be a polynomial. Then

$$g'(X) = n a_n X^{n-1} + (n-1) a_{n-1} X^{n-2} + \ldots + 2a_2 X + a_1$$

is called the *(formal) derivative* of $g(X)$.

The formal derivative of $g(X)$ is well-defined, that is, the formal derivatives of two polynomials with the same coefficients agree in all coefficients. Furthermore, we note that

$$\deg g(X) \leq n \implies \deg g'(X) \leq n-1$$

and that the derivative operation is linear, that is, $(g+h)'(X) = g'(X) + h'(X)$ and $(cg)'(X) = cg'(X)$ for two polynomials g and h and for a constant c.

The formal derivative of a polynomial is defined in a way so that for real (or complex) x the function $g'(x)$ is the derivative of $g(x)$ known from calculus. Since the formal derivative $g'(X)$ is again a polynomial, we can differentiate it further and get $g''(X)$, the *second derivative* of g. After differentiating $g(X)$ n times, we get the *n-th derivative* $g^{(n)}(X)$ *of* $g(X)$.

The product or *Leibniz rule* and the chain rule known from calculus apply:

Proposition 3.3 *Let $g(X)$ and $h(X)$ be two polynomials. Then*

$$(gh)'(X) = g'(X)h(X) + g(X)h'(X), \tag{3.16}$$

$$g(h)'(X) = g'(h(X)) \cdot h'(X). \tag{3.17}$$

Proof First, we prove the product rule (3.16): Both sides of (3.16) are linear in both g and h. Therefore, it is sufficient to verify the equality (3.16) for the case $g(X) = X^m$ and $h(X) = X^n$:

$$(X^{m+n})' = (m+n)X^{m+n-1} = mX^{m-1}X^n + nX^mX^{n-1} = (X^m)'X^n + X^m(X^n)'.$$

Now let's move on to the chain rule (3.17): Since both sides of (3.17) are linear in $g(X)$, we can again start from $g(X) = X^m$. Due to the already proven product rule, the left side of (3.17) then gives

$$(h^m)'(X) = h'(X) \cdot h^{m-1}(X) + h(X) \cdot (h^{m-1})'(X) = \cdots = mh'(X)h^{m-1}(X).$$

Since $g'(X) = mX^{m-1}$, this equals the right side of the equality to be proven (3.17). □

Polynomials can be formally differentiated without using the methods of calculus. The usual differentiation rules can be derived algebraically.

For the k-th derivative, it follows by induction from (3.16) that

$$(gh)^{(k)}(X) = \sum_{i+j=k} \binom{k}{i} \cdot g^{(i)}(X) \cdot h^{(j)}(X). \tag{3.18}$$

As a special case of (3.17), we note

$$g'(X-a) = g(X-a)'. \tag{3.19}$$

We can now prove the following proposition:

Proposition 3.4 *Let $f(X)$ be a normalized polynomial with algebraic coefficients. An algebraic number a is a k-*fold root of $f(X)$ *if and only if*

$$f(a) = f'(a) = \cdots = f^{(k-1)}(a) = 0,$$

but $f^{(k)}(a) \neq 0$.

Proof According to Theorem 3.1, it holds

$$f(X) = (X-a)^k \cdot g(X)$$

for a natural number k and a normalized polynomial g with $g(a) \neq 0$. (It is $k = 0$ if and only if a is not a root of $f(X)$.) Due to (3.18) and (3.19), we get

$$\begin{aligned} f^{(\ell)}(X) &= \sum_{i+j=\ell} \binom{\ell}{i} \cdot ((X-a)^k)^{(i)} \cdot g^{(j)} \\ &= \sum_{i+j=\ell} \binom{\ell}{i} \cdot k \cdot (k-1) \cdots (k-i+1) \cdot (X-a)^{k-i} \cdot g^{(j)} \\ &= \sum_{i+j=\ell} \binom{\ell}{i} \frac{k!}{(k-i)!} \cdot (X-a)^{k-i} \cdot g^{(j)} \end{aligned}$$

for $\ell \leq k$. In the case of $\ell < k$, in each term $i < k$, that is, each term of $f^{(\ell)}$, and thus also $f^{(\ell)}$, has a as a root, that is $f^{(\ell)}(a) = 0$ for $\ell < k$.

In the case of $\ell = k$ there is exactly one summand (which corresponds to $i = k$) that does not contain a factor $X - a$. This is $\frac{k!}{(k-k)!}\, g(X) = k! \cdot g(X)$ and does not have a as a root because $g(a) \neq 0$. Thus, $f^{(k)}(a) \neq 0$. □

So far, we have pursued the question of whether a polynomial equation

$$X^n + a_{n-1}X^{n-1} + \cdots + a_1 X + a_0 = 0$$

has solutions and if so, how many. The fundamental theorem of algebra gives us a positive answer to the solvability, and in the proof of Theorem 2.2 and of Theorem 3.1 there is a method to numerically determine all solutions (with multiplicities) $x_1, \ldots, x_n$ from the coefficients $a_0, a_1, \ldots, a_{n-1}$.

We now ask the reverse question, which may initially seem somewhat unnatural, but will guide us in the right direction in our search for solution formulas: Given the solutions (with multiplicities) $x_1, \ldots, x_n$, determine the coefficients $a_0, a_1, \ldots, a_{n-1}$ from them.

Given algebraic numbers $x_1, \ldots, x_n$, there is a (unique) normalized polynomial equation of degree n that has exactly these numbers (with multiplicities) as solutions.

This problem is not difficult to solve. Let's start with the case $n = 2$. According to Theorem 3.1 we have

$$X^2 + a_1 X + a_0 = (X - x_1) \cdot (X - x_2).$$

When expanded, the right-hand side is $X^2 - (x_1 + x_2)\, X + x_1 x_2$. Comparison of coefficients then gives the equations

$$a_1 = -(x_1 + x_2), \qquad a_0 = x_1 x_2,$$

that is, we have expressed the coefficients a_1 and a_0 in terms of the two solutions x_1 and x_2. Let's move on to the case $n = 3$. Here we have

$$\begin{aligned} X^3 + a_2 X^2 + a_1 X + a_0 &= (X - x_1) \cdot (X - x_2) \cdot (X - x_3) \\ &= X^3 - (x_1 + x_2 + x_3)\, X^2 + (x_1 x_2 + x_1 x_3 + x_2 x_3)\, X - x_1 x_2 x_3, \end{aligned}$$

so

$$a_2 = -(x_1 + x_2 + x_3), \qquad a_1 = x_1 x_2 + x_1 x_3 + x_2 x_3, \qquad a_0 = -x_1 x_2 x_3,$$

the coefficient a_2 is thus the sum of the solutions x_i up to a sign, the coefficient a_1 is the sum of all double products of the solutions and the coefficient a_0 is the triple product of the solutions up to a sign. For $n \geq 4$ we can obviously derive similar formulas in the same way.

In order to be able to write these in a concise form, we define:

Definition 3.3 The polynomial

$$e_k(X_1, \ldots, X_n) := \sum_{1 \leq i_1 < i_2 < \cdots < i_k \leq n} X_{i_1} \cdot X_{i_2} \cdots X_{i_k}, \qquad 0 \leq k \leq n$$

is called the *k-th elementary symmetric function* in $X_1, \ldots, X_n$.

Since $e_0(X_1, \ldots, X_n) = 1$ does not depend on the indeterminates at all, often only $e_1, \ldots, e_n$ are counted as elementary symmetric functions. With Definition 3.3, Vieta's theorem[2] reads as follows:

Theorem 3.2 *Let*

$$X^n + a_{n-1} X^{n-1} + \cdots + a_1 X + a_0 = 0$$

be a normalized polynomial equation with algebraic coefficients $a_0, \ldots, a_{n-1}$. *If* $x_1, \ldots, x_n$ *are its solutions with multiplicities, then*

$$a_{n-i} = (-1)^i e_i(x_1, \ldots, x_n),$$

for $i \in \{0, \ldots, n\}$, *where* $a_n = 1$.

Proof By Theorem 3.1, we have

$$X^n + a_{n-1}X^{n-1} + \cdots + a_1 X + a_0 = (X - x_1) \cdot (X - x_2) \cdots (X - x_n).$$

Multiplying out the right side and comparing coefficients then yields the assertion. □

The coefficients of a normalized polynomial equation are, up to sign, the elementary symmetric functions in the solutions.

3.3 The Discriminant

The elementary symmetric functions $e_k(X_1, \ldots, X_n)$ are called *symmetric* because they are invariant under permutations of the X_i: Let σ be an n-letter permutation, that is a rearrangement

$$\begin{pmatrix} 1 & 2 & \ldots & n \\ \sigma(1) & \sigma(2) & \ldots & \sigma(n) \end{pmatrix}$$

of the numbers $1, \ldots, n$ in the order $\sigma(1), \sigma(2), \ldots, \sigma(n)$ (each number from 1 to n appears exactly once among $\sigma(1), \sigma(2), \ldots, \sigma(n)$).

For an elementary symmetric function, we have

$$e_k(X_{\sigma(1)}, \ldots, X_{\sigma(n)}) = e_k(X_1, \ldots, X_n). \tag{3.20}$$

If, for example, $n = 3$ and

$$\sigma = \begin{pmatrix} 1\,2\,3 \\ 3\,1\,2 \end{pmatrix},$$

then (3.20) reads

$$e_k(X_3, X_1, X_2) = e_k(X_1, X_2, X_3).$$

If $f(X_1, \ldots, X_n)$ is an arbitrary polynomial in the n variables $X_1, \ldots, X_n$, we call f *symmetric* if the equality

$$f(X_{\sigma(1)}, X_{\sigma(2)}, \ldots, X_{\sigma(n)}) = f(X_1, X_2, \ldots, X_n)$$

holds for every n-letter permutation σ.

An example of a symmetric polynomial that is not an elementary symmetric function is for example $X_1^2 + X_2^2 + \cdots + X_n^2$.

Part of the statement of Vieta's theorem, Theorem 3.2, can then be formulated as follows: The coefficients of a normalized polynomial equation are symmetric polynomials in the solutions (with multiplicities) of the equation. The fact that the coefficients can be universally expressed in a symmetric way in terms the solutions is very natural: otherwise, it would depend on the order in which the solutions are numbered—but there is no obvious reason why the solutions of a polynomial equation should have a naturally defined order. We will further investigate these questions in Chap. 5 within the framework of Galois theory.

A polynomial in n variables is called symmetric if it does not change when the variables are interchanged.

We can express that a polynomial is symmetric as follows in a compact way: If $f(X_1, \ldots, X_n)$ is an arbitrary polynomial in n variables and σ is an n-letter permutation, we define a new polynomial $\sigma \cdot f$ in n variables by

$$(\sigma \cdot f)(X_1, \ldots, X_n) \coloneqq f(X_{\sigma(1)}, \ldots, X_{\sigma(n)}).$$

We say, $\sigma \cdot f$ is the result of the *operation* of σ on f. The polynomial f is obviously symmetric if and only if

$$\sigma \cdot f = f$$

holds for all n-letter permutations σ.

We may have the following correspondence in mind: Instead of permutations, consider rotations of the plane around an origin and instead of polynomials, geometric figures in the plane. Each rotation σ transforms a geometric figure f into a new geometric figure

$\sigma \cdot f$; a geometric figure f is exactly (rotationally) symmetric if $\sigma \cdot f = f$ holds for all rotations σ. In this picture, a circle whose center is the origin of the plane corresponds to a symmetric polynomial.

The elementary symmetric functions are the most important examples of symmetric polynomials. They are called *elementary* because every other symmetric function can be expressed in terms them. Consider, for example, the symmetric polynomial $X_1^2 + X_2^2$. This is not one of the elementary symmetric functions $X_1 + X_2$ or $X_1 X_2$, but we can describe it in terms of these two, because according to the first binomial formula:

$$X_1^2 + X_2^2 = (X_1 + X_2)^2 - 2(X_1 X_2)$$

The fact that this is generally possible is the statement of the so-called fundamental theorem on symmetric functions:

Theorem 3.3 *Let $f(X_1, \ldots, X_n)$ be a symmetric polynomial whose coefficients are algebraic (or integer, rational, real, complex) numbers. Then there is exactly one polynomial $H(Y_1, \ldots, Y_n)$ with algebraic (or integer, rational, real, complex) coefficients such that*

$$f(X_1, \ldots, X_n) = H(e_1(X_1, \ldots, X_n), \ldots, e_n(X_1, \ldots, X_n)).$$

Every symmetric polynomial is therefore uniquely a sum of multiples of products of elementary symmetric functions. The reversal is much more trivial: every polynomial expression in the elementary symmetric functions is again symmetric.

Every symmetric polynomial can be uniquely represented as a linear combination of products of elementary symmetric polynomials.

Proof Let d be a natural number with $\deg f(X_1, \ldots, X_n) \leq d$. In addition to the statement of Theorem 3.3, we will also show that $\deg H(X, X^2, \ldots, X^n) \leq d$.

We prove this by induction over n. In the case of $n = 0$, there is nothing to show, as f must necessarily be a constant and we can set $H = f$.

In the case of $n > 0$, we perform another induction for fixed n, this time over d. In the case of $d = 0$, there is nothing to show, as f must then again be a constant. So let $d > 0$. By induction hypothesis (of the induction over n), there exists due to $e_i(X_1, \ldots, X_{n-1}, 0) = e_i(X_1, \ldots, X_{n-1})$ for $i < n$ a polynomial $H_1(Y_1, \ldots, Y_{n-1})$ with

$$f(X_1, \ldots, X_{n-1}, 0) = H_1(e_1(X_1, \ldots, X_{n-1}, 0), \ldots, e_{n-1}(X_1, \ldots, X_{n-1}, 0)),$$

and it holds that $\deg H_1(X, X^2, \ldots, X^n) \leq d$. We set

$$f_1(X_1, \ldots, X_n) \coloneqq f(X_1, \ldots, X_n) - H_1(e_1(X_1, \ldots, X_n), \ldots, e_n(X_1, \ldots, X_n)).$$

Then $f_1(X_1, \ldots, X_n)$ is a symmetric polynomial with $\deg f_1(X_1, \ldots, X_n) \leq d$. If we set $X_n = 0$ in this definition, we get $f_1(X_1, \ldots, X_{n-1}, 0) = 0$, that is $f_1(X_1, \ldots, X_n)$ is divisible by X_n. But since $f_1(X_1, \ldots, X_n)$ is symmetric in $X_1, \ldots, X_n$, $f_1(X_1, \ldots, X_n)$ is also divisible by $X_1, \ldots, X_{n-1}$, hence by the expression $e_n(X_1, \ldots, X_n) = X_1 \cdots X_n$.

There exists therefore a polynomial $f_2(X_1, \ldots, X_n)$ with

$$f_1(X_1, \ldots, X_n) = e_n(X_1, \ldots, X_n) \cdot f_2(X_1, \ldots, X_n);$$

$f_2(X_1, \ldots, X_n)$ is a symmetric polynomial with $\deg f_2(X_1, \ldots, X_n) \leq d - n < d$.

By induction hypothesis (of the induction over d), there exists a polynomial $H_2(Y_1, \ldots, Y_n)$ with $\deg H_2(X, X^2, \ldots, X^{n-1}) \leq d - n$ such that

$$f_2(X_1, \ldots, X_n) = H_2(e_1(X_1, \ldots, X_n), \ldots, e_n(X_1, \ldots, X_n)),$$

thus

$$f(X_1, \ldots, X_n) = H(e_1(X_1, \ldots, X_n), \ldots, e_n(X_1, \ldots, X_n))$$

if we set $H(Y_1, \ldots, Y_n) = H_1(Y_1, \ldots, Y_{n-1}) + Y_n \cdot H_2(Y_1, \ldots, Y_n)$. Moreover, $\deg H(X, X^2, \ldots, X^n) \leq d$.

It remains to show the uniqueness of H. For this purpose, let $\tilde{H}(Y_1, \ldots, Y_n)$ be another polynomial with $f(X_1, \ldots, X_n) = \tilde{H}(e_1(X_1, \ldots, X_n), \ldots, e_n(X_1, \ldots, X_n))$, that is

$$(H - \tilde{H})(e_1(X_1, \ldots, X_n), \ldots, e_n(X_1, \ldots, X_n)) = 0.$$

Then it follows from the following Lemma 3.2, that $H - \tilde{H} = 0$, so $H = \tilde{H}$. □

Lemma 3.2 *For every natural number n the elementary symmetric functions $e_1(X_1, \ldots, X_n)$, $\ldots$, $e_n(X_1, \ldots, X_n)$ are* algebraically independent*, that is, if $H(Y_1, \ldots, Y_n)$ is a polynomial with*

$$H(e_1(X_1, \ldots, X_n), \ldots, e_n(X_1, \ldots, X_n)) = 0, \tag{3.21}$$

then it follows that $H = 0$.

Proof We conduct the proof—just like the proof of Theorem 3.3—by induction over n and d, where d is a natural number with $\deg H(X, X^2, \ldots, X^n) \leq d$.

In the case of $n = 0$, there is nothing to show. In the case of $n > 0$, we first consider the case $d = 0$, the induction start. Because $\deg H(X, X^2, \ldots, X^n) \leq d = 0$, we have that $H(Y_1, \ldots, Y_n)$ must be a constant, so that $H(Y_1, \ldots, Y_n) = 0$ follows directly from (3.21).

It remains to consider $d > 0$: We write $H(Y_1, \ldots, Y_n)$ in the form

$$H(Y_1, \ldots, Y_n) = H_0(Y_1, \ldots, Y_{n-1}) + Y_n \cdot H_1(Y_1, \ldots, Y_{n-1}, Y_n) \tag{3.22}$$

with uniquely determined polynomials $H_0(Y_1, \ldots, Y_{n-1})$ and $H_1(Y_1, \ldots, Y_n)$. In particular, $H_0(Y_1, \ldots, Y_{n-1}) = H(Y_1, \ldots, Y_{n-1}, 0)$.

If we substitute $Y_i = e_i(X_1, \ldots, X_n)$, we get

$$\begin{aligned} 0 &= H(e_1(X_1, \ldots, X_n), \ldots, e_n(X_1, \ldots, X_n)) \\ &= H_0(e_1(X_1, \ldots, X_n), \ldots, e_{n-1}(X_1, \ldots, X_n)) \\ &+ e_n(X_1, \ldots, X_n) \cdot H_1(e_1(X_1, \ldots, X_n), \ldots, e_n(X_1, \ldots, X_n)). \end{aligned} \tag{3.23}$$

We recall that $e_i(X_1, \ldots, X_{n-1}, 0) = e_i(X_1, \ldots, X_{n-1})$ for $i < n$ and also $e_n(X_1, \ldots, X_{n-1}, 0) = 0$.

By evaluating (3.23) at $X_n = 0$, we thus obtain

$$H_0(e_1(X_1, \ldots, X_{n-1}), \ldots, e_{n-1}(X_1, \ldots, X_{n-1})) = 0.$$

According to the induction hypothesis (the induction over n), this means $H_0(Y_1, \ldots, Y_{n-1}) = 0$. The Eq. (3.23) thus becomes

$$0 = e_n(X_1, \ldots, X_n) \cdot H_1(e_1(X_1, \ldots, X_n), \ldots, e_n(X_1, \ldots, X_n)).$$

We are allowed to cancel and conclude

$$0 = H_1(e_1(X_1, \ldots, X_n), \ldots, e_n(X_1, \ldots, X_n)).$$

Because $\deg H_1(X, X^2, \ldots, X_n) \leq \deg H(X, X^2, \ldots, X^n) - n < d$ we can apply the induction hypothesis (of the induction over d) to the polynomial $H_1(Y_1, \ldots, Y_n)$ and obtain $H_1(Y_1, \ldots, Y_n) = 0$; all in all due to (3.22) and $H_0(Y_1, \ldots, Y_{n-1}) = 0$ thus $H(Y_1, \ldots, Y_n) = 0$. □

Next, we want to combine Theorem 3.2 with Theorem 3.3: Let

$$X^n + a_{n-1}X^{n-1} + \cdots + a_1 X + a_0 = 0 \tag{3.24}$$

be a normalized polynomial equation (with algebraic coefficients) whose solutions (with multiplicities) are given by $x_1, \ldots, x_n$. If $f(X_1, \ldots, X_n)$ is a symmetric polynomial, then

there exists a polynomial $H(Y_0, \ldots, Y_{n-1})$ with

$$f(x_1, \ldots, x_n) = H(a_0, \ldots, a_{n-1}).$$

We have thus shown that every symmetric function in the solutions of a general normalized polynomial equation is a polynomial in the coefficients of this equation. We will illustrate this with an important example:

The expression

$$(-1)^{\binom{n}{2}} \prod_{i \neq j} (x_i - x_j)$$

is obviously symmetric in the solutions $x_1, \ldots, x_n$. Thus, there exists exactly one polynomial Δ in the coefficients $a_0, \ldots, a_{n-1}$ with

$$\Delta = (-1)^{\binom{n}{2}} \prod_{i \neq j} (x_i - x_j).$$

The expression Δ is called the *discriminant* of the Eq. (3.24) or discriminant of the polynomial $f(X) = X^n + a_{n-1}X^{n-1} + \cdots + a_1 X + a_0$.

Since the product of the pairwise differences of the solutions is zero if and only if the equation has a multiple solution, we can infer an important property of the discriminant Δ: The discriminant of a normalized polynomial equation is zero if and only if the polynomial equation has multiple solutions. Since Δ is a polynomial in the *known* coefficients of the Eq. (3.24), we do not need to know the solutions to decide whether there are only simple or also multiple solutions!

The discriminant is a specific polynomial in the coefficients of an equation. The discriminant vanishes if and only if multiple solutions exist. Since the discriminant can be determined without knowledge of the solutions, we can determine the presence of solutions with multiplicity without calculating the solutions themselves.

We illustrate this with quadratic and cubic equations:

Example 3.1 Let

$$X^2 + pX + q = 0$$

be a normalized quadratic equation (with algebraic coefficients) with solutions x_1 and x_2. According to Theorem 3.2, $p = -(x_1 + x_2)$ and $q = x_1 x_2$. The discriminant is

$$\Delta = -(x_1 - x_2)(x_2 - x_1) = (x_1 - x_2)^2,$$

symmetric in x_1 and x_2 and therefore a polynomial in p and q: If we expand the discriminant, we get

$$\Delta = x_1^2 - 2x_1x_2 + x_2^2 = (x_1 + x_2)^2 - 4x_1x_2 = p^2 - 4q.$$

A quadratic equation therefore has exactly two different solutions if and only if its coefficients satisfy the inequality $p^2 - 4q \neq 0$—a fact that follows directly from the validity of the well-known solution formula for quadratic equations.

Example 3.2 Let

$$X^3 + pX + q = 0$$

be a depressed cubic equation (with algebraic coefficients) whose solutions are x_1, x_2 and x_3. According to Theorem 3.2, $x_1+x_2+x_3 = 0$, $x_1x_2+x_2x_3+x_1x_3 = p$ and $-x_1x_2x_3 = q$.

The discriminant of the equation is

$$\Delta = (x_1 - x_2)^2(x_1 - x_3)^2(x_2 - x_3)^2. \tag{3.25}$$

According to Theorem 3.3, there is a universal polynomial in p and q that represents Δ. And indeed, a longer calculation shows that

$$\Delta = -4p^3 - 27q^2. \tag{3.26}$$

(This postulated equality can be systematically checked by expressing p and q on the right side of (3.26) through x_1, x_2, x_3 and then in both the right side of (3.26) and (3.25) the solution x_3 by means of the relationship $x_3 = -x_1 - x_2$ is expressed through x_1 and x_2. Afterwards, everything is to be expanded, and the results are to be compared.

However, with a trick, it is possible to reach the goal faster: We know that Δ is of degree 6 in x_1, x_2 and x_3. Thus, Δ, as a polynomial in p and q, must be a linear combination of terms of the form p^3 and q^2—all other products made from p and q have the wrong degree in x_1, x_2 and x_3. The coefficients of this linear combination can be determined using concrete example polynomials.)

Although we have not yet developed a formula to solve (depressed) cubic equations (apart from Theorem 2.2), we can already decide based on the coefficients p and q whether such an equation has a multiple solution or not.

3.4 Transcendence of π and the Impossibility of Squaring the Circle

Coming towards the end of this chapter, we want to use the results obtained so far to solve a first classical problem of geometry, the problem of the *squaring of the circle*: Given a circle, construct a square with the same area as the circle using only a compass and straightedge. If the radius of the circle is r, then the area of the circle is known to be πr^2. If the side length of the square to be constructed is a, then its area is a^2.

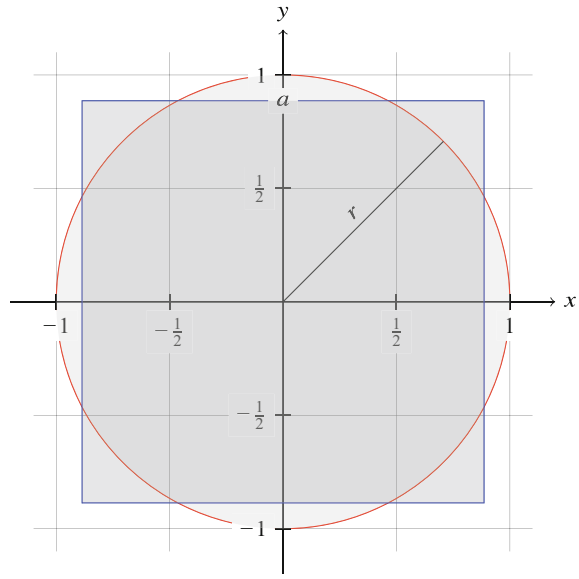
> The squaring of the circle means determining the side length a of a square from the radius r of a circle with compass and straightedge, such that the square has the same area as the circle.

Since a square can always be constructed for a given side, the task is therefore equivalent to constructing a line of length a from a line of length r using only a compass and straightedge, such that $\pi r^2 = a^2$, i.e.,

$$a = r\sqrt{\pi}.$$

We can identify the Gaussian number plane with the drawing plane as in Fig. 3.1 so that a radius of the circle is the line from 0 to 1 in the Gaussian number plane, i.e., $r = 1$. The task is then to construct a point with distance $\sqrt{\pi}$ from the origin using a

Fig. 3.1 Squaring the circle



compass and straightedge. After the problem was posed in antiquity in the fifth century BC, mathematicians in the following centuries were neither able to provide a construction rule, nor to show that such a construction is fundamentally impossible. Only by the end of the nineteenth century, the following theorem was known:

Theorem 3.4 *The squaring of the circle is impossible with compass and straightedge.*

Proof If the squaring of the circle were possible, the construction of the number $\sqrt{\pi}$ would also be possible. To show Theorem 3.4, one must show that $\sqrt{\pi}$ is not a constructible number. In fact, we even show that $\sqrt{\pi}$ is not an algebraic number: If $\sqrt{\pi}$ were an algebraic number, then its square π would also be an algebraic number. However, this is not the case, as follows from the following theorem. □

Theorem 3.5 (Carl Louis Ferdinand von Lindemann,[3] 1882) *The number π is a transcendental number.*

Before this theorem was proven, it was only known since 1761 by Johann Heinrich Lambert[4] that π is an irrational number. But since many irrational numbers can be constructed with compass and straightedge—for example, we have learned about $\sqrt{2}$—,this did not say anything about the possibility of the squaring of the circle.

The number π (and thus also $\sqrt{\pi}$) is transcendental and thus not algebraic. Therefore, π and $\sqrt{\pi}$ are not constructible numbers because every constructible number is algebraic. Therefore, the squaring of the circle is impossible.

Since we want the proof to be as elementary as possible with the tools introduced so far, we will prepare it with two auxiliary lemmas. For this purpose, we use (only in this section) the following notations: Let $P(X)$ be a polynomial with complex coefficients. Then we set

$$P^*(X) := \sum_{k\geq 0} P^{(k)}(X).$$

Here, the infinite sum is effectively a finite one since $P^{(k)}$ can only be unequal to zero if $\deg P \leq k$.

For $P(X) = X^\ell$, for example, we have that

$$P^*(X) = \ell! \sum_{m=0}^{\ell} \frac{X^m}{m!}.$$

If $P(X) = a_n X^n + a_{n-1} X^{n-1} + \cdots + a_1 X + a_0$, we define, for every complex number x, the expression

$$\overline{P}(x) \coloneqq |a_n| \cdot |x|^n + |a_{n-1}| \cdot |x|^{n-1} + \cdots + |a_1| \cdot |x| + |x_0|.$$

According to the triangle inequality, $|P(x)| \leq \overline{P}(x)$. If $Q(X)$ is another polynomial, the triangle inequality also implies that $\overline{P \cdot Q}(x) \leq \overline{P}(x) \cdot \overline{Q}(x)$.

This brings us to the first lemma:

Lemma 3.3 *Let x be a complex number, t an integer, $Q(X)$ a polynomial and p a natural number. Then*

$$\left(\frac{1}{(p-1)!} \cdot Q(tX) \cdot (tX - tx)^p \right)^* (x) = p \cdot R(tx)$$

for a polynomial $R(X)$, whose coefficients are integer polynomial expressions in the coefficients of $Q(X)$.

Proof of Lemma 3.3 It is

$$\begin{aligned}
&\left(\frac{1}{(p-1)!} \cdot Q(tX) \cdot (tX - tx)^p \right)^* (x) \\
&= \frac{1}{(p-1)!} \sum_{\ell=0}^{\infty} (Q(tX) \cdot (tX - tx)^p)^{(\ell)}(x) \\
&= \frac{1}{(p-1)!} \sum_{k=0}^{\infty} \sum_{q=0}^{\infty} \binom{k+q}{q} \cdot t^k Q^{(k)}(tx) \cdot ((tX - tx)^p)^{(q)}(x) \\
&= p \sum_{k=0}^{\infty} \binom{k+p}{p} \cdot t^{k+p} \cdot Q^{(k)}(tx),
\end{aligned}$$

and the right side is p times a polynomial expression, whose coefficients are integer polynomial in the coefficients of $Q(X)$. □

The second lemma is:

Lemma 3.4 *Let $P(X)$ be a polynomial with complex coefficients and x a complex constant. Then*

$$|P^*(x) - \mathrm{e}^x P^*(0)| \leq |x| \mathrm{e}^{|x|} \overline{P}(x).$$

Proof of Lemma 3.4 Due to the linearity of $P^*(X)$ in $P(X)$ and the triangle inequality, it is sufficient to verify the inequality to be proven for the case $P(X) = X^\ell/\ell!$. In this case,

$$P^*(X) = \sum_{m=0}^{\ell} \frac{X^m}{m!},$$

in particular $P^*(0) = 1$ and thus

$$\begin{aligned} |P^*(x) - \mathrm{e}^x P^*(0)| &= \left| \sum_{m=0}^{\ell} \frac{x^m}{m!} - \sum_{m\geq 0} \frac{x^m}{m!} \right| = \left| - \sum_{m\geq 0} \frac{x^{m+\ell+1}}{(m+\ell+1)!} \right| \\ &\leq \sum_{m\geq 0} \frac{|x|^{m+\ell+1}}{(m+\ell+1)!}, \end{aligned}$$

where the last inequality follows from the triangle inequality.

The last line can be further estimated due to $m!\,\ell! = m!\,\ell!\,1! \leq (m+\ell+1)!$ as follows:

$$\sum_{m\geq 0} \frac{|x|^{m+\ell+1}}{(m+\ell+1)!}, \leq \sum_{m\geq 0} \frac{|x||x|^m|x|^\ell}{m!\,\ell!} = |x|\mathrm{e}^{|x|}\frac{|x|^\ell}{\ell!} = |x|\mathrm{e}^{|x|}\overline{P}(x),$$

with which everything is proven. □

Proof of Theorem 3.5 [5] Let $f(X)$ be a polynomial with rational coefficients such that $f(\pi) = 0$. We need to show that $f(X) = 0$. To do this, we assume that $f(X) \neq 0$, and want to lead this to a contradiction.

We can assume that f is a normalized polynomial. The polynomial

$$g(X) := f(\mathrm{i}X)\, f(-\mathrm{i}X)$$

has at least $\pi\,\mathrm{i}$ and $-\pi\,\mathrm{i}$ as roots and a priori coefficients of the form $a + b\,\mathrm{i}$ with $a, b \in \mathbf{Q}$. Since the complex conjugate polynomial of $g(X)$ coincides with $g(X)$ due to

$$\overline{g(X)} = \overline{f(\mathrm{i}X) f(-\mathrm{i}X)} = f(-\mathrm{i}X) f(\mathrm{i}X) = g(X),$$

the coefficients of $g(X)$ are actually rational numbers.

Let $x_1, \ldots, x_m$ be the roots (with multiplicities) of $g(X)$, that is

$$g(X) = (X - x_1) \cdots (X - x_m).$$

We can assume that $x_1 = \pi\mathrm{i}$ and $x_2 = -\pi\,\mathrm{i}$.

Since $1 + e^{x_1} = 1 + e^{\pi i} = 1 + (-1) = 0$, in fact

$$(1 + e^{x_1}) \cdots (1 + e^{x_m}) = 0.$$

If we expand the left side using the formula $e^x\, e^{x'} = e^{x+x'}$, we get

$$1 + \sum_{j_1} e^{x_{j_1}} + \sum_{j_1 < j_2} e^{x_{j_1} + x_{j_2}} + \sum_{j_1 < j_2 < j_3} e^{x_{j_1} + x_{j_2} + x_{j_3}} + \cdots + e^{x_1 + \cdots + x_m} = 0, \tag{3.27}$$

where each summation indices runs through the values $1, \ldots, m$.

The coefficients of the polynomial

$$h(X) := \prod_{r=0}^{m} \prod_{j_1 < \cdots < j_r} (X - x_{j_1} - \cdots - x_{j_r})$$

are symmetric in the x_j. According to Theorem 3.3 they are therefore polynomials in the elementary symmetric functions of the x_j. According to Theorem 3.2, the elementary symmetric functions in the x_j are in turn, up to sign, the coefficients of the polynomial $g(X)$, which are all rational numbers. It follows that $h(X)$ is a normalized polynomial with rational coefficients.

By construction, the roots (with multiplicity) of $h(X)$ are just all finite sums of the form $x_{j_1} + \cdots + x_{j_r}$ with $1 \le j_1 < \cdots < j_r \le m$ and $0 \le r \le m$. We can therefore write the polynomial in the form

$$h(X) = c\, X^q\, t(X)$$

where c is a rational number, q is a natural number and $t(X) = t_n X^n + \cdots + t_1 X + t_0$ is a polynomial with integer coefficients and $t_n, t_0 \neq 0$. Since in $h(X)$ a factor corresponding to the empty sum 0 (the factor to $r = 0$) occurs, we have $q \ge 1$.

If $y_1, \ldots, y_n$ are the roots (with multiplicity) of $t(X)$, we can therefore write (3.27) in the form

$$0 = \sum_{i=1}^{q} e^0 + \sum_{i=1}^{n} e^{y_i} = q + \sum_{i=1}^{n} e^{y_i}$$

so $q = -\sum_{i=1}^{n} e^{y_i}$. Due to Theorem 3.1, we have

$$t(X) = t_n\, (X - y_1)\, (X - y_2) \cdots (X - y_n). \tag{3.28}$$

We denote by y the maximum of the real numbers $|y_1|, \ldots, |y_n|$. Let p be a natural number that will be chosen later. We then set

$$P(X) := \frac{t_n^{np-1} X^{p-1} t(X)^p}{(p-1)!}.$$

According to Lemma 3.4,

$$\begin{aligned} \left| q P^*(0) + \sum_{i=1}^{n} P^*(y_i) \right| &= \left| \sum_{i=1}^{n} \left(P^*(y_i) - \mathrm{e}^{y_i} P^*(0) \right) \right| \\ &\le \sum_{i=1}^{n} |y_i| \mathrm{e}^{y_i} \overline{P}(y_i) \le n y \mathrm{e}^{y} \overline{P}(y). \end{aligned} \tag{3.29}$$

According to the triangle inequality,

$$\overline{P}(y) \le \frac{|t_n|^{np-1} |y|^{p-1} \overline{t}(y)^p}{(p-1)!}.$$

We denote the right side of this inequality with a_p. There exists a $C < 1$ with

$$\frac{a_{p+1}}{a_p} = \frac{|t_n|^n |y| \overline{t}(y)}{p} < C$$

for sufficiently large p. This implies the existence of a real number A with $\overline{P}(y) \le A\, C^p$ for large p. From this and from (3.29), it follows that

$$\left| q P^*(0) + \sum_{i=1}^{n} P^*(y_i) \right| < 1 \tag{3.30}$$

for sufficiently large p. We will lead this to a contradiction by showing for a suitable choice of p that the left side is a non-vanishing integer.

We write $P(X)$ in the form

$$P(X) = \sum_{\ell \ge 0} \frac{g_\ell}{\ell!} X^\ell,$$

so

$$g_\ell = \begin{cases} 0 & \text{for } \ell < p-1, \\ t_n^{np-1} t_0^p & \text{for } \ell = p-1 \text{ and} \\ p\, \tilde{g}_\ell & \text{for } \ell \ge p \text{ and an integer } \tilde{g}_\ell. \end{cases}$$

If we choose for p a sufficiently large prime number with $p > q$, $p > |t_0|$, $p > |t_n|$ (at this point the theorem about the infinity of prime numbers obviously comes in), we have that

$$q\,P^*(0) = q\sum_{\ell\geq 0} g_\ell = q t_n^{np-1} t_0^p + qp\sum_{\ell\geq p}\tilde{g}_\ell$$

is an integer, not divisible by p. Thus, it is enough to show, that the sum $\sum_{i=1}^n P^*(y_i)$ is an integer divisible by p, because then the left side of (3.30) must be a non-vanishing integer, which would provide the contradiction.

Let an $i \in \{1,\ldots,n\}$ be fixed. Due to (3.28), it follows that

$$\begin{aligned} P(X) &= \frac{t_n^{np-1} X^{p-1}\, t(X)^p}{(p-1)!} \\ &= \frac{t_n^{np-1} X^{p-1}\cdot t_n^p\cdot (X-y_1)^p\cdots(X-y_n)^p}{(p-1)!} \\ &= \frac{(t_n X)^{p-1}\cdot (t_n X - t_n y_1)^p\cdots(t_n X - t_n y_n)^p}{(p-1)!}. \end{aligned}$$

According to Lemma 3.3, it is

$$P^*(y_i) = p\cdot D_i(t_n y_1,\ldots,t_n y_n),$$

where $D_i(Y_1,\ldots,Y_n)$ is a universal polynomial with integer coefficients that is symmetric in the variables $Y_1,\ldots,Y_{i-1},Y_{i+1},\ldots,Y_n$.

Thus, it is

$$\sum_{i=1}^n P^*(y_i) = p\sum_{i=1}^n D_i(t_n y_1,\ldots,t_n y_n) =: p\,D(t_n y_1,\ldots,t_n y_n). \tag{3.31}$$

Here, $D(Y_1,\ldots,Y_n)$ is a polynomial symmetric in $Y_1,\ldots,Y_n$ with integer coefficients. Thus, $D(t_n y_1,\ldots,t_n y_n)$ can be written as a polynomial with integer coefficients in the elementary symmetric functions of the $t_n y_i$ according to Theorem 3.3.

We recall that the y_i are the roots of the polynomial $t(X) = t_n X^n + t_{n-1}X^{n-1} + \ldots + t_1 X + t_0$. It follows that $t_n y_1,\ldots,t_n y_n$ are precisely the roots of the polynomial

$$s(X) = X^n + t_n t_{n-1} X^{n-1} + \ldots + t_n^{n-1} t_1 X + t_n^n t_0$$

Since the polynomial $s(X)$ is normalized, the elementary symmetric functions in the $t_n y_i$ are precisely the coefficients of $s(X)$ up to sign, according to Theorem 3.2, and all of these are integers. Thus, $D(t_n y_1,\ldots,t_n y_n)$ is an integer expression. Consequently, the right (and

thus also the left) side of (3.31) is divisible by p, which was to be shown to bring about the contradiction. □

The proof for the transcendence of π crucially involves the fundamental theorem of elementary symmetric functions. We also use the infinity of the set of prime numbers.

Summary

- A **polynomial** is a formal expression in one or more variables over a **coefficient ring**.
- We can **evaluate** polynomials by substituting concrete numbers for the variable. A **root** of a polynomial $f(X)$ is a number x, for which the evaluation $f(x) = 0$ holds. A polynomial equation is nothing but the question of the roots of a given polynomial.
- Let $t_1, \ldots, t_n$ be algebraic numbers. A complex number z is called **polynomial** in $t_1, \ldots, t_n$, if there exists a polynomial $f(X_1, \ldots, X_n)$ with rational coefficients such that $z = f(t_1, \ldots, t_n)$, and **rational** in $t_1, \ldots, t_n$, if there exist two polynomials $p(X_1, \ldots, X_n)$ and $q(X, \ldots, X_n)$ with rational coefficients, such that $z = p(t_1, \ldots, t_n)/q(t_1, \ldots, t_n)$. It turns out that the terms polynomial and rational are equivalent.
- If $X_1, \ldots, X_n$ are indeterminates, there are certain polynomials in $X_1, \ldots, X_n$, the so-called **elementary symmetric functions** $e_1, \ldots, e_n$, which are symmetric in $X_1, \ldots, X_n$. Furthermore, every other **symmetric polynomial** can be uniquely written as a polynomial in the elementary symmetric functions.
- According to **Vieta's theorem**, the coefficients of a polynomial are, up to sign, the elementary symmetric functions evaluated on all zeros of the polynomial in $\overline{\mathbf{Q}}$. If the polynomial has rational coefficients, then all symmetric polynomials on the zeros take rational values.
- The **discriminant** of a polynomial $f(X)$ indicates whether this polynomial has a multiple root. The discriminant of $f(X)$ is a symmetric polynomial in the zeros of $f(X)$. According to Vieta's theorem, the discriminant is thus a polynomial in the coefficients of $f(X)$.

(continued)

- The number π, which satisfies the relationship $1 + e^{\pi i} = 0$, is not a solution of a polynomial equation with rational coefficients and is therefore not an algebraic number, but transcendental. The proof is achieved with the help of the theory of elementary symmetric functions and Vieta's theorem.
- Since every constructible complex number must be algebraic, the number π cannot be constructible. Thus, the **squaring of the circle** is impossible.

Exercises

Polynomials

3.1 Let $f(X)$ and $g(X)$ be two polynomials with $\deg f(X) \leq n$ and $\deg g(X) \leq m$. Show that $\deg(f(X) + g(X)) \leq \max\{n, m\}$ and $\deg(f(X) \cdot g(X)) \leq n + m$.

3.2 Give an example of a polynomial $f(X)$ and two algebraic numbers x and y such that $f(x \cdot y) \neq f(x) \cdot f(y)$.

3.3 Is $X + \sqrt{2}$ a divisor of $X^3 - 2X$?

3.4 A polynomial $f(X)$ with integer coefficients divides another polynomial $g(X)$ with integer coefficients. Show that for every integer n the integer $f(n)$ is a divisor of the integer $g(n)$.

3.5 Does the polynomial $X^7 + 11X^3 - 33X + 22$ have a divisor of the form $(X - a)(X - b)$, where a and b are rational numbers?

3.6 Let $p(X)$, $q(X)$, $f(X)$ and $g(X)$ be polynomials with algebraic coefficients. Let the polynomial $d(X)$ divide the two polynomials $f(X)$ and $g(X)$. Show that $d(X)$ also divides the polynomial $p(X) \cdot f(X) + q(X) \cdot g(X)$.

3.7 Let $f(X) = 3X^4 - X^3 + X^2 - X + 1$ and $g(X) = X^3 - 2X + 1$. Provide polynomials $q(X)$ and $r(X)$ such that $f(X) = q(X)g(X) + r(X)$, so that $q(X)$ and $r(X)$ are polynomials with rational coefficients and $\deg r(X) < \deg g(X)$.

3.8 To what extent can a polynomial in the two variables X and Y be considered as a polynomial in one variable Y, whose coefficients are polynomials in X? How can this statement be generalized to polynomials with more than two variables?

3.9 Write $\frac{1}{\sqrt{2}+5\sqrt{3}}$ as a polynomial expression with rational coefficients in $\sqrt{2}$ and $\sqrt{3}$.

3.10 Let t be a complex number such that $\mathbf{Q}[t] = \mathbf{Q}(t)$. Show that t is algebraic.

Vieta's Theorem

3.11 Provide a normalized polynomial equation of the third degree that has 1 as a double solution, 2 as a simple solution and no further solutions.

3.12 Why are both sides of (3.16) linear in $g(X)$ as well as in $h(X)$, and why is it therefore sufficient to check the Eq. (3.16) only for $g(X) = X^m$ and $h(X) = X^n$?

3.13 Let $f(X)$ be a polynomial with algebraic coefficients with $\deg f \leq n$ for a natural number n. Let $x_0, \ldots, x_n$ be pairwise different algebraic numbers with $f(x_0) = \cdots = f(x_n) = 0$. Show that $f(X)$ is the zero polynomial.

3.14 Let $f(X)$ and $g(X)$ be two polynomials with algebraic coefficients and $\deg f$, $\deg g \leq n$ for a natural number n. Let $x_0, \ldots, x_n$ be algebraic numbers with $f(x_0) = g(x_0), \ldots, f(x_n) = g(x_n)$. Show that $f(X) = g(X)$ holds in this case.

3.15 Let $x_0, \ldots, x_n$ be pairwise different algebraic numbers. Let $i \in \{0, \ldots, n\}$. Show that exactly one polynomial $f(X)$ of degree at most n with algebraic coefficients and

$$f(x_j) = \begin{cases} 1 & \text{for } j = i \text{ and} \\ 0 & \text{for } j \in \{0, \ldots, i-1, i+1, \ldots, n\} \end{cases}$$

exists.

3.16 Let $x_0, \ldots, x_n$ be pairwise different algebraic numbers. Let $y_0, \ldots, y_n$ be further algebraic numbers. Show that exactly one polynomial $f(X)$ of degree at most n with algebraic coefficients and $f(x_i) = y_i$ for $i \in \{0, \ldots, n\}$ exists.

3.17 Let $g(X)$ and $h(X)$ be two polynomials. Justify by induction why

$$(gh)^{(k)}(X) = \sum_{i+j=k} \binom{k}{i} g^{(i)}(X)\, h^{(j)}(X)$$

follows from Proposition 3.3.

3.18 Let $f(X)$ be a polynomial. Show that $f^{(n+1)} = 0 \iff \deg f \leq n$ for every natural number n.

3.19 Let $f(X)$ be a polynomial and x a complex number. Show that the expansion of $f(X)$ according to $X - x$ is given by the *Taylor's*[6] *formula*

$$f(X) = \sum_{k=0}^{\infty} \frac{f^{(k)}(x)}{k!} (X - x)^k.$$

3.20 Write down the second elementary symmetric function in the five variables X, Y, Z, U, and V explicitly.

3.21 Let $X^4 + a_3X^3 + a_2X^2 + a_1X + a_0 = 0$ be a normalized polynomial equation of fourth degree, whose solutions with multiplicities are x_1, x_2, x_3 and x_4. Express the coefficients a_0, a_1, a_2 and a_3 explicitly as polynomials in the x_i.

3.22 Use Vieta's formula for $n = 2$ to derive the well-known solution formula for normalized quadratic equations.

The Discriminant

3.23 Let $f(X, Y, Z, W) \coloneqq XY + ZW + XYZW$. How many 4-letter permutations σ are there such that $\sigma \cdot f = f$?

3.24 Show that $e_k(\underbrace{1, \ldots, 1}_{n}) = \binom{n}{k}$.

3.25 Write $X^2 + Y^2 + Z^2$ as a polynomial in the elementary symmetric functions in X, Y, and Z.

3.26 Write $X_1^2 + X_2^2 + \cdots + X_n^2$ as a polynomial in the elementary symmetric functions in $X_1, \ldots, X_n$.

3.27 Let $X^3 + pX + q = 0$ be a depressed cubic equation. Show that its discriminant is given by $-4p^3 - 27q^2$.

3.28 Let $X^3 + pX + q = 0$ be a depressed cubic equation with integer coefficients p and q. Show that the equation has three different solutions when q is odd.

3.29 Calculate the discriminant of a general cubic equation $X^3 + aX^2 + bX + c = 0$.

3.30 Show that $X^3 - 5X^2 + 3X + 9 = 0$ has at most two different solutions.

3.31 Show that the discriminant of a polynomial equation $X^n + a_{n-1}X^{n-1} + \cdots + a_1X + a_0 = 0$ with solutions $x_1, \ldots, x_n$ is given by the square of *Vandermonde's*[7] *determinant*

$$\det \begin{pmatrix} 1 & \ldots & 1 \\ x_1 & \ldots & x_n \\ \vdots & & \vdots \\ x_1^{n-1} & \ldots & x_n^{n-1} \end{pmatrix}.$$

3.32 Let $X^n + a_{n-1}X^{n-1} + \cdots + a_1X + a_0 = 0$ be a normalized polynomial equation with rational coefficients. Show that it has at least one non-real root if its discriminant is negative.

3.33 Let $f(X)$ and $g(X)$ be two normalized polynomials with rational coefficients and with roots (with multiplicities) $x_1, \ldots, x_n$ and $y_1, \ldots, y_m$, respectively. Show that the expression $R = \prod_{i,j}(x_i - y_j)$ is a polynomial in the elementary symmetric functions of the coefficients of $f(X)$ and the elementary symmetric functions of the coefficients of $g(X)$.

3.34 Let $X^2 + aX + b = 0$ and $X^2 + cX + d = 0$ be two quadratic equations. Provide a polynomial expression in a, b, c, and d that vanishes if and only if the two equations have a common solution.

Transcendence of π and the Impossibility of Squaring the Circle

3.35 Let (z_n) be a convergent sequence of complex numbers with limit z. If the z_n are algebraic, is z necessarily also algebraic?

3.36 Is $\sqrt[3]{\pi}$ an algebraic number? Is π^3 algebraic?

3.37 Is the following problem solvable? Given a circle, construct a regular triangle with the same area using only a compass and straightedge.

3.38 Give a sequence of pairwise different transcendental numbers.

3.39 Let n be a natural number. Provide an algorithm for finding a prime number $p > n$ and prove that it is correct.

3.40 Let $a_0, \ldots, a_m$ be complex numbers with $a_m \mathrm{e}^m + a_{m-1}\mathrm{e}^{m-1} + \cdots + a_1 \mathrm{e} + a_0 = 0$. For a natural number p we further define the polynomial

$$P(X) = \frac{X^{p-1}(X-1)^p \cdots (X-m)^p}{(p-1)!}.$$

Show that the expression

$$\left|a_0 P^*(0) + a_1 P^*(1) + \cdots + a_m P^*(m)\right|$$

approaches zero for large p.

3.41 Show that e is a transcendental number, proceeding in a manner very similar to the proof of Theorem 3.5.

3.42 Let

$$x := \sum_{k=0}^{\infty} 10^{-k!}.$$

Show that x is a convergent sum, and give its first digits. The number x is called *Liouville's constant*.

3.43 Show using elementary methods that Liouville's constant is a transcendental number.

Endnotes

[1] The reason why we define *the* degree only for normalized polynomials is that it generally cannot be decided constructively for general a_n (from any coefficient domain) whether $a_n = 0$ or $a_n \neq 0$. Constructively, the degree for an arbitrary, non-normalized, polynomial is therefore not always well-defined. On the other hand, $\deg f \leq n$ is a valid judgment.

[2] François Viète, 1540–1603, French lawyer and mathematician.

[3] Carl Louis Ferdinand von Lindemann, 1852–1939, German mathematician.

[4] Johann Heinrich Lambert, 1728–1777, Swiss mathematician, physicist and astronomer.

[5] We essentially follow Oskar Perron's (1880–1975, German mathematician) proof in [1].

[6] Brooke Taylor, 1685–1731, British mathematician.

[7] Alexandre-Théophile Vandermonde, 1735–1796, French mathematician, musician, and chemist.

Reference

1. Perron, O.: Irrationalzahlen. Walter de Gruyter, Berlin (1960)

On the Impossibility of Doubling the Cube and Trisecting the Angle

4

There is a natural order in the domain of algebraic numbers.

Abstract

As we saw in the last chapter, we can ask for factorizations of polynomials (into polynomials of lower degree), just as we can ask for possible factorizations of integers.

In this context, we call a polynomial *irreducible* if it does not allow such a factorization. In other words, the irreducible polynomials play the role of prime numbers in the ring of polynomials. Every linear polynomial $X - a$ must be irreducible, because already for reasons of degree it cannot possess a factorization into polynomials of lower degree. Due to the fundamental theorem of algebra, the linear polynomials are in turn the only irreducible ones, if we assume the algebraic numbers as the ring of coefficients.

Over the rational numbers, the theory is more complicated, but also more interesting. Here, there are non-irreducible polynomials of higher degree like $X^2 - 1 = (X - 1) \cdot (X + 1)$, as well as irreducible ones like $X^2 + 1$.

Given this, a natural question is how to determine whether a polynomial, say over the rational numbers, is irreducible. To this end, we provide a numerical method by which we can definitely determine whether such a polynomial is irreducible or not. This implies that every polynomial has an essentially unique decomposition into a product of irreducible polynomials, comparable to the prime factor decomposition of integers.

Even though the numerical method we provide always works, there are simpler criteria for irreducibility for many cases. We present the Eisenstein criterion and another method that is based on irreducibility modulo a prime number. In particular,

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M. Nieper-Wißkirchen, *Elementary Galois Theory*,
Mathematics Study Resources 3, https://doi.org/10.1007/978-3-662-66643-2_4

we compare the irreducibility of polynomials with rational coefficients with the irreducibility of polynomials with integer coefficients.

An algebraic number z is defined as a complex number that is a root of a polynomial with rational coefficients. It turns out that among these there is exactly one irreducible normalized one, the so-called *minimal polynomial* of z. We simply call the degree of this minimal polynomial the *degree* of z.

With the help of this degree concept, we finally show at the end of the chapter that neither the doubling of the cube nor the trisection of the angle is possible with compass and straightedge alone.

4.1 Separability

In this section, we want to show that we can rewrite any polynomial equation (with algebraic coefficients) into one that has the same pairwise different solutions as the original equation, but all with multiplicity one only. Since the rewritten equation usually has a smaller degree than the original equation, we can consider it as a simplification of the original equation, at least if we are only interested in the pairwise different solutions, but not in their multiplicities.

To get there, we need the concept of the greatest common divisor of two polynomials. We already know this concept from the ring of integers: For any two given integers n and m there always exists a positive integer d that divides the numbers n and m, so that any further common divisor of n and m is also a divisor of d. Division with remainder for polynomials gives us a corresponding statement for normalized polynomials:

Proposition 4.1 *Let $f(X)$ and $g(X)$ be two polynomials with algebraic coefficients. Then there exists a normalized polynomial $d(X)$ with algebraic coefficients (where we explicitly allow $d(X) = 0$) that divides both $f(X)$ and $g(X)$ and for which polynomials $p(X)$ and $q(X)$ with*

$$d(X) = p(X) \cdot f(X) + q(X) \cdot g(X) \tag{4.1}$$

exist.

From (4.1) it follows that every (normalized) polynomial, which is a divisor of $f(X)$ and $g(X)$, must also be a divisor of $d(X)$, that is, the polynomial $d(X)$, whose existence is given by Proposition 4.1, is a greatest common divisor of $f(X)$ and $g(X)$. Furthermore, a greatest common divisor is uniquely determined. If the constant polynomial 1 is a greatest common divisor of $f(X)$ and $g(X)$, we call $f(X)$ and $g(X)$ *coprime*.

If $f(X)$ and $g(X)$ are two polynomials, they have a greatest common divisor $d(X)$, a uniquely determined normalized polynomial (where $d(X) = 0$ is allowed).

The method underlying the proof of Proposition 4.1 for the construction of $d(X)$ is the so-called Euclidean algorithm:

Proof We can assume without loss of generality that $\deg f(X) \geq \deg g(X)$. Let n be a natural number with $\deg g < n$. We do the proof by induction over n. In the case of $n = 0$, necessarily $g(X) = 0$, so we can set $d(X) := f(X) = 1 \cdot f(X) + 0 \cdot g(X)$.

It remains to show the induction step for $n > 0$. We can therefore assume $g(X) \neq 0$, that is, we have $g(X) = c\tilde{g}(X)$ with a normalized polynomial $\tilde{g}(X)$ and an algebraic number $c \neq 0$.

We perform division with remainder and obtain

$$f(X) = h(X) \cdot \tilde{g}(X) + r(X) \tag{4.2}$$

for two polynomials $h(X)$ and $r(X)$ with $\deg r(X) < \deg \tilde{g}(X) < n$. We can apply the induction hypothesis to the pair $\tilde{g}(X)$ and $r(X)$ and obtain a common divisor $d(X)$ of $\tilde{g}(X)$ and $r(X)$ such that polynomials $\tilde{q}(X)$ and $p(X)$ with

$$d(X) = \tilde{q}(X) \cdot \tilde{g}(X) + p(X) \cdot r(X) \tag{4.3}$$

exist.

Since $d(X)$ divides the right side of (4.2), $d(X)$ also divides the left side, that is, $d(X)$ is also a divisor of $f(X)$. Moreover, from (4.3) and (4.2) we obtain the representation

$$\begin{aligned} d(X) &= \tilde{q}(X) \cdot \tilde{g}(X) + p(X) \cdot r(X) \\ &= \tilde{q}(X) \cdot \tilde{g}(X) + p(X) \cdot (f(X) - h(X)\tilde{g}(X)) \\ &= p(X) \cdot f(X) + (\tilde{q}(X) - p(X)h(X)) \cdot \tilde{g}(X) \\ &= p(X) \cdot f(X) + c^{-1} (\tilde{q}(X) - h(X)) \cdot g(X) \end{aligned}$$

for $d(X)$, completing the proof (with the setting $q(X) := c^{-1} (\tilde{q}(X) - h(X))$). □

If we look closely at the proof of Proposition 4.1, we realize that it works just as well for polynomials with rational coefficients as it does with algebraic coefficients, that is, Proposition 4.1 remains correct if we replace "algebraic coefficients" with "rational coefficients". (In contrast, the statement for integer coefficients is incorrect because we must be able to divide by any number not equal to zero in the proof.) Furthermore, we

note that a greatest common divisor of two polynomials with rational coefficients is also a greatest common divisor in the ring of polynomials with algebraic coefficients, that is, when we consider the polynomials as polynomials with algebraic coefficients.

Example 4.1 We want to calculate the greatest common divisor of the polynomials

$$f(X) = X^3 - 2X^2 + 2X - 4$$

and

$$g(X) = X^2 - 3X + 2$$

and follow the proof of Proposition 4.1. First, we note that $f(X)$ and $g(X)$ are normalized and that $\deg f(X) > \deg g(X)$. We perform division with remainder and obtain

$$X^3 - 2X^2 + 2X - 4 = (X + 1) \cdot (X^2 - 3X + 2) + 3X - 6.$$

The remainder is therefore $r(X) = 3X - 6 = 3(X - 2)$. In the next step (the induction hypothesis in the above proof), we must perform division with remainder of $g(X)$ by $\tilde{r}(X) = X - 2$. In this case, the division without remainder is possible; indeed

$$X^2 - 3X + 2 = (X - 1) \cdot (X - 2).$$

Consequently, $\tilde{r}(X) = X - 2$ is the greatest common divisor of $g(X)$ and $r(X)$ and thus also of $f(X)$ and $g(X)$.

The Euclidean algorithm works as well for integers as it does for polynomials if we replace divisibility of polynomials by divisibility of integers. That is, we can formulate:

Proposition 4.2 *Let a and b be two integers. Then there exists an integer $d \geq 0$ that is a common divisor of a and b and for which further integers r and s with*

$$d = r \cdot a + s \cdot b$$

exist. □

We can calculate a greatest common divisor for integers as well as for polynomials using the Euclidean algorithm.

Another way to calculate the greatest common divisor of two normalized polynomials is by considering zeros: If $f(X)$ and $g(X)$ are two polynomials, we can write them according to Theorem 3.1 in the forms

$$f(X) = (X - y_1)^{d_1} \cdot (X - y_2)^{d_2} \dots (X - y_k)^{d_k}$$

and

$$g(X) = (X - y_1)^{e_1} \cdot (X - y_2)^{e_2} \cdots (X - y_k)^{e_k}$$

where $y_1, \dots, y_k$ are pairwise different algebraic numbers and $d_1, \dots, d_k$ and $e_1, \dots, e_k$ are natural numbers (some of which can also be zero). Let for each $j \in \{1, \dots, k\}$ the natural number b_j be the minimum of d_j and e_j. We claim that then

$$d(X) = (X - y_1)^{b_1} \cdot (X - y_2)^{b_2} \cdots (X - y_k)^{b_k}$$

is the greatest common divisor of $f(X)$ and $g(X)$: Since $b_j \leq d_j, e_j$, we already have that $d(X)$ is a divisor of both $f(X)$ and $g(X)$.

If $h(X)$ is another normalized polynomial that is a common divisor of $f(X)$ and $g(X)$, then $h(X)$ has at most the roots $y_1, \dots, y_k$, that is, we can write

$$h(X) = (X - y_1)^{f_1} \cdot (X - y_2)^{f_2} \dots (X - y_k)^{f_k},$$

according to Theorem 3.1, for certain natural numbers $f_1, \dots, f_k$.

Since $h(X)$ is divisible by $(X - y_1)^{f_1}$, $f(X)$ must also be divisible by $(X - y_1)^{f_1}$, so $f_1 \leq d_1$ must apply. Analogously follows $f_j \leq d_j, e_j$, so $f_j \leq b_j$ for $j \in \{1, \dots, k\}$. Thus, $h(X)$ is a divisor of $d(X)$. Thus, $d(X)$ is the greatest common divisor of $f(X)$ and $g(X)$.

The greatest common divisor of two normalized polynomials $f(X)$ and $g(X)$ can therefore be characterized as the normalized polynomial whose pairwise different roots are the common roots of $f(X)$ and $g(X)$, so that the multiplicity of a root x is the minimum of the multiplicity of x as a root of $f(X)$ or $g(X)$.

The greatest common divisor of two polynomials can also be determined by means of the factorization into linear factors.

We want to apply these findings to a normalized polynomial $f(X)$ and its derivative. So let $f(X)$ be a normalized polynomial of degree $n > 0$. Then $\frac{1}{n} f'(X)$ is again a normalized polynomial. As before, we write

$$f(X) = (X - y_1)^{e_1} \cdot (X - y_2)^{e_2} \cdots (X - y_k)^{e_k}$$

with pairwise different algebraic numbers $y_1, \ldots, y_k$ and positive natural numbers $e_1, \ldots, e_k$, that is, $y_1, \ldots, y_k$ are the pairwise different roots of $f(X)$, whose respective multiplicities are given by $e_1, \ldots, e_k$.

According to Proposition 3.4, $e_1 - 1, \ldots, e_k - 1$ are the respective multiplicities of the roots $y_1, \ldots, y_k$ of $f'(X)$ (and thus also of $\frac{1}{n} f'(X)$). Consequently, the greatest common divisor of a polynomial $f(X)$ and its normalized formal derivative is

$$d(X) = (X - y_1)^{e_1 - 1} \cdot (X - y_2)^{e_2 - 1} \cdots (X - y_k)^{e_k - 1}. \tag{4.4}$$

Since $d(X)$ is a divisor of $f(X)$, there exists (exactly) one polynomial $f_0(X)$ with $f(X) = f_0(X)d(X)$. Looking at (4.4), we recognize that

$$f_0(X) = (X - y_1) \cdot (X - y_2) \cdots (X - y_k)$$

with pairwise different roots $y_1, \ldots, y_k$. We have thus constructed a polynomial that has the same distinct roots as $f(X)$, but all with multiplicity one, that is, $f_0(X)$ is separable according to the following definition:

Definition 4.1 Let $f(X)$ be a normalized polynomial with algebraic coefficients. The polynomial is called *separable* if all roots of $f(X)$ are simple roots.

Recalling the definition of the discrimant, we see that a polynomial is separable if and only if its discriminant does not vanish. If we are only interested in the pairwise different solutions of the equation

$$f(X) = 0,$$

we can instead solve the equation

$$f_0(X) = 0.$$

Let's also assume that $f(X)$ has not only algebraic, but even rational coefficients. What about the coefficients of $f_0(X)$? Since the determination of the greatest common divisor of two polynomials also works for polynomials with rational coefficients, the greatest common divisor $d(X)$ of $f(X)$ and $\frac{1}{n} f'(X)$ has rational coefficients.

We claim that from the equation $f(X) = f_0(X) \cdot d(X)$ it follows that $f_0(X)$ also has rational coefficients, and prove it as follows: First, we perform the division with remainder of polynomials with rational coefficients and obtain $f(X) = q(X) \cdot d(X) + r(X)$ with rational polynomials $q(X)$ and $r(X)$ and $\deg r(X) < \deg d(X)$. Since this representation is also unique for polynomials with algebraic coefficients and in this case we have $f(X) = f_0(X) \cdot d(X)$, it follows $r(X) = 0$ and $f_0(X) = q(X)$, in particular $f_0(X)$ therefore has rational coefficients.

(The generalization of this fact to several polynomials is: If the product of normalized polynomials with algebraic coefficients is a polynomial with rational coefficients and all factors except at most one have rational coefficients, then all factors have rational coefficients.)

We can thus transform any normalized polynomial equation with rational coefficients into an equivalent polynomial equation with rational coefficients whose solutions are all simple.

For every normalized polynomial equation, there is a unique normalized polynomial equation that has the same pairwise different solutions, but each with multiplicity 1.

Example 4.2 Let $f(X) = X^4 + 2X^2 + 1$. We want to calculate $f_0(X)$, i.e. a polynomial that has the same roots as $f(X)$, but with multiplicity one. For this, we form the normalized formal derivative

$$\frac{1}{4} f'(X) = X^3 + X = X \cdot (X^2 + 1).$$

The greatest common divisor of $f(X)$ and $\frac{1}{4} f'(X)$ is $d(X) = X^2 + 1$. Thus,

$$f_0 = \frac{X^4 + 2X^2 + 1}{X^2 + 1} = X^2 + 1.$$

The distinct roots of $f(X)$ are therefore the roots of $X^2 + 1$, i.e., i and $-$i.

4.2 Irreducible Polynomials

Consider a normalized polynomial $f(X)$ with rational coefficients and the associated polynomial equation

$$f(X) = 0. \tag{4.5}$$

Imagine that we express the polynomial $f(X)$ as a product $f(X) = g(X) \cdot h(X)$ of two normalized polynomials $g(X)$ and $h(X)$ of positive and degree and with rational coefficients. The Eq. (4.5) thus becomes

$$g(X) \cdot h(X) = 0.$$

Since the product of two algebraic numbers vanishes if and only if (at least) one of the factors vanishes, it follows that a solution of (4.5) is a solution of $g(X) = 0$ or of $h(X) = 0$ and conversely, every solution of $g(X) = 0$ or of $h(X) = 0$ is a solution of (4.5). It is to be expected that the equations $g(X) = 0$ and $h(X) = 0$ are easier to solve than (4.5), after all, the degree of the normalized polynomials $g(X)$ and $h(X)$ is strictly smaller than the degree of $f(X)$. In the following, we therefore want to study to what extent polynomials can be factored.

To get there, let's first consider a positive integer n. A *decomposition* of n is a product of the form $n = n_1 \cdots n_m$, where $n_1, \ldots, n_m$ are positive integers not equal to one. We consider two decompositions as equal if they only differ in the order of the factors. The number 12 allows for example the decompositions $12 = 2 \cdot 6$, $12 = 2 \cdot 2 \cdot 3$, $12 = 3 \cdot 4$ and $12 = 12$. The number 5 on the other hand only has the decomposition $5 = 5$. A positive integer p is obviously a prime number if and only if it allows only the decomposition $p = p$. It is therefore sensible to call a prime number an *irreducible* (positive integer) number. We can transfer the (un-)decomposability concept to normalized polynomials: A *decomposition* of a normalized polynomial $f(X)$ (with rational coefficients) is a product of the form $f(X) = f_1(X) \cdots f_m(X)$, where $f_1(X), \ldots, f_m(X)$ are non-constant normalized polynomials. Two decompositions that only differ in the order of the factors we again consider as equal. Then we can define:

Definition 4.2 A normalized polynomial $f(X)$ with rational coefficients is called *irreducible* (over the rational numbers) if it only allows the trivial decomposition $f(X) = f(X)$.

A polynomial over the rational numbers is irreducible if it is non-constant and if it cannot be written as a product of two non-constant polynomials over the rational numbers.

For example, the polynomial $X^2 - 1$ is not irreducible because we can write $X^2 - 1 = (X - 1) \cdot (X + 1)$. What about the polynomial $X^2 + 1$ on the other hand? Suppose it has a proper decomposition, that is $X^2 + 1 = f(X) \cdot g(X)$ for two non-constant normalized polynomials $f(X)$ and $g(X)$ with rational coefficients. Since we have $2 = \deg(X^2 + 1) = \deg f(X) + \deg g(X)$, $f(X)$ and $g(X)$ must both be polynomials of degree 1. Since they are

normalized, there exist rational numbers a and b with $f(X) = X - a$ and $g(X) = X - b$. Assuming that X^2+1 has a proper decomposition, we thus have $X^2+1 = (X-a)\cdot(X-b)$. The right-hand side has the rational numbers a and b as roots. The left-hand side, however, has no roots in the rational numbers, so we arrive at a contradiction. Thus, $X^2 + 1$ is irreducible.

By considering roots as in this example, we can for determine for each polynomial whether it is irreducible: Let $f(X)$ be a non-constant, normalized polynomial with rational coefficients, which we want to examine for irreducibility. Let $x_1, \ldots, x_n$ be its roots with multiplicities, so

$$f(X) = (X - x_1) \cdots (X - x_n). \tag{4.6}$$

(This is only a decomposition over the rational numbers if all roots x_1 to x_n are rational numbers. In general, it is just a decomposition over the algebraic numbers.)

Furthermore, let

$$f(X) = g(X) \cdot h(X), \tag{4.7}$$

where $g(X)$ and $h(X)$ are normalized polynomials with algebraic coefficients. Furthermore, assume that $g(X)$ has rational coefficients, which also means $h(X)$ has rational coefficients.

Let $y_1, \ldots, y_r$ be the roots with multiplicities of $g(X)$ and $z_1, \ldots, z_s$ be the roots with multiplicities of $h(X)$. Then we have

$$f(X) = g(X) \cdot h(X) = (X - y_1) \cdots (X - y_r) \cdot (X - z_1) \cdots (X - z_s).$$

Since this representation of $f(X)$ according to Theorem 3.1 is unique up to order, the comparison with (4.6), shows that we can choose the order of $x_1, \ldots, x_n$ such that $x_1 = y_1$, $\ldots, x_r = y_r, x_{r+1} = z_1, \ldots, x_n = z_s$. In other words, we have

$$g(X) = (X - x_1) \cdots (X - x_r), \qquad h(X) = (X - x_{r+1}) \cdots (X - x_n).$$

Consequently, every product representation of the form (4.7) corresponds to a partition of the roots of $f(X)$ (with multiplicities) into two parts $x_1, \ldots, x_r$ and $x_{r+1}, \ldots, x_n$ such that $(X - x_1) \cdots (X - x_r)$ (and thus also $(X - x_{r+1}) \cdots (X - x_n)$) is a polynomial with rational coefficients.

Let's call a subsequence $x_{i_1}, \ldots, x_{i_k}$, $1 \leq i_1 < i_2 < \cdots < i_k \leq n$, of the roots (with multiplicities) of $f(X)$ a *combination of roots with multiplicities* and such combination *proper*, if $k \notin \{0, n\}$, so we can formulate the following criterion for the irreducibility of a polynomial $f(X)$ with rational coefficients:

The polynomial $f(X)$ is irreducible if and only if there is no proper combination x_{i_1}, $\ldots, x_{i_k}$, $1 \le i_1 < i_2 < \cdots < i_k \le n$ of the roots (with multiplicities) of $f(X)$ such that

$$(X - x_{i_1}) \cdots (X - x_{i_k}) \tag{4.8}$$

is a polynomial with rational coefficients. Since according to Theorem 3.2 the coefficients of (4.8) up to sign are the elementary symmetric functions in the x_{i_j}, we can reformulate the criterion:

The polynomial $f(X)$ is irreducible if and only if there is no proper combination x_{i_1}, $\ldots, x_{i_k}$, $1 \le i_1 < i_2 < \cdots < i_k \le n$ of the roots (with multiplicities) of $f(X)$ such that the elementary symmetric functions

$$e_j(x_{i_1}, \ldots, x_{i_k}), \quad 1 \le j \le k, \tag{4.9}$$

evaluated in the x_{i_j} are all rational numbers.

Can we use this to decide whether a normalized polynomial over the rational numbers is irreducible? Is there an algorithm for this problem? Since the expressions (4.9) are always algebraic numbers, it would suffice to have a procedure that determines whether a given algebraic number is a rational number. In fact, we are going to give such a procedure:

Let x be an algebraic number that we want to test whether it is a rational number. If $x = 0$, then x is obviously rational, so we can assume $x \neq 0$. Since x is algebraic, there exist integers $b_0, \ldots, b_n$ with $b_n x^n + b_{n-1} x^{n-1} + \cdots + b_1 x + b_0 = 0$. We can assume that $b_n \neq 0$ (otherwise we simply omit the term $b_n x^n$). Similarly, we can assume that $b_0 \neq 0$ (otherwise we omit the term b_0 and divide by $x \neq 0$). If x is rational, the equation

$$b_n X^n + b_{n-1} X^{n-1} + \cdots + b_1 X + b_0 = 0 \tag{4.10}$$

must in particular have a rational solution, that is a solution of the form

$$y = \frac{p}{q} \tag{4.11}$$

with integers p and q, where $p \neq 0$ and $q > 0$.

We can further assume that the fraction p/q is fully reduced, that is, that p and q have no common prime divisor. If we then substitute y into (4.10) and multiply by q^n, we get the equation

$$b_n p^n + b_{n-1} p^{n-1} q + \cdots + b_1 p q^{n-1} + b_0 q^n = 0$$

as a condition for whether y is a solution of (4.10).

If we transform (4.10) into

$$b_n p^n = -b_{n-1} p^{n-1} q - \cdots - b_1 p q^{n-1} - b_0 q^n$$

we recognize that $b_n p^n$ must be divisible by q if y is a solution of (4.10). Since p has no common prime factor with q, b_n must be divisible by q. Since $b_n \neq 0$, only finitely many integer values for q are possible. We can also transform (4.10) into

$$b_0 q^n = -b_n p^n - b_{n-1} p^{n-1} q - \cdots - b_1 p q^{n-1}$$

We thus find that $b_0 q^n$ must be divisible by p if y is a solution of (4.10). Since q has no common prime factor with p, b_0 must be divisible by p. Since $b_0 \neq 0$, only finitely many integer values for p are possible. Consequently, there are only finitely many rational numbers y that could be a solution of (4.10).

To find all rational solutions of a polynomial equation with integer coefficients, it is sufficient to test finitely many rational numbers.

The algebraic number x is thus rational if and only if it matches one of these finitely many rational numbers y. And the equality of two algebraic numbers is decidable. Thus, in total, we have found an algorithm that allows us to decide whether a given polynomial with rational coefficients is irreducible over the rational numbers.

In the case that we want to decide whether a normalized polynomial $f(X)$ with integer coefficients is irreducible, we can computationally simplify the approach: The roots with multiplicity $x_1, \ldots, x_n$ are by definition all algebraic integers. Since the sum and product of algebraic integers are again algebraic integers, the elementary symmetric functions in a combination of the $x_1, \ldots, x_n$ are again algebraic integers.

To test whether $f(X)$ is irreducible, we need to find out whether certain of these algebraic integers x are rational numbers. Since we can assume that $b_n = 1$ in (4.10) for algebraic integers, the q in (4.11) as a divisor of b_n is necessarily equal to 1, that is, we only need to test all integer divisors of $b_0 \neq 0$ to see if they match x. If we can already see from a decimal expansion of x that x is not an integer, we do not need to test anything in particular, but can directly exclude that x is a rational number.

A normalized polynomial with integer coefficients is irreducible over the rational numbers if and only if for all proper combinations of the roots, the elementary symmetric functions are not simultaneously integers.

We want to demonstrate the procedure using two examples.

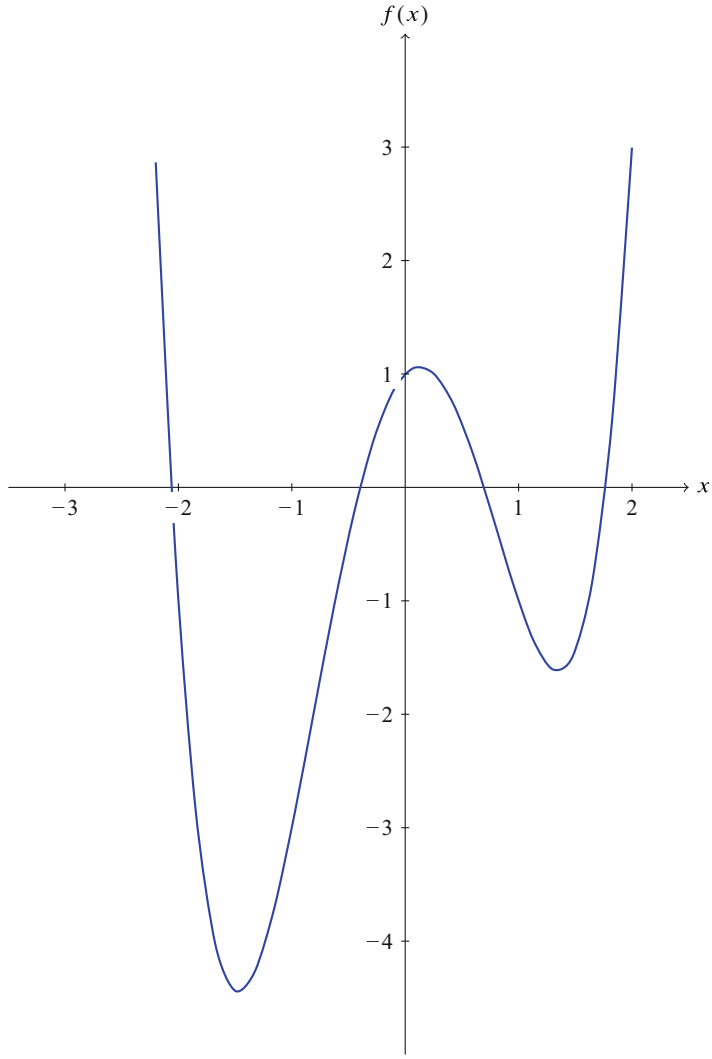


Fig. 4.1 The real values of the polynomial $f(X) = X^4 - 4X^2 + X + 1$

Example 4.3 Let

$$f(X) = X^4 - 4X^2 + X + 1$$

as in Fig. 4.1. We can calculate the roots of $f(X)$ using the proof method of Theorem 3.1, or we ask a numerical analyst. As roots of $f(X)$ we obtain the algebraic numbers

$$x_1 = -2.061\ldots, \quad x_2 = 1.764\ldots, \quad x_3 = 0.694\ldots, \quad x_4 = -0.396\ldots.$$

To determine whether $f(X)$ is irreducible, we need to look at the elementary symmetric functions of a one- or two-element combination of the four solutions (a three-element combination corresponds to a one-element combination by considering the complement). The solutions $x_1, \ldots, x_4$ themselves are the elementary symmetric functions in all one-element combinations. Obviously, none of these is an integer. For the two-element combinations we have

$$\begin{aligned} e_1(x_1, x_2) &= x_1 + x_2 = -0.297\ldots, & e_2(x_1, x_2) &= x_1 \cdot x_2 = -3.64\ldots \\ e_1(x_1, x_3) &= x_1 + x_3 = -1.368\ldots, & e_2(x_1, x_3) &= x_1 \cdot x_3 = -1.430\ldots \\ e_1(x_1, x_4) &= x_1 + x_4 = -2.458\ldots, & e_2(x_1, x_4) &= x_1 \cdot x_4 = 0.817\ldots. \end{aligned}$$

(All other two-element combinations are complements of these.) Also here at least one obviously non-integer number appears in each row, which means that the polynomial $f(X)$ does not allow a factor of degree 2 over the rational numbers. This proves that $f(X)$ is irreducible.

Example 4.4 Let

$$f(X) = X^4 - 7X^3 + 4X^2 + 27X - 15$$

be the polynomial in Fig. 4.2. Approximations of the roots of $f(X)$ are

$$x_1 = 5.449\ldots, \qquad x_2 = 2.791\ldots, \qquad x_3 = -1.791\ldots, \qquad x_4 = 0.551\ldots.$$

As in Example 4.3, the fact that none of these four numbers can be an integer implies that $f(X)$ does not split off a linear factor. This brings us to the two-element combinations. Let's look at which coefficients might possibly have an integer as a sum. This is obviously the case for x_1 and x_4, because we have $e_1(x_1, x_4) = x_1 + x_4 = 6.000\ldots$ (rounded to three places). The second elementary symmetric function in the two roots, $e_2(x_1, x_4) = x_1 \cdot x_4 = 3.000\ldots$, also looks like an integer. We can verify this by using Proposition 4.1 to calculate the remainder of the division of $f(X)$ by the polynomial

$$g(X) = X^2 - 6X + 3 \approx X^2 - 6.000\ldots\, X + 3.000\ldots$$

We obtain

$$f(X) = h(X) \cdot g(X)$$

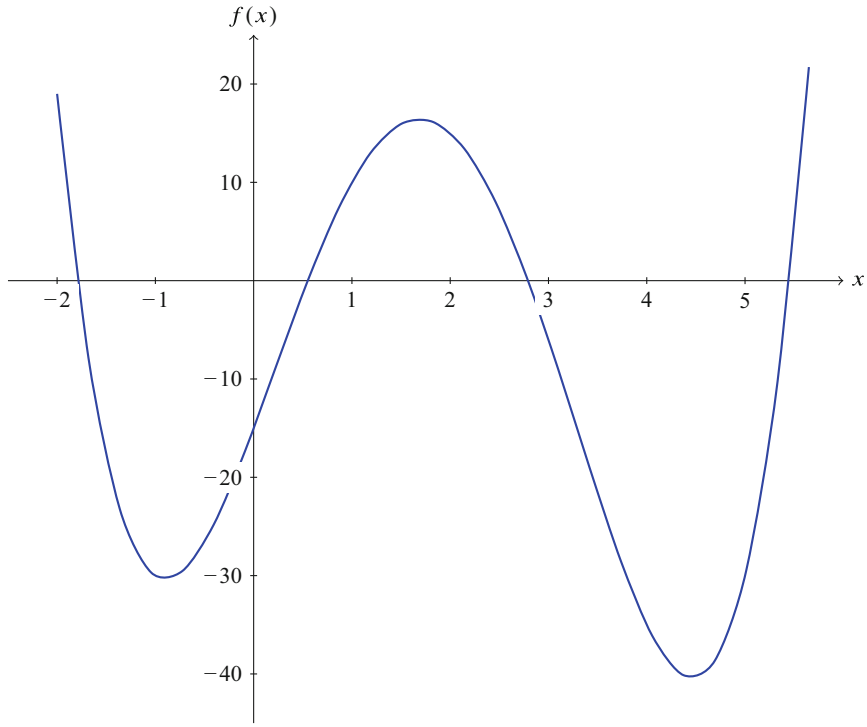


Fig. 4.2 The real values of the polynomial $f(X) = X^4 - 7X^3 + 4X^2 + 27X - 15$

with $h(X) = X^2 - X - 5$, so there is no remainder. Thus, we have found the decomposition of $f(X)$ into irreducible polynomials, namely

$$X^4 - 7X^3 + 4X^2 + 27X - 15 = (X^2 - 6X + 3) \cdot (X^2 - X - 5).$$

We noted above that the irreducible integers are precisely the prime numbers. From school, we know that every positive integer can be uniquely (except for order) written as a product of prime numbers. Is there an analogy for polynomials, can we thus write normalized polynomials with rational coefficients uniquely (except for order) as a product of irreducible polynomials? The answer to this question is yes. In analogy to the integers, we call such a decomposition again the *prime factor decomposition*.

We first show the existence of such a decomposition for a normalized polynomial $f(X)$ with rational coefficients. If $f(X)$ is irreducible, then $f(X) = f(X)$ is a decomposition into irreducible polynomials, and we are done.

Otherwise, $f(X)$ is not irreducible, and we can write $f(X) = g(X) \cdot h(X)$ for two normalized polynomials with rational coefficients and $\deg g(X), \deg h(X) < \deg f(X)$.[1] If then $g(X)$ and $h(X)$ are irreducible, then we are done again, otherwise we further decompose $g(X)$ and/or $h(X)$, and so on.

Since the degree of the involved polynomials always gets smaller, this procedure must eventually come to an end (induction over the degree of $f(X)$ is the keyword to formally write down this argument).

The uniqueness of the decomposition then follows from:

Proposition 4.3 *Let $f_1(X), \ldots, f_n(X)$ and $g_1(X), \ldots, g_m(X)$ be irreducible normalized polynomials with rational coefficients such that*

$$f_1(X) \cdots f_n(X) = g_1(X) \cdots g_m(X). \tag{4.12}$$

Then the factors $f_1(X), \ldots, f_n(X)$ coincide with the factors $g_1(X), \ldots, g_m(X)$ with multiplicities and up to order, that is, $m = n$, and there exists a permutation σ of n letters with $g_i(X) = f_{\sigma(i)}(X)$ for all $i \in \{1, \ldots, n\}$.

Every polynomial has a unique decomposition into irreducible factors over the rational numbers, up to order.

The proof of Proposition 4.3 is achieved with the following important theorem, Abel's irreducibility theorem.

Theorem 4.1 *Let $f(X)$ and $g(X)$ be two normalized polynomials with rational coefficients that have an algebraic number x as a common root. If $f(X)$ is irreducible, then $f(X)$ divides the polynomial $g(X)$.*

If $g(X)$ is also irreducible, it follows in particular $f(X) = g(X)$, that is, two irreducible normalized polynomials with rational coefficients are already equal if they have a common root.

Proof of Theorem 4.1 Let $d(X)$ be a greatest common divisor of $f(X)$ and $g(X)$, that is, a normalized polynomial which divides $f(X)$ and $g(X)$ and for which polynomials $p(X)$ and $q(X)$ exist according to Proposition 4.1 with

$$d(X) = p(X)f(X) + q(X)g(X). \tag{4.13}$$

If we set $X = x$ in (4.13), we get

$$d(x) = p(x)f(x) + q(x)g(x) = p(x) \cdot 0 + q(x) \cdot 0 = 0.$$

In particular, $d(X)$ has a root, namely x. Thus, $d(X)$ is not the constant polynomial one. Since $d(X)$ is a divisor of $f(X)$ and $f(X)$ is irreducible by assumption, necessarily $d(X) = f(X)$.

The assertion to be proved follows from the fact that $d(X)$ divides the polynomial $g(X)$. □

From Theorem 4.1 we can infer (proof below):

Corollary 4.1 *Let $f(X)$ be a normalized irreducible polynomial with rational coefficients. If $f(X)$ is then a divisor of a product $g(X) \cdot h(X)$ of two polynomials $g(X)$ and $h(X)$ with rational coefficients, then $f(X)$ divides at least one of the two factors $g(X)$ and $h(X)$.*

In this sense, a normalized irreducible polynomial behaves like a prime number in the integers: If a prime number divides a product, then the prime number also divides at least one of the two factors. We therefore also call a normalized polynomial $f(X) \neq 1$ with this property *prime*. The statement of Corollary 4.1 is therefore that every normalized irreducible polynomial is prime. (The converse, namely that every prime polynomial is also irreducible is much easier to see.)

Proof of Corollary 4.1 Let x be a root of $f(X)$. Since $f(X) \cdot k(X) = g(X) \cdot h(X)$ for some polynomial $k(X)$ with rational coefficients according to the assumption, x is also a root of $g(X) \cdot h(X)$, so $g(x) = 0$ or $h(x) = 0$. Without loss of generality, let $g(x) = 0$. Since $f(X)$ is irreducible, it follows from Theorem 4.1 that $f(X)$ is a divisor of $g(X)$. □

The statement of Corollary 4.1 can easily be generalized to more than two factors. We use this for the proof we still have to do:

Proof of Proposition 4.3 We do the proof by induction over n. In the case of $n = 0$, the left side of (4.12) equals one, so it is of polynomial degree zero. This must therefore also apply to the right side of (4.12). But this means that $m = 0$, which completes the proof of the assertion for $n = 0$.

Since $f_n(X)$ is irreducible, $f_n(X)$ must divide one of the factors $g_1(X), \ldots, g_m(X)$ according to Corollary 4.1. Without loss of generality, we can assume that $j = m$ (otherwise we rearrange the $g_j(X)$ beforehand). Since $f_n(X)$ and $g_m(X)$ are both irreducible, only $f_n(X) = g_m(X)$ is possible. We are therefore allowed to reduce (4.12) by these polynomials and obtain

$$f_1(X) \cdots f_{n-1}(X) = g_1(X) \cdots g_{m-1}(X),$$

to which we can apply the induction hypothesis. Thus, the polynomials $f_1(X), \ldots, f_{n-1}(X)$ coincide with the polynomials $g_1(X), \ldots, g_{m-1}(X)$ in multiplicity and up to

order. Since $f_n(X) = g_m(X)$ as well, the induction step is completed and the proof is thus provided. □

Abel's irreducibility theorem has the following important consequence for irreducible polynomials:

Proposition 4.4 *Let $f(X)$ be an irreducible normalized polynomial over the rational numbers. Then $f(X)$ is separable.*

Proof Let $d(X)$ be the greatest common divisor of $f(X)$ and $\frac{1}{n}f'(X)$, where n is the degree of $f(X)$. In Sect. 4.1, we saw that $f_0(X) = f(X)/d(X)$ is a separable normalized polynomial that (up to multiplicities) has the same roots as $f(X)$. According to Theorem 4.1, $f(X)$ is therefore a divisor of $f_0(X)$. Since by construction $f_0(X)$ is a divisor of $f(X)$, it follows $f(X) = f_0(X)$, so $f(X)$ is separable. □

Over the rational numbers, an irreducible normalized polynomial is always separable, so it has no double roots.

The uniqueness of the prime factor decomposition for polynomials allows in particular the following definition:

Definition 4.3 Let $f(X)$ be a normalized polynomial with rational coefficients. Let further $p(X)$ be an irreducible polynomial. The multiplicity with which the factor $p(X)$ appears in the prime factor decomposition of $f(X)$ is called the *valuation* of $f(X)$ at the point $p(X)$. (By definition, the valuation is zero if $p(X)$ does not appear as a factor in the prime factor decomposition at all.)

For the valuation of $f(X)$ at the point $p(X)$ we also write

$$\operatorname{ord}_{p(X)} f(X).$$

Example 4.5 If x is a rational root of $f(X)$, then the valuation of $f(X)$ at $(X - x)$ is the multiplicity of the root x.

Before we conclude this section, we want to give another proof for the uniqueness of prime factorization that does not rely on the fundamental theorem of algebra, i.e., the existence of roots of non-constant polynomials.

For this, we need an alternative proof for Corollary 4.1: Let $f(X)$ be a normalized irreducible polynomial, which is a divisor of a product $g(X) \cdot h(X)$. Let $f(X)$ not be a

divisor of $g(X)$. We need to show that $f(X)$ is a divisor of $h(X)$. The greatest common divisor of $f(X)$ and $g(X)$ cannot be $f(X)$. Since $f(X)$ is irreducible, it must therefore be 1. Thus, there exist polynomials $p(X)$ and $q(X)$ with rational coefficients such that

$$1 = p(X) \cdot f(X) + q(X) \cdot g(X).$$

If we multiply the equation by $h(X)$, we get

$$h(X) = p(X) \cdot f(X) \cdot h(X) + q(X) \cdot g(X) \cdot h(X).$$

Both terms on the right side are divisible by $f(X)$. Thus, $f(X)$ also divides $h(X)$, the left side.

4.3 Irreducibility over the Integers

In Sect. 4.2 we learned a constructive method which decides whether a given polynomial $f(X)$ is irreducible or not. This general method is, however, generally laborious to carry out, because it requires knowledge of the roots of $f(X)$ and thus generally, the numerical approximation according to the proof method of Theorem 2.2 is necessary.

In this section we want to learn a series of methods, which represent simpler criteria for the indecomposability of a polynomial (the reverse task, determining the decomposability of a polynomial is usually easier, because it is enough to specify any proper factor, and it is not necessary to prove that no such factor exists). With all the following criteria, however, it should be noted that they cannot be applied to every polynomial, that is, a polynomial may be irreducible and yet it is not possible to show this with any of the following criteria.

We start with a criterion by Carl Friedrich Gauss. For this we first need some definitions for non-zero polynomials $f(X) = a_n X^n + a_{n-1} X^{n-1} + \cdots + a_1 X + a_0$ with integer coefficients $a_0, \ldots, a_n$.

If $a_n \neq 0$, we call a_n the *leading coefficient* of $f(X)$ and n the *degree* of $f(X)$, written $\deg f(X) = n$.

The greatest common divisor of $a_0, \ldots, a_n$ is called the *content* of $f(X)$. For example, the content of $2X^2 - 6X$ is 2 and the content of $5X^3 - X + 35$ is 1. If, as in the last example, the content is one, the polynomial is called *primitive*. For any polynomial $f(X)$ with integer coefficients, we see that the content c of $f(X)$ is the positive integer for which $c^{-1} f(X)$ is a primitive polynomial.

This observation allows us to extend the concept of content to non-vanishing polynomials $f(X)$ with rational coefficients. Namely, the *content* c of $f(X)$ is the positive rational number for which $c^{-1} f(X)$ is a primitive polynomial with integer coefficients. This polynomial constructed from $f(X)$ will be denoted by $\tilde{f}(X)$ throughout this section.

For illustration, let's consider $f(X) = X^3 - \frac{5}{3}X + \frac{2}{15}$. If we multiply this polynomial by 15 (the common denominator), we obtain the integer polynomial $15X^3 - 25X + 2$, whose

content is the greatest common divisor of 15, −25 and 2, which is 1. Thus, the content of $f(X)$ is given by $\frac{1}{15}$.

Obviously, the content of a polynomial is an integer only if the polynomial has only integer coefficients, and it is one exactly if and only if the polynomial is primitive with integer coefficients. The content of a normalized polynomial is always the inverse of an integer, namely the inverse of the common denominator of all reduced coefficient fractions.

The so-called Gaussian lemma (more precisely, this is a variant of a series of statements that are subsumed under this name) makes a statement about the content of the product of two polynomials:

Lemma 4.1 *Let $g(X)$ and $h(X)$ be two polynomials with rational coefficients. Then the content of the product $g(X) \cdot h(X)$ of the polynomials is the product of the contents of the polynomials $g(X)$ and $h(X)$.*

In particular, the product of primitive polynomials is again primitive, and if the product of two polynomials with integer coefficients is primitive, then the factors must also be primitive.

Proof The cases $g(X) = 0$ or $h(X) = 0$ are trivial, so we may exclude them in the following. By dividing $g(X)$ and $h(X)$ by their respective contents if necessary, we can assume that $g(X)$ and $h(X)$ are primitive integer polynomials, in particular, they have content 1. We then have to show that the content of $g(X) \cdot h(X)$ is also equal to one.

Let $g(X) = a_n X^n + a_{n-1} X^{n-1} + \cdots + a_1 X + a_0$ and $h(X) = b_m X^m + b_{m-1} X^{m-1} + \cdots + b_1 X + b_0$, so that we have

$$g(X) \cdot h(X) = \sum_{\ell \geq 0} \sum_{i+j=\ell} a_i b_j \, X^\ell .$$

We do the proof by induction over $n + m$. Let c be the content of $g(X) \cdot h(X)$, and let d be the greatest common divisor of c and a_n. Then d divides all coefficients, and thus also the content, of the polynomial $(g(X) - a_n X^n) \cdot h(X) = g(X) \cdot h(X) - a_n X^n \cdot h(X)$. By induction hypothesis, the content of this polynomial is the product of the contents of $g(X) - a_n X^n$ and $h(X)$. Since the content of $h(X)$ is one, d must therefore divide the content of $g(X) - a_n X^n$. But this means that d also divides the content of $g(X)$. Since $g(X)$ is primitive, it follows that $d = 1$.

By swapping the roles of $g(X)$ and $h(X)$, we show that the greatest common divisor of c and b_m is also equal to one. Let's abbreviate the greatest common divisor of two numbers x and y for the moment with $\gcd(x, y)$. Then the following calculation applies:

$$\begin{aligned} \gcd(c, a_n b_m) &= \gcd(\gcd(c, c b_m), a_n b_m) = \gcd(c, \gcd(c b_m, a_n b_m)) \\ &= \gcd(c, b_m \ \gcd(c, a_m)) = \gcd(c, b_m) = 1. \end{aligned}$$

Since c by definition is also a divisor of $a_n b_m$, $c = 1$. □

Under the multiplication of polynomials, their contents multiply as well.

For the following corollary, which is also often referred to as the Gaussian lemma, we need another definition that extends the concept of irreducibility to (primitive) polynomials with integer coefficients: We call a primitive polynomial $f(X)$ with integer coefficients *irreducible* if every decomposition $f(X) = g(X) \cdot h(X)$ into two integer polynomials is of the form that $g(X)$ or $h(X)$ is a constant.

Corollary 4.2 *A normalized polynomial $f(X)$ with rational coefficients is irreducible if and only if the primitive polynomial $\tilde{f}(X)$ with integer coefficients is irreducible (over the integers).*

If the polynomial $f(X)$ in Corollary 4.2 is already defined over the integers, then it is automatically primitive, since it is normalized. A normalized polynomial with integer coefficients is therefore irreducible as a polynomial over the rational numbers if and only if it is irreducible as a polynomial over the integers.

Proof If $f(X) = g(X) \cdot h(X)$ is a decomposition of $f(X)$ into normalized polynomials with rational coefficients, then division by the product of the contents of $g(X)$ and $h(X)$ according to Lemma 4.1 gives the decomposition $\tilde{f}(X) = \tilde{g}(X) \cdot \tilde{h}(X)$ of $\tilde{f}(X)$ into primitive polynomials with integer coefficients. If $\tilde{f}(X)$ is therefore irreducible, then $f(X)$ must also be irreducible.

Conversely, let $\tilde{f}(X) = \tilde{g}(X) \cdot \tilde{h}(X)$ be a decomposition of $\tilde{f}(X)$ into primitive polynomials with integer coefficients. We notice that we get $f(X)$ from $\tilde{f}(X)$ by dividing $\tilde{f}(X)$ by its leading coefficient. This leading coefficient of $\tilde{f}(X)$ is the product of the leading coefficients of $\tilde{g}(X)$ and $\tilde{h}(X)$. If we divide the decomposition by this product, we thus obtain a decomposition of the form $f(X) = g(X) \cdot h(X)$, where $g(X)$ and $h(X)$ are normalized polynomials with rational coefficients. If $f(X)$ is therefore irreducible, then $\tilde{f}(X)$ must also be irreducible. □

In the last section, we saw how we can determine whether a polynomial with integer coefficients is irreducible by approximating the roots to a few decimal places—for this, the elementary symmetric functions in combinations of the roots with multiplicity had to be compared with integers.

We now show that with the help of the proof method of Corollary 4.2 we can specify a slightly extended, very similar procedure that works for normalized polynomials with arbitrary, not necessarily integer coefficients: Let $f(X)$ be such a normalized polynomial

with rational coefficients. The content of $f(X)$ is the inverse of a positive integer d. According to our considerations in Sect. 4.2, every normalized factor $g(X)$ of $f(X)$ with rational coefficients is of the form

$$g(X) = (X - x_{i_1}) \cdots (X - x_{i_k}),$$

$1 \leq i_1 < i_2 \cdots < i_k \leq n$, where $x_1, \ldots, x_n$ are the roots of $f(X)$ with multiplicities. For such a $g(X)$, there exists a normalized polynomial $h(X)$ with rational coefficients, such that $f(X) = g(X) \cdot h(X)$. The inverses of the contents of $g(X)$ and $h(X)$ are integers and according to Lemma 4.1 in particular divisors of d. If we multiply $g(X)$ with d, we obtain an integer multiple of $\tilde{g}(X)$, thus a polynomial $d \cdot g(X)$ with integer coefficients.

The coefficients of $d \cdot g(X)$ are given by the elementary symmetric functions in $x_{i_1}, \ldots, x_{i_k}$ multiplied with d. To determine whether $f(X)$ has nontrivial factors, it is therefore sufficient to check all proper combinations $x_{i_1}, \ldots, x_{i_k}$, $1 \leq i_1 < i_2 \cdots < i_k \leq n$, of the roots with multiplicities of $f(X)$ to see if the expressions

$$d \cdot e_j(x_{i_1}, \ldots, x_{i_k})$$

for $j \in \{1, \ldots, k\}$ are integers.

Example 4.6 We want to investigate whether the polynomial

$$f(X) = X^3 - \frac{3}{2}X^2 + X - \frac{6}{5}$$

is irreducible over the rational numbers. Let x_1, x_2, x_3 be the three roots with multiplicities of $f(X)$. The content of $f(X)$ is $\frac{1}{10}$, the inverse of the content is $d = 10$.

Since $f(X)$ is of degree 3, it is sufficient to investigate one-element combinations of the roots of $f(X)$, that is, whether $10 \cdot x_1$, $10 \cdot x_2$ or $10 \cdot x_3$ is an integer. A numerical root determination yields to four decimal places

$$x_1 = 1.3985\ldots, \quad x_2 = 0.0507 \cdots + 0.9249\ldots \text{ i},$$
$$x_3 = 0.0507 \cdots - 0.9249\ldots \text{ i},$$

that is, none of the numbers is a tenth of an integer. Thus, $f(X)$ is irreducible over the rational numbers.

How can we directly apply Corollary 4.2 to determine whether a normalized polynomial over the rational numbers is irreducible? Obviously we need criteria for when a primitive polynomial over the integers is irreducible. One such criterion is the so-called Eisenstein's[2] criterion, named after a student of Carl-Friedrich Gauss, of whom he said to be, alongside

Archimedes[3] and Isaac Newton[4] the only mathematician who had made history (although Ferdinand Eisenstein would not have earned the admiration of his teacher for this result alone):

Proposition 4.5 *Let $f(X) = a_n X^n + a_{n-1} X^{n-1} + \cdots + a_1 X + a_0$ be a primitive polynomial with integer coefficients. Let p be a prime number, such that $a_0, \ldots, a_{n-1}$ are divisible by p and a_0 is not divisible by p^2. Then $f(X)$ is irreducible.*

Before we proceed to the proof, we want to give an example for the application of Proposition 4.5:

Example 4.7 Consider the normalized polynomial

$$f(X) = X^5 - \frac{2}{3}X^4 + \frac{8}{3}X^2 - 2.$$

This is irreducible over the rational numbers according to Corollary 4.2, if and only if the primitive polynomial

$$\tilde{f}(X) = 3X^5 - 2X^4 + 8X^2 - 6$$

is irreducible over the integers. Since this is the case according to the Eisenstein criterion, Proposition 4.5, for the prime number $p = 2$, $f(X)$ is consequently an irreducible polynomial over the rational numbers.

Proof of Proposition 4.5 First of all, we note that a_n cannot be divisible by p, otherwise p would be a common divisor of all coefficients of $f(X)$ and $f(X)$ would not be primitive.

Next, we consider a possible decomposition $f(X) = g(X) \cdot h(X)$ with integer polynomials

$$g(X) = \sum_{i \geq 0} b_i X^i, \qquad h(X) = \sum_{j \geq 0} c_j X^j.$$

Since $a_0 = b_0 \cdot c_0$ is divisible by p, but not by p^2 and p is a prime number, either b_0 or c_0 must be divisible by p. Without loss of generality, let p be a divisor of b_0, but not of c_0 (otherwise we swap the roles of $g(X)$ and $h(X)$). We show that $h(X)$ must necessarily be constant. To do this, we consider the next coefficient of $f(X)$, namely $a_1 = b_0 c_1 + b_1 c_0$, from which we can derive

$$b_1 c_0 = a_1 - b_0 c_1.$$

The right side is (at least for $n > 1$) divisible by p which means that $b_1 c_0$ is also divisible by p. Since p is a prime number and c_0 is not divisible by p, b_1 must therefore be divisible by p. In the next step, we then consider

$$b_2 c_0 = a_2 - b_0 c_2 - b_1 c_1.$$

The right side is again (at least for $n > 2$) divisible by p, so we can analogously conclude that b_2 is also divisible by p, and so on. If we proceed inductively, we find that $b_0, \ldots, b_{n-1}$ are all divisible by p. Now we consider the coefficient a_n:

$$a_n = b_0 c_n + b_1 c_{n-1} + \cdots + b_{n-1} c_1 + b_n c_0.$$

As noted above, a_n is not divisible by p. Therefore, the right side must also not be divisible by p. Since the first n terms of the right side are divisible by p, b_n must not be divisible by p. In particular, $b_n \neq 0$. But since $\deg f \leq n$, $h(X)$ must therefore be a constant polynomial, which was to be shown. □

There are irreducible polynomials over the integers that do not fulfill the Eisenstein criterion. But if a polynomial fulfills the Eisenstein criterion, it is irreducible. The Eisenstein criterion is therefore only a sufficient, not a necessary condition for irreducibility of polynomials over the integers.

The scope of the Eisenstein criterion is larger than it might seem at first glance: What about the irreducibility of the polynomial $f(X) = X^4 + X^3 + X^2 + X + 1$?

It is a primitive polynomial over the integers, however, the coefficient to the constant term is 1, which is not divisible by any prime number, so we cannot directly apply Proposition 4.5 for any prime number. On the other hand, trying to find any linear factor or factorization into two quadratic polynomials won't succeed either.

A trick helps us here: We consider instead of $f(X)$ the polynomial

$$\begin{aligned} g(X) = f(X+1) &= (X+1)^4 + (X+1)^3 + (X+1)^2 + (X+1) + 1 \\ &= X^4 + 4X^3 + 6X^2 + 4X + 1 + X^3 + 3X^2 + 3X + 1 \\ &\quad + X^2 + 2X + 1 + X + 1 + 1 \\ &= X^4 + 5X^3 + 10X^2 + 10X + 5. \end{aligned}$$

For this, we can obviously apply Proposition 4.5, namely to the prime number $p = 5$, that is, $g(X) = f(X+1)$ is an irreducible primitive polynomial over the integers. What does this mean for $f(X)$?

If $f(X) = f_1(X) \cdot f_2(X)$ were a decomposition of $f(X)$, then $g(X) = f(X+1) = f_1(X+1) \cdot f_2(X+1)$ would be a decomposition of $g(X)$. But the polynomial $g(X)$ is not irreducible. Thus, we have shown that $f(X)$ must be irreducible. We can generalize this conclusion: Let $f(X)$ and $h(X)$ be normalized polynomials over the rational numbers. If $h(X)$ is non-constant and $f(h(X))$ is irreducible, then $f(X)$ is also irreducible.

A polynomial $f(X)$ is irreducible if and only if $f(X+a)$ is irreducible for some number a.

4.4 Irreducibility Modulo a Prime Number

Another irreducibility criterion, we should get to know is the reduction modulo an integer n. For this, we recall the following terminology: Two integers a and b are called *congruent* modulo n, written

$$a \equiv b \pmod{n},$$

if the difference between a and b is divisible by n, that is, in the case $n \neq 0$, if a and b have the same remainder when divided by n.

For example, 5 and -7 are congruent modulo 2, but not modulo 5 or modulo 8. Modulo 0, every number is congruent only to itself; modulo 1, all numbers are congruent to each other.

Obviously, $a \equiv 0$ modulo n holds if and only if a is a multiple of n, that is, n is a divisor of the number a. (Because of this fact, it becomes clear that this terminology can be a tool in proofs like Proposition 4.5, where many arguments are based on divisibility properties.)

The congruence of two numbers (modulo n) is an equivalence relation, that is, in detail: Every number a is congruent to itself (reflexivity), if a is congruent to b, then b is also congruent to a (symmetry), and if a is congruent to b and b is congruent to a c—of course, modulo the same integer n—,then a is also congruent to c (transitivity).

The relation of congruence is compatible with arithmetic operations, that is, if $a \equiv b$ modulo n and $a' \equiv b'$ modulo n, then $a + a' \equiv b + b'$ modulo n and $a \cdot a' \equiv b \cdot b'$ modulo n. An important rule known from arithmetic in the domain of integers does not apply, however, namely the cancellation rule in the following form:

If $a \cdot b = 0$ in the domain of integers, it follows that $a = 0$ or $b = 0$. On the other hand, $a \cdot b \equiv 0$ modulo n does not generally imply $a \equiv 0$ or $b \equiv 0$: For example, $2 \cdot 3 \equiv 0$ modulo 6, but neither $2 \equiv 0$ nor $3 \equiv 0$ modulo 6.

The reason is that 6 is not a prime number. We can formulate the following rule: Let p be a prime number. Then

$$a \cdot b \equiv 0 \implies (a \equiv 0) \vee (b \equiv 0) \pmod{p}.$$

To understand this, we simply unravel the definitions: If the product of two integers is divisible by a prime number p, then (at least) one of the two factors is divisible by p.

Finally, we need the following statement about the existence of multiplicative inverses modulo n:

Proposition 4.6 *Let n be an integer. For each integer a, there exists an integer b with*

$$a \cdot b \equiv 1 \pmod{n}$$

if a and n are coprime.

In the case that n is a prime number, such a b exists if and only if a is not a multiple of n. In the case that b exists for a given a, we call *a invertible modulo n* or a *unit* modulo n and b an *inverse modulo n of a*.

(We have not chosen the indefinite article "a" without reason: If b is an inverse modulo n, then $b + n$ is also an inverse modulo n. Inverses modulo n are therefore generally not unique. However, any two choices of b are congruent modulo n.)

Proof Let $a \cdot b \equiv 1 \pmod{n}$, that is, $a \cdot b - 1$ is a multiple of n. Let d be a common divisor of a and n, thus also a common divisor of a and $a \cdot b - 1$. Since d also divides $a \cdot b$, the divisibility of 1 by d follows. Thus, $d \in \{1, -1\}$, consequently, a and n are coprime under the premise $a \cdot b \equiv 1 \pmod{n}$.

Conversely, let a and n be coprime, that is, 1 is a greatest common divisor of a and n. According to the Euclidean algorithm for integers, Proposition 4.2, there exist integers r and s with $1 = r \cdot a + s \cdot n$. It follows that $a \cdot r \equiv 1 \pmod{n}$, so we can choose $b = r$. □

We extend the congruence arithmetic from integers to polynomials: Considering two polynomials $f(X)$ and $g(X)$ with integer coefficients, we say $f(X)$ and $g(X)$ are *congruent* modulo an integer n if all their corresponding coefficients are congruent modulo n to each other. The rules of congruence arithmetic transfer one-to-one to this case.

For simplicity, we consider a prime number $n = p$ in the following. (In this case, we do not have the problem that the product of two integers can vanish modulo n, even if neither of the two factors vanishes modulo n.)

A normalized polynomial $f(X)$ with integer coefficients is called *irreducible modulo p*, if for every pair $g(X)$ and $h(X)$ of normalized polynomials with integer coefficients and for which

$$f(X) \equiv g(X) \cdot h(X) \pmod{p} \tag{4.14}$$

holds, one of the two polynomials $g(X)$ or $h(X)$ is congruent to a constant modulo p.

Since the congruence relation is compatible with the basic arithmetic operations, the irreducibility of the polynomial $f(X)$ modulo p implies the irreducibility of any other normalized polynomial with integer coefficients modulo p that is modulo p congruent to $f(X)$.

Let $f(X)$ be a polynomial with integer coefficients that is not necessarily normalized, but whose leading coefficient a is not divisible by p. According to Proposition 4.6, there exists an integer b with $ab \equiv 1$ modulo p. The product $bf(X)$ is thus congruent modulo p to a normalized polynomial $\bar{f}(X)$. Since b is uniquely defined up to addition of multiples of p, $\bar{f}(X)$ is well-defined up to congruence modulo p.

This leads to the following definition: The polynomial $f(X)$ is called *irreducible*, if the normalized polynomial $\bar{f}(X)$ is irreducible.

Example 4.8 Let $p = 3$. Let $f(X) = 2X^4 + X^3 + X + 2$. Then $b = 2$ is an inverse of $a = 2$ modulo 3, because $2 \cdot 2 \equiv 1$ modulo 3. By definition, $f(X)$ is therefore irreducible modulo 3 if the normalized polynomial

$$\bar{f}(X) = X^4 + 2X^3 + 2X + 1 \equiv 2 \cdot f(X) \pmod 3$$

is irreducible modulo 3.

(A similar definition could also be introduced for polynomials with rational numbers as coefficients: Let $f(X) = a_n X^n + a_{n-1} X^{n-1} + \cdots + a_1 X + a_0$ be a polynomial with rational coefficients and $a_n \neq 0$. Then $f(X)$ is called irreducible, if the normalized polynomial $\bar{f}(X) = a_n^{-1} f(X)$ is irreducible.)

We can effectively determine whether a normalized polynomial $f(X)$ with integer coefficients is irreducible modulo the prime number p: If $\deg f(X) = d$, then $f(X)$ is irreducible if and only if for every two normalized polynomials $g(X)$ and $h(X)$ with $\deg g(X)$, $\deg h(X) < d$ that are not congruent to a constant polynomial modulo p the Eq. (4.14) does not hold.

A priori, we have to consider infinitely many polynomials $g(X)$ and $h(X)$. With a little consideration, however, we can narrow down our search space: Every polynomial $g(X)$ is modulo p congruent to (exactly) one polynomial $\tilde{g}(X)$ whose coefficients all come from the set $\{0, \ldots, p-1\}$. According to the rules of congruence arithmetic,

$$f(X) \equiv g(X) \cdot h(X) \pmod p \iff f(X) \equiv \tilde{g}(X) \cdot h(X) \pmod p,$$

that is, it is sufficient to test the finitely many non-constant polynomials $g(X)$ with $\deg g(X) < d$, whose coefficients all come from the set $\{0, \ldots, p-1\}$. Similarly, it is sufficient to consider finitely many polynomials $h(X)$. Thus, we can determine in finitely many steps whether $f(X)$ is irreducible modulo p.

We want to illustrate this procedure using an example:

Example 4.9 Let $f(X) = X^4 + 15X^3 + 7$. We want to determine whether $f(X)$ is irreducible modulo $p = 5$. For this, it is sufficient to consider the congruence

$$X^4 + 15X^3 + 7 \equiv g(X) \cdot h(X) \pmod 5 \tag{4.15}$$

for the cases $g(X) = X - a$ and $g(X) = X^2 + bX + c$ with $a, b, c \in \{0, \ldots, 4\}$. (The case $g(X) = X^3 + aX^2 + bX + c$ is symmetric to the linear case by swapping $g(X)$ and $h(X)$.)

In the case $g(X) = X - a$, a is a root of $g(X)$. From (4.15) it would follow that

$$f(a) = a^4 + 15a^3 + 7 \equiv a^4 + 2 \equiv 0 \pmod 5$$

for an $a \in \{0, \ldots, 4\}$. A simple test, however, rules this out. Thus, (4.15) cannot be fulfilled for linear $g(X)$.

It remains to consider the case $g(X) = X^2 + bX + c$. Then $h(X)$ is of the form $h(X) = X^2 + eX + f$ with $e, f \in \{0, \ldots, 4\}$. If (4.15) holds, a comparison of coefficients between both sides yields the following:

$$\begin{aligned} e + b &\equiv 15 \equiv 0 \pmod 5 \\ f + be + c &\equiv 0 \pmod 5 \\ bf + ce &\equiv 0 \pmod 5 \\ cf &\equiv 7 \equiv 2 \pmod 5 \end{aligned}$$

It follows in order, everything modulo 5: $e \equiv -b$, $(c + f) \equiv b^2$, $(f - c) \cdot b \equiv 0$ and $cf \equiv 2$. Since 5 is a prime number, the penultimate congruence yields $f \equiv c$ or $b \equiv 0$.

In the first case, the last equation implies $f^2 \equiv 2$. The only squares modulo 5 are $0 \equiv 0^2$, $1 \equiv 1^2 \equiv 4^2$ and $4 \equiv 2^2 \equiv 3^2$. Thus, this case cannot occur.

In the second case, $b \equiv 0$, the third equation implies that $c \equiv -f$, so $f^2 \equiv -2 \equiv 3$ from the last equation. This is again impossible.

Thus, (4.15) can never be fulfilled, so $f(X)$ is irreducible modulo 5.

Since we can effectively test irreducibility modulo a prime number p, the following proposition gives us, in connection with Corollary 4.2, an important criterion for irreducibility of polynomials with rational coefficients:

Proposition 4.7 *Let $f(X)$ be a primitive polynomial with integer coefficients. Let there exists a prime number p such that the leading coefficient of $f(X)$ is not divisible by p and that $f(X)$ is irreducible modulo p. Then $f(X)$ is irreducible as a primitive polynomial over the integers.*

Proof Let $f(X) = g(X) \cdot h(X)$ for two integer polynomials $g(X)$ and $h(X)$. We have to show that $g(X)$ or $h(X)$ is a constant. Let a be an inverse of the leading coefficient of $g(X)$ modulo p, and let b be an inverse of the leading coefficient of $h(X)$ modulo p. Then $ag(X)$ modulo p congruent to a normalized polynomial $\bar{g}(X)$ and $bh(X)$ modulo p congruent to a normalized polynomial $\bar{h}(X)$.

Let $\bar{f}(X)$ be a normalized polynomial with integer coefficients that is congruent to $abf(X)$ modulo p. By assumption, $\bar{f}(X)$ is irreducible modulo p. It follows that

$$\bar{f}(X) \equiv \bar{g}(X) \cdot \bar{h}(X) \pmod{p}.$$

Thus, $\bar{g}(X)$ or $\bar{h}(X)$ is congruent to a constant modulo p. Let's first consider the case that $\bar{g}(X)$ is congruent to a constant modulo p. This also means that $g(X)$ is congruent to a constant modulo p, and we can write $g(X) = pk(X) + c$, where c is an integer. Since the highest coefficient of $g(X)$ is not divisible by p, $k(X)$ must be a constant polynomial, that is, $g(X)$ is itself constant.

The second case, that $\bar{h}(X)$ is congruent to a constant, implies analogously that $h(X)$ is constant. □

A normalized polynomial with integer coefficients that is irreducible modulo any prime number p is also irreducible over the integers and thus, according to the Gaussian lemma, also irreducible over the rational numbers.

Example 4.10 From Example 4.9 we deduce that $f(X) = X^4 + 15X^3 + 7$ is irreducible modulo 5. Since the leading coefficient of $f(X)$, namely 1, is not divisible by 5 it follows with Proposition 4.7, that $f(X)$ is an irreducible polynomial over the integers. Due to Corollary 4.2, $f(X)$ is also irreducible over the rational numbers.

The application of Proposition 4.7 depends on choosing a suitable n. Let's consider again $f(X) = X^4 + 15X^3 + 7$ and reduce modulo $n = 3$. Then we have

$$X^4 + 15X^3 + 7 \equiv X^4 + 1 \equiv (X^2 + X - 1) \cdot (X^2 - X - 1) \pmod{3},$$

as can be quickly proven by expanding the right hand side. So $X^4 + 15X^3 + 7$ is not irreducible modulo 3 (and yet irreducible as a polynomial over the integers).

In this section, we have defined when a polynomial with integer coefficients is irreducible modulo a prime number p. Since we can determine for each normalized polynomial with integer coefficients whether it is irreducible modulo p or not, the existence of a prime factorization, follows immediately, just as in the case of polynomials over the rational numbers. That means: For every normalized polynomial $f(X)$ with

integer coefficients there exist normalized polynomials $f_1(X), \ldots, f_n(X)$ with integer coefficients, such that

$$f(X) \equiv f_1(X) \cdots f_n(X) \pmod{p}$$

and the $f_1, \ldots, f_n$ are all irreducible modulo p.

We may ask whether this prime factorization is unique (up to the order of the factors). Here we can certainly only expect uniqueness modulo p (because we can replace each factor $f_i(X)$ by one congruent to it modulo p). And then the uniqueness follows as in the case of polynomials with rational coefficients, i.e., Proposition 4.3:

The proof of Proposition 4.3 essentially incorporates two things: On the one hand, the statement from Corollary 4.1, which we proved at the end of Sect. 4.2 using Proposition 4.1, and on the other hand, the cancellation rule (3.3). If we go through the proofs of Proposition 4.1 and (3.3) again, we find that they also work within congruence arithmetic modulo p.

For the proof of Proposition 4.1 (and the underlying Proposition 3.1 about polynomial division) we need the fact that an integer n that is not congruent to 0 modulo p has an inverse modulo p, i.e., we can divide by n modulo p.

At this point, the reader may have already guessed that no uniqueness of a prime factorization of any kind can be expected when we calculate modulo an integer that is not a prime number. For example, we have

$$(X+2) \cdot (X+3) \equiv X^2 + 5X + 6 \equiv X^2 + 5X \equiv X \cdot (X+5) \pmod{6}.$$

The uniqueness of prime factorizations is not a matter of course.

4.5 The Degree of Algebraic Elements

In the following, we want to develop a measure for the "complexity" of an algebraic number. By definition, an algebraic number is a root of a normalized polynomial with rational coefficients. An obvious possibility is to measure the complexity of an algebraic number by the degree of this normalized polynomial.

A rational number a is, for example, a root of $X - a$, i.e., a polynomial of degree 1. On the other hand, $\sqrt{2}$ is a root of $X^2 - 2$, a polynomial of the second degree, and $\sqrt{2+\sqrt{2}}$ is a root of $X^4 - 4X^2 + 2 = 0$, a polynomial of the fourth degree. So we could say, $\sqrt{2}$ is more complicated than any rational number and $\sqrt{2+\sqrt{2}}$ is again more complicated than $\sqrt{2}$. But this naive definition has a weakness. So is $\sqrt{2}$ also a root of $X^6 - 2X^4$. Is the measure for the complexity of $\sqrt{2}$ now 2 (because $\sqrt{2}$ is a root of a polynomial

of the second degree) or 6 (because $\sqrt{2}$ is a root of a polynomial of the sixth degree)? Obviously, we must select a specific polynomial among all possible polynomials that have an algebraic number as a root, before we can take its degree as a measure. The following important proposition helps with this:

Proposition 4.8 *Let a be an algebraic number. Then there exists exactly one irreducible normalized polynomial $f(X)$ with rational coefficients that has a as a root.*

Every algebraic number is the solution of exactly one normalized polynomial equation with rational coefficients, whose degree is minimal.

The uniquely determined polynomial $f(X)$ is called the *minimal polynomial* of a over the rational numbers. The degree of $f(X)$ is called the *degree of a over the rational numbers*. (Thus, the degree of an algebraic number takes on the meaning of the complexity of a mentioned in the introductory sentences.) We write

$$[\mathbf{Q}(a) : \mathbf{Q}]$$

for the degree of a over the rational numbers.

Proof of Proposition 4.8 Since a is an algebraic number, by definition there exists a polynomial $g(X)$ with rational coefficients and $g(a) = 0$. This polynomial has a decomposition $g(X) = g_1(X) \cdots g_n(X)$ into irreducible polynomials. From $g(a) = 0$, it follows $g_1(a) \cdots g_n(a) = 0$. Thus, at least one of the factors must vanish. Without loss of generality, we have $g_1(a) = 0$. If we set $f(X) = g_1(X)$, we have thus shown the existence of an irreducible normalized polynomialich that has a as a root.

The uniqueness follows because two irreducible normalized polynomials with a common root, in this case a, must coincide (this is immediately inferred from Theorem 4.1).

□

According to the Abel's irreducibility theorem, the minimal polynomial of an algebraic number a is thus the common divisor of all normalized polynomials that have a as a root. In particular, the minimal polynomial is the polynomial of smallest degree with rational coefficients that has a as a root. According to Proposition 4.4, it follows that the minimal polynomial of an algebraic number over the rational numbers is always separable. As we will see in the second volume, this property can be attributed to the rational numbers, namely the rational numbers form a so-called *perfect field*.

Let's return to our polynomials from the introductory sentences to this section: The polynomial $X^2 - 2$ is irreducible according to Eisenstein's criterion, Proposition 4.5 (we

consider the prime number 2). Thus, $X^2 - 2$ is the minimal polynomial of $\sqrt{2}$ over the rational numbers. Consequently, we have

$$[\mathbf{Q}(\sqrt{2}) : \mathbf{Q}] = 2.$$

The polynomial $X^4 - 4X^2 + 2$ is also irreducible according to Eisenstein's criterion (again with the prime number 2). It is thus also the minimal polynomial of its root $\sqrt{2+\sqrt{2}}$. Consequently, we have

$$\left[\mathbf{Q}\left(\sqrt{2+\sqrt{2}}\right) : \mathbf{Q}\right] = 4.$$

We recall that an algebraic number a is called an integral algebraic number if a is a root of a normalized polynomial $g(X)$ with integer coefficients.

Let $f(X)$ be the minimal polynomial of a over the rational numbers. In particular, we have $g(X) = f(X) \cdot h(X)$, where $h(X)$ is another normalized polynomial with rational coefficients. The contents of $f(X)$ and $h(X)$ are inverses of integers, and their product is the content of $g(X)$, which is equal to one since $g(X)$ is normalized and has integer coefficients. Consequently, the content of $f(X)$ (and of $h(X)$) is also equal to one, $f(X)$ is thus a normalized polynomial with integer coefficients.

Thus, we obtain the following statement: An algebraic number is an algebraic integer if and only if its minimal polynomial only has integer coefficients. We can thus effectively determine whether a given algebraic number is an algebraic integer. In particular, we can now prove the existence of non-integral algebraic numbers:

Example 4.11 Consider the complex number $a = \frac{1}{2} + \sqrt{3}$. We have $3 = (a - \frac{1}{2})^2 = a^2 - a + \frac{1}{4}$. Thus, a is a root of the normalized polynomial

$$f(X) = X^2 - X - \frac{11}{4}.$$

The primitive polynomial $\tilde{f}(X) = 4X^2 - 4X - 11$ is irreducible (for example, it has no root modulo 5 and is thus irreducible modulo 5), and thus $f(X)$ is also irreducible. Consequently, $f(X)$ is the minimal polynomial of a. The complex number a is thus an algebraic number of degree 2 over the rational numbers, which is not an algebraic integer since $f(X)$ does not only have integer coefficients.

For the degree of an algebraic number, we want to give an important interpretation. To do this, we repeat the concept of linear dependence: Namely, we call complex numbers $x_1, \ldots, x_n$ *linearly dependent over the rational numbers* if rational numbers $a_1, \ldots, a_n$ with

$$\sum_{i=1}^{n} a_i x_i = 0$$

exist such that not all $a_1, \ldots, a_n$ are simultaneously zero.[5]

Proposition 4.9 *Let x be an algebraic number. The degree of x over the rational numbers is the smallest natural number d such that $1, x, x^2, \ldots, x^d$ are linearly dependent over the rational numbers.*

Proof Let n be the degree of the minimal polynomial $f(X) = X^n + a_{n-1}X^{b-1} + \cdots + a_1 X + a_0$ of x over the rational numbers. Since $f(x) = 0$, we have the nontrivial linear relationship

$$a_0 + a_1 x + \cdots + a_{n-1}x^{n-1} + x^n = 0$$

between the numbers $1, x, x^2, \ldots, x^n$. Thus, $1, x, x^2, \ldots, x^n$ are linearly dependent over the rational numbers.

If

$$b_0 + b_1 x + \cdots + b_{d-1}x^{d-1} + b_d x^d = 0$$

is another nontrivial linear relationship with rational coefficients $b_0, \ldots, b_d$, we first choose $d' \leq d$, such that $b_{d'} \neq 0$ and $b_j = 0$ for $d' < j \leq d$. Thus, the normalized polynomial

$$g(X) = X^{d'} + \frac{b_{d'-1}}{b_{d'}} X^{d'-1} + \cdots + \frac{b_1}{b_{d'}} X + \frac{b_0}{b_{d'}}$$

of degree d' has the algebraic number x as a root. Consequently, the minimal polynomial $f(X)$ of x is a divisor of $g(X)$, so we have $n \leq d' \leq d$. Thus, n is the smallest natural number such that $1, x, x^2, \ldots, x^n$ are linearly dependent over the rational numbers. □

If n is the degree of the algebraic number x, then we can represent x^n as a rational linear combination of $1, x, x^2, \ldots, x^{n-1}$, in particular as a linear combination of powers with smaller exponents. If we multiply this linear combination with x^d, d a natural number, then we see that we can also represent x^{n+d} as a rational linear combination of powers of x with smaller exponent (than $n+d$). By replacing x^{n+d-1} in the obtained linear combination again by a linear combination of powers of x with smaller exponents etc., we finally obtain the representation of x^{n+d} as a rational linear combination of $1, x, x^2, \ldots, x^{n-1}$.

For an algebraic number x, we have denoted with $\mathbf{Q}(x) = \mathbf{Q}[x]$ the set of polynomial expressions in x over the rational numbers, that is, the set of algebraic numbers of the form $a_m x^m + a_{m-1}x^{m-1} + \cdots + a_1 x + a_0$, where the $a_0, \ldots, a_m$ are rational numbers. Our considerations above result in all these numbers being rational linear combinations of $1, x, x^2, \ldots, x^{n-1}$. On the other hand, $1, x, x^2, \ldots, x^{n-1}$ are linearly independent. We can therefore say that these elements form a *basis of* $\mathbf{Q}(x)$ *over the rational numbers*. Our knowledge from Proposition 4.9 can be briefly written in the form

$$\dim_{\mathbf{Q}} \mathbf{Q}(x) = [\mathbf{Q}(x) : \mathbf{Q}].$$

If x is an algebraic number of degree n over the rational numbers, then algebraic numbers rational in x form a vector space of dimension n over the rational numbers.

Every rational algebraic number in x can be uniquely written as a rational linear combination of $1, x, x^2, \ldots, x^{n-1}$.

4.6 The Primitive Element Theorem

In Sect. 4.5, we saw that the minimal polynomial of the algebraic number $x = \sqrt{2+\sqrt{2}}$ over the rational numbers is given by

$$f(X) = X^4 - 4X^2 + 2,$$

that is, $f(X)$ is the normalized polynomial of smallest degree with rational coefficients that has the algebraic number x as a root.

However, if we allow $\sqrt{2}$ (and all rational expressions in $\sqrt{2}$) in addition to the rational numbers, there is a normalized polynomial of smaller degree that has x as a root, namely

$$g(X) = X^2 - 2 - \sqrt{2}. \tag{4.16}$$

We can therefore say that the concept of the minimal polynomial is obviously a "relative" concept that depends on the underlying coefficient domain (rational numbers or numbers rational in $\sqrt{2}$). In Sect. 4.5, we examined the concept of the minimal polynomial over the rational numbers. In the following, we want to do this for more general coefficient domains.

For this, we first choose an algebraic number y such as $\sqrt{2}$. The allowed coefficients are then numbers rational in y. Since y is algebraic, such a number can always be written as a polynomial in y with rational coefficients, i.e., as an expression of the form $a_n y^n + a_{n-1}y^{n-1} + \cdots + a_1 y + a_0$ with rational numbers $a_0, \ldots, a_n$. The allowed coefficient field is therefore $\mathbf{Q}(y) = \mathbf{Q}[y]$. In the case of $y = 0$ (or y being any other rational number), we

return to the previously considered coefficient field of rational numbers. For brevity, we agree on the following terminology:

A *polynomial over* y is a polynomial whose coefficients are all elements of $\mathbf{Q}(y)$, i.e., numbers rational in y. The set of polynomials over y is denoted by $\mathbf{Q}(y)[X]$. The polynomial $g(X)$ from (4.16) is, for example, a polynomial over $\sqrt{2}$.

To generalize the considerations that led to the minimal polynomial over the rational numbers to the more general coefficient domain $\mathbf{Q}(y)$, a so-called *number field*, we repeat some things in this more general context:

Since the domain of numbers rational in y is closed under the basic arithmetic operations within the algebraic numbers, the proof of Proposition 4.1 can be directly transferred from polynomials with algebraic coefficients to polynomials over y, and we obtain:

Proposition 4.10 *Let $f(X)$ and $g(X)$ be two polynomials over y. Then there exists a normalized polynomial $d(X)$ over y (where we explicitly allow $d(X) = 0$) that divides both $f(X)$ and $g(X)$ and for which polynomials $p(X)$ and $q(X)$ over y with*

$$d(X) = p(X) \cdot f(X) + q(X) \cdot g(X)$$

exist. □

As in the case of polynomials with rational coefficients, the following statements can be derived from Proposition 4.10: Let

$$f(X) = f_1(X) \cdot f_2(X) \cdots f_n(X),$$

where $f(X)$, $f_2(X)$, …, $f_n(X)$ are polynomials over y and $f_1(X)$ is a polynomial with algebraic coefficients. Then $f_1(X)$ is also a polynomial over y.

If $f(X)$ is a polynomial over y, and we denote by $f_0(X)$ the separable polynomial with the same roots (without multiplicities) as $f(X)$, then $f_0(X)$ is also a polynomial over y.

We continue with extending some definitions to polynomials over y: Let $g(X)$ be a normalized polynomial over y. A *decomposition* of $g(X)$ over y is a product of the form $g(X) = g_1(X) \cdots g_m(X)$, where $g_1(X)$, …, $g_m(X)$ are non-constant normalized polynomials over y. A polynomial $g(X)$ over y is called *irreducible* over y if it only allows the trivial decomposition $g(X) = g(X)$ over y.

For example, the irreducible polynomial $f(X) = X^4 - 4X^2 + 2$ over the rational numbers is not irreducible over $\sqrt{2}$, as it allows the decomposition

$$f(X) = (X^2 - 2 - \sqrt{2}) \cdot (X^2 - 2 + \sqrt{2})$$

over $\sqrt{2}$.

Even if $f(X)$ is irreducible over the rational numbers, it is generally not irreducible over an extension of the coefficient domain.

We have established in Sect. 4.2 that a polynomial $f(X)$ with rational coefficients is irreducible over the rational numbers if and only if there is no proper combination $x_{i_1}, \ldots, x_{i_k}$, $1 \leq i_1 < i_2 \cdots < i_k \leq n$ of the roots (with multiplicities) of $f(X)$, such that the elementary symmetric functions

$$e_j(x_{i_1}, \ldots, x_{i_k}), \quad 1 \leq j \leq k$$

evaluated in the x_j are all rational numbers. The considerations that led to this are just as feasible for polynomials over y, so that a normalized polynomial $g(X)$ is irreducible over y if and only if there is no proper combination of roots with multiplicities of $g(X)$ whose elementary symmetric functions are all rational in y. To derive from this an effective decision procedure about the irreducibility of polynomials over y, we need, analogously to the case of polynomials over the rational numbers, a procedure to determine whether a given algebraic number x is rational in y.

This task is essentially solved with the following theorem of the primitive element:

Theorem 4.2 *Let x and y be two algebraic numbers. Then there exists an algebraic number z such that z is rational in x and y and both x and y are rational in z.*

The statement of the theorem is thus that an algebraic number z with

$$\mathbf{Q}(z) = \mathbf{Q}(x, y)$$

exists. The element z is called a *primitive element* for x and y. The statement of the theorem can easily be extended to more than two elements.

If $x_1, \ldots, x_n$ are algebraic numbers, then there exists an algebraic number z rational in $x_1, \ldots, x_n$ such that $x_1, \ldots, x_n$ are each rational in z.

The relationship between x, y, and z is illustrated by a diagram as in Fig. 4.3. The lines in the diagram represent extensions of coefficient domains: So every rational number in x and also in y is rational. Furthermore, every number rational in x is a number rational in z, and every rational number in y is also a rational number in z.

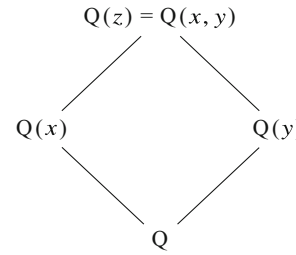


Fig. 4.3 The primitive element theorem

Proof Let λ initially be any rational number. We make the ansatz

$$z = x + \lambda y.$$

Obviously, z is then rational in x and y. We then have to consider for which choices of λ x and y can be expressed through z, in turn. It is sufficient to show that y is rational in z, because due to $x = z - \lambda y$, it would then follow that x is also rational in z.

Let $f(X)$ be the minimal polynomial of x over the rational numbers and $g(X)$ the minimal polynomial of y over the rational numbers. With S we want to denote the set of rational numbers of the form

$$\frac{x' - x}{y - y'}$$

where x' is any root of $f(X)$ and y' is a root of $g(X)$ different from y.

We claim that for all rational numbers λ outside of S, the number z is primitive for x and y. To this end, we consider the polynomial

$$h(X) = f(z - \lambda X)$$

over z. Since $x = z - \lambda y$ holds by construction of z, $h(X)$ has the number y as a root. Let $d(X)$ be the greatest common divisor of $g(X)$ and $h(X)$, which can be calculated with the Euclidean algorithm. The coefficients of $d(X)$ are numbers rational in z.

As a common root of $g(X)$ and $h(X)$, y is also a root of $d(X)$. Since $g(X)$ as a minimal polynomial over the rational numbers is separable, the roots of $g(X)$ are pairwise different.[6] Thus, the roots of the divisor $d(X)$ are also pairwise different.

Suppose, $\deg d(X) \geq 2$. Then there exists an algebraic number $y' \neq y$ with $d(y') = 0$. This number y' is therefore a simultaneous root of $g(X)$ and $h(X)$. Set $x' := z - \lambda y'$. Then $f(x') = f(z - \lambda y') = h(y') = 0$. Thus, we have

$$\frac{x' - x}{y - y'} = \frac{z - \lambda y' - x}{y - y'} = \frac{x + \lambda y - \lambda y' - x}{y - y'} = \lambda \in S,$$

a contradiction to the choice of λ.

Therefore, $d(X)$ must be a normalized polynomial over z of degree 1, i.e., $d(X) = X - a$ with a number a rational in z. Because $d(y) = 0$, it follows that $y = a$, so y is a number rational in z. □

The proof of the theorem about the primitive element is based on the fact that the minimal polynomial of one of the two involved algebraic numbers x and y is separable.

Example 4.12 We want to construct a primitive element for $x = \sqrt{2}$ and $y = \sqrt{3}$. The minimal polynomial of x over the rational numbers is $f(X) = X^2 - 2$; the minimal polynomial of y over the rational numbers is $g(X) = X^2 - 3$. The roots of $f(X)$ are $\sqrt{2}$ and $-\sqrt{2}$; the roots of $g(X)$ are $\sqrt{3}$ and $-\sqrt{3}$. The set defined in the proof of Theorem 4.2 consists in this example only of the number 0 (all other possible coefficients are irrational). Thus, we can choose $\lambda = 1$, for example, and find that $\sqrt{2} + \sqrt{3}$ is primitive to $\sqrt{2}$ and $\sqrt{3}$.

How can we use the theorem of the primitive element to determine whether an algebraic number x is rational in y? For this, we first choose a primitive element z for x and y. Both the numbers rational in y and the numbers rational in z have bases over the rational numbers, the lengths of which are given by the degrees of the algebraic numbers y and z. Moreover, by construction of z, every number rational in y is a number rational in z. Due to the statements known from linear algebra about the length of bases, it must therefore always be the case that $[\mathbf{Q}(y) : \mathbf{Q}] \leq [\mathbf{Q}(z) : \mathbf{Q}]$.

In the case $[\mathbf{Q}(y) : \mathbf{Q}] = [\mathbf{Q}(z) : \mathbf{Q}]$, the bases have the same length, so $\mathbf{Q}(z) = \mathbf{Q}(y)$. Since x is rational in z, it is in this case therefore also rational in y.

In the case $[\mathbf{Q}(y) : \mathbf{Q}] < [\mathbf{Q}(z) : \mathbf{Q}]$, there must be numbers rational in z that are not rational in y. And therefore z cannot be rational in y. Because $z \in \mathbf{Q}(x, y)$, x also cannot be rational in y, because otherwise $\mathbf{Q}(x, y) = \mathbf{Q}(y)$.

Therefore, a simple comparison of the degrees of y and z over the rational numbers determines whether x is a number rational in y or not. In particular, we now have a procedure for determining or refuting the irreducibility of a polynomial over y.

Example 4.13 We want to determine whether $x = \sqrt{2}$ is a number rational in $y = \sqrt{3}$. To do this, we first write down the minimal polynomial of y: It is $g(X) = X^2 - 3$. Thus, y has degree 2 over the rational numbers.

We have already seen that $z = \sqrt{2} + \sqrt{3}$ is a primitive element for x and y. We claim that

$$h(X) = X^4 - 10X^2 + 1 = 0$$

is the minimal polynomial of z over the rational numbers: A brief calculation shows that $x_1 = z$ is a root of $h(X)$. The other roots are $x_2 = \sqrt{2} - \sqrt{3}$, $x_3 = -\sqrt{2} + \sqrt{3}$ and $x_4 = -\sqrt{2} - \sqrt{3}$. A calculation to a few decimal places shows that there is no proper combination of roots whose elementary symmetric functions are all integers. Thus, the primitive polynomial $h(X)$ is irreducible over the integers and therefore also irreducible as a polynomial over the rational numbers.

Since $\deg h(X) = 4$, it follows that

$$4 = [\mathbf{Q}(\sqrt{2} + \sqrt{3}) : \mathbf{Q}] > [\mathbf{Q}(\sqrt{3}) : \mathbf{Q}] = 2.$$

Thus, $\mathbf{Q}(\sqrt{2} + \sqrt{3}) = \mathbf{Q}(\sqrt{2}, \sqrt{3})$ is strictly greater than $\mathbf{Q}(\sqrt{3})$, so $\sqrt{2}$ is not a number rational in $\sqrt{3}$, and in particular cannot be expressed as a polynomial in $\sqrt{3}$ with rational coefficients.

In the case that the coefficient domain consists of the rational numbers, we have inferred from the existence of the irreducibility test that every polynomial with rational coefficients has a prime factor decomposition over the rational numbers. The same applies to polynomials over y. We have inferred the uniqueness of this decomposition in the case of rational coefficients from the Abel's irreducibility theorem Theorem 4.1, whose proof can be literally transferred to polynomials over y. That is, we have:

Theorem 4.3 *Let $f(X)$ and $g(X)$ be two normalized polynomials over an algebraic number y that have an algebraic number x as a common root. If $f(X)$ is irreducible over y, then $f(X)$ divides the polynomial $g(X)$.* □

From this we infer the uniqueness of the prime factor decomposition of polynomials over y:

Proposition 4.11 *Let $f_1(X), \ldots, f_n(X)$ and $g_1(X), \ldots, g_m(X)$ be irreducible normalized polynomials over y such that*

$$f_1(X) \cdots f_n(X) = g_1(X) \cdots g_m(X).$$

Then the factors $f_1(X), \ldots, f_n(X)$ coincide with the factors $g_1(X), \ldots, g_m(X)$ with multiplicities and up to order, that is, $m = n$, and there exists a permutation σ of n letters with $g_i(X) = f_{\sigma(i)}(X)$ for all $i \in \{1, \ldots, n\}$. □

Example 4.14 The polynomial

$$f(X) = X^3 - 2X^2 + X - 2$$

has the prime factor decomposition

$$f(X) = (X^2 + 1) \cdot (X - 2)$$

over the rational numbers (the equality of both sides is confirmed by explicit calculation; also, both factors are irreducible). Since over i, the first factor, $X^2 + 1 = (X - i) \cdot (X + i)$, decomposes, the prime factor decomposition of $f(X)$ over i is given by

$$f(X) = (X - i) \cdot (X + i) \cdot (X - 2)$$

This example thus demonstrates that the prime factor decomposition of a polynomial can very well depend on which coefficient range we allow.

From the above generalizations, the following follows analogously to the procedure as in Sect. 4.5:

Proposition 4.12 *Let x be an algebraic number. Then there exists exactly one irreducible normalized polynomial $f(X)$ over y that has x as a root.* □

This uniquely determined polynomial $f(X)$ we call the *minimal polynomial* of x over y. The degree of $f(X)$ is called the *degree* of x over y. We write

$$[\mathbf{Q}(x, y) : \mathbf{Q}(y)]$$

for the degree of x over y. In the case that y is rational in x, i.e. $\mathbf{Q}(x, y) = \mathbf{Q}(x)$, we also write

$$[\mathbf{Q}(x) : \mathbf{Q}(y)]$$

for the degree of x over y. As in the case of rational coefficients, we can show that the minimal polynomial over y is always separable.

The terms irreducibility, prime factor decomposition, and minimal polynomial can be defined not only over the rational numbers, but also relative to a more general number field.

Let's return to our example from the beginning of the section. There we have found that

$$g(X) = X^2 - 2 - \sqrt{2}$$

is a polynomial over $y = \sqrt{2}$ that has $x = \sqrt{2 + \sqrt{2}}$ as a root.

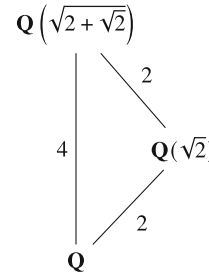


Fig. 4.4 Relative degree of algebraic elements

We claim that $g(X)$ is irreducible over $\sqrt{2}$. If $g(X)$ were not irreducible, $g(X)$ would decompose into two linear factors over $\sqrt{2}$, in particular, $\sqrt{2+\sqrt{2}}$ would be a number rational in $\sqrt{2}$. Since $\sqrt{2} = (\sqrt{2+\sqrt{2}})^2 - 2$, also $\sqrt{2}$ is a number rational in $\sqrt{2+\sqrt{2}}$. We would thus have $\mathbf{Q}(\sqrt{2}) = \mathbf{Q}\left(\sqrt{2+\sqrt{2}}\right)$. But this cannot be, since the dimension of the left vector space over the rational numbers is the degree of $\sqrt{2}$ over the rational numbers, i.e., 2, while the corresponding dimension of the left side is the degree of $\sqrt{2+\sqrt{2}}$ over the rational numbers, which we determined to be 4 at the beginning of the section.

Consequently, $g(X)$ must be irreducible over $\sqrt{2}$ and is thus the minimal polynomial of $\sqrt{2+\sqrt{2}}$ over $\sqrt{2}$. The degree of $\sqrt{2+\sqrt{2}}$ over $\sqrt{2}$ is thus 2. We illustrate this with a diagram as in Fig. 4.4, where the numbers next to the lines indicate the relative degrees.

At the end of Sect. 4.5, we gave an interpretation of the degree of an algebraic number x over the rational numbers, namely as dimension of $\mathbf{Q}(x)$ over the rational numbers: So we have seen, that $1, x, x^2, \ldots, x^{n-1}$ form a basis of $\mathbf{Q}(x)$ over $\mathbf{Q}$, if n is the degree of the minimal polynomial of x over the rational numbers.

The same considerations remain true, when we move to the relative situation: Let

$$g(X) = X^n + a_{n-1}X^{n-1} + \cdots + a_1 X + a_0$$

be the minimal polynomial of x over an algebraic number y.

In particular, the $a_0, \ldots, a_{n-1}$ are numbers rational in y, and n is the degree of x over y. From the fact that $g(X)$ as a polynomial over y and with zero x has minimal degree, it follows that $1, x, x^2, \ldots, x^{n-1}$ are linearly independent over y. This means, if

$$b_0 + b_1 x + b_2 x^2 + \cdots + b_{n-1}x^{n-1} = 0$$

holds for numbers $b_0, \ldots, b_{n-1}$ rational in y, then already $b_0 = b_1 = \cdots = b_{n-1} = 0$. Let z be any number rational in x and y, i.e., any element of $\mathbf{Q}(x, y) = \mathbf{Q}[x, y]$. We can write this in the form

$$z = \sum_{i,j} c_{ij} x^i y^j \tag{4.17}$$

with rational coefficients c_{ij}. Due to the relation $x^n = -a_0 - a_1 x - \cdots - a_{n-1} x^{n-1}$, which follows from $g(x) = 0$, we can replace in (4.17) powers of x with an exponent that is greater or equal to n by linear combinations of $1, x, x^2, \ldots, x^{n-1}$ whose coefficients are rational in y. This allows us to write z as a linear combination of $1, x, x^2, \ldots, x^{n-1}$ over y. Thus, $1, x, x^2, \ldots, x^{n-1}$ is a linearly independent generating system of all numbers rational in x and y over y, i.e., a basis, the length of which is given by the degree of x over y. We can express this insight compactly by

$$[\mathbf{Q}(x, y) : \mathbf{Q}(y)] = \dim_{\mathbf{Q}(y)} \mathbf{Q}(x, y)$$

4.7 The Degree Formula and the Impossibility of Doubling the Cube and Trisecting the Angle

In this section, we want to use the insights gained in this chapter to show that both the task of doubling the cube and that of trisecting the angle are unsolvable with only a compass and straightedge.

The problem of cube doubling is the one illustrated in Fig. 4.5: Given the length of a side of a cube, construct only with compass and straightedge the length of the side of a cube with double volume. If the side length of the given cube is given by a, then its volume is known to be given by a^3. Let's call the side length of the cube to be constructed b, so $b^3 = 2a^3$ should hold. We can solve this equation for b and get

$$b = a\sqrt[3]{2}.$$

Similar to how we solved the problem of squaring of the circle, we make an identification of the drawing plane with the Gaussian number plane in such a way that the given stretch of length a in the drawing plane coincides with the stretch from 0 to

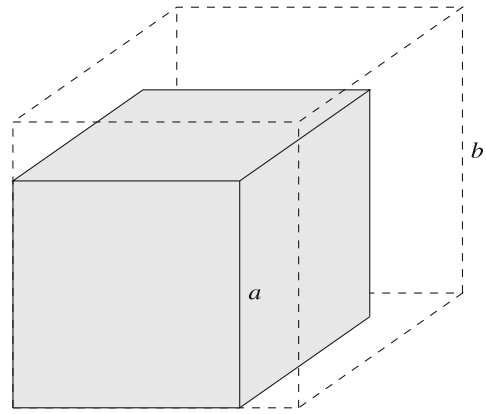


Fig. 4.5 Doubling of the cube

1 in the Gaussian number plane, i.e., $a = 1$. The task is then to construct a point with distance $\sqrt[3]{2}$ from the origin using only a compass and straightedge. If we could do this, we could, in particular, also construct such a point on the positive part of the real axis; the real number $\sqrt[3]{2}$ would thus be constructible. To be able to prove the impossibility of doubling the cube, it is therefore sufficient to show that $\sqrt[3]{2}$ is not a constructible number. If $\sqrt[3]{2}$ were transcendental, this would be immediately clear, as constructible numbers are algebraic according to Theorem 2.1. However, $\sqrt[3]{2}$ as a solution to the equation

$$X^3 - 2 = 0$$

is itself an algebraic number, so there is no such simple proof as in the case of Theorem 3.4 (simple in the relative sense that the transcendence of π is known!). To prove that certain algebraic numbers are not constructible, we must look more closely at the characterization of constructible numbers according to Theorem 2.1. According to this theorem, an algebraic number x is constructible if and only if there is a sequence of algebraic numbers $x_1, \ldots, x_{n-1}, x_n = x$ such that for all $i \in \{1, \ldots, n\}$ the algebraic number x_i or x_i^2 is rational in $x_1, \ldots, x_{i-1}$.

From this description we want to conclude:

Lemma 4.2 *Let x be a constructible number. Then x is an algebraic number whose degree over the rational numbers is a power of two, i.e., of the form 2^n, $n \in \mathbf{N}_0$.*

The degree of a constructible number is always a power of two. Thus, any algebraic number whose degree is not a power of two is not constructible.

Once we have proven Lemma 4.2, we have solved the problem of doubling the cube (in the negative):

Theorem 4.4 *Doubling the cube with compass and straightedge is impossible.*

Proof of Theorem 4.4 If doubling the cube were possible, $\sqrt[3]{2}$ would be a constructible number. The degree of $\sqrt[3]{2}$ over the rational numbers is 3, since $f(X) = X^3 - 2$ as an irreducible polynomial (for example by the Eisenstein criterion with $p = 2$) is the minimal polynomial of $\sqrt[3]{2}$ over the rational numbers. So, an algebraic number would be constructible whose degree over the rational numbers is not a power of two, a contradiction to Lemma 4.2. □

To be able to prove Lemma 4.2, we need the following important relationship between the degrees of algebraic elements, which is generally known as the "degree formula":

Theorem 4.5 *Let x be an algebraic number. Let y be another algebraic number that is rational in x (i.e. $\mathbf{Q}(x, y) = \mathbf{Q}(x)$). The following formula holds for the degrees of x and y over the rational numbers and the degree of x over y:*

$$[\mathbf{Q}(x) : \mathbf{Q}] = [\mathbf{Q}(x) : \mathbf{Q}(y)] \cdot [\mathbf{Q}(y) : \mathbf{Q}]. \tag{4.18}$$

More precisely, the following statement holds: If $u_1, \ldots, u_m$ form a basis of $\mathbf{Q}(y)$ over $\mathbf{Q}$ and $v_1, \ldots, v_n$ form a basis of $\mathbf{Q}(x)$ over $\mathbf{Q}(y)$, then $u_1 v_1, \ldots, u_m v_n$ form a basis of $\mathbf{Q}(x)$ over $\mathbf{Q}$.

Proof Since the degrees in (4.18) are given by the lengths of bases, it is sufficient to prove the second part of Theorem 4.5.

Let z be a number rational in x. By assumption, there exist rational numbers $b_1, \ldots, b_n$ in y such that

$$z = \sum_{j=1}^{n} b_j v_j. \tag{4.19}$$

Furthermore, there exist rational numbers $a_{1j}, \ldots, a_{mj}$ such that

$$b_j = \sum_{i=1}^{m} a_{ij} u_i$$

for all $j \in \{1, \ldots, n\}$. If we substitute (4.20) into (4.19), we obtain

$$z = \sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij} u_i v_j. \tag{4.20}$$

Consequently, the products $u_1 v_1, \ldots, u_m v_n$ form a system of generators over the rational numbers for all numbers rational in x.

It remains to show that $u_1 v_1, \ldots, u_m v_n$ are linearly independent over the rational numbers. To do this, we make the assumption

$$\sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij} u_i v_j = 0 \tag{4.21}$$

for rational numbers $a_{11}, \ldots, a_{mn}$ and must show that $a_{11} = \cdots = a_{mn} = 0$.

To do this, we write (4.21) in the form

$$\sum_{j=1}^{n}\left(\sum_{i=1}^{m} a_{ij}u_i\right)v_j = 0.$$

Since $v_1, \ldots, v_n$ are linearly independent over the numbers rational in y, it follows

$$\sum_{i=1}^{m} a_{ij}u_i = 0$$

for all $j \in \{1, \ldots, n\}$. From the linear independence of $u_1, \ldots, u_m$ over the rational numbers, it finally follows $a_{ij} = 0$ for all $i \in \{1, \ldots, m\}$. □

Example 4.15 One application of the degree formula is to determine the irreducibility of polynomials, as will be illustrated in the following example:

Let

$$x = \sqrt[3]{1 + \mathrm{i}\sqrt{3}}.$$

Then x satisfies the equation $f(X) = 0$ with

$$f(X) = X^6 - 2X^3 + 4$$

as can be confirmed by substitution and expansion. Since $f(X + 1) = X^6 + 6X^5 + 15X^4 + 18X^3 + 9X^2 + 3$ is irreducible according to the Eisenstein criterion (for the prime number $p = 3$), $f(X)$ is also irreducible over the rational numbers and thus the minimal polynomial of x over the rational numbers. Consequently, $[\mathbf{Q}(x) : \mathbf{Q}] = 6$.

We want to calculate the minimal polynomial of x over $y = \mathrm{i}\sqrt{3}$. The algebraic number y itself has degree 2 over the rational numbers, because its minimal polynomial is $X^2 + 3$. First, we note that

$$g(X) = X^3 - 1 - \mathrm{i}\sqrt{3}$$

is a polynomial over y that has x as a root. We want to use the degree formula to show that this polynomial over y is irreducible, thus the minimal polynomial of x over y: If $g(X)$ were not the minimal polynomial, there would be a polynomial of lower degree over y that has x as a root, so we would have $[\mathbf{Q}(x) : \mathbf{Q}(y)] < 2$. On the other hand, according to the degree formula, however,

$$[\mathbf{Q}(x) : \mathbf{Q}(y)] = [\mathbf{Q}(x) : \mathbf{Q}]/[\mathbf{Q}(y) : \mathbf{Q}] = 6/2 = 3,$$

contradiction.

In a "tower" of number fields, the total degree is the product of the relative degrees.

The degree formula immediately implies the following important result:

Corollary 4.3 *Let x be an algebraic number. Let y be an algebraic number that is rational in x. Then the degree of y over the rational numbers is a divisor of the degree of x over the rational numbers.* □

With this, we can finally give the still outstanding proof of Lemma 4.2, that is, we show that the degree of every constructible element x over the rational numbers is a power of two.

Proof of Lemma 4.2 Due to Theorem 2.1, there exist algebraic numbers $x_1, \ldots, x_{n-1}$, $x_n = x$, such that for all $i \in \{1, \ldots, n\}$ the element x_i or its square is rational in $x_1, \ldots, x_{i-1}$. Let z_i be a primitive element to $x_1, \ldots, x_i$. We want to show by induction over i that the degree of z_i over the rational numbers is a power of two. Once we have this, Lemma 4.2 will follow since the degree of the rational element x_n in z_n due to Corollary 4.3 is a divisor of the degree of z_n, thus also a power of two.

The induction start $i = 0$ is trivial, because z_0 is any rational number and thus has degree 1, thus a power of two, over the rational numbers. For the induction step to some $i > 0$, we use that that z_i is a rational expression in z_{i-1} and x_i. Since x_i^2 is a rational expression in z_{i-1}, 1 and x_i generate the numbers rational in z_i over the numbers rational in z_{i-1}. Thus, the degree d of z_i over z_{i-1} is either 1 or 2. According to Theorem 4.5, it is

$$[\mathbf{Q}(z_i) : \mathbf{Q}] = [\mathbf{Q}(z_i) : \mathbf{Q}(z_{i-1})] \cdot [\mathbf{Q}(z_{i-1}) : \mathbf{Q}] = d \cdot [\mathbf{Q}(z_{i-1}) : \mathbf{Q}].$$

According to the induction hypothesis, the last factor is a power of two. Thus the degree of z_i over the rational numbers is also a power of two. □

We can immediately exclude the constructibility of algebraic numbers if their degree is not a power of two. The converse, that every algebraic number whose degree is a power of two, is constructible, is however false. For a counterexample, we consider the polynomial

$$f(X) = X^4 + 8X + 12.$$

This normalized polynomial is irreducible over the rational numbers, because it is irreducible over the integers, as we can see as follows: First, $f(X)$ has no root because

none of the potential divisors of 12 is a root, as can be confirmed by evaluating. This leaves only a decomposition of the form

$$f(X) = (X^2 + aX + b) \cdot (X^2 + cX + d)$$

with integers a, b, c, d. Comparing coefficients yields among other things $a + c = 0$, $ac + b + d = 0$ and $bd = 12$. This would give us $c = -a$, so $b + d = -ac = a^2$ would be a square in the integers. However, for all possible choices of b and d that satisfy $bd = 12$, $b + d$ is not a square.

Let x be an algebraic number that is a root of $f(X)$. Then $f(X)$ is the minimal polynomial of x over the rational numbers, so the degree of x over the rational numbers is given by 4, a power of two. Yet we claim, however, that x is not a constructible number. To this end, we assume that x is constructible and want to lead this to a contradiction. From the proof of Lemma 4.2, it follows that there exists an algebraic number y of degree 2 over the rational numbers, such that the minimal polynomial $g(X)$ of x over y has degree 2.

Since y is a root of a normalized polynomial of degree 2, y is a rational expression in a square root of a rational number according to the known solution formula for quadratic equations. We can assume without loss of generality that y itself is a square root of a rational number, say $y^2 = e$ with e rational. The coefficients of $g(X)$ and more generally the numbers rational in y are all of the form $a + by$ with rational numbers a and b, since y is a root of a normalized polynomial of degree 2 over the rational numbers. We can therefore write the minimal polynomial of x over y as

$$g(X) = X^2 + (a + by)X + c + dy$$

where a, b, c, and d are rational numbers.

We have $g(x) = 0$. If we bring the terms linear in y to one side and then square, we get $x^4 + 2ax^3 + (a^2 + 2c)x^2 + 2acx + c^2 = b^2ex^2 + 2bdex + d^2e$, so

$$x^4 + 2ax^3 + (a^2 + 2c - b^2e)x^2 + (2ac - 2bde)x + c^2 - d^2e = 0.$$

The minimal polynomial of x over the rational numbers, i.e., $f(X)$, is unique, so by comparing coefficients we conclude $a = 0$ and thus

$$2c - b^2e = 0,$$
$$-2bde = 8$$

and

$$c^2 - d^2e = 12$$

Since due to the second relation neither b nor d vanish, $z = -4\frac{b}{d}$ is a well-defined invertible rational number. It follows $e = \frac{z}{b^2}$, from which the first relation gives $2c = z$ and the third gives $c^2 = 12 + d^2 e = 12 + 16z^{-1}$.

Substituting $2c = z$ into the last equality, we get $z^2 = 4c^2 = 48 + 64z^{-1}$, so

$$z^3 - 48z - 64 = 0. \tag{4.22}$$

None of the integer divisors of 64 is a root of $X^3 - 48X - 64$, so no rational z can exist that satisfies (4.22), contradiction. Therefore, x is not a constructible number.

There are algebraic numbers whose degree is a power of two, but which are nevertheless not constructible. Thus, the degree being a power of two is only a necessary, but not a sufficient condition for the constructibility of an algebraic number.

At the end of this chapter, we would like to briefly describe the problem of trisecting an angle and also provide the proof of impossibility, which is very similar to the proof of the impossibility of doubling the cube. The problem of angle trisection is as follows: Given an arbitrary angle α, construct an angle β with $3\beta = \alpha$ using only a compass and straightedge.

The impossibility of angle trisection does not mean that no angle can be trisected—since an angle of 30° can obviously be constructed, the angle of 90° can obviously be trisected—,but that there is no method that provides a trisection for every angle. To prove the impossibility of angle trisection, it is therefore sufficient to specify an angle whose trisection is impossible. We want to do this for the angle $\alpha = 60°$, which itself can be constructed with a compass and straightedge. Suppose we could construct $\beta = \frac{\alpha}{3} = 20°$ as in Fig. 4.6. Then the line segment $x = \cos\beta$ would also be constructible by dropping a

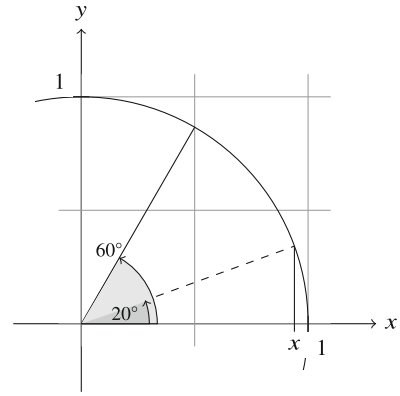


Fig. 4.6 Trisecting an angle of 60°

perpendicular, and thus $y = 2x$ would also be constructible. From the addition theorems of cosine, it follows

$$\cos 3\beta = 4\cos^3 \beta - 3\cos \beta.$$

Since $\cos 3\beta = \cos 60° = \frac{1}{2}$, y thus satisfies the cubic equation

$$y^3 - 3y - 1 = 0.$$

The polynomial $f(X) = X^3 - 3X - 1$ is irreducible over the rational numbers, since $f(X+1) = X^3 + 3X^2 - 3$ is irreducible according to the Eisenstein criterion (for the prime number $p = 3$). Therefore, $f(X)$ is the minimal polynomial of y over the rational numbers, so y has degree 3 over the rational numbers. Consequently, y is not a constructible number.

We have thus proven:

Theorem 4.6 *Trisecting the general angle with a compass and straightedge is impossible.* □

If it were possible to trisect an angle, we could construct an angle of 20° from the constructible angle of 60°. However, its cosine is an algebraic number of degree 3 and is therefore not constructible. Therefore, the general trisection of an angle cannot be possible.

Summary

- Any two polynomials have a **greatest common divisor**.
- The greatest common divisor can be calculated using the **Euclidean algorithm**.
- A polynomial is called **separable** if it does not have multiple roots in the ring of algebraic numbers, i.e., if its discriminant does not vanish. A polynomial is separable if and only if it is coprime to its **formal derivative**. It is sufficient to study separable polynomial equations if we are not interested in the multiplicities of the solutions..
- A polynomial is called **irreducible** over some ring of coefficients if it cannot be written as a product of polynomials of lower degree. Over the rational numbers, there is an effective numerical test for irreducibility.

(continued)

- If a primitive polynomial fulfills the **Eisenstein criterion**, it is irreducible.
- If a normalized integer polynomial is irreducible **modulo a prime number**, it is also irreducible over the rational numbers.
- Every algebraic number x is the root of exactly one normalized irreducible polynomial with rational coefficients. This polynomial is called the **minimal polynomial** of x. The degree of the minimal polynomial is the **degree** of the algebraic number. The rational numbers are exactly the algebraic numbers of degree 1 over the rational numbers.
- If y is an algebraic number, we can also consider the minimal polynomial and degree of x **over** y. Here, the coefficients of the minimal polynomial of x over y may be numbers rational in y.
- According to the **degree formula**, the degree of x over the rational numbers is equal to the degree of x over y multiplied by the degree of y over the rational numbers.
- This shows that the degree of a constructible number over the rational numbers is always a **power of two**, because a square root x from an algebraic number y is either of degree 1 or degree 2 over y.
- For the **doubling of the cube** with compass and straightedge, the construction of $\sqrt[3]{2}$ is necessary. However, this number has degree 3 over the rational numbers, as it is the root of the irreducible polynomial $X^3 - 2$.
- The cosine of $20°$ is the root of an irreducible polynomial of degree 3 over the rational numbers and is therefore not constructible. Since $\cos 60°$ is constructible, it follows that **angle trisection** with compass and straightedge is generally not possible.

Exercises

Separability

4.1 Let

$$f(X) = X^3 - 2X^2 + 2X - 4$$

and

$$g(X) = X^2 - 3X + 2$$

be as in Example 4.1. Give polynomials $p(X)$ and $q(X)$ with $(X - 2) = p(X) \cdot f(X) + q(X) \cdot g(X)$.

4.2 Provide a proof of Proposition 4.2.

4.3 Let $f(X)$ and $g(X)$ be two polynomials with algebraic coefficients. Show that exactly one normalized polynomial $d(X)$ (the case $d(X) = 0$ is explicitly allowed) exists that is a greatest common divisor of $f(X)$ and $g(X)$.

4.4 Let $f(X)$ and $g(X)$ be two polynomials with algebraic coefficients. Define and construct the *least common multiple of* $f(X)$ *and* $g(X)$.

4.5 Let $f(X)$ be a normalized polynomial over the rational numbers. Show that $f(X)$ is separable if and only if the greatest common divisor of $f(X)$ and $f'(X)$ is the constant polynomial 1.

4.6 Provide a normalized polynomial equation of minimal degree over the rational that has the same solutions (but without multiplicities) as the equation

$$X^7 + X^6 - 4X^4 - 4X^3 + 4X + 4 = 0.$$

4.7 Let a be an algebraic number. Let $f(X) = X^3 + 2a^2X - a + 5$. Construct a polynomial equation with rational coefficients that is satisfied by a if and only if $f(X)$ is not a separable polynomial.

Irreducible Polynomials

4.8 Is a normalized polynomial of degree 1 with rational coefficients always irreducible?

4.9 Let x and y be two algebraic numbers with $x \cdot y = 0$. Show that then $x = 0$ or $y = 0$ (or both).

4.10 Show that normalized polynomials of degree 2 and 3 over the rational numbers are irreducible if and only if they have no rational root.

4.11 Show that the polynomial

$$f(X) = X^3 - \frac{3}{2}X^2 + X - \frac{6}{5}$$

has no rational root.

4.12 Let $f(X)$ and $g(X)$ be two normalized polynomials with rational coefficients. Provide a method for calculating the greatest common divisor of $f(X)$ and $g(X)$ that uses the decomposition of $f(X)$ and $g(X)$ into irreducible polynomials.

4.13 Let $f(X) \neq 1$ be a normalized polynomial over the rational numbers that is prime, that is: If $f(X)$ divides a product $g(X) \cdot h(X)$ of polynomials $g(X)$ and $h(X)$ with rational coefficients, then $f(X)$ divides at least one of the factors $g(X)$ and $h(X)$. Show that $f(X)$ is an irreducible polynomial.

4.14 Let $f(X)$ be a normalized irreducible polynomial with rational coefficients. Let $g_1(X), \ldots, g_n(X)$ be further polynomials with rational coefficients. Let $f(X)$ be a divisor of the product $g_1(X) \cdots g_n(X)$. Show that then there exists an $i \in \{1, \ldots, n\}$ such that $f(X)$ is a divisor of $g_i(X)$.

4.15 Show that the valuation behaves like a logarithm. This means the following: Let $p(X)$ be an irreducible normalized polynomial over the rational numbers. Then

$$\operatorname{ord}_{p(X)} 1 = 0$$

and

$$\operatorname{ord}_{p(X)}(f(X) \cdot g(X)) = \operatorname{ord}_{p(X)} f(X) + \operatorname{ord}_{p(X)} g(X)$$

hold for arbitrary normalized polynomials $f(X)$ and $g(X)$.

Irreducibility over the Integers

4.16 Numerically determine the roots of $f(X) = X^4 - 10X^2 + 1$ to a few decimal places, and use this information to show that $f(X)$ is irreducible over the rational numbers.

4.17 Let $f(X)$ be a non-vanishing polynomial with rational coefficients. Show that the content of $f(X)$ is an integer if and only if $f(X)$ has integer coefficients.

4.18 Let $f(X)$ be a normalized polynomial with rational coefficients. Show that the content of $f(X)$ is the inverse of the leading coefficient of $\tilde{f}(X)$.

4.19 Let $f(X)$ be a polynomial with rational coefficients. Show that the content of $f(X)$ is the inverse of an integer when $f(X)$ is normalized. Does the converse also hold?

4.20 Let $f(X) = X^6 + X^5 + X^4 + X^3 + X^2 + X + 1$. Show that $f(X)$ is irreducible.

4.21 Let $f(X) = X^5 + X^4 + X^3 + X^2 + X + 1$. Show that $f(X)$ is not irreducible.

4.22 Provide an example that the condition in Eisenstein's irreducibility criterion, Proposition 4.5, that p^2 is not a divisor of a_0 is necessary.

4.23 Provide an example that the condition in Eisenstein's irreducibility criterion, Proposition 4.5, that p is a divisor of all $a_0, \ldots, a_{n-1}$, is necessary.

4.24 Let $f(X)$ and $g(X)$ be two normalized polynomials of positive degree with rational coefficients such that $f(g(X))$ is irreducible. Is $g(X)$ also irreducible?

Irreducibility Modulo a Prime Number

4.25 Let a, a', b and b' be four integers such that the congruences $a \equiv a'$ and $b \equiv b'$ (all modulo some integer n) hold. Verify explicitly that $a + b \equiv a' + b'$ modulo n.

4.26 Let a be an integer with $a \equiv 1$ modulo 3. For which exponents n is then $a^n \equiv 2$ modulo 3?

4.27 Calculate two inverses of 6 modulo 35.

4.28 Let n be a nonnegative integer and a an integer that is coprime to n. Let b and b' be two integers with $ab \equiv ab' \equiv 1 \pmod{n}$. Show that $b \equiv b' \pmod{n}$.

4.29 Let $f(X)$ be a normalized polynomial with integer coefficients. Let p be a prime number such that $f(X)$ modulo p is not irreducible. Is $f(X)$ also not irreducible?

4.30 Let $b_0, \ldots, b_m$ be integers, different from zero. Show that there are only finitely many polynomials $g(X)$ with integer coefficients and $\deg g(X) \leq m$ such that for all $i \in \{0, \ldots, m\}$ the integer $g(i)$ is a divisor of b_i.

4.31 Let $f(X)$ be a primitive polynomial of degree n with integer coefficients. Let $f(i) \neq 0$ for all integers i with $0 \leq i \leq \frac{n}{2}$.

Let

$$f(X) = g(X) \cdot h(X) \tag{4.23}$$

be a factorization, where $g(X)$ and $h(X)$ are polynomials with integer coefficients. Without loss of generality, we assume that $\deg g(X) \leq \frac{n}{2}$ (otherwise we swap both factors).

Argue that for all integers i with $0 \leq i \leq \frac{n}{2}$, the integer $g(i)$ is a divisor of $f(i)$. Infer from this that only finitely many integer polynomials $g(X)$ with (4.23) can exist. Derive a method from this to determine whether $f(X)$ is irreducible over the integers or not.

How can this method be extended to all primitive polynomials of degree n with integer coefficients?

(This method was first proposed by Leopold Kronecker,[7] who was the first to point out the necessity of a method to determine the irreducibility of a polynomial with integer coefficients.)

The Degree of Algebraic Elements

4.32 Let x be an algebraic number. Show that x has degree 1 over the rational numbers if and only if x is rational.

4.33 Give an algebraic number of degree 7 over the rational numbers.

4.34 Show that for every positive natural number n, there exists an algebraic number of degree n over the rational numbers.

4.35 Let x and y be algebraic numbers, whose degrees over the rational numbers are n and m respectively. Show that the degrees of $x + y$ and xy are at most $n \cdot m$.

4.36 Show that the *golden ratio*

$$\phi = \frac{1 + \sqrt{5}}{2} \tag{4.24}$$

is an algebraic integer, even though in the representation (4.24) there are denominators that obviously cannot be cancelled out.

4.37 Give an algebraic number that is not an algebraic integer.

4.38 Let a and d be two integers. Let d be positive. Show that $a + \sqrt{d}$ is an algebraic integer, and calculate its degree depending on a and d.

4.39 Given the algebraic number $x = \sqrt{2} + \sqrt[3]{2}$. Give a natural number n and a vanishing nontrivial linear combination of $1, x, x^2, \ldots, x^n$ with rational coefficients.

4.40 Let z be the solution of the equation

$$X^n + a_{n-1}X^{n-1} + \cdots + a_1 X + a_0 = 0,$$

where the $a_0, \ldots, a_{n-1}$ are algebraic numbers. Give an upper bound for the degree of z in terms of n and the degrees of $a_0, \ldots, a_{n-1}$ over the rational numbers.

4.41 Let x be a solution of the equation

$$X^4 - 2X^3 + 12X - 10 = 0.$$

Express x^6 as a linear combination of $1, x, x^2$ and x^3 with rational coefficients.

4.42 Let $f(X) = X^n + a_{n-1}X^{n-1} + a_1X + a_0$ be the minimal polynomial of the algebraic number x over the rational numbers. Multiplication with x induces a linear mapping

$$\varphi\colon \mathbf{Q}(x) \to \mathbf{Q}(x), \quad y \mapsto x \cdot y.$$

Write down the representation matrix A of this endomorphism with respect to the basis $1, x, x^2, \ldots, x^{n-1}$ of $\mathbf{Q}(x)$ over $\mathbf{Q}$. Show that $f(X)$ is the minimal polynomial of A.

The Primitive Element Theorem

4.43 Give two complex numbers that are linearly independent over the real numbers, but linearly dependent over the complex numbers.

4.44 Let x be a number rational in $\sqrt{3}$. Is a square root of x generally also a number rational in $\sqrt{3}$?

4.45 Find a primitive element for i and $\sqrt[3]{2}$.

4.46 Express $\sqrt{2}$ and $\sqrt{3}$ as polynomials in $\sqrt{2} + \sqrt{3}$ with rational coefficients.

4.47 Show with elementary methods using the ansatz $\sqrt{2} = a + b\sqrt{3}$ with rational numbers a and b that $\sqrt{2}$ is not a number rational in $\sqrt{3}$.

4.48 Let $x_1, \ldots, x_n$ be algebraic numbers. Show that an algebraic number z exists that is rational in $x_1, \ldots, x_n$, and so that each $x_1, \ldots, x_n$ is rational in z, that is

$$\mathbf{Q}(z) = \mathbf{Q}(x_1, \ldots, x_n).$$

4.49 Let $f(X)$ be a polynomial with rational coefficients. Show that an algebraic number y exists such that $f(X)$ completely factors into linear factors over y.

4.50 Calculate the degree of $\sqrt{2} + \mathrm{i}$ over the rational numbers, over $\sqrt{2}$ and over i.

4.51 Provide a polynomial with rational coefficients that is irreducible over the rational numbers, but factors into two irreducible polynomials over $\sqrt{2}$ and into four irreducible polynomials over $\sqrt{2} + \mathrm{i}$.

4.52 Let ζ be a solution of the polynomial equation $X^4 + X^3 + X^2 + X + 1 = 0$. Show that ζ is a number rational in $\alpha := \mathrm{e}^{\frac{\pi \mathrm{i}}{5}}$, and provide a basis of $\mathbf{Q}(\alpha)$ over $\mathbf{Q}(\zeta)$.

The Degree Formula and the Impossibility of Doubling the Cube and Trisecting the Angle

4.53 Let x be an algebraic number, let y be an algebraic number that is rational in x, and let z be an algebraic number that is rational in y. How can the degree of x over z be calculated from the degree of x over y and the degree of y over z?

4.54 Let x be an algebraic number, let y be an algebraic number that is rational in x, and let z be an algebraic number that is rational in y. Show that the degree of y over z is a divisor of the degree of x over z.

4.55 Let x, y and z be three algebraic numbers such that both x and y are rational in z and such that y is rational in z. Show that

$$[\mathbf{Q}(z) : \mathbf{Q}(x)] \cdot [\mathbf{Q}(x) : \mathbf{Q}] = [\mathbf{Q}(z) : \mathbf{Q}(y)] \cdot [\mathbf{Q}(y) : \mathbf{Q}],$$

and give a diagram for illustration.

4.56 Let x be an algebraic number. Let $f(X)$ be a normalized, nonlinear polynomial with rational coefficients that is irreducible over the rational numbers. Show under the assumption that the degree of $f(X)$ is coprime to the degree of x over the rational numbers, that no number rational in x is a root of $f(X)$.

4.57 Let x be an algebraic number. Let y be an algebraic number that is rational in x. Let $f(X)$ be a normalized, nonlinear polynomial over y that is irreducible over y. Show under the assumption that the degree of $f(X)$ is coprime to the degree of x over y that no rational number in x is a root of $f(X)$.

4.58 Is the following statement true or false?

Let x and y be algebraic numbers, and let z be a primitive element to x and y. Then the degree of z over the rational numbers is a divisor of the product of the degrees of x and y over the rational numbers.

4.59 Show that the problem of angle quartering with compass and straightedge is solvable.

4.60 Show that the problem of angle quintisection with compass and straightedge is not solvable.

4.61 Show that the construction of a regular nonagon with compass and straightedge is impossible.

4.62 Let α be an angle such that $\cos\alpha$ is a constructible number. Show that α can be trisected with compass and straightedge if and only if the polynomial

$$f(X) = 4X^3 - 3X + \cos\alpha$$

is not irreducible over $\cos\alpha$.

4.63 Show that an algebraic number z is constructible if and only if a sequence $1 = z_0, z_1, \ldots, z_{n-1}, z_n = z$ of algebraic numbers exists such that for all $i \in \{1, \ldots, n\}$, the number z_{i-1} is rational in z_i and z_i has degree 2 over z_{i-1}.

Endnotes

[1]This is, as we have seen, constructively possible.

[2]Ferdinand Gotthold Max Eisenstein, 1823–1852, German mathematician.

[3]Archimedes of Syracuse, ca. 287–212 BC, Greek mathematician, physicist, engineer, inventor, and astronomer.

[4]Isaac Newton, 1643–1727, English physicist, mathematician, astronomer, natural philosopher, and theologian.

[5]In the language of abstract linear algebra we thus consider **C** as a **Q**-vector space. A system of complex numbers are linearly dependent over the rational numbers in our sense if they form a linearly dependent system of vectors in the **Q**-vector space **C**.

[6]We therefore need the separability of the minimal polynomial *of one* of the two numbers x and y.

[7]Leopold Kronecker, 1823–1891, German mathematician.

On the Constructability of Regular n-Gons

5

Symmetries are not only beautiful, but they also allow us to understand equations.

Abstract

Numbers like $\sqrt{2}$ and $-\sqrt{2}$ are algebraically indistinguishable (over the rational numbers), as both numbers have the same minimal polynomial, namely $X^2 - 2$. We call such numbers *Galois conjugates* of each other. We show that two Galois conjugate numbers not only have the same minimal polynomial, but that even every polynomial equation that is satisfied by one number is also satisfied by its Galois conjugates and vice versa.

In general, we call polynomial relationships between several algebraic numbers *algebraic relations*. Of particular interest are the algebraic relations between the roots of a polynomial. If we number the roots, we call a permutation of the roots a *(Galois) symmetry* if all algebraic relations between these roots are preserved under this permutation.

All symmetries together form the Galois group of the roots. From the definition of the Galois group, it is not immediately clear how we can effectively calculate it. This is done using *Galois resolvents*, and we provide a complete procedure in this chapter.

By being able to assign a group to each (the roots of each) polynomial, we can in turn draw conclusions about the polynomial and its roots from the group structure. Therefore, in this chapter, we look at some very general statements about groups, such as the class equation. We apply these theorems to the Galois groups of the so-called cyclotomic polynomials. As an application, we provide a complete classification of the regular n-gons that can be constructed with compass and straightedge.

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5.1 Galois Conjugates

Let x_1 be an algebraic number. As we have seen in the last chapter, there exists exactly one normalized irreducible polynomial $f(X)$ with rational coefficients which has x_1 as a root, namely the minimal polynomial of x_1. This polynomial usually has other roots besides x_1, which are all pairwise different as minimal polynomials over the rational numbers are separable.

Let's call these other roots $x_2, \ldots, x_n$. We have the decomposition

$$f(X) = (X - x_1) \cdots (X - x_n)$$

over the algebraic numbers. Since x_1 uniquely determines its minimal polynomial, the algebraic numbers $x_2, \ldots, x_n$ are also determined by x_1, which justifies giving them their own name: We call $x_2, \ldots, x_n$ the *Galois conjugates* of x_1.

Obviously, there are $n - 1$ conjugates if x_1 is of degree n over the rational numbers. (Often we will count x_1 itself as a Galois conjugate of x_1 so that we get exactly n Galois conjugates in this count. This definition is symmetric in $x_1, \ldots, x_n$: So if y is a Galois conjugate of x, then x is also a Galois conjugate of y.)

In the case that x_1 is a rational number, there are no further Galois conjugates. In the case that x_1 is of degree 2 over the rational numbers, x_1 satisfies an equation of the form

$$X^2 + pX + q = 0, \tag{5.1}$$

that is, we have $x_1 = \frac{1}{2}(-p + \delta)$, where δ is one of the square roots of $\Delta := p^2 - 4q$. The algebraic number x_1 has exactly one other Galois conjugate, namely $x_2 = \frac{1}{2}(-p - \delta)$, the other solution of (5.1). The transition to the Galois conjugate here thus corresponds to the choice of the other square root in the well-known solution formula for quadratic equations.

> Algebraic numbers x do not occur alone; they have Galois conjugates. And there are exactly as many algebraic numbers in the Galois conjugation class of x as the degree of x as an algebraic number indicates.

If $z = a + b\,\mathrm{i}$ is a complex number with real part a and imaginary part b, then its complex conjugate is known to be given by $\overline{z} = a - b\,\mathrm{i}$. We can also consider the complex conjugation as Galois conjugation, at least for the case that a and b are both rational numbers: Namely, $z = a + b\,\mathrm{i}$ and $\overline{z} = a - b\,\mathrm{i}$ are both roots of the normalized polynomial

$$f(X) = (X - z) \cdot (X - \overline{z}) = (X - a - b\,\mathrm{i}) \cdot (X - a + b\,\mathrm{i}) = X^2 - 2aX + a^2 + b^2$$

with rational coefficients. If z is not a rational number, i.e. $b \neq 0$, then $f(X)$ has no root in the rational numbers, and as a polynomial of second degree is therefore irreducible. Consequently, $f(X)$ is the minimal polynomial of z, and the complex conjugate $\overline{z}$ is also the Galois conjugate of z.

In some cases, the concept of complex conjugation coincides with that of Galois conjugation.

Example 5.1 Finally, let's consider a slightly more complicated example for Galois conjugates. The complex number $x_1 = \sqrt{2} + \sqrt{3}$ is algebraic as it is a root of the polynomial

$$f(X) = X^4 - 10X^2 + 1.$$

As we have seen in Example 4.13, $f(X)$ is irreducible, so it is the minimal polynomial of x_1 over the rational numbers. Thus, x_1 has three other Galois conjugates, the remaining roots of $f(X)$. These are

$$x_2 = \sqrt{2} - \sqrt{3}, \qquad x_3 = -\sqrt{2} + \sqrt{3}, \qquad x_4 = -\sqrt{2} - \sqrt{3}, \tag{5.2}$$

each of which can be easily verified by substituting into $f(X)$.

The order of the remaining roots, which we have chosen in (5.2), was initially an arbitrary order. For example, we could have taken $-\sqrt{2} + \sqrt{3}$ as x_2 and $\sqrt{2} - \sqrt{3}$ as x_3. This observation, as simple as it may seem at the moment, is extremely important, as it draws attention to the question of whether there are any relationships between the roots of a polynomial (and which ones might suggest a certain order). We will pursue this question, i.e. the question of symmetries among the roots, in Sect. 5.2. We present there the decisive progress of Galois' theory compared to those of his predecessors or contemporaries.

In the following, we want to formulate some important statements about Galois conjugates, which we will use later. Let t be an algebraic number.

Proposition 5.1 *If t' is a Galois conjugate of t and if $f(X)$ is a polynomial with rational coefficients that has t as a root, then it also has t' as a root.*

We can also interpret this proposition in such a way that we cannot distinguish Galois conjugates by means of polynomial equations with rational coefficients.

Proof Let $g(X)$ be the minimal polynomial of t over the rational numbers, that is, $g(X)$ is irreducible and $g(t) = 0$. According to Theorem 4.1, $f(X)$ is a multiple of $g(X)$, so $f(X) = g(X)h(X)$ for another polynomial $h(X)$. As a Galois conjugate t' of t, t' satisfies the equation $g(t') = 0$, so $f(t') = g(t') \cdot h(t') = 0 \cdot h(t') = 0$. □

Every algebraic number t has relatives, its Galois conjugates. The Galois conjugates of t are exactly those algebraic numbers that are solutions to exactly those polynomial equations over the rational numbers for which t is also a solution.

Lemma 5.1 *Let $x_1, \ldots, x_n$ be the solutions (with multiplicities) of a polynomial equation with rational coefficients. If $V(X_1, \ldots, X_n)$ is a polynomial with rational coefficients, then the Galois conjugates of $t = V(x_1, \ldots, x_n)$ are all of the form $t' = V(x_{\sigma(1)}, \ldots, x_{\sigma(n)})$, where σ is an n-letter permutation.*[1]

Proof Let

$$h(X) \coloneqq \prod_{\sigma \in S_n} (X - V(x_{\sigma(1)}, \ldots, x_{\sigma(n)})),$$

where the product runs over all n-letter permutations. The coefficients of this polynomial are symmetric functions in the solutions $x_1, \ldots, x_n$ of a polynomial equation with rational coefficients, according to Theorem 3.2 thus polynomials in the coefficients of this equation, thus rational numbers.

The factor of $h(X)$ that belongs to the identical permutation $\left(\begin{smallmatrix} 1 & 2 & \cdots & n \\ 1 & 2 & \cdots & n \end{smallmatrix}\right)$ is the factor $X - V(x_1, \ldots, x_n) = X - t$. Thus, t is a root of $h(X)$. Since $h(X)$ is a polynomial with rational coefficients, we can apply Proposition 5.1 and find that every Galois conjugate t' of t is also a root of $h(X)$. Thus, t' must be a root of a factor of $h(X)$, thus a factor of the form $X - V(x_{\sigma(1)}, \ldots, x_{\sigma(n)})$. Consequently, $t' = V(x_{\sigma(1)}, \ldots, x_{\sigma(n)})$. □

Using this lemma, we can conclude that Galois conjugates of primitive elements are again primitive elements. More precisely:

Proposition 5.2 *Let $x_1, \ldots, x_n$ be the solutions (with multiplicities) of a polynomial equation with rational coefficients. If t is a primitive element to $x_1, \ldots, x_n$, then every Galois conjugate t' of t is also a primitive element for $x_1, \ldots, x_n$.*

Proof As a primitive element for $x_1, \ldots, x_n$, t is of the form $t = V(x_1, \ldots, x_n)$, where $V(X_1, \ldots, X_n)$ is a polynomial with rational coefficients. According to Lemma 5.1, $t' = V(x_{\sigma(1)}, \ldots, x_{\sigma(n)})$ for an n-letter permutation σ. Since $x_1, \ldots, x_n$ in t are rational, t' is in particular a polynomial in t with rational coefficients, thus also rational in t. It follows,

that every number rational in t' is also a number rational in t. Since the degrees of t and t' over the rational numbers are the degree of the same minimal polynomial, there cannot be more numbers rational in t than in t'. Consequently, every number rational in t is also rational in t'. Thus, t' is also a primitive element for $x_1, \ldots, x_n$. □

5.2 The Galois Group of an Equation

In the last section, we considered the Galois conjugates $x_2, \ldots, x_n$ of an initially chosen algebraic number x_1. In the following we want to eliminate the special role of the algebraic number x_1. We simply consider any polynomial $f(X)$ over the rational numbers and ask for symmetries between the roots. (This subsumes the above considerations by choosing $f(X)$ as the minimal polynomial of a previously given algebraic number.) But in order to be able to distinguish the individual roots from each other we restrict ourselves in the following to the case that $f(X)$ is a separable polynomial, so there are no multiple roots. (In fact, many of the following statements are simply wrong for non-separable polynomials.)

Before we examine the roots of a separable polynomial for symmetries in more detail, we want to develop an understanding of symmetries using the planar and spatial Euclidean geometries. Let's consider the planar line in Fig. 5.1. Under a symmetry of this line, we understand a motion of the plane (made from a rotation and shift) that transforms the line back into itself. Such a motion is already determined by the motion of the two endpoints x_1 and x_2 of the line in Fig. 5.1. A symmetry of the line immediately catches the eye, namely the one that rotates the line by 180° so that the two endpoints swap their positions. This symmetry can be written as

$$x_1 \mapsto x_2,$$
$$x_2 \mapsto x_1$$

Using permutations, we can write this even shorter. If

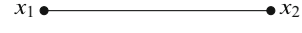
$$\sigma = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix},$$

then the symmetry of the line that swaps the two endpoints is given by

$$x_i \mapsto x_{\sigma(i)}, \qquad i \in \{1, 2\}.$$

The motion of the line that is thus defined by σ, we call the *operation* of σ on the line.

Fig. 5.1 A line in the plane



Somewhat more succinctly, we will simply say that the permutation σ is a symmetry of the line. Since every symmetry of the line is given by a permutation of the endpoints x_1 and x_2, we can identify each symmetry of the line with a certain 2-letter permutation. What other 2-letter permutations besides σ are there? Only one more, namely the so-called identical permutation

$$\mathrm{id} = \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix}.$$

This corresponds to the trivial permutation of the endpoints of the line, namely the one that does not move x_1 and x_2. The corresponding motion is the trivial one, which moves nothing at all and is therefore also a symmetry of the line.

We thus find that all (i.e., both) 2-letter permutations are symmetries of the line in Fig. 5.1. We call this totality of permutations the *symmetry group G* of the line and write

$$G = \left\{ \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \right\}.$$

We say that the symmetry group of the line is the *full symmetric group* S_2.

Next, we want to study the symmetries of a planar equilateral triangle as in Fig. 5.2. A motion of the triangle that maps the triangle back onto itself is again given by the induced permutation of its corners, that is, we need to examine all 3-letter permutations to see if they are symmetries of the equilateral triangle.

The 3-*letter* symmetric group S_3 has exactly 6 elements, that is, there are exactly six 3-letter permutations, namely:

$$\begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2, \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}. \tag{5.3}$$

Through motions in the plane, namely rotation by 0°, 120° and 240° around the center of the triangle, the first three permutations in the listing (5.3) can be realized.

The remaining three permutations correspond to exchanges of two vertices of the triangle while keeping the third fixed. Since this would involve a change of orientation,

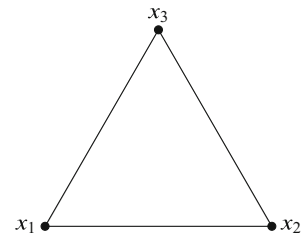


Fig. 5.2 An equilateral triangle in the plane

such permutations cannot be achieved by rotations or translations, that is, motions of the Euclidean plane. Thus, the last three permutations in (5.3) are not symmetries of the triangle in Fig. 5.2. We have determined its symmetry group G to be

$$G = \left\{ \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} \right\}, \tag{5.4}$$

which is not the full symmetric group S_3.

This symmetry group G has two special names that we want to describe briefly. First, we recall the concept of the *signum* or *sign* $\operatorname{sgn} \sigma$ of a permutation σ:

An *inversion* of a permutation σ is a pair (i, j) of natural numbers with $1 \leq i < j \leq n$ and $\sigma(i) > \sigma(j)$ (that is, σ swaps the relative order of i and j). Then we set

$$\operatorname{sgn} \sigma := \begin{cases} 1, & \text{if } \sigma \text{ has an even number of inversions, and} \\ -1, & \text{if } \sigma \text{ has an odd number of inversions.} \end{cases}$$

A permutation with $\operatorname{sgn} \sigma = 1$ is called an *even permutation*, a permutation with $\operatorname{sgn} \sigma = -1$ is called an *odd permutation*.

We assign a signum to each permutation, either the number $+1$ or -1.

Applying these terms to the symmetry group G of the equilateral triangle, we see that the symmetries consist exactly of the even permutations in S_3.

The set of all even permutations within the symmetric group S_n of n-letter permutations is denoted by the symbol A_n and is called the *alternating group*. We can therefore say that the symmetry group of the equilateral triangle in Fig. 5.2 is given by the alternating group A_3.

The alternating group consists precisely of the permutations with positive signum, i.e., the even permutations.

We can approach the symmetry group of the equilateral triangle also from another angle: For each number $k \in \{0, \ldots, n-1\}$ we define an n-letter permutation σ_k by

$$\sigma : i \mapsto \begin{cases} i + k, & \text{if } i + k \leq n \text{ and} \\ i + k - n, & \text{if } i + k > n. \end{cases}$$

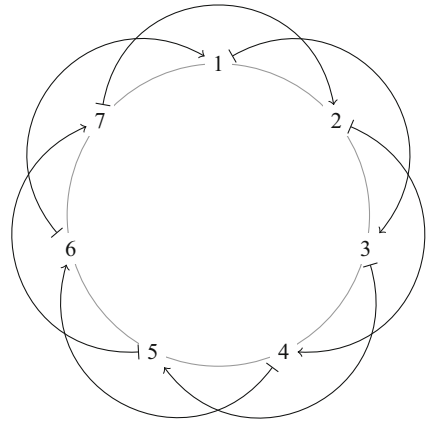


Fig. 5.3 The cyclic permutation $\sigma_2 \in S_7$

These n permutations $\sigma_0, \ldots, \sigma_{n-1}$ are combined into the so-called *cyclic group* C_n. (In Fig. 5.3 we have illustrated σ_2 in the case $n = 7$.) In the case of $n = 3$, C_3 consists exactly of the three elements that form the symmetry group of the triangle in (5.4), so we have in particular $C_3 = A_3$.

The cyclic group C_n consists of n elements.

At this point, it is appropriate to introduce another notation for permutations. Namely, we write the permutation $\sigma_2 \in S_7$ from Fig. 5.3 also in the form

$$\sigma_2 = (1, 3, 5, 7, 2, 4, 6).$$

The notation on the right side means that 1 is mapped to 3, 3 to 5, etc., and finally 4 to 6 and 6 to 1. This so-called cycle notation is also useful for other permutations: For example, $(1, 2, 3) \in S_4$ is the 4-letter permutation that maps 1 to 2, 2 to 3, 3 to 1, and finally 4 to itself. Another example is $(3, 6) \in S_n$. This is a permutation that maps 3 to 6, 6 to 3, and the remaining elements to themselves.

For further illustration, let's consider two more examples:

Example 5.2 If we consider the plane square in Fig. 5.4, we realize after some thought that the only symmetries of the square are the rotations by 0°, 90°, 180°, and 270° around the center. Thus, exactly the permutations

$$G = \left\{ \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix} \right\}$$

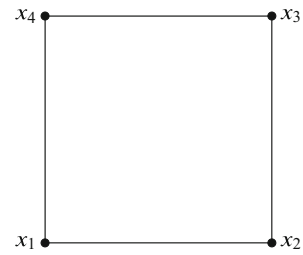


Fig. 5.4 A square in the plane

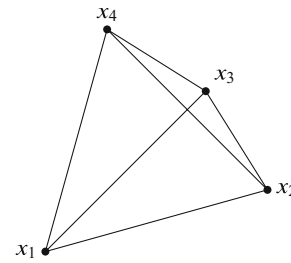


Fig. 5.5 A tetrahedron

form the symmetry group of the plane square. Since obviously $G = \mathrm{C}_4$, the symmetry group of the square is a cyclic group.

Example 5.3 The last example is a spatial one, the tetrahedron in Fig. 5.5. Symmetries of the tetrahedron are motions of Euclidean space that map the tetrahedron back onto the tetrahedron. Such a motion is uniquely determined by the induced permutation of the four vertices $x_1, \ldots, x_4$ so that we can again specify the symmetry group of the tetrahedron through permutations. Not all 4-letter permutations are certainly eligible. Any odd permutation, for example, that swaps two corners and fixes the other two, would correspond to a reversal of orientation of the tetrahedron, so it cannot be induced by a motion. The same applies to all other odd permutations. In contrast, it can be explicitly shown that each of the twelve even 4-letter permutations can be realized as a symmetry of the tetrahedron—we leave the going through all the permutations at this point however to the reader. It follows that the symmetry group of the tetrahedron in Fig. 5.5 is the alternating group A_4.

The symmetry group of an equilateral triangle is $\mathrm{C}_3 = \mathrm{A}_3$, that of a square is C_4. The symmetry group of the regular tetrahedron is A_4.

The symmetry groups of all the above examples from plane and spatial geometry have the following common properties: The identical symmetry (the one that corresponds to

the identical permutation) is always an element of the symmetry groups. If σ and τ are symmetries, then the composition of the transformations given by σ and τ is again a symmetry. Every symmetry is a reversible operation, and the reversal of a symmetry is again a symmetry.

In the formalism of n-letter permutations we define: If σ and τ are two n-letter permutations, then $\tau \circ \sigma$ is the *composition* of σ and τ, that is, the n-letter permutation, which maps $i \in \{1, \ldots, n\}$ to $\tau(\sigma(i))$.

Furthermore, σ^{-1} is the *inverse permutation* to σ, that is, the unique permutation with $\sigma^{-1} \circ \sigma = \mathrm{id}$, where id denotes the identical n-letter permutation. Then we can formulate:

Definition 5.1 Let n be a natural number. Let $\sigma_1, \ldots, \sigma_m$ be pairwise different n-letter permutations. Then $G \coloneqq \{\sigma_1, \ldots, \sigma_m\}$ is called a *finite subgroup* of the symmetric group S_n if $\mathrm{id} \in G$ and if $\tau \circ \sigma \in G$ and $\sigma^{-1} \in G$ for all $\sigma, \tau \in G$.

In this sense, the symmetry groups of the considered plane and spatial figures are all subgroups of corresponding symmetric groups.

We can invert and compose permutations as well as form the identical permutation. These operations do not lead out of subgroups.

Subgroups of the symmetric groups describe symmetries (of finite objects).

What do the symmetries of line, triangle, square and tetrahedron have to do with the roots of a separable polynomial $f(X)$ with rational coefficients? As the naming of the vertices in the geometric examples suggests, these vertices should correspond to the roots. In the case of geometric figures, we have considered permutations of the vertices. Now we want to consider permutations of the roots. In the geometric case, we called a permutation a symmetry if it preserves the geometry of the figure (i.e., it comes from a motion). What must a permutation of the roots of a separable polynomial $f(X)$ preserve for us to meaningfully call it a symmetry of the roots?

To this end, we introduce the following notation: Let $x_1, \ldots, x_n$ be algebraic numbers. An *algebraic relation* between $x_1, \ldots, x_n$ (over the rational numbers) is an equality of the form

$$H(x_1, \ldots, x_n) = 0$$

where $H(X_1, \ldots, X_n)$ is a polynomial with rational coefficients.

For example, $H(x_1, x_2, x_3) = 0$ for $H(X_1, X_2, X_3) = X_1X_2 - X_3$ is an algebraic relation between $\sqrt{2}$, $\sqrt{3}$ and $\sqrt{6}$.

If we swap the given algebraic numbers among each other, algebraic relations are usually not preserved. In our example with $\sqrt{2}$, $\sqrt{3}$ and $\sqrt{6}$, $H(\sqrt{3}, \sqrt{2}, \sqrt{6}) = 0$ is another algebraic relation, but not $H(\sqrt{6}, \sqrt{2}, \sqrt{3}) = 0$ (as this is actually a false statement).

With the help of the concept of the algebraic relation between the roots $x_1, \ldots, x_n$ of a normalized separable polynomial $f(X)$ (with rational coefficients), we can define: A *symmetry of the roots* of $f(X)$ is an n-ary permutation σ such that for every algebraic relation $H(x_1, \ldots, x_n) = 0$ between the roots, $H(x_{\sigma(1)}, \ldots, x_{\sigma(n)}) = 0$ is also an algebraic relation.

A permutation is therefore exactly a symmetry of the roots of $f(X)$ when it preserves every algebraic relation that exists between the roots.

A Galois symmetry of the roots of a separable polynomial preserves all algebraic relations between the roots.

We can imagine the algebraic relations between the $x_1, \ldots, x_n$ as the edges and faces between the vertices in our geometric examples: If vertices or faces are connected via edges, every symmetry operation must preserve this edge. In the case of the square as in Fig. 5.4, for example, any permutation that maps x_1 and x_2 to x_1 and x_3 is not a symmetry operation, since there is an edge between x_1 and x_2, but not between x_1 and x_3.

As a shorthand, we also simply call H an algebraic relation between the $x_1, \ldots, x_n$ if $H(x_1, \ldots, x_n) = 0$, and write $\sigma \cdot H$ for $H(X_{\sigma(1)}, \ldots, X_{\sigma(n)})$. Furthermore, we write $\sigma \cdot x_i = x_{\sigma(i)}$ for a symmetry σ and $i \in \{1, \ldots, n\}$. We say, $\sigma \cdot x_i$ is the result of the *action* of σ on the root x_i.

Let's illustrate the definition with the example of the polynomial

$$f(X) = X^4 - 10X^2 + 1 = 0$$

with the roots

$$x_1 = \sqrt{2} + \sqrt{3}, \quad x_2 = \sqrt{2} - \sqrt{3}, \quad x_3 = -\sqrt{2} + \sqrt{3}, \quad x_4 = -\sqrt{2} - \sqrt{3}. \tag{5.5}$$

A symmetry of these roots is a 4-letter permutation (of which there are a total of $4! = 24$) that preserves all algebraic relations $H(x_1, \ldots, x_4) = 0$ between the roots.

We can easily exclude some permutations as symmetries. Consider the permutation

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix},$$

which maps x_1 to x_2, x_2 to x_3, x_3 to x_4 and x_4 to x_1. Through $H(x_1, x_2, x_3, x_4) = 0$ with $H(X_1, \ldots, X_4) := (X_1 + X_2)^2 - 8$, an algebraic relation between $x_1, \ldots, x_4$ is given. On the other hand,

$$H(x_{\sigma(1)}, \ldots, x_{\sigma(4)}) = H(x_2, x_3, x_4, x_1) = (x_2 + x_3)^2 - 8 = -8,$$

that is, $\sigma \cdot H$ is not an algebraic relation between $x_1, \ldots, x_4$. By definition, σ is therefore not a symmetry of the roots, as it does not preserve all algebraic relations between the roots.

If we want to prove conversely that a certain permutation σ is a symmetry of the roots, the task appears much more difficult, as we must then show for *every* algebraic relation $H(x_1, \ldots, x_4) = 0$ that $\sigma \cdot H$ is again an algebraic relation.

Since we have to check a priori infinitely many relations whether they are preserved, we can only conclude without further consideration that the identical permutation is a symmetry. In the example, in the case of the permutation σ that swaps x_1 and x_2 and also x_3 and x_4 we would not find any algebraic relation that is not preserved. However, it would not be enough to test ten or twenty of these relations, but we would have to be sure that all others are preserved as well.

So, to precisely determine the symmetries of the roots among all permutations, we obviously need another crucial idea. Galois had such an idea, and the following central proposition goes back to his works:

Proposition 5.3 *Let $f(X)$ be a normalized separable polynomial over the rational numbers whose different roots are given by $x_1, \ldots, x_n$. Let further $t = V(x_1, \ldots, x_n)$ be a primitive element to $x_1, \ldots, x_n$, where $V(X_1, \ldots, X_n)$ is a polynomial with rational coefficients. Let $h_1(X), \ldots, h_n(X)$ be polynomials with rational coefficients with $x_i = h_i(t)$ for $i \in \{1, \ldots, n\}$.*

Then for each n-letter permutation σ that is a symmetry of the roots of $f(X)$ there is exactly one Galois conjugate t' of t (where the case $t' = t$ is explicitly allowed here) such that

$$\sigma \cdot x_1 = h_1(t'), \quad \ldots, \quad \sigma \cdot x_n = h_n(t'). \tag{5.6}$$

Conversely, for each Galois conjugate t' of t ($t' = t$ again possible) there is exactly one symmetry σ so that (5.6) is fulfilled.

Additionally, under the induced bijection between the symmetries σ and the Galois conjugates t', it holds that

$$t' = V(\sigma \cdot x_1, \ldots, \sigma \cdot x_n).$$

(The existence of the polynomials $h_1, \ldots, h_n$ in the proposition follows from the fact that $x_1, \ldots, x_n$ are rational in t.)

The number of Galois conjugates of a primitive element to the roots $x_1, x_2, \ldots, x_n$ of a polynomial corresponds to the number of symmetries of these roots.

The set of n-letter permutations that are symmetries of the roots of $f(X)$ and can be determined by the proposition is called the *Galois group of the roots* $x_1, \ldots, x_n$ of $f(X)$.

From the knowledge of the Galois conjugates of a primitive element to the roots of a polynomial, we can determine the Galois group of the roots.

We write

$$\sigma \in \mathrm{Gal}_{\mathbf{Q}}(x_1, \ldots, x_n)$$

if σ is a symmetry of the roots $x_1, \ldots, x_n$ of the normalized separable polynomial $f(X)$ with rational coefficients. From Proposition 5.3, we learn that the number of symmetries between the roots of a separable polynomial is given by the degree of a primitive element for the roots over the rational numbers.

If σ and τ are symmetries of the roots of $f(X)$, then $\tau \circ \sigma$ also preserves all algebraic relations between these roots, because if $H(x_1, \ldots, x_n) = 0$ is such a relation, then $H(\sigma \cdot x_1, \ldots, \sigma \cdot x_n) = 0$ is also such and thus also $H(\tau \cdot \sigma \cdot x_1, \ldots, \tau \cdot \sigma \cdot x_n) = (\tau \circ \sigma) \cdot H(x_1, \ldots, x_n) = 0$.

Trivially, the identity permutation is also a symmetry.

We can also show, that σ^{-1} is a symmetry of the roots of $f(X)$: Let $H(x_1, \ldots, x_n) = 0$ be an algebraic relation. Suppose, $h := \sigma^{-1} H(x_1, \ldots, x_n) = H(\sigma^{-1} \cdot x_1, \ldots, \sigma^{-1} \cdot x_n) \neq 0$. Then h has a multiplicative inverse, and we can write h^{-1} in the form $h^{-1} = U(x_1, \ldots, x_n)$ for a polynomial $U(X_1, \ldots, X_n)$ with rational coefficients. Consequently, $H(\sigma^{-1}x_1, \ldots, \sigma^{-1}x_n) \cdot U(x_1, \ldots, x_n) - 1 = 0$ is an algebraic relation that is preserved by σ, that is $H(x_1, \ldots, x_n) \cdot U(\sigma \cdot x_1, \ldots, \sigma \cdot x_n) - 1 = 0$. But this is a contradiction to $H(x_1, \ldots, x_n) = 0$. Therefore, $h = 0$ must be, so σ^{-1} is also a symmetry. We have thus proven that the Galois group is a subgroup of the symmetric group S_n.

The Galois group of the roots of a separable polynomial of degree n is a subgroup of the symmetric group S_n. It therefore contains the identity and is closed under inversion and composition.

Before we prove the proposition, we want to briefly apply it to our example with the roots (5.5). It is $t = \sqrt{2}+\sqrt{3}$ a primitive element to $\sqrt{2}+\sqrt{3}$, $\sqrt{2}-\sqrt{3}$, $-\sqrt{2}+\sqrt{3}$ and $-\sqrt{2}-\sqrt{3}$. Because $\sqrt{2} = \frac{1}{2}(t^3 - 9t)$ and $\sqrt{3} = \frac{1}{2}(11t - t^3)$, it follows

$$\begin{aligned}\sqrt{2}+\sqrt{3} &= t,\\ \sqrt{2}-\sqrt{3} &= t^3 - 10t,\\ -\sqrt{2}+\sqrt{3} &= -t^3 + 10t,\\ -\sqrt{2}-\sqrt{3} &= -t.\end{aligned}$$

In the notations of Proposition 5.3 and (5.5) we therefore have $V(X_1, X_2, X_3, X_4) = X_1$ and $h_1(X) = X$, $h_2(X) = X^3 - 10X$, $h_3(X) = -X^3 + 10X$, $h_4(X) = -X$.

The Galois conjugates to t are (including t) given by $t_1 = \sqrt{2}+\sqrt{3}$, $t_2 = \sqrt{2}-\sqrt{3}$, $t_3 = -\sqrt{2}+\sqrt{3}$ and $t_4 = -\sqrt{2}-\sqrt{3}$ since they are all roots of the minimal polynomial of t, namely $f(X)$. (The fact that they coincidentally coincide with the roots whose symmetry we want to calculate is a coincidence, due to our specific example.)

We calculate

$$\begin{aligned}&h_1(t_1) = x_1, \quad h_2(t_1) = x_2, \quad h_3(t_1) = x_3, \quad h_4(t_1) = x_4,\\ &h_1(t_2) = x_2, \quad h_2(t_2) = x_1, \quad h_3(t_2) = x_4, \quad h_4(t_2) = x_3,\\ &h_1(t_3) = x_3, \quad h_2(t_3) = x_4, \quad h_3(t_3) = x_1, \quad h_4(t_3) = x_2,\\ &h_1(t_4) = x_4, \quad h_2(t_4) = x_3, \quad h_3(t_4) = x_2, \quad h_4(t_4) = x_1.\end{aligned}$$

From this we can read off according to Proposition 5.3 that exactly the following four permutations are symmetries of the roots of $f(X) = X^4 - 10X^2 + 1$:

We thus obtain

$$\begin{aligned}&\mathrm{Gal}_{\mathbf{Q}}(\sqrt{2}+\sqrt{3}, \sqrt{2}-\sqrt{3}, -\sqrt{2}+\sqrt{3}, -\sqrt{2}-\sqrt{3})\\ &\qquad = \left\{\begin{pmatrix}1&2&3&4\\1&2&3&4\end{pmatrix}, \begin{pmatrix}1&2&3&4\\2&1&4&3\end{pmatrix}, \begin{pmatrix}1&2&3&4\\3&4&1&2\end{pmatrix}, \begin{pmatrix}1&2&3&4\\4&3&2&1\end{pmatrix}\right\}.\end{aligned}$$

These permutations therefore preserve, according to Proposition 5.3, whose proof we still have to provide, all algebraic relations between the four roots in (5.5).

In order not to make the proof of Proposition 5.3 too lengthy, we outsource an important fact to a lemma:

Lemma 5.2 *Let $x_1, \ldots, x_n$ be roots of a polynomial with rational coefficients. Let $t = V(x_1, \ldots, x_n)$ be a primitive element to $x_1, \ldots, x_n$, where $V(X_1, \ldots, X_n)$ is a polynomial with rational coefficients. Let $h_1(X), \ldots, h_n(X)$ be polynomials with rational coefficients with $x_i = h_i(t)$ for $i \in 1, \ldots, n$.*

For each Galois conjugate t' of t it then holds that

$$t' = V(h_1(t'), \ldots, h_n(t')),$$

that is, t' is in particular determined by its values for the polynomials $h_1(X), \ldots, h_n(X)$.

Proof of Lemma 5.2 The polynomial

$$g(X) = V(h_1(X), \ldots, h_n(X)) - X$$

has rational coefficients and t as a root. Thus, according to Proposition 5.1, it also has every Galois conjugate t' of t as a root. □

Proof of Proposition 5.3 First, let σ be an n-letter permutation that preserves every algebraic relation between $x_1, \ldots, x_n$. We need to show that there is a Galois conjugate t' of t such that $x_{\sigma(i)} = h_i(t')$ for all $i \in \{1, \ldots, n\}$.

For this, we consider the minimal polynomial $g(X)$ of t over the rational numbers, that is $g(X) = \prod_{j=1}^{m}(X - t_j)$, where $t_1, \ldots, t_m$ are all Galois conjugates (including t itself) of t. We number the t_j in such a way that $t_1 = t$.

According to Lemma 5.1, for each $j \in \{1, \ldots, m\}$ there is an n-letter permutation τ_j with $t_j = V(x_{\tau_j(1)}, \ldots, x_{\tau_j(n)})$, that is, we have the equality

$$g(X) = \prod_{j=1}^{m}(X - V(x_{\tau_j(1)}, \ldots, x_{\tau_j(n)})) \tag{5.7}$$

of polynomials with rational coefficients. For the factor $j = 1$, because $t = V(x_1, \ldots, x_n)$, we can assume that τ_1 is the identical permutation.

Since two polynomials are equal if they match in all coefficients, the equality (5.7) is nothing other than a sequence of algebraic relations between $x_1, \ldots, x_n$, one for each pair

of coefficients on both sides. These algebraic relations are preserved by the permutation σ, as we have assumed it as a symmetry of $x_1, \ldots, x_n$. Therefore, it follows

$$g(X) = \prod_{j=1}^{m} (X - V(\sigma \cdot x_{\tau_j(1)}, \ldots, \sigma \cdot x_{\tau_j(n)})). \tag{5.8}$$

The first factor (for $j = 1$) is (since we could assume τ_1 as identical permutation) $X - V(\sigma \cdot x_1, \ldots, \sigma \cdot x_n)$. Thus, $V(\sigma \cdot x_1, \ldots, \sigma \cdot x_n)$ is a root of the right side of (5.8), and therefore also a root of the left side.

The roots of the left side are precisely the Galois conjugates of t, so there exists exactly one Galois conjugate t' of t with

$$t' = V(\sigma \cdot x_1, \ldots, \sigma \cdot x_n). \tag{5.9}$$

By $x_i = h_i(t) = h_i(V(x_1, \ldots, x_n))$, an algebraic relation between the x_i is given. This is preserved by σ, so we finally have

$$\sigma \cdot x_i = h_i(V(\sigma \cdot x_1, \ldots, \sigma \cdot x_n)) = h_i(t'), \qquad i \in \{1, \ldots, n\}.$$

Next, we need to prove the uniqueness of t', given σ. Let t'' be another Galois conjugate with $\sigma \cdot x_i = h_i(t'')$ for all $i \in \{1, \ldots, n\}$; in particular $h_i(t') = h_i(t'')$ for all $i \in \{1, \ldots, n\}$. From the above Lemma 5.2, it follows that $t' = t''$.

Now let t' be a Galois conjugate of t (including the possibility $t' = t$). We need to show that there is (exactly) one symmetry σ of the roots of $f(X)$ with $x_{\sigma(i)} = h_i(t')$ for $i \in \{1, \ldots, n\}$. To do this, we first show that $h_1(t'), \ldots, h_n(t')$ is a permutation of $x_1, \ldots, x_n$.

For each $i \in \{1, \ldots, n\}$, $f(h_i(X))$ is a polynomial with rational coefficients that has t as a root because $x_i = h_i(t)$. Consequently, t' is also a root of $f(h_i(X))$ according to Proposition 5.1, that is, $h_i(t')$ is a root of $f(X)$ and thus equal to one of the n algebraic numbers $x_1, \ldots, x_n$.

If we further assume that $h_i(t') = h_j(t')$ for $i, j \in \{1, \ldots, n\}$, then t' is a root of $h_i(X) - h_j(X)$, a polynomial with rational coefficients. Thus, t is also a root of $h_i(X) - h_j(X)$ according to Proposition 5.1, so we have $x_i = h_i(t) = h_j(t) = x_j$. Since $x_1, \ldots, x_n$ are pairwise different, $i = j$ follows from our assumption $h_i(t') = h_j(t')$.

We thus find that $h_1(t'), \ldots, h_n(t')$ is a permutation of $x_1, \ldots, x_n$, so there is (exactly) one n-letter permutation σ with $h_i(t') = x_{\sigma(i)}$ for $i \in \{1, \ldots, n\}$.

It remains to show that σ is a symmetry of the roots of $h(X)$. To do this, we consider an algebraic relation $H(X_1, \ldots, X_n)$ over the rational numbers between the roots $x_1, \ldots, x_n$ of $f(X)$. We thus have

$$H(x_1, \ldots, x_n) = H(h_1(t), \ldots, h_n(t)) = 0.$$

Thus, t is a root of the polynomial $H(h_1(X), \ldots, h_n(X))$ with rational coefficients. According to Proposition 5.1, t' is also a root of this polynomial. We thus have

$$H(x_{\sigma(1)}, \ldots, x_{\sigma(n)}) = H(h_1(t'), \ldots, h_n(t')) = 0,$$

that is, the permutation σ preserves the algebraic relation $H(X_1, \ldots, X_n)$.

The addition in Proposition 5.3 was proven in the meantime, namely by (5.9). □

5.3 On Invariants of the Galois Action

Let $x_1, \ldots, x_n$ be the roots of a normalized separable polynomial whose coefficients are rational numbers. Let z be an expression rational in $x_1, \ldots, x_n$, that is, we can write

$$z = F(x_1, \ldots, x_n) \tag{5.10}$$

for a polynomial $F(X_1, \ldots, X_n)$ with rational coefficients. Let σ be an element of the Galois group G of $x_1, \ldots, x_n$, that is a permutation of the roots that preserves all algebraic relations of the roots over the rational numbers. Just as σ acts on the roots through $\sigma \cdot x_i = x_{\sigma(i)}$, we want to let σ act on z by setting

$$\sigma \cdot z := F(\sigma \cdot x_1, \ldots, \sigma \cdot x_n). \tag{5.11}$$

However, at this point we must be careful because there is usually more than one polynomial $F(X_1, \ldots, X_n)$ with the property Eq. (5.10). Let $z = G(x_1, \ldots, x_n)$ for another polynomial $G(X_1, \ldots, X_n)$ with rational coefficients. So to show that the action in (5.11) is well-defined, we need to show that

$$F(\sigma \cdot x_1, \ldots, \sigma \cdot x_n) = G(\sigma \cdot x_1, \ldots, \sigma \cdot x_n). \tag{5.12}$$

This follows just from the definition of a symmetry of $x_1, \ldots, x_n$: So $H(X_1, \ldots, X_n) := F(X_1, \ldots, X_n) - G(X_1, \ldots, X_n)$ has only rational coefficients, and we have

$$H(x_1, \ldots, x_n) = F(x_1, \ldots, x_n) - G(x_1, \ldots, x_n) = z - z = 0,$$

that is, $H(X_1, \ldots, X_n) = 0$ is an algebraic relation of $x_1, \ldots, x_n$. As for every element σ of G we have

$$H(\sigma \cdot x_1, \ldots, \sigma \cdot x_n) = 0,$$

we have shown (5.12).

We say, the Galois group in $x_1, \ldots, x_n$ *acts* (or *operates*) by (5.11) on the numbers rational in $x_1, \ldots, x_n$. If τ is another symmetry of $x_1, \ldots, x_n$ besides σ, it obviously holds that

$$\tau \cdot (\sigma \cdot z) = (\tau \circ \sigma) \cdot z.$$

Also, we have $\mathrm{id} \cdot z = z$.

The Galois group of roots of a separable polynomial operates on all algebraic elements that are rational in the roots.

If σ is not a symmetry of the roots but only a general permutation of $x_1, \ldots, x_n$, we can only let σ act on formal polynomials in the $x_1, \ldots, x_n$, but not on the values of these polynomials, that is, we must read $\sigma \cdot H(x_1, \ldots, x_n)$ for a polynomial $H(X_1, \ldots, X_n)$ in the general case as $(\sigma \cdot H)(x_1, \ldots, x_n)$ and not as $H(\sigma \cdot x_1, \ldots, \sigma \cdot x_n)$. In the case, that σ is a symmetry, however, we have $(\sigma \cdot H)(x_1, \ldots, x_n) = H(\sigma \cdot x_1, \ldots, \sigma \cdot x_n)$.

The fundamental theorem of elementary symmetric functions tells us in conjunction with Vieta's Theorem that every polynomial formally symmetric in $x_1, \ldots, x_n$ and with rational coefficients takes a rational value on $x_1, \ldots, x_n$, that is, if $\sigma \cdot H = H$ holds for a polynomial $H(X_1, \ldots, X_n)$ and for all n-letter permutations σ, then $H(x_1, \ldots, x_n)$ is a rational number.

In the case that we only consider such permutations σ which are symmetries of the roots $x_1, \ldots, x_n$, we can look at the condition $\sigma \cdot H(x_1, \ldots, x_n) = H(x_1, \ldots, x_n)$ for all symmetries σ, that is, the condition that the value $H(x_1, \ldots, x_n)$ is *invariant* under the action of the Galois group of $x_1, \ldots, x_n$. Although this condition is obviously weaker than the condition that the polynomial H is formally symmetric in $x_1, \ldots, x_n$, it also follows under the weaker condition that the value $H(x_1, \ldots, x_n)$ is a rational number:

Proposition 5.4 *Let $x_1, \ldots, x_n$ be the roots of a normalized separable polynomial $f(X)$ with rational coefficients. Let z be a number rational in $x_1, \ldots, x_n$ which is invariant under the Galois group G of $x_1, \ldots, x_n$, that is, $\sigma \cdot z = z$ for all symmetries in G. Then z is already a rational number.*

The reverse direction, that a rational number is invariant under all elements of G, is trivial because we can simply consider a rational number as a constant polynomial in $x_1, \ldots, x_n$, which is invariant under the exchange of $x_1, \ldots, x_n$ for obvious reasons.

Proof Let $z = H(x_1, \ldots, x_n)$ where $H(X_1, \ldots, X_n)$ is a polynomial with rational coefficients. Let t be a primitive element to $x_1, \ldots, x_n$. Let $h_1(X), \ldots, h_n(X)$ be polynomials with rational coefficients with $x_i = h_i(t)$ for $i \in \{1, \ldots, n\}$. We set

$$h(T) \coloneqq H(h_1(T), \ldots, h_n(T)).$$

We then have to show that $z = h(t)$ is a rational number, that is, that the degree of z over $\mathbf{Q}$ is equal to one. According to the degree formula, we can just as well show

$$[\mathbf{Q}(t) : \mathbf{Q}(z)] = [\mathbf{Q}(t) : \mathbf{Q}] =: m,$$

which we will do in the following:

For this, let $t_1 = t, t_2, \ldots, t_m$ be the Galois conjugates of t. According to Proposition 5.3, for each $j \in \{1, \ldots, m\}$ there is a symmetry σ of $x_1, \ldots, x_n$ such that $h_i(t_j) = \sigma \cdot x_i$. By assumption, in particular, $h(t_j) = h(t) = z$ for all $j \in \{1, \ldots, m\}$.

Let $g(X)$ be the minimal polynomial of t over z, whose degree is at most m. The coefficients of $g(X)$ are polynomial expressions in z over the rational numbers, that is, we can write $g(X) = G(X, z)$, where $G(X, Y)$ is a polynomial over the rational numbers. Since t is a root of $g(X)$, t is also a root of $G(X, h(X))$, a polynomial with rational coefficients. Consequently, all t_j are roots of $G(X, h(X))$.

It follows that

$$g(t_j) = G(t_j, z) = G(t_j, h(t_j)) = 0.$$

Thus, $g(X)$ has the m different roots $t_1, \ldots, t_m$. Consequently, $\deg g(X) = m$, so $[\mathbf{Q}(t) : \mathbf{Q}(z)] = m$. □

The invariants under the operation of the Galois group of the roots of a separable polynomial over the rational numbers are precisely the rational numbers.

Corollary 5.1 *Let $x_1, \ldots, x_n$ be the roots of a normalized separable polynomial $f(X)$ with rational coefficients. If then $f(X)$ is irreducible over the rational numbers, the Galois group of $x_1, \ldots, x_n$ operates* transitively *on $x_1, \ldots, x_n$, that is, for every pair $i, j \in \{1, \ldots, n\}$ there exists (at least) one symmetry σ of $x_1, \ldots, x_n$ with $\sigma \cdot x_i = x_j$.*

Proof Set

$$g(X) \coloneqq \prod_{\sigma} (X - \sigma \cdot x_i),$$

where the product runs over all symmetries of $x_1, \ldots, x_n$. If we apply a symmetry τ of $x_1, \ldots, x_n$ to $g(X)$ (i.e., to the coefficients of $g(X)$), we obtain

$$\tau \cdot g(X) = \prod_{\sigma} (X - \tau \cdot \sigma \cdot x_i) = \prod_{\sigma} (X - (\tau \circ \sigma) \cdot x_i).$$

If σ runs through all symmetries, then $\sigma' = \tau \circ \sigma$ also runs through all symmetries (the reversal is given by $\sigma = \tau^{-1} \circ \sigma'$). Thus, $\tau \cdot g(X) = g(X)$.

The coefficients of $g(X)$ are therefore invariant under the symmetries; according to Proposition 5.4, $g(X)$ is therefore a normalized polynomial with rational coefficients. Since $f(X)$ is irreducible, $x_1, \ldots, x_n$ are Galois conjugate over the rational numbers. The number x_i is a root of $g(X)$ by construction (since id is a symmetry). Consequently, x_j is also a root of $g(X)$. Thus, there exists a symmetry σ with $\sigma \cdot x_i = x_j$. □

In the case of an irreducible polynomial, the Galois group operates transitively on the roots.

With this knowledge, we can easily calculate the Galois group G of the roots x_1, x_2, x_3 of an irreducible, normalized, separable polynomial $f(X)$ of third degree over the rational numbers.

We recall the discriminant

$$\Delta = (x_1 - x_2)^2 \cdot (x_1 - x_3)^2 \cdot (x_2 - x_3)^2$$

of $f(X)$, which is a (known) polynomial in the coefficients of $f(X)$ (in the case of the depressed cubic $f(X) = X^3 + pX + q$, we have $\Delta = -4p^3 - 27q^2$). We now make a case distinction.

In the first case, we assume that Δ a square in $\mathbf{Q}$, that is, there exists a rational number δ with

$$\delta = (x_1 - x_2) \cdot (x_1 - x_3) \cdot (x_2 - x_3). \tag{5.13}$$

Due to the separability of $f(X)$, $\delta \neq 0$.

The Eq. (5.13) is an algebraic relation of x_1, x_2 and x_3 over the rational numbers, so it must be preserved by every element of the Galois group G.

Therefore, G must not contain any odd permutations, as these change the sign of the right-hand side of (5.13). So we have $G \subset \mathrm{A}_3$. No proper subset of A_3 operates transitively on the roots (and contains id), so we consequently have $G = \mathrm{A}_3$ according to Corollary 5.1.

Now let's assume that Δ is not a square in the rational numbers, so that

$$\delta := (x_1 - x_2) \cdot (x_1 - x_3) \cdot (x_2 - x_3)$$

is not a rational number. According to Proposition 5.4, there must be a symmetry that does not leave δ invariant. This means that at least one odd permutation is in G. Since according to Corollary 5.1 the elements in G also operate transitively on x_1, x_2 and x_3, G must also contain at least one other element. Simple considerations (S_3 only has six elements, so there are not that many possibilities) show due to the closureness property of G under compositions that G then already is the full symmetric group S_3. In summary, we have

$$\mathrm{Gal}_{\mathbf{Q}}(x_1, x_2, x_3) = \begin{cases} A_3 & \text{for } \sqrt{\Delta} \in \mathbf{Q} \text{ and} \\ S_3 & \text{for } \sqrt{\Delta} \notin \mathbf{Q}. \end{cases}$$

For the Galois group of an irreducible polynomial of third degree there are only two possibilities, namely the alternating group A_3 or the full symmetric group S_3. We can distinguish these based on the discriminant. In the first case, it is a square, in the second case it is not.

5.4 Galois Resolvents

In order to effectively apply Proposition 5.3 to concrete polynomials, we need a good method to find a primitive element for the roots of an equation. We can use the proof method of Theorem 4.2 for this. An alternative method is the content of the following proposition, which goes back to Lagrange:

Proposition 5.5 *Let $x_1, \ldots, x_n$ be the roots of a normalized separable polynomial $f(X)$. If $V(X_1, \ldots, X_n)$ is a polynomial so that $t_\sigma = V(x_{\sigma(1)}, \ldots, x_{\sigma(n)})$ takes a different value for each n-ary permutation σ, then $t = V(x_1, \ldots, x_n)$ is a primitive element for $x_1, \ldots, x_n$.*

Such a polynomial $V(X_1, \ldots, X_n)$ is called a *Galois resolvent* of $f(X)$.

The applicability of Proposition 5.5 naturally depends on the answer to the question of whether we can effectively find a Galois resolvent. The fortunately positive answer is given by the following lemma:

Lemma 5.3 *Let $f(X)$ be a normalized separable polynomial with roots $x_1, \ldots, x_n$. Then there are integers $m_1, \ldots, m_n$ such that*

$$V(X_1, \ldots, X_n) = m_1 X_1 + \cdots + m_n X_n$$

is a Galois resolvent for $x_1, \ldots, x_n$.

In particular, we always find a Galois resolvent in such a way that it is linear (over the integers) in the roots.

Proof of Lemma 5.3 Let $V(X_1, \ldots, X_n) = m_1 X_1 + \cdots + m_n X_n$, where $m_1, \ldots, m_n$ are initially arbitrary integers.

We consider the polynomial

$$g(X) = \prod_{\sigma \in \mathrm{S}_n} (X - V(x_{\sigma(1)}, \ldots, x_{\sigma(n)})),$$

where the product runs over all n-letter permutations. The coefficients of $g(X)$ are symmetric expressions in the roots of $f(X)$, so polynomials in the coefficients of $f(X)$. Thus, $g(X)$ is a polynomial with rational coefficients.

Obviously, $V(X_1, \ldots, X_n)$ is a Galois resolvent of $f(X)$ if and only if $g(X)$ has no multiple roots, i.e., when the discriminant Δ of $g(X)$ does not vanish.

This discriminant depends polynomially on the m_i. More precisely, there exists a polynomial $\Delta(Y_1, \ldots, Y_n)$ with rational coefficients and with $\Delta = \Delta(m_1, \ldots, m_n)$, namely

$$\Delta(Y_1, \ldots, Y_n) = \prod_{\sigma, \tau} ((Y_1 x_{\sigma(1)} + \cdots + Y_n x_{\sigma(n)}) - (Y_1 x_{\tau(1)} + \cdots + Y_n x_{\tau(n)}))^2, \qquad (5.14)$$

where the product runs over all unordered pairs σ, τ of n-letter permutations with $\sigma \neq \tau$. Since $x_1, \ldots, x_n$ are pairwise different, for $\sigma \neq \tau$ there always exists an $i \in \{1, \ldots, n\}$ with $\sigma(i) \neq \tau(i)$ and thus $x_{\sigma(i)} - x_{\tau(i)} \neq 0$. Consequently, each individual factor of the representation (5.14) is different from zero, and thus $\Delta(Y_1, \ldots, Y_n)$ is a non-vanishing polynomial. In particular, we can choose $m_1, \ldots, m_n$ such that $\Delta(m_1, \ldots, m_n) \neq 0$. □

Galois resolvents can always be constructed. With their help, we find primitive elements and ultimately the totality of all Galois symmetries.

This leaves:

Proof of Proposition 5.5 We first choose an enumeration $\sigma_1, \ldots, \sigma_N$ of all different n-letter permutations, where we can assume without loss of generality that σ_1 is the identical permutation. For $j \in \{1, \ldots, N\}$, let $t_j = V(x_{\sigma_j(1)}, \ldots, x_{\sigma_j(n)})$.

By assumption, the algebraic numbers $t_1, \ldots, t_N$ are all pairwise different. Further, we abbreviate $y_j \coloneqq x_{\sigma_j(1)}$. We consider

$$\begin{aligned} y_1 + \cdots + \quad y_N &=: b_0, \\ t_1 y_1 + \cdots + \quad t_N y_N &=: b_1, \\ t_1^2 y_1 + \cdots + \quad t_N^2 y_N &=: b_2, \\ &\vdots \\ t_1^{N-1} y_1 + \cdots + t_N^{N-1} y_N &=: b_{N-1}. \end{aligned} \tag{5.15}$$

The numbers $b_0, \ldots, b_{N-1}$ are each symmetric expressions in $x_1, \ldots, x_n$ and thus polynomials in the coefficients of $f(X)$, hence rational numbers.

We can read (5.15) as an inhomogeneous linear system of equations if we consider $y_1, \ldots, y_N$ as unknowns. The determinant of the coefficient matrix

$$\begin{pmatrix} 1 & \ldots & 1 \\ t_1 & \ldots & t_N \\ t_1^2 & \ldots & t_N^2 \\ \vdots & & \vdots \\ t_1^{N-1} & \ldots & t_N^{N-1} \end{pmatrix}$$

is known to be given by the Vandermonde determinant

$$D = \prod_{1 \le i < j \le N} (t_j - t_i).$$

It is $D \neq 0$ since, by assumption, $t_1, \ldots, t_N$ are pairwise different. Let further be

$$D_1 = \det \begin{pmatrix} b_0 & 1 & \ldots & 1 \\ b_1 & t_2 & \ldots & t_N \\ b_2 & t_2^2 & \ldots & t_N^2 \\ \vdots & & \vdots & \\ b_{N-1} & t_2^{N-1} & \ldots & t_N^{N-1} \end{pmatrix}.$$

According to Cramer's rule for the solution of an inhomogeneous system of equations it is

$$x_1 = y_1 = \frac{D_1}{D} = \frac{D_1 \cdot D}{D^2}.$$

The denominator D^2 of the right side is symmetric in $t_1, \ldots, t_N$ and thus also in $x_1, \ldots, x_n$, hence a rational number. The signs of D_1 and D each change upon swapping two elements from $t_2, \ldots, t_n$, that is, $D_1 D$ remains invariant under swapping two elements from $t_2, \ldots, t_N$, thus is symmetric in $t_2, \ldots, t_N$. This allows $D_1 D$ to be written as a polynomial with rational coefficients in t_1 and the elementary symmetric functions in $t_2, \ldots, t_N$.

We will show in the following, in the form of Lemma 5.4, that every elementary symmetric function in $t_2, \ldots, t_N$ is a polynomial with integer coefficients in the elementary symmetric functions in $t_1, \ldots, t_N$ and in t_1. Consequently, $D_1 D$ is a polynomial with rational coefficients in the elementary symmetric functions in $t_1, \ldots, t_N$ and in t_1. The elementary symmetric functions in $t_1, \ldots, t_N$ are symmetric in $x_1, \ldots, x_n$, hence rational numbers. Thus, $D_1 D$ and thus also $x_1 = \frac{D_1 \cdot D}{D^2}$ is a polynomial in $t_1 = V(x_1, \ldots, x_n)$ with rational coefficients. Quite analogously, it follows that also $x_2, \ldots, x_n$ are polynomials in $V(x_1, \ldots, x_n)$ with rational coefficients. Therefore, $V(x_1, \ldots, x_n)$ is a primitive element to $x_1, \ldots, x_n$ as it is itself rational in $x_1, \ldots, x_n$. □

Lemma 5.4 *Let $n \geq 1$. The elementary symmetric functions $e_1(X_2, \ldots, X_n)$, $\ldots$, $e_{n-1}(X_2, \ldots, X_n)$ are polynomials in the elementary symmetric functions $e_1(X_1, \ldots, X_n), \ldots, e_n(X_1, \ldots, X_n)$ and in X_1 with integer coefficients.*

Proof By definition of the elementary symmetric functions, the following relations hold:

$$\begin{aligned}
e_1(X_1, \ldots, X_n) &= e_1(X_2, \ldots, X_n) &&+ X_1, \\
e_2(X_1, \ldots, X_n) &= e_2(X_2, \ldots, X_n) &&+ X_1 \cdot e_1(X_2, \ldots, X_n), \\
&\vdots \\
e_{n-1}(X_1, \ldots, X_n) &= e_{n-1}(X_2, \ldots, X_n) &&+ X_1 \cdot e_{n-2}(X_2, \ldots, X_n).
\end{aligned}$$

Rearranging these equations yields

$$\begin{aligned}
e_1(X_2, \ldots, X_n) &= e_1(X_1, \ldots, X_n) &&- X_1, \\
e_2(X_2, \ldots, X_n) &= e_2(X_1, \ldots, X_n) &&- X_1 \cdot e_1(X_2, \ldots, X_n), \\
&\vdots \\
e_{n-1}(X_2, \ldots, X_n) &= e_{n-1}(X_1, \ldots, X_n) &&- X_1 \cdot e_{n-2}(X_2, \ldots, X_n).
\end{aligned}$$

From this, recursive formulas for the representation of the $e_j(X_2, \ldots, X_n)$ as polynomials in the $e_i(X_1, \ldots, X_n)$ and X_1 can be derived. □

We want to use the insights of this section to work through another example for the calculation of the Galois group of the roots of a separable polynomial:

Example 5.4 Consider the polynomial $f(X) = X^4 - 4X^3 - 4X^2 + 8X - 2$. This polynomial has as roots approximately the four numbers

$$x_1 = 4.51\ldots, \qquad x_2 = 0.85\ldots, \qquad x_3 = 0.31\ldots, \qquad x_4 = -1.67\ldots,$$

so in particular, it is separable.

Next, we need a primitive element t for $x_1, \ldots, x_4$. According to Proposition 5.5, it is sufficient to find a Galois resolvent. The proof of Lemma 5.3 contains a method to find such a resolvent. However, it is often faster to simply guess a Galois resolvent: For this, we make the linear ansatz $t = V(x_1, x_2, x_3, x_4)$ with

$$V(X_1, X_2, X_3, X_4) = -X_2 + X_3 - 2X_4.$$

To prove that this is indeed a Galois resolvent, we have to calculate the values of $V(x_{\sigma(1)}, x_{\sigma(2)}, x_{\sigma(3)}, x_{\sigma(4)})$ for all permutations. These are a total of 24 values to be calculated.

A quick numerical calculation yields

$$\begin{aligned}
V(x_1,x_2,x_3,x_4) &= 2.82\ldots, & V(x_1,x_2,x_4,x_3) &= -3.14..., \\
V(x_1,x_3,x_2,x_4) &= 3.88..., & V(x_1,x_3,x_4,x_2) &= -3.68..., \\
V(x_1,x_4,x_2,x_3) &= 1.89..., & V(x_1,x_4,x_3,x_2) &= 0.30..., \\
V(x_2,x_1,x_3,x_4) &= -0.86..., & V(x_2,x_1,x_4,x_3) &= -6.82..., \\
V(x_2,x_3,x_1,x_4) &= 7.55..., & V(x_2,x_3,x_4,x_1) &= -11.02..., \\
V(x_2,x_4,x_1,x_3) &= 5.56..., & V(x_2,x_4,x_3,x_1) &= -7.04..., \\
V(x_3,x_1,x_2,x_4) &= -0.32..., & V(x_3,x_1,x_4,x_2) &= -7.88..., \\
V(x_3,x_2,x_1,x_4) &= 7.02..., & V(x_3,x_2,x_4,x_1) &= -11.55..., \\
V(x_3,x_4,x_1,x_2) &= 4.50..., & V(x_3,x_4,x_2,x_1) &= -6.51..., \\
V(x_4,x_1,x_2,x_3) &= -4.30..., & V(x_4,x_1,x_3,x_2) &= -5.89..., \\
V(x_4,x_2,x_1,x_3) &= 3.04..., & V(x_4,x_2,x_3,x_1) &= -9.56..., \\
V(x_4,x_3,x_1,x_2) &= 2.51..., & V(x_4,x_3,x_2,x_1) &= -8.50...,
\end{aligned} \tag{5.16}$$

and these numbers are pairwise different, which means, $t = -x_2 + x_3 - 2x_4$ is a Galois resolvent. Next, we need the minimal polynomial $g(X)$ for t. Since t, as an integer linear combination of algebraic integers, is again an algebraic integer, $g(X)$ has integer coefficients.

It is $g(X) = (X - t_1) \cdots (X - t_m)$ where $t_1, \ldots, t_m$ are the Galois conjugates of t, where we set $t_1 = t$. According to Lemma 5.1, each t_i is of the form $V(x_{\sigma(1)}, \ldots, x_{\sigma(n)})$, so one

of the numbers from (5.16). To determine the minimal polynomial of t, we only need to find a minimal number of factors of the form $X - V(x_{\sigma(1)}, x_{\sigma(2)}, x_{\sigma(3)}, x_{\sigma(4)})$ among which the factor $X - t = X - V(x_1, x_2, x_3, x_4)$ occurs, so that the product becomes a polynomial with integer coefficients. It's best to let a computer do this, which gives us the following minimal polynomial:

$$\begin{aligned} g(X) &= \prod_{j=1}^{8} (X - V(x_{\tau_j(1)}, x_{\tau_j(2)}, x_{\tau_j(3)}, x_{\tau_j(4)})) \\ &= X^8 + 16X^7 - 40X^6 - 1376X^5 - 928X^4 \\ &\quad + 34048X^3 + 22208X^2 - 253184X + 72256. \end{aligned}$$

The involved permutations are given by

$$\tau_1 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 4 \end{pmatrix}, \quad \tau_2 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 2 & 1 & 4 \end{pmatrix}, \quad \tau_3 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 1 & 4 & 3 & 2 \end{pmatrix}, \quad \tau_4 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix},$$
$$\tau_5 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix}, \quad \tau_6 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix}, \quad \tau_7 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}, \quad \tau_8 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 3 & 2 & 1 \end{pmatrix}.$$

In particular, $m = 8$, and we have eight Galois conjugates. According to Proposition 5.3 each Galois conjugate t' corresponds exactly to one symmetry σ of the roots. For this symmetry,

$$t' = V(x_{\sigma(1)}, x_{\sigma(2)}, x_{\sigma(3)}, x_{\sigma(4)}). \tag{5.17}$$

Since $V(X_1, X_2, X_3, X_4)$ is a Galois resolvent, in our case there is only one permutation for each Galois conjugate $t' = t_i$ that fulfills (5.17), namely $\sigma = \tau_i$. Thus, τ_i must be one of the eight symmetries of the roots $x_1, \ldots, x_4$. We therefore have

$$\mathrm{Gal}_{\mathbf{Q}}(x_1, x_2, x_3, x_4) = \{\tau_1, \tau_2, \tau_3, \tau_4, \tau_5, \tau_6, \tau_7, \tau_8\}.$$

How this sample calculation can be generalized to other separable polynomials should be clear. The most important insight from the above example, we formulate again in a general form:

Proposition 5.6 *Let $x_1, \ldots, x_n$ be the roots of a normalized separable polynomial $f(X)$ with rational coefficients. Let $V(X_1, \ldots, X_n)$ be a Galois resolvent to $x_1, \ldots, x_n$. The*

symmetries of the roots of $x_1, \ldots, x_n$ are then exactly those n-letter permutations $\tau_1, \ldots, \tau_m$, so that

$$g(X) = (X - V(x_{\tau_1(1)}, \ldots, x_{\tau_1(n)})) \cdots (X - V(x_{\tau_m(1)}, \ldots, x_{\tau_m(n)}))$$

is the minimal polynomial to $t = V(x_1, \ldots, x_n)$. □

(Since t must be a root of $g(X)$, we can assume without restriction that $\tau_1 = \mathrm{id}$.)

With the help of the minimal polynomial of a Galois resolvent t, we can determine all elements of the Galois group. These correspond to the Galois conjugates of t.

We conclude this section with a slightly more general example, which can be applied to any irreducible cubic polynomial over the rational numbers:

Example 5.5 Let x_1, x_2 and x_3 be the roots of a normalized irreducible polynomial $f(X)$ with rational coefficients. We claim that $V(X_1, X_2, X_3) = X_1 - X_2$ is a Galois resolvent for x_1, x_2 and x_3, so in particular $t = x_1 - x_2$ is primitive to x_1, x_2 and x_3. To this end, we need to show that $x_{\sigma(1)} - x_{\sigma(2)}$ takes six different values for the six permutations σ from S_3: Let two permutations σ and τ be given with

$$x_{\sigma(1)} - x_{\sigma(2)} = x_{\tau(1)} - x_{\tau(2)} \tag{5.18}$$

We need to show that then $\sigma = \tau$ holds.

Since $\sigma(1), \sigma(2), \tau(1), \tau(2) \in \{1, 2, 3\}$ and $\sigma(1) \neq \sigma(2)$ and $\tau(1) \neq \tau(2)$, $\tau(1) = \sigma(1)$ or $\tau(1) = \sigma(2)$ or $\tau(2) = \sigma(1)$ or $\tau(2) = \sigma(2)$ holds.

In the case $\tau(1) = \sigma(1)$, we obtain from (5.18) that $x_{\sigma(2)} = x_{\tau(2)}$. Since x_1, x_2 and x_3 are pairwise different due to the separability of the irreducible polynomial $f(X)$, it follows that $\tau(2) = \sigma(2)$ and thus $\sigma = \tau$.

In the case $\tau(1) = \sigma(2)$, it follows from (5.18) that $2x_{\tau(1)} = x_{\sigma(1)} + x_{\tau(2)}$. If further $\sigma(1) = \tau(2)$, we would have $2x_{\tau(1)} = 2x_{\tau(2)}$, thus $x_{\tau(1)} = x_{\tau(2)}$ in contradiction to the fact that x_1, x_2 and x_3 are pairwise different. Thus, $\sigma(1) \neq \tau(2)$, that is, we have an equation of the form $2x_i = x_j + x_k$, where $i, j, k \in \{1, 2, 3\}$ are pairwise different. This implies $x_i = \frac{1}{3}(x_1 + x_2 + x_3)$. The right-hand side is rational according to Vieta's theorem, so x_i is also rational. This would mean that $f(X)$ has a rational root. This contradicts the irreducibility of $f(X)$. Consequently, the case $\tau(1) = \sigma(2)$ cannot occur at all.

The remaining cases $\tau(2) = \sigma(1)$ or $\tau(2) = \sigma(2)$ can be handled analogously.

Irreducible polynomials of degree 3 have a natural Galois resolvent, namely $X_1 - X_2$.

5.5 Lagrange's Theorem and the Class Equation

The virtue of Galois theory consist in the fact that we can reduce statements about the solutions of a polynomial equation to statements about the Galois group of these solutions. However, this is only meaningful if we know something about Galois groups or more generally about subgroups G of the symmetric groups S_n. We want to collect statements about these in this section. Since it will generally not be important for us from which permutation group G a subgroup is, we want to simply use the term *group* for such a subgroup of a permutation group.

Let G be a group, i.e., a subgroup of the S_n for a natural number n, that is, $G = \{\sigma_1, \ldots, \sigma_m\}$, where $\sigma_1, \ldots, \sigma_m$ are pairwise different n-letter permutations and G contains the identical permutation and is closed with respect to composition and the formation of inverses. Since the composition is obviously associative, i.e.,

$$(\sigma \circ \tau) \circ \varrho = \sigma \circ (\tau \circ \varrho) \tag{5.19}$$

for three n-letter permutations σ, τ and ϱ, we can simply write $\sigma \circ \tau \circ \varrho$ for the expression in (5.19). By id, we have denoted the identical permutation. This fulfills the property $\mathrm{id} \circ \sigma = \sigma$ for every permutation σ.

We have defined σ^{-1} as the inverse permutation to σ, that is, σ^{-1} fulfills the equation $\sigma^{-1} \circ \sigma = \mathrm{id}$. Due to the existence of these inverse permutations, we have the following cancellation rule:

If $\tau \circ \sigma = \tau \circ \sigma'$ holds for permutations σ, σ' and τ, we can compose this equation from the left with τ^{-1}. Because $\tau^{-1} \circ \tau = \mathrm{id}$, it follows that $\sigma = \sigma'$.

It is easy to see that the inverse permutation also fulfills $\sigma \circ \sigma^{-1} = \mathrm{id}$. But this can also be systematically derived: We have

$$\sigma^{-1} \circ \sigma \circ \sigma^{-1} = \mathrm{id} \circ \sigma^{-1} = \sigma^{-1}.$$

Multiplying by $(\sigma^{-1})^{-1}$ from the left yields $\sigma \circ \sigma^{-1} = \mathrm{id}$. From this we can formally conclude that also $\sigma \circ \mathrm{id} = \sigma$ (so far we have only used $\mathrm{id} \circ \sigma = \sigma$) because

$$\sigma \circ \mathrm{id} = \sigma \circ \sigma^{-1} \circ \sigma = \mathrm{id} \circ \sigma = \sigma.$$

Finally, we want to systematically derive two more equalities. We claim that $(\sigma^{-1})^{-1} = \sigma$. But by definition of the inverse permutation this is equivalent to $\sigma \circ \sigma^{-1} = \mathrm{id}$, which we have just calculated.

For two permutations σ and τ also $(\tau \circ \sigma)^{-1} = \sigma^{-1} \circ \tau^{-1}$ holds because $(\sigma^{-1} \circ \tau^{-1}) \circ (\tau \circ \sigma) = \mathrm{id}$ due to the associativity of the composition. So much for the elementary "arithmetic" of the composition of permutations.

We call two elements σ and σ' of G *conjugate*, if there is another element τ of G such that

$$\sigma' = \tau \circ \sigma \circ \tau^{-1}. \tag{5.20}$$

If we visualize τ as a renumbering of the numbers $1, \ldots, n$, then σ' corresponds to the permutation σ after renumbering. This is quite analogous to the equation $A' = SAS^{-1}$ of linear algebra for the representation matrix of an endomorphism after basis change. Conjugacy is an *equivalence relation*, i.e., a reflexive, symmetric and transitive relation.

In detail, this means: Every element σ is conjugate to itself (choose $\tau = \mathrm{id}$ in (5.20)). If σ is conjugate to σ', then σ' is also conjugate to σ, because composition with τ^{-1} from the left and τ from the right in (5.20) yields $\tau^{-1} \circ \sigma' \circ \tau = \sigma$. This corresponds to an equation of the form (5.20) (with τ^{-1} instead of τ) due to $(\tau^{-1})^{-1} = \tau$.

Finally, the following holds: If σ is conjugate to σ' and σ' is conjugate to σ'', then σ is also conjugate to σ'': Because if $\sigma' = \tau \circ \sigma \circ \tau^{-1}$ and $\sigma'' = \tau' \circ \sigma' \circ \tau'^{-1}$, it follows that

$$\sigma'' = \tau' \circ \sigma' \circ \tau'^{-1} = \tau' \circ \tau \circ \sigma \circ \tau^{-1} \circ \tau'^{-1} = (\tau' \circ \tau) \circ \sigma \circ (\tau' \circ \tau)^{-1}.$$

A permutation σ which is only conjugate to itself is a permutation with

$$\tau \circ \sigma \circ \tau^{-1} = \sigma$$

for all elements τ of G, that is, σ *commutes* with all permutations from G. We then say that σ lies in the *center* of G, written

$$\sigma \in \mathrm{Z}(G).$$

The permuation id is always an element in the center. If there are further elements of G in the center, we say G has a *nontrivial center*.

In general, permutations do not commute. In the center of a group are exactly those permutations that commute with all other permutations.

Example 5.6 Let $n \geq 3$. We claim that the full symmetric group S_n has a trivial center. Let σ be an element in the center of S_n. We want to show that σ is the identity. Suppose σ is not the identity. Then there is an $i \in \{1, \ldots, n\}$ with $j := \sigma(i) \neq i$.

Now we make a case distinction. First, we consider the case that $\sigma(j) = i$. In this case, let τ be the n-letter permutation that swaps i and an element $k \in \{1, \ldots, n\} \setminus \{i, j\}$ and keeps the remaining numbers from $\{1, \ldots, n\}$ fixed. Then we have

$$(\tau \circ \sigma \circ \tau^{-1})(k) = \tau(\sigma(\tau^{-1}(k))) = \tau(\sigma(i)) = \tau(j) = j,$$

but $\sigma(k) \neq j$ (because $\sigma(i) = j$). Thus, $\tau \circ \sigma \circ \tau^{-1} \neq \sigma$, so σ does not commute with τ, contradiction.

Finally, we consider the case that $k := \sigma(j) \neq i$. Then we choose τ as the n-letter permutation that swaps i and j and keeps the remaining numbers from $\{1, \ldots, n\}$ fixed. In this case, we have

$$(\tau \circ \sigma \circ \tau^{-1})(i) = \tau(\sigma(\tau^{-1}(i))) = \tau(\sigma(j)) = \tau(k) = k,$$

but $\sigma(i) = j \neq k$. Thus, $\tau \circ \sigma \circ \tau^{-1} \neq \sigma$, again a contradiction.

The elements in the center of G form a subgroup of S_n: Suppose $\sigma, \sigma' \in Z(G)$ and $\tau \in G$ is arbitrary, then $\tau \circ (\sigma \circ \sigma') \circ \tau^{-1} = \tau \circ \sigma \circ \tau^{-1} \circ \tau \circ \sigma' \circ \tau^{-1} = \sigma \circ \sigma'$ and $\tau \circ \sigma^{-1} \circ \tau^{-1} = (\tau \circ \sigma \circ \tau^{-1})^{-1} = \sigma^{-1}$. Composition and inversion of elements from the center therefore do not lead out of the center.

We want to derive a statement in the following that guarantees us a non-trivial center as soon as we know something about the number of elements in G. The number of permutations contained in G is called *order* of G, written

$$[G : 1].$$

The order of a group is another name for the number of its elements.

Example 5.7 Since there are exactly $n!$ different n-letter permutations, we have $[S_n : 1] = n!$.

If σ is an n-letter permutation, we denote with

$$[G : G_\sigma]$$

the number of elements from G (including σ) that are conjugate to σ.

If $\sigma_1, \ldots, \sigma_r$ run through a representative system of the conjugacy classes of G, that is, every permutation from G is conjugate to exactly one permutation σ_j with $j \in \{1, \ldots, r\}$, then we obviously have

$$[G : 1] = \sum_{j=1}^{r} [G : G_{\sigma_j}]. \tag{5.21}$$

Since every element from the center is only conjugate to itself, each element from the center appears exactly once among $\sigma_1, \ldots, \sigma_r$. For an element σ from the center, it also holds $[G : G_\sigma] = 1$. Conjugacy classes of elements that are not in the center consist of at least two permutations and are called *non-central*.

We can thus bring (5.21) into the following form, which is known under the name *class equation*:

Proposition 5.7 *Let G be a group. The order of G is calculated as*

$$[G : 1] = [\mathrm{Z}(G) : 1] + \sum_{\sigma} [G : G_\sigma], \tag{5.22}$$

where the sum runs over a representative system of the non-central conjugacy classes of G. □

We have illustrated the class equation in Fig. 5.6. The points in Fig. 5.6 correspond to the elements of G, the cells correspond to the conjugacy classes of group elements. The points that are in a cell with exactly one point correspond to the central group elements. The class equation simply states that the total number of points is the sum of the number of points in cells with one point and the sum of the numbers of points in cells with at least two points.

The class equation is a relationship between the number of elements of a group and the size of its conjugacy classes. The number of elements in the center is exactly the number of one-element conjugacy classes.

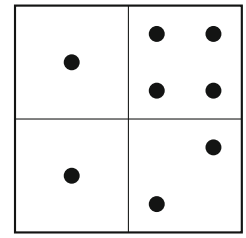


Fig. 5.6 Illustration of the class equation

The Eq. (5.22) is interesting because we can make statements about the number of elements conjugate to an element $\sigma \in G$. For this we use that $[G : G_\sigma]$ divides the order of G, which we can see this as follows:

We say an element τ is in the *centralizer* of σ, written

$$\tau \in G_\sigma$$

if $\tau \circ \sigma \circ \tau^{-1} = \sigma$. With G_σ we therefore denote the set of all elements τ that commute with σ. It can again be easily verified that G_σ is a group itself. We claim that

$$[G : 1] = [G : G_\sigma] \cdot [G_\sigma : 1]. \tag{5.23}$$

We prove this formula directly in greater generality. Namely, we can replace G_σ by any subgroup of G (that is, by a subgroup of S_n, whose elements are all already in G).

What then should $[G : H]$ be? For this we define that two elements τ and τ' are *congruent* modulo H, written

$$\tau \equiv \tau' \pmod{H},$$

if $\tau'^{-1} \circ \tau \in H$, which is equivalent to there being a ϱ from H with $\tau = \tau' \circ \varrho$. It can again be easily verified that congruence modulo a subgroup H is an equivalence relation.

With $[G : H]$ we denote the number of congruence classes modulo H, i.e., the size of a representative system for the congruence classes. This number is called the *index* of H in G.

Why does this definition in the case of $H = G_\sigma$ coincide with the original definition of $[G : G_\sigma]$ as the number of elements conjugate to σ (including σ)?

To see this, we consider the mapping that assigns to each $\tau \in G$ the element $\tau \circ \sigma \circ \tau^{-1}$, which is conjugate to σ. Obviously, this mapping is surjective onto the elements conjugate to σ. Since $\tau \circ \sigma \circ \tau^{-1} = \tau' \circ \sigma \circ \tau'^{-1}$ is equivalent to $\tau'^{-1} \circ \tau \circ \sigma \circ \tau^{-1} \circ \tau' = \sigma$, i.e., to $\tau'^{-1} \circ \tau \in G_\sigma$, i.e., to τ and τ' being congruent modulo G_σ, the mapping $\tau \mapsto \tau \circ \sigma \circ \tau^{-1}$ induces a bijection between a representative system for the congruence classes and the system of elements conjugate to σ. The formula (5.23) then follows from the following theorem of Lagrange:

Proposition 5.8 *Let G be a group. For every subgroup H of G,*

$$[G : 1] = [G : H] \cdot [H : 1]. \tag{5.24}$$

The Eq. (5.24) explains the initially perhaps unusual-looking notation $[G : 1]$ for the order of a group. It also reminds a bit of the degree formula for algebraic elements. With the help of Galois theory, we will later see that this is no coincidence.

Proof We choose a representative system $R = \{\tau_1, \ldots, \tau_r\}$ for the congruence classes modulo H, that is, for each $\tau \in G$ there exists exactly one $j \in \{1, \ldots, r\}$ such that τ is congruent to τ_j modulo R.

We then claim that

$$R \times H \to G, \quad (\tau_j, \varrho) \mapsto \tau_j \circ \varrho \tag{5.25}$$

is a bijection: If $\tau_i \circ \varrho = \tau_j \circ \varrho'$ for $i, j \in \{1, \ldots, r\}$ and $\varrho, \varrho' \in H$, then $\tau_j^{-1} \circ \tau_i = \varrho' \circ \varrho^{-1}$.

The right side is an element from H. Thus, τ_i and τ_j are congruent modulo H, by definition, $\tau_i = \tau_j$. From this follows $\mathrm{id} = \varrho' \circ \varrho^{-1}$, thus $\varrho' = \varrho$. Consequently, (5.25) is injective.

For surjectivity, we consider any element τ in G. Then there exists a $j \in \{1, \ldots, r\}$ with $\tau_j^{-1} \circ \tau =: \varrho \in H$. Consequently, $\tau = \tau_j \circ \varrho$ is in the image of (5.25). □

The order of a subgroup is always a divisor of the group order.

Using (5.23) we can show:

Lemma 5.5 *Let G be a group, whose order is a prime power greater than one. Then G has a nontrivial center.*

A group, whose order is a prime power, we call a *p-group*. (Here, the trivial power 1 is also allowed.)

Proof Let $[G : 1] = p^r$ for a prime number p and a natural number $r > 1$. By the class equation, we have

$$p^r = [G : 1] = [\mathrm{Z}(G) : 1] + \sum_{\sigma} [G : G_\sigma].$$

Since the summands $[G : G_\sigma] > 1$ are all divisors of $[G : 1]$ and thus multiples of p, $[\mathrm{Z}(G) : 1] = p^r - \sum_{\sigma} [G : G_\sigma]$ is also a multiple of p. Thus, $\mathrm{Z}(G)$ must consist of more than one element. □

In particular, $\mathrm{Z}(G)$ is itself again a p-group.

Nontrivial p-groups have a nontrivial center.

In the situation of Lemma 5.5 we can even say more about the center of G. For this, we define the *order* of an element σ in G:

The order of σ is the smallest positive natural number d with $\sigma^d = \mathrm{id}$. This number d is well-defined: We need to show that there is a positive natural number e with $\sigma^e = \mathrm{id}$. To do this, we consider the sequence

$$\sigma, \sigma^2, \sigma^3, \ldots$$

of permutations. Since there are only finitely many n-letter permutations, there must be positive natural numbers $i < j$ with $\sigma^i = \sigma^j$. Composition with $\sigma^{-i} = (\sigma^{-1})^i$ from the left yields $\mathrm{id} = \sigma^{j-i}$; this means that the identity must appear at some point in the above sequence.

For the order of σ, we also write

$$\mathrm{ord}\sigma.$$

We further write $\langle\sigma\rangle$ for the subgroup generated by σ, i.e., the system consisting of the pairwise different elements $\mathrm{id}, \sigma, \sigma^2, \ldots, \sigma^{\mathrm{ord}\sigma-1}$. Since $\sigma^{\mathrm{ord}\sigma} = \mathrm{id}$ and thus $\sigma^{-1} = \sigma^{\mathrm{ord}\sigma-1}$, we can write

$$\langle\sigma\rangle = \{\ldots, \sigma^{-2}, \sigma^{-1}, \mathrm{id}, \sigma, \sigma^2, \ldots\}$$

In particular, we recognize that $\langle\sigma\rangle$ is indeed a subgroup, with $\mathrm{ord}\sigma$ many elements, i.e.,

$$\mathrm{ord}\sigma = [\langle\sigma\rangle : 1].$$

From Lagrange's theorem, Proposition 5.8, it follows:

Proposition 5.9 *The order of an element σ of G is a divisor of the order of G.* □

The order of every element in a group is a divisor of the group order.

With this, we can prove the following sharpening of (5.5):

Proposition 5.10 *Let p be a prime number and G a p-group of order $[G : 1] > 1$. Then there is an element of order p in the center of G.*

Proof Let $[G : 1] = p^r$ for a positive natural number r. Let σ be a non-trivial element in the center of G. Since the order of σ divides the group order p^r, we have $\mathrm{ord}\sigma = p^s$ for a

positive natural number $s \leq r$. Thus, $\sigma^{p^{s-1}}$ is an element in the center of G that has order p. □

In every non-trivial p-group, there is an element of order p.

5.6 Cyclotomic Polynomials

We call the equation

$$X^n - 1 = 0 \tag{5.26}$$

the n-th cyclotomic equation because its n solutions are the n-th roots of unity ζ_n^k, $k = 0, \ldots, n$, which divide the unit circle into n equal sections. (We remind of Fig. 2.4.)

A better understanding of (5.26) will eventually provide us with a criterion for when a regular n-gon (whose vertices we can identify with the n-th roots of unity) is constructible with compass and straightedge. For example, the Galois group of the polynomial $X^n - 1$ will lead to a better understanding of (5.26); this calculation will be our first example for the computation of the Galois group of a whole series of polynomials (and not just concretely given single polynomials).

First, we note that the normalized polynomial $f(X) = X^n - 1$ is a separable polynomial: The roots, which we want to consider in the order

$$x_1 = \zeta_n, \quad x_2 = \zeta_n^2, \quad \ldots, \quad x_{n-1} = \zeta_n^{n-1}, \quad x_n = \zeta_n^n = 1 \tag{5.27}$$

are pairwise different.

(An alternative proof for separability uses the derivative $f'(X) = nX^{n-1}$: because we have $1 = \frac{1}{n} f'(X) \cdot X - f(X)$, the polynomials $f(X)$ and $f'(X)$ are thus coprime.)

However, what does not hold (at least in the case $n \neq 1$) is that $f(X)$ is irreducible. For example, we have

$$f(X) = X^n - 1 = (X - 1) \cdot (X^{n-1} + X^{n-2} + \cdots + X + 1).$$

But also the right factor on the right hand is generally not irreducible, so we generally still have not found a decomposition of $f(X)$ into irreducible factors.

To get there, we first define for each natural number d another normalized polynomial, namely

$$\Phi_d(X) = \prod_{\substack{i\in\{1,\dots,d\}\\(d,i)=1}} (X - \zeta_d^i),$$

the so-called d-th *cyclotomic polynomial*. Here we write $(d, i) = 1$ for two integers d and i if d and i are coprime. The polynomial $\Phi_d(X)$ is thus composed of as many factors as there exist coprime positive natural numbers less than or equal to d. We write this number as $\varphi(d)$.

(The function that assigns the number $\varphi(n)$ to each natural number n is called *Euler's φ-function*.) So we have

$$\deg \Phi_d(X) = \varphi(d).$$

The first result of this section is the following:

Proposition 5.11 *For each natural number n, the following equality of polynomials holds:*

$$X^n - 1 = \prod_{d|n} \Phi_d(X). \tag{5.28}$$

Here, the product runs over all divisors (1 and n included) of n.

Proof The polynomial on the left side is a normalized polynomial of degree n. The polynomial on the right side of the equality to be proven is also normalized and has the degree $\sum_{d|n} \varphi(d)$. According to the definition of Euler's φ-function, this is the number of pairs (d, k'), where d is a divisor of n and $k' \in \{1, \dots, n\}$ is a natural number coprime to d.

We claim that these pairs are in bijection with the set $\{1, \dots, n\}$:

Let $k \in \{1, \dots, n\}$. Let ℓ be its greatest common divisor with n. Then $k' = \frac{k}{\ell}$ is coprime to $d = \frac{n}{\ell}$. and furthermore, $k' \in \{1, \dots, d\}$. Conversely, if d is a divisor of n, say $n = \ell \cdot d$ for an $\ell \in \{1, \dots, n\}$ and $k' \in \{1, \dots, d\}$ is a number coprime to d, then $k = \ell \cdot k'$ is a natural number from $\{1, \dots, n\}$.

Thus, $\sum_{d|n} \varphi(d) = n$. Consequently, the polynomials on both sides of (5.28) are of the same degree. Since the polynomial $X^n - 1$ has n different roots ζ_n^k with $k \in \{1, \dots, n\}$, it is sufficient for the proof of equality, that the polynomial on the right side also has ζ_n^k as roots.

For this, let ℓ again be the greatest common divisor of n and k. If we then set $d = \frac{n}{\ell}$ and $k' = \frac{k}{\ell}$, then

$$\zeta_n^k = \zeta_n^{\ell \cdot k'} = (\zeta_n^\ell)^{k'} = \zeta_d^{k'},$$

because $(d, k') = 1$, is therefore a root of $\Phi_d(X)$ and thus also of the right side $\prod_{d|n} \Phi_d(X)$. □

We can use Proposition 5.11 to calculate all cyclotomic polynomials recursively. To do this, we simply bring (5.28) into the form

$$\Phi_n(X) = \frac{X^n - 1}{\prod\limits_{\substack{d|n \\ d<n}} \Phi_d(X)}.$$

Performing polynomial division thus leads to the calculation of $\Phi_n(X)$ recursively. Because $\Phi_1(X) = X - 1$, for example

$$\Phi_2(X) = \frac{X^2 - 1}{\Phi_1(X)} = \frac{X^2 - 1}{X - 1} = X + 1.$$

From this follows, for example,

$$\Phi_4(X) = \frac{X^4 - 1}{\Phi_1(X) \cdot \Phi_2(X)} = \frac{X^4 - 1}{(X - 1) \cdot (X + 1)} = X^2 + 1.$$

There is a recursive calculation formula for the cyclotomic polynomials.

We will see that (5.28) represents the decomposition of $X^n - 1$ into irreducible polynomials over the rational numbers. For this, we first have to show, that the $\Phi_d(X)$ are actually polynomials over the rational numbers. We even show:

Lemma 5.6 *For every natural number d, $\Phi_d(X)$ is a primitive polynomial with integer coefficients.*

Proof We can prove this by induction over d: The induction start is $d = 0$. In this case, $\Phi_d(X) = 1$, the constant polynomial 1. For the induction step, we apply Proposition 5.11 to $X^d - 1$ and get:

$$X^d - 1 = \prod_{d'|d} \Phi_{d'}(X) = \Phi_d(X) \cdot \prod_{\substack{d'|d \\ d'<d}} \Phi_{d'}(X).$$

On the left side is a primitive polynomial with integer coefficients. On the right side, according to the induction assumption, all factors (except possibly one) are primitive polynomials with integer coefficients. Therefore, the remaining factor $\Phi_d(X)$ must also be a primitive polynomial with integer coefficients, according to Lemma 4.1. □

Finally, we come to the main result of this section:

Theorem 5.1 *Let d be a natural number. The cyclotomic polynomial $\Phi_d(X)$ is irreducible over the rational numbers.*

The cyclotomic polynomials are the minimal polynomials of the primitive roots of unity. The polynomials of the form $X^n - 1$ are products of cyclotomic polynomials.

Thus, (5.28) is the decomposition of $X^n - 1$ into irreducible polynomials over the rational numbers. We will prove Theorem 5.1 with the help of the following lemma:

Lemma 5.7 *Let p be a prime number coprime to d. If ζ is Galois conjugate to ζ_d, then ζ^p is also Galois conjugate to ζ_d.*

Proof of Theorem 5.1 Let $f(X)$ be the minimal polynomial of ζ_d over the rational numbers. We want to show that $\Phi_d(X) = f(X)$. To do this, we show that every root of $\Phi_d(X)$ is also a root of $f(X)$.

The roots of $\Phi_d(X)$ are all of the form ζ_d^k, where $k \in \{1, \ldots, d\}$ and k is coprime to d. We then write $k = p_1 \cdots p_m$, where $p_1, \ldots, p_m$ are prime numbers, which (necessarily) are coprime to d.

According to Lemma 5.7, $\zeta_d^{p_1}$ is Galois conjugate to ζ_d. Again according to Lemma 5.7, $\zeta_d^{p_1 p_2} = (\zeta_d^{p_1})^{p_2}$ is also Galois conjugate to ζ_d, etc. Finally, we find that $\zeta_d^{p_1 \cdots p_m} = \zeta_d^k$ is Galois conjugate to ζ_d. According to Proposition 5.1, ζ_d^k is also a root of $f(X)$. □

Before we present the proof of Lemma 5.7, we make some preliminary considerations: Let p be a prime number. If $k \in \{1, \ldots, p-1\}$ is a natural number, then the binomial coefficient

$$\binom{p}{k} = \frac{p \cdot (p-1) \cdots (p-k+1)}{k!}$$

is divisible by p, since the factor p does not appear in the prime factorization of the denominator.

We can also write this result as

$$\binom{p}{k} \equiv 0 \pmod{p}, \qquad k \in \{1, \dots, p-1\}$$

From this, we can immediately conclude that the binomial theorem modulo p has a particularly simple form, namely

$$(X+Y)^p \equiv \sum_{k=0}^{p} \binom{p}{k} X^{p-k} Y^k \equiv \binom{p}{0} X^p + \binom{p}{p} Y^p \equiv X^p + Y^p \pmod{p} \tag{5.29}$$

as a congruence of two polynomials in X and Y.

This result can be easily extended to more than one summand: Suppose $g(X) = a_n X^n + a_{n-1} X^{n-1} + \cdots + a_1 X + a_0$ is a polynomial with integer coefficients, then we get

$$\begin{aligned} g(X)^p &\equiv (a_n X^n + a_{n-1} X^{n-1} + \cdots + a_1 X + a_0)^p \\ &\equiv (a_n X^n)^p + (a_{n-1} X^{n-1})^p + \cdots + (a_1 X)^p + a_0^p \\ &\equiv a_n^p (X^p)^n + a_{n-1}^p (X^p)^{n-1} + \cdots + a_1^p X^p + a_0^p \pmod{p}. \end{aligned} \tag{5.30}$$

We can further simplify this expression. Namely, for every integer a it holds that

$$a^p \equiv a \pmod{p}.$$

This statement is the so-called Fermat's little theorem, which can also be derived similar to (5.29):

$$a^p \equiv (\underbrace{1 + \cdots + 1}_{a})^p \equiv \underbrace{1^p + \cdots + 1^p}_{a} \equiv a \pmod{p}.$$

This simplifies (5.30) to

$$g(X)^p \equiv g(X^p) \pmod{p}.$$

Modulo a prime number p, we can push p-th powers into sums and products and more generally into polynomials with integer coefficients.

Proof of Lemma 5.7 Let $f(X)$ be the minimal polynomial of ζ_d over the rational numbers. This has ζ as a root. We need to show that it also has ζ^p as a root. Since $\Phi_d(X)$ is a polynomial with root ζ_d, $f(X)$ must be a divisor of $\Phi_d(X)$. Since Φ_d is normalized and

primitive with integer coefficients, it follows from Lemma 4.1 that $f(X)$ also has integer coefficients.

$f(X)$ is a divisor of $X^d - 1$ for the same reasons, so we can write

$$X^d - 1 = f(X) \cdot g(X)$$

for another normalized and primitive polynomial $g(X)$ with integer coefficients.

Since ζ is a root of $X^d - 1$, ζ is a d-th root of unity. With ζ, ζ^p is also a d-th root of unity, thus a root of $X^d - 1$. Therefore, ζ^p is a root of $f(X)$ or of $g(X)$. We assume that $g(\zeta^p) = 0$, and want to lead this to a contradiction:

Since under the assumption that the polynomial $g(X^p)$ has the algebraic number ζ as a root, the minimal polynomial of ζ must be a divisor of this polynomial, so

$$g(X^p) = f(X) \cdot h(X)$$

for another normalized primitive polynomial $h(X)$ with integer coefficients. From our previous considerations on arithmetic modulo p, it follows $g(X)^p \equiv f(X) \cdot h(X)$ modulo p, that is, $f(X)$ is a divisor of $g(X)^p$ modulo p. Let $q(X)$ be a modulo p irreducible factor of $f(X)$, which then also divides $g(X)^p$. Due to the uniqueness of the prime factor decomposition of polynomials modulo p, it follows that $q(X)$ also divides $g(X)$ modulo p. Thus, $q(X)^2$ is a divisor of $X^d - 1 = f(X) \cdot g(X)$ modulo p, that is,

$$X^d - 1 \equiv q(X)^2 \cdot r(X) \pmod{p} \tag{5.31}$$

for another normalized polynomial $r(X)$ with integer coefficients. If we differentiate both sides of (5.31), we get

$$dX^{d-1} \equiv q(X) \cdot (2q'(X) \cdot r(X) + q(X) \cdot r'(X)) \pmod{p}.$$

Since d is coprime to p, d is invertible modulo p, so we get that $q(X)$ modulo p is a divisor of X^{d-1}. Due to the uniqueness of prime factorization, it must therefore be the case that $q(X) \equiv X$ modulo p. From (5.31) it would then follow that

$$0 \equiv 0^d - 1 \equiv -1 \pmod{p},$$

which is an obvious contradiction. □

The cyclotomic polynomial Φ_d is irreducible modulo a prime number p if p is not a prime divisor of d.

With the knowledge about the decomposition of $X^n - 1$ into irreducible polynomials, we can easily determine the Galois group of the roots of $X^n - 1$ over the rational numbers.

The roots are given in (5.27). Since $x_2, \ldots, x_n$ are powers of x_1, $x_1 = \zeta_n$ is a primitive element for the roots of $X^n - 1$, i.e., the n-th roots of unity.

The minimal polynomial of ζ_n is $\Phi_n(X)$, because $\Phi_n(X)$ has x_1 as a root and is normalized and irreducible over the rational numbers. The Galois conjugates of ζ_n are therefore all $\varphi(n)$ roots of unity of the form ζ_n^d, where $d \in \{1, \ldots, n\}$ is coprime to n. (These n-th roots of unity are called *primitive* n-th roots of unity.)

According to Proposition 5.3, the Galois group of $X^n - 1$ has exactly $\varphi(n)$ different elements, namely for each Galois conjugate ζ_n^d exactly one symmetry σ_d of the roots $x_1, \ldots, x_n$.

If we define $h_i(X) := X^i$ for $i \in \{1, \ldots, n\}$, then $\zeta_n^i = x_i = h_i(t)$ for all $i \in \{1, \ldots, n\}$. Therefore, for each $d \in \{1, \ldots, n\}$ coprime to n, the symmetry d acts through

$$\sigma_d \cdot \zeta_n^i = h_i(\zeta_n^d) = (\zeta_n^d)^i = \zeta_n^{di}.$$

We can therefore describe the Galois group of the roots of $X^n - 1$ by

$$\mathrm{Gal}_{\mathbf{Q}}(\zeta_n, \zeta_n^2, \ldots, \zeta_n^{n-1}, \zeta_n^n) = \left\{ \zeta_n^i \mapsto \zeta_n^{di} \mid d \in \{1, \ldots, n\}, (d, n) = 1 \right\}$$

The elements of the Galois group of the cyclotomic polynomials operate by exponentiation on the primitive roots of unity.

In the case that $n = p$ is a prime number, we can analyze the structure of the Galois group of the roots of $X^n - 1$ even more explicitely. For this, we need a definition from number theory:

Specifically, we call an integer g a *primitive root* modulo p if every integer d that is not 0 modulo p (that is, d is a unit modulo p) is a power of g modulo p, i.e., modulo p all units modulo p appear in the sequence

$$g, g^2, g^3, g^4, \ldots .$$

Since $g^p \equiv g$ modulo p according to Fermat's little theorem, all units modulo p obviously must be congruent modulo p to an element from $g, g^2, \ldots, g^{p-1}$. Since there are exactly $p - 1$ different units modulo p, this is exactly the case when $g, g^2, \ldots, g^{p-1}$ modulo p are pairwise different.

If we fix a primitive root modulo p, then every unit modulo p is a power of this primitive root.

With the help of the polynomial arithmetic we have developed, it is not difficult to show the existence of primitive roots modulo p:

Lemma 5.8 *For every prime number p there exists a primitive root g modulo p.*

Proof According to Fermat's little theorem, for all units d modulo p, $d^p \equiv d$ modulo p, so $d^{p-1} \equiv 1$ modulo p, since d is invertible module p. We call the smallest positive natural number a with $d^a \equiv 1$ modulo p the *order of d modulo p*. (So $a \leq p-1$.)

Let e be the least common multiple of all orders modulo p of the units modulo p. We consider the prime factor decomposition $e = p_1^{f_1} \cdot p_2^{f_2} \cdots p_s^{f_s}$ with pairwise different prime numbers. By definition of e as the least common multiple, there must be, for each $1 \leq i \leq s$, a unit g_i modulo p whose order modulo p is a multiple of $p_i^{f_i}$. A suitable power of g_i is then a unit modulo p of the order $p_i^{f_i}$ modulo p. Without loss of generality, we can assume that g_i is such a unit modulo p.

It follows that $g \coloneqq g_1 \cdot g_2 \cdots g_s$ is a unit modulo p whose order $\tilde{e}$ is a divisor of e. Suppose, $\tilde{e} \cdot p_i$ is a divisor of e. Then in particular $g^{p_i^{-1}e} \equiv 1$ modulo p; but $g^{p_i^{-1}e} \equiv g_i^{p_i^{f_i-1}}$ modulo p, and the right side cannot be congruent 1 modulo p by definition of the order of g_i. It follows, that $\tilde{e}$ cannot be a proper divisor of e, so $\tilde{e} = e$.

It remains to show that $e = p-1$ because then, g, g^2, …, g^{p-2} are all different elements, so they necessarily run through all units modulo p. Since e must be a divisor of $p-1$, we already know that $e \leq p-1$. It remains to show $e \geq p-1$: Modulo p, the polynomial $X^e - 1$ has the numbers $1, \ldots, p-1$ as roots. Since these numbers are also pairwise different modulo p, there are the modulo p pairwise different linear factors $X-1$, …, $X-(p-1)$ in the prime factor decomposition of $X^e - 1$ modulo p. Consequently, $e \geq p-1$ must hold. □

Through the proof, it becomes obvious why primitive roots modulo p are named as such: They are $(p-1)$-th roots of unity modulo p, and every other $(p-1)$-th root of unity modulo p is a power of such a primitive root modulo p.

Interestingly, these $(p-1)$-th roots of unity modulo p help us understand the Galois group of the p-th roots of unity over the rational numbers. To see this, we recall the definition of the cyclic group C_n. The permutation

$$\sigma_1 = \begin{pmatrix} 1 & 2 & \ldots & n-1 & n \\ 2 & 3 & \ldots & n & 1 \end{pmatrix} \in \mathrm{C}_n$$

has the property that every other permutation in C_n is of the form

$$\sigma_1^k = \underbrace{\sigma_1 \circ \cdots \circ \sigma_1}_{k}$$

for $k \in \{0, \ldots, n-1\}$. (Here we set $\sigma_1^0 := \mathrm{id}$ in analogy to the power notation $x^0 = 1$ for numbers x.) We therefore call σ_1 a *generator* of C_n.

In generalization, we call a Galois group *cyclic* if it contains a symmetry σ such that every other symmetry is an integral power of σ. (Here we define a negative power by $\sigma^{-n} = (\sigma^{-1})^n$.)

We want to show that the Galois group of the p-th roots of unity is cyclic. To do this, we choose a primitive root g modulo p. We consider the element σ of the Galois group of $\zeta_p, \zeta_p^2, \ldots, \zeta_p^p$ given by

$$\sigma : \zeta_p^i \mapsto \zeta_p^{ig}.$$

For an integer k,

$$\sigma^k : \zeta_p^i \mapsto \zeta_p^{ig^k}.$$

Since g has been chosen as a primitive root modulo p, g, g^2, …, g^{p-1} run through all units of d modulo n. Thus, for each $d \in \{1, \ldots, p-1\}$, there is a $k \in \{1, \ldots, p-1\}$ with $g^k \equiv d$ modulo p. Since $\zeta_p^p = 1$, we have $\zeta_p^{g^k i} = \zeta_p^{di}$. Consequently, $\sigma, \sigma^2, \ldots, \sigma^{p-1}$ run through all elements of the Galois group of the p-th roots of unity, meaning, this Galois group is indeed cyclic.

Let p be a prime number. The Galois group of the roots of the p-th cyclotomic polynomial $\Phi_p(X) = X^{p-1} + X^{p-2} + \cdots + X + 1$ is cyclic of order $\phi(p) = p - 1$.

5.7 On the Constructability of Regular n-Gons

The aim of this section is to prove the following result of Carl-Friedrich Gauss with our knowledge about cyclotomic polynomials:

Theorem 5.2 *Let n be a positive natural number. The regular n-gon can be constructed with compass and straightedge if and only if n is of the form*

$$n = 2^r p_1 \cdots p_s$$

where r is a natural number and $p_1, \ldots, p_s$ are pairwise different Fermat primes.

Here, a prime number p is called a *Fermat prime* if p is of the form

$$p = F_k = 2^{2^k} + 1, \qquad k \in \mathbf{N}_0.$$

Remark 5.1 The sequence of numbers F_k starts with

$$F_0 = 3, \quad F_1 = 5, \quad F_2 = 17, \quad F_3 = 257, \quad F_4 = 65537, \quad F_5 = 4294967297.$$

Pierre de Fermat observed that $F_0, \ldots, F_4$ are primes and conjectured that all further F_k, $k \geq 5$ are primes. However, Leonhard Euler found in 1732 the divisor 631 of the number F_5 and thus refuted Pierre de Fermat's conjecture. To this day, besides $F_0, \ldots, F_4$ no further Fermat primes have been found. However, it is also not proven that there are no more.

In particular, the regular 17-gon is constructible.

Why was Pierre de Fermat interested in primes of this special form, i.e., primes which are one less than a power of two, whose exponent is again a power of two? It seems more obvious to first examine numbers of the form $2^n + 1$, where n is any natural number, to see if they are prime.

Suppose n contains an odd divisor $q > 2$, i.e., $n = p \cdot q$ for another positive natural number q. According to the known summation formula for the geometric sum, we then have

$$\begin{aligned} 2^n + 1 = (2^p)^q + 1 = 1 - (-2^p)^q = (1 + 2^p) \\ \cdot (1 - 2^p + (2^p)^2 - \cdots + (2^p)^{q-1}). \end{aligned} \tag{5.32}$$

For the first factor, $1 + 2^p \geq 3$. For the second factor, which must also be positive, $1 - 2^p + (2^p)^2 - \cdots + (2^p)^{q-1} \equiv 1 - 2^p \not\equiv 1 \pmod{(2^p)^2}$. Thus, (5.32) is a nontrivial factorization of $2^n + 1$. Consequently, $2^n + 1$ can only be a prime if n itself is a power of two. This is why Fermat primes are of this special form.

The only primes of the form $2^n + 1$ have to be of the form $2^{2^k} + 1$, i.e., Fermat primes.

To provide a proof for Theorem 5.2, we identify our drawing plane with the Gaussian number plane so that the center of the n-gon to be constructed coincides with the complex number 0 and one vertex with the complex number 1. We recall that a regular n-gon is constructible if and only if a primitive n-th root of unity is constructible (from which it follows that all other n-th roots of unity are constructible as powers of the primitive one).

Therefore, for the proof of Theorem 5.2 we need to decide for which n a primitive n-th root of unity is constructible. For the rest of this section, we want to call a natural number n *constructible* if a a primitive n-th root of unity is constructible.

To prove Theorem 5.2, we start with some simple lemmas:

Lemma 5.9 *If n is a constructible natural number, and if m is a (nonnegative) divisor of n, then m is also constructible.*

Proof of Lemma 5.9 Let $n = d \cdot m$ for a natural number d. By assumption, $\zeta_n = \mathrm{e}^{\frac{2\pi\mathrm{i}}{n}}$ is constructible. Therefore,

$$\zeta_m = \zeta_n^d = \mathrm{e}^{\frac{2\pi\mathrm{i}}{m}}$$

is also constructible. □

Lemma 5.10 *If n and m are two coprime natural numbers, then their product $n \cdot m$ is also constructible.*

Proof of Lemma 5.10 Let r and s be two integers with $1 = rn + sm$. By assumption, $\zeta_n = \mathrm{e}^{\frac{2\pi\mathrm{i}}{n}}$ and $\zeta_m = \mathrm{e}^{\frac{2\pi\mathrm{i}}{m}}$ are constructible. Therefore,

$$\zeta_{nm} = \zeta_n^s \cdot \zeta_m^r = \mathrm{e}^{\frac{2\pi\mathrm{i}}{nm}}$$

is also constructible. □

For the constructibility of regular n-gons, we only need to consider the case where n is a prime power.

Proof of Theorem 5.2 Due to Lemmas 5.9 and 5.10, it is sufficient to prove Theorem 5.2 for the case where n is a prime power.

Consider the case of the prime number 2. We need to show that for every natural number r, the natural number 2^r is constructible. This follows from the fact that ζ_{2^r} is the $(r-1)$-fold iterated square root of -1 and we can construct square roots with compass and straightedge (in fact, we only need to extract square roots from numbers on the unit circle, for which bisecting angles is sufficient).

Next, consider the case of an odd prime number p that is not a Fermat prime. We need to show that for every positive natural number r, the number p^r is not constructible (according to the above, even $r = 1$ would suffice).

To do this, we calculate the degree of ζ_{p^r} over the algebraic numbers. This is given by the degree of the cyclotomic polynomial $\Phi_{p^r}(X)$, which in turn is $\varphi(p^r)$. Since the positive natural numbers d with $1 \leq d \leq p^r$ that have a common divisor with p^r are just given by the p^{r-1} multiples of p in this interval, it is

$$\varphi(p^r) = p^r - p^{r-1} = p^{r-1} \cdot (p-1).$$

$p-1$ is not a power of two of the form 2^m, otherwise, due to the primality of p according to our previous considerations, m would again have to be a power of two, so p would be a Fermat prime. Therefore, the degree of ζ_{p^r} is not a power of two over the rational numbers, and ζ_{p^r} is therefore not constructible according to Lemma 4.2.

What remains is the case where p is a Fermat prime, i.e., a prime number of the form $p = 2^{2^k} + 1$ for a natural number k. We need to show that the natural number p is constructible, but that for all $r > 1$ the number p^r is not constructible. (Due to Lemma 5.10, it is sufficient to assume $r = 2$.) Following the previous considerations on the Euler φ-function, we have

$$\varphi(p^r) = p^{r-1} \cdot (p-1).$$

In the case of $r > 1$, this is not a power of two, so according to Lemma 4.2 it is not constructible. In the case $r = 1$, we have $\varphi(p) = p - 1 = 2^{2^k}$, which is a power of two. Then the constructibility of ζ_p follows from the following Proposition 5.12. □

Proposition 5.12 *Let x be an algebraic number such that all Galois conjugates of x are rational in x. Then x is constructible if and only if the degree of x over the rational numbers is a power of two.*

Proof of Proposition 5.12 One direction is a consequence of Lemma 4.2: If the degree of x is not a power of two, then x is certainly not constructible. So let x be an algebraic number such that all Galois conjugates in x are rational and for which $[\mathbf{Q}(x) : \mathbf{Q}] = 2^n$ for a natural number n.

We want to show by induction on n that x is constructible. The base case $n = 0$ is trivial. So let $n > 0$. Let $x_1, x_2, \ldots, x_{2^n}$ be the Galois conjugates (including x) of x. Since x is primitive to $x_1, \ldots, x_{2^n}$, the Galois group G to $x_1, \ldots, x_{2^n}$ has exactly 2^n elements. According to Proposition 5.10, there is a symmetry σ in the center of G which is of order 2, that is, $\sigma \neq \mathrm{id}$, but $\sigma^2 = \mathrm{id}$. Since $\sigma \neq \mathrm{id}$, there exists an $i \in \{1, \ldots, 2^n\}$ with $\sigma(x_i) \neq x_i$.

We can assume without loss of generality that $x_1, \ldots, x_{2^n}$ are numbered in such a way that $\sigma \cdot x_1 = x_2$. Now let $i \in \{1, \ldots, 2^n\}$ be arbitrary. Then there exists (exactly) one symmetry τ in G with $x_i = \tau \cdot x_1$. Since σ as an element in the center commutes with all symmetries, we have

$$\sigma \cdot x_i = \sigma \cdot (\tau \cdot x_1) = \tau \cdot (\sigma \cdot x_1) = \tau \cdot x_2.$$

Since τ is a permutation of the roots, $x_i = \tau \cdot x_1 \neq \tau \cdot x_2$. Consequently, $x_j = \sigma \cdot x_i \neq x_i$. Because $\sigma \cdot x_j = \sigma^2 \cdot x_i = x_i$, we can pair the $x_1, \ldots, x_{2^n}$ into pairs x_i, x_j such that σ swaps x_i and x_j. We can therefore assume without loss of generality that the $x_1, \ldots, x_{2^n}$ are numbered such that

$$\sigma \cdot x_i = \begin{cases} x_{i+1} & \text{for } i \text{ odd and} \\ x_{i-1} & \text{for } i \text{ even} \end{cases}$$

and $x_1 = x$.

We choose y_1 as a primitive element to $x_1 + x_2$ and $x_1 x_2$. In particular, we can write $y_1 = r(x_1 + x_2, x_1 x_2) = s(x_1, x_2)$ for polynomials $r(Y_1, Y_2)$ and $s(X_1, X_2)$ with rational coefficients. Here, $s(X_1, X_2)$ is symmetric in X_1 and X_2. For $j \in \{2, \ldots, 2^{n-1}\}$ let $y_j \coloneqq s(x_{2j-1}, x_{2j})$. Then we set

$$g(X) \coloneqq (X - y_1) \cdots (X - y_{2^{n-1}}).$$

If τ is a symmetry in G, then $\tau \cdot g(X)$ (we let here τ act on the coefficients of $g(X)$) is given by

$$\begin{aligned} \tau \cdot g(X) &= \tau \cdot ((X - s(x_1, x_2)) \cdots (X - s(x_{2^n-1}, x_{2^n}))) \\ &= (X - s(\tau \cdot x_1, \tau \cdot x_2)) \cdots (X - s(\tau \cdot x_{2^n-1}, \tau \cdot x_{2^n})) \\ &= (X - s(x_1, x_2)) \cdots (X - s(x_{2^n-1}, x_{2^n})) = g(X), \end{aligned}$$

where the last equality is due to the fact that τ commutes with σ, permuting the pairs made from x_{2j-1} and x_{2j}, but preserving their totality.

According to Proposition 5.4, the coefficients of $g(X)$ are thus rational numbers. Since $\deg g(X) = 2^{n-1}$ and y_1 is a root of $g(X)$, the degree of y_1 over the rational numbers is therefore less than or equal to 2^{n-1}—due to the degree formula ($y_1 = x_1 + x_2$ is rational in x), therefore a power of two.

We claim that y_j for $j \in \{1, \ldots, 2^{n-1}\}$ is rational in y_1. To see this, we write $x_{2j-1} = h_j(x_1)$ for a polynomial $h_j(X)$ with rational coefficients. Thus,

$$\begin{aligned} y_j &= s(x_{2j-1}, x_{2j}) = s(h_j(x_1), \sigma \cdot h_j(x_1)) = s(h_j(x_1), h_j(\sigma \cdot x_1)) \\ &= s(h_j(x_1), h_j(x_2)). \end{aligned} \tag{5.33}$$

The right side of (5.33) is symmetric in x_1 and x_2 and thus a polynomial in the elementary symmetric functions $x_1 + x_2$ and $x_1 x_2$. Since y_1 is primitive to $x_1 + x_2$ and $x_1 x_2$, it follows, that y_j is a polynomial in y_1 with rational coefficients.

With this, we can apply the induction hypothesis and find that y_1 is constructible. Finally, we consider the polynomial

$$f(X) = (X - x_1) \cdot (X - x_2) = X^2 - (x_1 + x_2)X + x_1 x_2.$$

The coefficients of this polynomial are symmetric expressions in x_1 and x_2 and thus rational in y_1. Since $x = x_1$ is a root of this quadratic polynomial, the degree of x_1 over y_1 is less than or equal to two. From the constructibility of y_1, the constructibility of x_1 follows. □

Assuming that an algebraic number x is primitive for all its Galois conjugates, a necessary and sufficient condition for the constructibility of x is given by its degree, namely, that the degree is a power of two.

Summary

- Algebraic numbers are called **Galois conjugates** when they have the same minimal polynomial. Galois conjugates each satisfy the same polynomial equations.
- If $x_1, \ldots, x_n$ are algebraic numbers, an **algebraic relation** between them is a polynomial $H(X_1, \ldots, X_n)$ such that $H(x_1, \ldots, x_n) = 0$.
- A permutation of the roots $x_1, \ldots, x_n$ of a separable polynomial is an element of the **Galois group** of $x_1, \ldots, x_n$ if this permutation preserves all algebraic relations.
- The elements of the Galois group correspond bijectively to the Galois conjugates of a **primitive element** of $x_1, \ldots, x_n$.
- The **Galois resolvent** allows to effectively determine such primitive elements and their Galois conjugates and thus the Galois group.
- The elements of the Galois groups act not only on the roots $x_1, \ldots, x_n$, but on all numbers rational in $x_1, \ldots, x_n$. The **invariants** of this action are exactly the rational numbers.
- Every nontrivial group of prime order p has a nontrivial **center**, according to the **class equation**, as follows from the **Lagrange's theorem**. This also implies the existence of a group element of order p.
- If the order of the Galois group of a polynomial is a **power of two**, it thus has an element of order 2. This proves that the roots of a polynomial are constructible

(continued)

with compass and straightedge if and only if the order of the corresponding Galois group is a power of two.
- A **regular n-gon** is constructible if and only if n is the product of a power of two and pairwise distinct Fermat primes. For exactly these n, the order of the Galois group of the n-th **cyclotomic polynomial** is a power of two.

Exercises

Galois Conjugates

5.1 Give two algebraic numbers that are not Galois conjugates of each other.

5.2 Let t be an algebraic number. Justify why the product of t with all its Galois conjugates is a rational number. What about the sum?

5.3 Let $f(X)$ and $g(X)$ be the minimal polynomials of two algebraic numbers x and y. Show that x and y are conjugates if and only if the greatest common divisor of $f(X)$ and $g(X)$ is a non-constant polynomial.

5.4 Let x, y and z be three algebraic numbers. Let x be Galois conjugate to y and y are Galois conjugate to z. Show that x is Galois conjugate to z.

5.5 How many Galois conjugates does $x_1 = \sqrt[3]{1+\sqrt{2}}$ have?

5.6 Let p and q be two different prime numbers. Calculate the Galois conjugates of $\sqrt{p}+\sqrt{q}$.

5.7 Show that Proposition 5.1 characterizes the Galois conjugates of an algebraic number t, that is, an algebraic number t' is a Galois conjugate to t (the case $t' = t$ is included here) if and only if every polynomial $f(X)$ with rational coefficients that has t as a root also has t' as a root.

5.8 Let $f(X)$ be a polynomial with rational coefficients. Let $x := f(t)$ where t is an algebraic number. Let t' be another algebraic number and $x' := f(t')$. Show that x and x' are Galois conjugates if t and t' are Galois conjugates.

5.9 Show by an example that Lemma 5.1 becomes false if $x_1, \ldots, x_n$ are not assumed to be solutions (with multiplicities) of a polynomial equation with rational coefficients, but can be any algebraic numbers.

The Galois Group of an Equation

5.10 Let n be a natural number. Provide a method to list all n-letter permutations, and show that there are a total of $n!$ of them.

5.11 Let n be a natural number. Let σ be the n-letter permutation that maps 1 to 2, 2 to 3, ... and finally $n-1$ to n and n to 1. Calculate its signum depending on n.

5.12 Calculate the symmetry group of a regular pentagon in the plane.

5.13 Calculate the symmetry group of a regular n-gon in the plane.

5.14 Calculate the symmetry group of the octahedron.

5.15 List all possible subgroups of the symmetric group S_3.

5.16 Show that the alternating group A_n is a subgroup of the symmetric group S_n.

5.17 Show that the cyclic group C_n is a subgroup of the symmetric group S_n.

5.18 Let σ and τ be symmetries of the roots $x_1, \ldots, x_n$ of a normalized separable polynomial. Show that $\tau \cdot (\sigma \cdot x) = (\tau \circ \sigma) \cdot x$.

5.19 Let σ and τ be two n-letter permutations. Let $H(X_1, \ldots, X_n)$ be an algebraic relation between the roots $x_1, \ldots, x_n$ of a normalized separable polynomial. Show that $\tau \cdot (\sigma \cdot H) = (\tau \circ \sigma) \cdot H$.

5.20 Let $x_1, \ldots, x_n$ be the roots of a normalized separable polynomial $f(X)$ over the rational numbers. Let x_1 and x_2 not be Galois conjugate. Show that no symmetry σ of the roots $x_1, \ldots, x_n$ of $f(X)$ exists such that $x_2 = \sigma \cdot x_1$.

5.21 Let $f(X) = g_1(X) \cdots g_m(X)$ where $f(X)$ and $g_1(X), \ldots, g_m(X)$ are normalized polynomials with rational coefficients. Let $f(X)$ be separable. Show that every symmetry of the roots of $f(X)$ permutes the roots of $g_i(X)$ only among themselves.

5.22 Let $f(X)$ be a quadratic polynomial over the rational numbers with roots x_1 and x_2 which we assume to be different. Calculate $\mathrm{Gal}_{\mathbf{Q}}(x_1, x_2)$ depending on the discriminant of $f(X)$.

5.23 Let $f(X)$ be an irreducible normalized cubic polynomial over the rational numbers with roots x_1, x_2 and x_3. Assume that x_1 is a primitive element for x_1, x_2 and x_3. Show that

the Galois group for the roots x_1, x_2 and x_3 of $f(X)$ has exactly three elements. Specify these elements.

5.24 Let $f(X)$ be an irreducible normalized cubic polynomial over the rational numbers with roots x_1, x_2 and x_3. Assume that x_1 is not a primitive element for x_1, x_2 and x_3. Show that the Galois group for the roots x_1, x_2 and x_3 of $f(X)$ has exactly six elements.

5.25 Let $f(X)$ be a normalized separable polynomial of degree n. Let t be a primitive element over the rational numbers for all roots of $f(X)$. Show that the degree of t is at most $n!$.

5.26 Let $x_1, \ldots, x_n$ be the roots of a normalized separable polynomial $f(X)$ with rational coefficients. Let $y_1, \ldots, y_n$ be the roots of the same polynomial in (possibly) different order. How can $\mathrm{Gal}_{\mathbf{Q}}(y_1, \ldots, y_n)$ be described by $\mathrm{Gal}_{\mathbf{Q}}(x_1, \ldots, x_n)$?

About Invariants of the Galois Action

5.27 Let $x_1, \ldots, x_n$ be the roots of a normalized separable polynomial with rational coefficients. Let z and w be two numbers rational in $x_1, \ldots, x_n$. Show for every symmetry σ of $x_1, \ldots, x_n$ that

$$\sigma \cdot (z + w) = \sigma \cdot z + \sigma \cdot w$$

and

$$\sigma \cdot (zw) = (\sigma \cdot z)\,(\sigma \cdot w)$$

hold.

5.28 Let $x_1, \ldots, x_n$ be the roots of a normalized separable polynomial $f(X)$ with rational coefficients. Show that $f(X)$ is irreducible if and only if the Galois group for $x_1, \ldots, x_n$ acts transitively on $x_1, \ldots, x_n$.

5.29 Let $x_1, \ldots, x_n$ be the roots of a normalized separable polynomial $f(X)$ with rational coefficients. Show that the Galois group for $x_1, \ldots, x_n$ is a subgroup of the alternating group A_n if and only if the discriminant of $f(X)$ has a square root in the rational numbers.

5.30 Show that the polynomial $f(X) = X^3 - 3X + 1$ is irreducible and separable over the rational numbers. Calculate the Galois group of the roots of $f(X)$.

5.31 Show that the polynomial $f(X) = X^3 + 3X + 1$ is irreducible and separable over the rational numbers. Calculate the Galois group of the roots of $f(X)$.

Galois Resolvents

5.32 Let $f(X_1, \ldots, X_n)$ be a non-vanishing polynomial with rational coefficients. Show that there exist integers $m_1, \ldots, m_n$ such that $f(m_1, \ldots, m_n) \neq 0$.

5.33 Provide a Galois resolvent for the polynomial $f(X) = X^2 + X + 1$.

5.34 Why is a Galois resolvent only defined for separable polynomials?

5.35 Let $x_1, \ldots, x_n$ be the roots of a normalized separable polynomial $f(X)$ with rational coefficients. Let C be a natural number such that

$$n \cdot \left| \frac{x_i - x_j}{x_k - x_\ell} \right| \leq C$$

for all $i, j, k, \ell \in \{1, \ldots, n\}$ with $k \neq \ell$. Show that

$$V(X_1, \ldots, X_n) = X_1 + C \cdot X_2 + C^2 \cdot X_3 + \ldots + C^{n-1} X_n$$

is a Galois resolvent for $f(X)$.

5.36 Let $f(X)$ be a normalized separable polynomial with n roots. Show that it is always possible to find a Galois resolvent $V(X_1, \ldots, X_n)$ of $f(X)$ that does not depend on X_1, i.e., is a polynomial just in $X_2, \ldots, X_n$.

5.37 Express the polynomial $X_1X_2X_3 - X_2X_3 + X_1^3X_2^2X_3^2$ as a polynomial in X_1 and in the elementary symmetric functions of X_1, X_2 and X_3.

5.38 Let $H(X_1, \ldots, X_n)$ be a polynomial that is symmetric in $X_3, \ldots, X_n$. Show that $H(X_1, \ldots, X_n)$ can be written as a polynomial in X_1, in X_2, and in the elementary symmetric functions of $X_1, \ldots, X_n$. Is this representation unique?

5.39 Calculate a Galois group of the polynomial $f(X) = X^3 - 3X - 4$ over the rational numbers using the method of the Galois resolvent.

5.40 Let $f(X)$ be a normalized separable polynomial with roots $x_1, \ldots, x_n$. Construct a finite sequence of algebraic relations $H_1(X_1, \ldots, X_n), \ldots, H_m(X_1, \ldots, X_n)$ between $x_1, \ldots, x_n$ over the rational numbers such that an n-letter permutation is a symmetry of the

roots $x_1, \ldots, x_n$ if and only if σ preserves the algebraic relations $H_1(X_1, \ldots, X_n), \ldots, H_m(X_1, \ldots, X_n)$.

5.41 Provide a primitive element for the three roots of the equation $X^3 - 7 = 0$.

5.42 Let $x_1, \ldots, x_n$ be the roots of a normalized separable polynomial over the rational numbers. Let $V(X_1, \ldots, X_n)$ be a polynomial with rational coefficients such that $V(x_{\sigma(1)}, \ldots, x_{\sigma(n)})$ takes pairwise different values on all permutations σ that lie in the Galois group of $x_1, \ldots, x_n$ over the rational numbers. Show that then $t = V(x_1, \ldots, x_n)$ is a primitive element to $x_1, \ldots, x_n$, but generally not a Galois resolvent.

The Lagrange Theorem and the Class Equation

5.43 Let G be a group. Let $\sigma \in G$. Show that the centralizer G_σ of σ is a subgroup of G.

5.44 Let G be a group. If i is a negative integer, we set $\sigma^i = (\sigma^{-1})^{-i}$. Also, $\sigma^0 = \mathrm{id}$. Show that $\sigma^i \circ \sigma^j = \sigma^{i+j}$ for any two integers i and j.

Why is it sufficient to prove this statement for the case that G is a full permutation group?

5.45 Let G be a group. Let H be a subgroup of G. Show that congruence modulo H is an equivalence relation on the elements of G.

5.46 Let G be a group. Let H be a subgroup of G and let K be a subgroup of H. Why is K also a subgroup of G?

5.47 Let G be a group. Let H be a subgroup of G and let K be a subgroup of H. Show that then

$$[G : K] = [G : H] \cdot [H : K]$$

holds.

5.48 Is there a subgroup with 80 elements in the permutation group S_5?

5.49 Let $f(X)$ be a normalized separable polynomial over the rational numbers. Let t be a primitive element to the roots (with multiplicity) $x_1, \ldots, x_n$ of $f(X)$. Show that the degree of t over the rational numbers is a divisor of $n!$.

5.50 Let G be a group. Let $\sigma \in G$ of order n. Let m be an integer. Show that the order of σ^m is given by $\frac{n}{d}$, where d is the greatest common divisor of n and m.

5.51 Let n be a natural number. Calculate the orders of the elements of the cyclic group C_n.

Cyclotomic Polynomials

5.52 Give the factorization of $X^3 + X^2 + X + 1$ into irreducible polynomials over the rational numbers.

5.53 Calculate the cyclotomic polynomial $\Phi_3(X)$.

5.54 Calculate the cyclotomic polynomial $\Phi_6(X)$.

5.55 Calculate the cyclotomic polynomial $\Phi_9(X)$.

5.56 Let A be a square matrix over the rational numbers, such that A^n is the identity matrix. Show that the characteristic polynomial of A is a product of cyclotomic polynomials of the form $\Phi_d(X)$, where d is a divisor of n.

5.57 Let A be a square matrix with 4 rows and columns over the rational numbers that is not is the identity matrix. Show that the only prime numbers p for which A^p can be the identity matrix are 2, 3, and 5.

5.58 Let p be a prime number. Show that

$$\Phi_p(X) = \frac{X^p - 1}{X - 1} = X^{p-1} + X^{p-2} + \cdots + X + 1.$$

5.59 Let p be a prime number. Show that

$$\Phi_p(X + 1) = X^{p-1} + \binom{p}{1} X^{p-2} + \cdots + \binom{p}{p-2} X + \binom{p}{p-1}.$$

5.60 Show by using the Eisenstein criterion and not the theory of this chapter that $X^{p-1} + X^{p-2} + \cdots + X + 1$ is irreducible over the rational numbers.

5.61 Let m be a natural number. Show that

$$X^{2m} + X^m + 1 = \frac{X^{3m} - 1}{X^m - 1} = \prod_{\substack{d \mid 3m \\ d \nmid m}} \Phi_d(X).$$

5.62 Let m be a natural number. Show that the polynomial $X^{2m} + X^m + 1$ is irreducible over the rational numbers if and only if $m = 3^k$ for a natural number k.

5.63 Let p be a prime number. Let r be a positive natural number. Show that

$$\Phi_{p^r}(X) = \Phi_p(X^{p^{r-1}}).$$

5.64 Let $n = p_1^{r_1} \cdots p_s^{r_s}$ be the prime factorization of a natural number, where we assume $r_1, \ldots, r_s > 0$. Show that

$$\Phi_n(X) = \Phi_{p_1 \cdots p_s}\left(X^{p_1^{r_1-1} \cdots p_s^{r_s-1}}\right).$$

5.65 Let $n > 1$ be an odd natural number. Show that $\Phi_{2n}(X) = \Phi_n(-X)$.

5.66 Let p be a prime number and n a natural number coprime to p. Show that

$$\Phi_{pn}(X) = \frac{\Phi_n(X^p)}{\Phi_n(X)}.$$

5.67 Let p be a prime number and n a positive integer multiple of p. Show that $\Phi_{pn}(X) = \Phi_n(X^p)$.

5.68 Let p be a prime number. Show that the binomial coefficient $\binom{p^2}{p}$ is divisible by p, but not by p^2.

5.69 Let d and n be natural numbers. Let $X = \{\zeta_n, \zeta_n^2, \ldots, \zeta_n^{n-1}, \zeta_n^n\}$ be the set of the n-th roots of unity. Show that

$$\sigma_d \colon X \to X, \quad \zeta \mapsto \zeta^d$$

is a bijection of X if and only if d is coprime to n.

5.70 List all primitive roots modulo 11.

5.71 Let p be a prime number. Give the prime factorization of $X^{p-1} - 1$ modulo p.

5.72 Let n be a natural number. List all generators of C_n.

On the Constructability of Regular n-Gons

5.73 List all $n \in \{1, \ldots, 100\}$ for which a regular n-gon can be constructed with compass and straightedge.

5.74 Provide a construction rule for the regular 15-gon.

5.75 Let n be a natural number. Show that

$$F_{n+1} = 2 + F_n F_{n-1} \cdots F_0.$$

5.76 Show that F_m and F_n are coprime for $m \neq n$. Conclude that there are infinitely many prime numbers.

5.77 A *Mersenne prime*[2] is a prime number of the form $M_n = 2^n - 1$, where n is a natural number. Show that M_n can only be a Mersenne prime if n itself is a prime number. Show even that M_n is divisible by M_d when d is a positive divisor of n.

5.78 Let p be a prime number. Let x be an algebraic number of degree p^n such that its Galois conjugates $x_1 = x, x_2, \ldots, x_{p^n}$ are all rational in x. Show that the Galois group of $x_1, \ldots, x_{p^n}$ has an element σ of order p in the center. Let y be a primitive element of $e_1(x_1, \sigma \cdot x_1, \ldots, \sigma^{p-1} \cdot x_1), \ldots, e_p(x_1, \sigma \cdot x_1, \ldots, \sigma^{p-1} \cdot x_1)$. Show, that y has degree p^{n-1} over the rational numbers.

5.79 Let n be a natural number, and let $f(X)$ be an irreducible, normalized, separable polynomial of degree 2^n. Let $x_1, \ldots, x_{2^n}$ be the solutions of $f(X)$, whose Galois group has order 2^n. Does it then follow that x_1 is a constructible number?

Endnotes

[1] This does not mean, however, that every element of the form $t' = V(x_{\sigma(1)}, \ldots, x_{\sigma(n)})$ is a Galois conjugate of t.

[2] Martin Mersenne, 1588–1648, French theologian, mathematician and music theorist.

On the Solvability of Polynomial Equations

6

Sometimes adopting a relative standpoint also illuminates the perspective on the absolute.

Abstract

Up to this point, we have used terms such as minimal polynomial, Galois conjugacy, and Galois group for polynomials with rational coefficients. It turns out that the theory becomes much more powerful when we also consider extensions of the rational numbers as the coefficient domain. We call this view the *relative*, while we refer to the rational number case as the *absolute*.

For example, $X^4 - 2$ is the minimal polynomial of a fourth root $\sqrt[4]{2}$ of 2 over the rational numbers. On the other hand, $X^2 - \sqrt{2}$ is the minimal polynomial of the same algebraic number, but this time over the number field $\mathbf{Q}(\sqrt{2})$, consisting of all algebraic numbers that are rational in $\sqrt{2}$. Accordingly, the degree of $\sqrt[4]{2}$ over $\mathbf{Q}$ is exactly 4, but over $\mathbf{Q}(\sqrt{2})$ it is only 2.

Just as the degree decreases, there are fewer Galois conjugates, and the Galois group becomes smaller the larger we choose the number field. We can use this observation to better understand the absolute case over the rational numbers: First, we look at the relative case over suitable extensions of the original coefficient domain. Then we successively reduce the number field, so that the Galois group successively increases until we find the Galois group over the rational numbers in the limit.

Extensions of the number field correspond exactly to subgroups of the Galois group, and we can specify an exact bijective correspondence in the form of the fundamental theorem of Galois theory. With the help of this main theorem, we reduce difficult questions such as the solvability of equations to simpler questions about finite groups.

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After we show that the roots of the cyclotomic polynomials (and thus the vertices of an equilateral n-gon when suitably drawn in the complex number plane) are given by radical expressions, we can deduce Galois' result, a necessary and sufficient condition for the solvability of polynomial equations by radicals.

It turns out that the general equation of degree five or higher is not solvable by radicals. There also special equations of degree five and higher that are not solvable by radicals. At the end of the chapter, we derive the solution formulas for equations of degree three and four, known since the beginning of the modern era, but using Galois' theory.

6.1 Relative Galois Groups

We want to introduce a new way of speaking that will save us a lot of writing in the following. Let $y_1, \ldots, y_n$ be algebraic numbers. We recall that instead of saying that a number x is rational in $y_1, \ldots, y_n$, we can write $x \in \mathbf{Q}(y_1, \ldots, y_n)$. Let $z_1, \ldots, z_m$ be another sequence of algebraic numbers. If every number rational in $y_1, \ldots, y_n$ is also rational in $z_1, \ldots, z_m$, we can describe this fact by $\mathbf{Q}(y_1, \ldots, y_n) \subseteq \mathbf{Q}(z_1, \ldots, z_m)$. If also every number rational in $z_1, \ldots, z_m$ is rational in $y_1, \ldots, y_n$, we write accordingly $\mathbf{Q}(y_1, \ldots, y_n) = \mathbf{Q}(z_1, \ldots, z_m)$. In this case, we call the sequences $y_1, \ldots, y_n$ and $z_1, \ldots, z_m$ *equivalent* (over the rational numbers).

We can also view $\mathbf{Q}(y_1, \ldots, y_n)$ as a symbol for the equivalence class of the sequence $y_1, \ldots, y_n$.

Example 6.1 That an algebraic number t is primitive for $y_1, \ldots, y_n$, we can also express by saying that t is equivalent to $y_1, \ldots, y_n$ (over the rational numbers).

Since such an equivalence class $\mathbf{Q}(y_1, \ldots, y_n)$ stands for the entirety of the numbers rational in $y_1, \ldots, y_n$; we also call $\mathbf{Q}(y_1, \ldots, y_n)$ a *(finitely generated) extension* (of the number field) of the rational numbers or simply a *number field*. We usually denote such extensions with large Latin letters $K, L, \ldots, E, F, \ldots$.

The theorem of the primitive element, Theorem 4.2, can be reformulated as: Every number field K is of the form $\mathbf{Q}(t)$ for an algebraic number t.

Let K be a number field over the rational numbers, so $K = \mathbf{Q}(t_1, \ldots, t_k)$ for algebraic elements $t_1, \ldots, t_k$. We call $t_1, \ldots, t_n$ *generators* of K.

Let $y_1, \ldots, y_n$ be further algebraic elements. If x is a number rational in $y_1, \ldots, y_n$ and $t_1, \ldots, t_k$, we also write

$$x \in K(y_1, \ldots, y_n).$$

If $z_1, \ldots, z_m$ are further algebraic numbers, and every number rational in $y_1, \ldots, y_n$ and $t_1, \ldots, t_k$ is also rational in $z_1, \ldots, z_m$ and $t_1, \ldots, t_k$ and vice versa, we write

$$K(y_1, \ldots, y_n) = K(z_1, \ldots, z_m)$$

and call the sequences $y_1, \ldots, y_n$ and $z_1, \ldots, z_m$ *equivalent over* K.

Using the symbol $E := K(y_1, \ldots, y_n)$ we denote the equivalence class of $y_1, \ldots, y_n$ over K. We call E a *(finitely generated) extension* of K or a *number field* over K. The elements $y_1, \ldots, y_n$ are called *generators* of E over K. Furthermore, K is called an *intermediate extension* of E over the rational numbers $\mathbf{Q}$, since every algebraic number in K (i.e., every number rational in $t_1, \ldots, t_m$) is already in E.

According to the theorem of the primitive element, $E = K(t)$ for an algebraic number t. The degree of t over K is the dimension of the K-vector space E, which we simply call the *degree* of E over K and express in formulas as $[E : K]$. If L is an intermediate extension of E over K (that is, L is a number field with $K \subseteq L \subseteq E$), then according to the degree formula we have

$$[E : K] = [E : L] \cdot [L : K].$$

In towers of number fields, the degree multiplies.

We initially defined the degree of an algebraic element x as the degree of the minimal polynomial of x over the rational numbers. Later we relativized this concept by defining the degree of x over a number field K as the degree of the polynomial of minimal degree, for which we allow numbers from K as coefficients.

For the Galois group, which we have so far defined for the roots of separable polynomials with rational coefficients, we now want to do something very similar.

Let K be a number field. If x is an algebraic number, we first define the *Galois conjugates* of x over K as the other roots of the minimal polynomial of x over K. If x' is a Galois conjugate over K to x, then x' is also a Galois conjugate to x over the rational numbers, because the minimal polynomial of x over K is, according to Abel's irreducibility theorem for polynomials over K, a divisor of the minimal polynomial of x over the rational numbers. In particular, the number of Galois conjugates of x can therefore not increase if we expand the number field from the rational numbers to K.

Example 6.2 The Galois conjugates of $x = \sqrt{2}+\sqrt{3}$ over the rational numbers are known to be given by $\pm\sqrt{2} \pm \sqrt{3}$. We want to determine which of these algebraic numbers are Galois conjugate to x over $K = \mathbf{Q}(\sqrt{2})$. To do this, we note that

$$g(X) := (X - \sqrt{2})^2 - 3 = X^2 - 2\sqrt{2} - 1$$

is the minimal polynomial of x over K (this has x as a root and has the correct degree, because $[\mathbf{Q}(\sqrt{2}+\sqrt{3}) : K] = 2$). Besides x, the other root of $g(X)$ is given by $\sqrt{2}-\sqrt{3}$, that is, $\sqrt{2}-\sqrt{3}$ is (apart from x itself) the only Galois conjugate of $\sqrt{2}+\sqrt{3}$ over K.

If x and y are Galois conjugate over a number field K, they are, in general, no longer Galois conjugate over an extension L of K.

For this relative concept of Galois conjugates, we can derive very similar statements as in the absolute situation over the rational numbers: If t is an algebraic number, then the following proposition can be proven in the same way as Proposition 5.1:

Proposition 6.1 *If t' is a Galois conjugate to t over K, and if $f(X)$ is a polynomial over K which has t as a root, then it also has t' as a root.* □

Also Lemma 5.1 and Proposition 5.2 can be generalized accordingly:

Lemma 6.1 *Let $x_1, \ldots, x_n$ be the solutions (with multiplicities) of a polynomial equation over K. If $V(X_1, \ldots, X_n)$ is a polynomial over K, the Galois conjugates of $t = V(x_1, \ldots, x_n)$ over K are all of the form $t' = V(x_{\sigma(1)}, \ldots, x_{\sigma(n)})$, where σ is an n-ary permutation.* □

Proposition 6.2 *Let $x_1, \ldots, x_n$ be the solutions (with multiplicities) of a polynomial equation over K. If t is a primitive element for $x_1, \ldots, x_n$ over K, then every Galois conjugate t' of t is a primitive element for $x_1, \ldots, x_n$ over K.* □

Here, t is called a *primitive element* for $x_1, \ldots, x_n$ over K, if we have $K(t) = K(x_1, \ldots, x_n)$, i.e., if t and $x_1, \ldots, x_n$ are equivalent over K. A primitive element for $x_1, \ldots, x_n$ over K is for example given by a primitive element for $x_1, \ldots, x_n, y_1, \ldots, y_m$ (over the rational numbers), when $K = \mathbf{Q}(y_1, \ldots, y_m)$.

In particular, we know how to calculate primitive elements over any number field.

Next, we come to the concept of a Galois group over K. For this, we consider a normalized separable polynomial $f(X)$ over K with roots $x_1, \ldots, x_n$.

An *algebraic relation* between $x_1, \ldots, x_n$ over K is then a polynomial $H(X_1, \ldots, X_n)$ over K such that $H(x_1, \ldots, x_n) = 0$.

A *symmetry* of $x_1, \ldots, x_n$ over K is an n-ary permutation σ such that for every algebraic relation $H(x_1, \ldots, x_n) = 0$ between $x_1, \ldots, x_n$ over K, $\sigma \cdot H$ is also an algebraic relation over K.

All symmetries of $x_1, \ldots, x_n$ over K form the *Galois group* of $x_1, \ldots, x_n$ over K. As with the Galois group over the rational numbers, the Galois group over K is also a subgroup of the symmetric group S_n.

We write

$$\mathrm{Gal}_K(x_1, \ldots, x_n)$$

for the Galois group of $x_1, \ldots, x_n$ over K.

In the following, we are going to vary the base number field K to learn about the involved Galois groups. For instance, it is clear that the Galois group of $x_1, \ldots, x_n$ over K is a subgroup of the Galois group of $x_1, \ldots, x_n$ over the rational numbers because every symmetry over K is also a symmetry over the rational numbers since every algebraic relation over the rational numbers is in particular also an algebraic relation over K. However, we can generalize this fact: If K is an intermediate extension of a number field L, then

$$\mathrm{Gal}_L(x_1, \ldots, x_n) \subset \mathrm{Gal}_K(x_1, \ldots, x_n),$$

which means, every symmetry over L is in particular a symmetry over K.

The larger the base number field, the smaller the Galois group.

We obtain the smallest possible symmetry group when we choose $L = K(x_1, \ldots, x_n)$. Then $H_1(X_1, \ldots, X_n) = X_1 - x_1, \ldots, H_n(X_1, \ldots, X_n) = X_n - x_n$ are all algebraic relations over L, which due to $x_i \neq x_j$ for $i \neq j$ are preserved by no permutation other than the identical one. We thus have

$$\mathrm{Gal}_{K(x_1,\ldots,x_n)}(x_1, \ldots, x_n) = \{\mathrm{id}\}.$$

As in the case of the Galois group over the rational numbers, the question how we can determine whether an n-letter permutation of $x_1, \ldots, x_n$ is a symmetry over K or not also arises in the case of the relative situation over K.

The positive answer to this is that we can prove the relative version of Proposition 5.3 just like Proposition 5.3 using Lemma 6.1:

Proposition 6.3 *Let $f(X)$ be a normalized separable polynomial over K, whose different roots are given by $x_1, \ldots, x_n$. Let further $t = V(x_1, \ldots, x_n)$ be a primitive element for $x_1, \ldots, x_n$ over K, where $V(X_1, \ldots, X_n)$ is a polynomial over K. Let $h_1(X), \ldots, h_n(X)$ be polynomials over K with $x_i = h_i(t)$ for $i \in \{1, \ldots, n\}$.*

Then for each n-ary permutation σ that is a symmetry of the roots of $f(X)$ over K, there is exactly one Galois conjugate t' over K of t (where the case $t' = t$ is explicitly allowed here) such that

$$\sigma \cdot x_1 = h_1(t'), \quad \ldots, \quad \sigma \cdot x_n = h_n(t').$$

Conversely, for each Galois conjugate t' over K of t ($t' = t$ again possible), there is exactly one symmetry σ over K so that (5.6) is fulfilled.

Additionally, under the induced bijection between the symmetries σ over K and the Galois conjugates t', we have the equality

$$t' = V(\sigma \cdot x_1, \ldots, \sigma \cdot x_n). \qquad \square$$

(The existence of the polynomials $h_1, \ldots, h_n$ in the proposition follows from the fact that $x_1, \ldots, x_n$ lie in $K(t)$.)

If σ is a symmetry of $x_1, \ldots, x_n$ over K and $z = F(x_1, \ldots, x_n)$ is an expression rational in $x_1, \ldots, x_n$ over K (that is $F \in K[X_1, \ldots, X_n]$), then

$$\sigma \cdot z := F(\sigma \cdot x_1, \ldots, \sigma \cdot x_n)$$

is well-defined, since all algebraic relations between $x_1, \ldots, x_n$ over K are preserved by σ. Thus, we have an action of the Galois group of $x_1, \ldots, x_n$ over K, which is defined on the elements of $K(x_1, \ldots, x_n)$. The invariants of this action are exactly the elements lying in K: This is a direct generalization of Proposition 5.4, which can also be proven in a completely analogous way:

The construction of the Galois group also works relatively over a number field.

Proposition 6.4 *Let $x_1, \ldots, x_n$ be the roots of a normalized separable polynomial $f(X)$ over K. Let z be an algebraic number in $K(x_1, \ldots, x_n)$ that is invariant under the Galois group G of $x_1, \ldots, x_n$ over K, that is, for which $\sigma \cdot z = z$ holds for all symmetries σ in G. Then z is already an element in K.* $\square$

The invariants under the action of the Galois group are exactly the elements of the base number field.

Corollary 6.1 *Let $x_1, \ldots, x_n$ be the roots of a normalized separable polynomial $f(X)$ over K. If $f(X)$ over K is irreducible, the Galois group of $x_1, \ldots, x_n$ over K operates* transitively *on $x_1, \ldots, x_n$, that is, for each pair $i,\ j \in \{1, \ldots, n\}$ there exists (at least) one symmetry σ of $x_1, \ldots, x_n$ over K with $\sigma \cdot x_i = x_j$.* □

Finally, we want to investigate when a symmetry σ of the roots $x_1, \ldots, x_n$ of a normalized separable polynomial over K is also a symmetry over L where L is an intermediate extension of $K(x_1, \ldots, x_n)$ over K.

By definition of an intermediate extension, L stands for an equivalence class of numbers $z_1, \ldots, z_k$ rational in $x_1, \ldots, x_n$ over K, so $L = K(z_1, \ldots, z_m) \subseteq K(x_1, \ldots, x_n)$. If we write $z_i = g_i(x_1, \ldots, x_n)$, where $g_1(X_1, \ldots, X_n), \ldots, g_k(X_1, \ldots, X_n)$ are polynomials over K, then $z_i - g_i(x_1, \ldots, x_n) = 0$ is certainly an algebraic relation of $x_1, \ldots, x_n$ over L.

If σ is thus an element of the Galois group of $x_1, \ldots, x_n$ over L, σ must preserve these relations, that is, we have

$$0 = z_i - g_i(\sigma \cdot x_1, \ldots, \sigma \cdot x_n) = z_i - \sigma \cdot g_i(x_1, \ldots, x_n),$$

from which, because $z_i = g_i(x_1, \ldots, x_n)$, it follows that $\sigma \cdot z_i = z_i$. If therefore $\sigma \in \mathrm{Gal}_K(x_1, \ldots, x_n)$ is an element of $\mathrm{Gal}_L(x_1, \ldots, x_n)$, the generators $z_1, \ldots, z_k$ of L over K must be invariant under the action of σ. The reverse statement also holds:

Let σ be an element of the Galois group $\mathrm{Gal}_K(x_1, \ldots, x_n)$ of $x_1, \ldots, x_n$ over K which preserves all generators $z_1, \ldots, z_k$ of L over K, that is, for which $\sigma \cdot z_i = z_i$ for $i \in \{1, \ldots, k\}$. Since every element from L is a polynomial in $z_1, \ldots, z_k$ over K and the action of the Galois group is compatible with the basic arithmetic operations, it follows directly that then every element z from L is invariant under such a symmetry σ.

If then $F(x_1, \ldots, x_n) = 0$ is any algebraic relation of $x_1, \ldots, x_n$ over L, then $\sigma \cdot F(x_1, \ldots, x_n) = F(\sigma \cdot x_1, \ldots, \sigma \cdot x_n)$, since the coefficients of $F(X_1, \ldots, X_n)$ are invariant under σ. Because $\sigma \cdot 0 = 0$, we thus have $F(\sigma \cdot x_1, \ldots, \sigma \cdot x_n) = 0$. The permutation σ thus preserves all algebraic relations of $x_1, \ldots, x_n$ over L and is thus an element of the Galois group of $x_1, \ldots, x_n$ over L. Summarized in a formula, we can thus write:

$$\mathrm{Gal}_{K(z_1,\ldots,z_k)}(x_1, \ldots, x_n) = \{\sigma \in \mathrm{Gal}_K(x_1, \ldots, x_n) \mid \sigma \cdot z_1 = z_1, \ldots, \sigma \cdot z_k = z_k\}.$$

The Galois group over an intermediate number field consists exactly of those symmetries, under whose action the elements of the intermediate number field remain invariant.

6.2 The Fundamental Theorem of Galois Theory

In the last section, we have retraced Galois theory from Chap. 5 in a relative version over any number field K (and for the choice $K = \mathbf{Q}$ we get back the absolute situation, in which the coefficients are only rational numbers). This generalization will be extremely important, even if we only want to investigate the Galois group G of the roots $x_1, \ldots, x_n$ of a separable polynomial $f(X)$ over the rational numbers:

Let K be a number field adapted to the situation of the respective problem (for example $K = \mathbf{Q}(x_1)$). The Galois group H of $x_1, \ldots, x_n$ over K is a subgroup of G (usually smaller than G), since there are generally fewer symmetries over K (in the case $K = \mathbf{Q}(x_1)$ such symmetries must, for example, map x_1 to itself). The idea is that H as a smaller group should be simpler than G. To investigate G, we can thus investigate H and then consider how many more symmetries are added when we move from H to G.

Conversely, we can ask whether every subgroup H of G appears as a Galois group in this way. And indeed, the answer to this question is positive, as the following proposition shows. For this proposition and the following, we fix a number field K.

Proposition 6.5 *Let $x_1, \ldots, x_n$ be the roots of a separable polynomial $f(X)$ over K, and let t be a primitive element for $x_1, \ldots, x_n$ over K. Let further $H = \{\sigma_1, \ldots, \sigma_m\}$ be an m-element subgroup of the Galois group of $x_1, \ldots, x_n$ over K. Let*

$$L = K(e_1(\sigma_1(t), \ldots, \sigma_m(t)), \ldots, e_m(\sigma_1(t), \ldots, \sigma_m(t))).$$

Then the Galois group of $x_1, \ldots, x_n$ over L is given by H.

The numbers in L are precisely the numbers in $K(x_1, \ldots, x_n)$ that are invariant under the action of all symmetries σ from H, that is, for which $\sigma \cdot x = x$ holds.

Every subgroup H of a Galois group corresponds to an intermediate extension L. If H is of order m, then L is generated over the underlying number field by m numbers that we can explicitly specify.

Before we prove Proposition 6.5, we make some preliminary considerations: Let σ be an element in H. Since H is closed under compositions, $\sigma \circ \sigma_1, \ldots, \sigma \circ \sigma_m$ are again elements in H. If $\sigma \circ \sigma_i = \sigma \circ \sigma_j$, then by composing with σ^{-1} from the left we see that $\sigma_i = \sigma_j$. Thus, there exists an m-ary permutation τ with $\sigma \circ \sigma_i = \sigma_{\tau(i)}$ for $i \in \{1, \ldots, m\}$.

Let t be any element that is rational in $x_1, \ldots, x_n$. Then we set

$$t_1 := \sigma_1 \cdot t, \qquad \ldots, \qquad t_m := \sigma_m \cdot t,$$

so $\sigma \cdot t_i = (\sigma \circ \sigma_i) \cdot t = t_{\tau(i)}$. In particular, the series $\sigma \cdot t_1, \ldots, \sigma \cdot t_m$ is just a permutation of $t_1, \ldots, t_m$. Consequently, $e_1(t_1, \ldots, t_m), \ldots, e_m(t_1, \ldots, t_m)$ are invariant under σ, so $\sigma \cdot e_j(t_1, \ldots, t_m) = e_j(t_1, \ldots, t_m)$.

Proof Let

$$g(X) := \prod_{i=1}^{m} (X - \sigma_i \cdot t). \tag{6.1}$$

Since the roots of $g(X)$ are given by $\sigma_1 \cdot t, \ldots, \sigma_m \cdot t$, $g(X)$ is a polynomial over L according to Vieta's theorem and the fundamental theorem about the elementary symmetric functions. Since one σ_i must be the identical permutation, t is a root of $g(X)$. The minimal polynomial of t over L is therefore a divisor of $g(X)$. Thus, the degree of t over L is less than or equal to the degree of $g(X)$, so $[L(t) : L] \leq m$.

Since t is primitive to $x_1, \ldots, x_n$ over L, the number of elements of the Galois group $\tilde{H}$ of $x_1, \ldots, x_n$ over L is given by the degree of t over L, that is, $\tilde{H}$ has at most m elements. To show $\tilde{H} = H$, it is therefore sufficient to show that every element σ of H is a symmetry of $x_1, \ldots, x_n$ over L. According to our considerations at the end of Sect. 6.1, it is sufficient to show that σ leaves the generators $e_1(t_1, \ldots, t_m), \ldots, e_m(t_1, \ldots, e_m)$ of L over K invariant. But we have already seen this.

It remains to prove the addendum. But this is proposition 6.4: The generators of L are invariant under every symmetry from H, and every number in $K(x_1, \ldots, x_n)$ that is invariant under every symmetry in H is in L. □

From the proof of Proposition 6.5 we learn, in particular, that (6.1), being a polynomial of degree m, must be the minimal polynomial of t over L.

An important consequence of Proposition 6.5 is the existence of an intermediate extension consisting of the numbers that are invariant under a finite set of group elements:

Corollary 6.2 *Let $x_1, \ldots, x_n$ be roots of a normalized separable polynomial over K. If $\sigma_1, \ldots, \sigma_m$ are symmetries of $x_1, \ldots, x_n$ over K, then there exists an intermediate extension L of $K(x_1, \ldots, x_n)$ over K that is invariant under $\sigma_1, \ldots, \sigma_m$ (that is, $\sigma_1 \cdot z = \cdots = \sigma_m \cdot z = z$ for all $z \in L$) and such that all numbers in $K(x_1, \ldots, x_n)$ that are invariant under $\sigma_1, \ldots, \sigma_m$, are in L.*

If t is a number in $K(x_1, \ldots, x_n)$ and $\sigma_1 \cdot t = \cdots = \sigma_m \cdot t = t$, we write

$$t \in K(x_1, \ldots, x_n)^{\sigma_1, \ldots, \sigma_m}.$$

The statement of Proposition 6.5 can thus be written as

$$L = K(x_1, \ldots, x_n)^{\sigma_1, \ldots, \sigma_m}.$$

Before we proceed with the proof of Corollary 6.2, we need another notation, namely the following:

If $\sigma_1, \ldots, \sigma_m$ are n-ary permutations, we denote by

$$\langle \sigma_1, \ldots, \sigma_m \rangle$$

the smallest subgroup of S_n that contains $\sigma_1, \ldots, \sigma_m$. This is obviously the set of all those permutations that can be formed from $\sigma_1, \ldots, \sigma_m$ by repeated composition and inversion and the identity.

We therefore call $\sigma_1, \ldots, \sigma_m$ *generators* of $\langle \sigma_1, \ldots, \sigma_m \rangle$.

(To determine $\langle \sigma_1, \ldots, \sigma_m \rangle$, we proceed step by step: First we consider $X_0 = \{\sigma_1, \ldots, \sigma_m\}$. If X_0 is already a subgroup, then $X = \langle \sigma_1, \ldots, \sigma_m \rangle$, and we are done. Otherwise, there is another group element σ that can be expressed by $\sigma_1, \ldots, \sigma_m$ by means of composition and inversion. In the next step we then consider $X_1 = X_0 \cup \{\sigma\}$. Either X_1 is a subgroup and thus equal to $\langle \sigma_1, \ldots, \sigma_m \rangle$, or we can construct a X_2 from it, etc. This process ends at some point since there are only finitely many n-ary permutations available.)

Proof A number x is invariant under $\sigma_1, \ldots, \sigma_m$ if and only if x is invariant under every permutation in $\langle \sigma_1, \ldots, \sigma_m \rangle$ because if x is invariant under the permutations σ and τ, it is also invariant under the permutations id, $\tau \circ \sigma$ and σ^{-1}. Thus we can assume without loss of generality that $\sigma_1, \ldots, \sigma_m$ themselves already form a subgroup H of the Galois group of $x_1, \ldots, x_n$ over K.

According to Proposition 6.5, there exists an intermediate extension L of $K(x_1, \ldots, x_n)$ over K such that the Galois group of $x_1, \ldots, x_n$ over L is precisely given by H. According to Proposition 6.5, the numbers in $K(x_1, \ldots, x_n)$ which are invariant under H are precisely the elements from L. □

Let L be an intermediate extension of $K(x_1, \ldots, x_n)$ over K so that every symmetry in the subgroup H is also a symmetry of $x_1, \ldots, x_n$ over L. It follows that in particular every number in L is invariant under all symmetries from H, so $L \subseteq K(x_1, \ldots, x_n)^H$.

Conversely, let L be any intermediate extension of $K(x_1, \ldots, x_n)$ over K, such that $L \subseteq K(x_1, \ldots, x_n)^H$. Then in particular the generators of L over K are invariant under the actions of the elements in H. According to our considerations at the end of Sect. 6.1, it follows that every element in the subgroup H is also a symmetry of $x_1, \ldots, x_n$ over L.

For these two considerations, we note that it is not essential that H is a subgroup. Thus, we have proven:

Proposition 6.6 *Let $x_1, \ldots, x_n$ be the roots of a normalized separable polynomial over K. For every intermediate extension L of $K(x_1, \ldots, x_n)$ over K and every finite subset H of symmetries of $x_1, \ldots, x_n$ over K, the following holds: Every number in L is invariant under all symmetries from H if and only if every symmetry in H is a symmetry of $x_1, \ldots, x_n$ over L.* □

Expressed in formulas, we have

$$L \subseteq K(x_1, \ldots, x_n)^H \iff H \subseteq \mathrm{Gal}_L(x_1, \ldots, x_n).$$

We can further tighten this connection between intermediate extensions and symmetries and arrive at the so-called fundamental theorem of Galois theory, which is essentially just a reformulation of the things already proven:

Theorem 6.1 *Let K be a number field. Let $x_1, \ldots, x_n$ be the roots of a normalized separable polynomial over K. Let $G = \mathrm{Gal}_K(x_1, \ldots, x_n)$ be the Galois group of $x_1, \ldots, x_n$ over K. Then the following holds:*

For each subgroup H of G there exists exactly one intermediate extension L of $K(x_1, \ldots, x_n)$ over K such that the Galois group of $x_1, \ldots, x_n$ over L is precisely H.

Theorem 6.1 obviously defines a bijection between the intermediate extensions L of $K(x_1, \ldots, x_n)$ over K and the subgroups of the Galois group of $x_1, \ldots, x_n$ over K.

If L is the intermediate extension associated with the subgroup H, we can characterize L according to Proposition 6.5 by the fact that the numbers in L are exactly those numbers in $K(x_1, \ldots, x_n)$ that are invariant under all symmetries under H.

Proof We have already proven the existence of L, as claimed in theorem 6.1, with proposition 6.5. It remains to show the uniqueness.

Let L' be another intermediate extension of $K(x_1, \ldots, x_n)$ over K, such that $\mathrm{Gal}_{L'}(x_1, \ldots, x_n) = \mathrm{Gal}_L(x_1, \ldots, x_n)$. In particular, it follows that the numbers in L' are invariant under all symmetries of $x_1, \ldots, x_n$ over L. According to proposition 6.4, this implies $L' \subseteq L$.

For the same reasons (with swapped roles), it follows that $L \subseteq L'$. □

There exists an inclusion-reversing bijective correspondence between the intermediate extensions and the subgroups of a Galois group.

Both the intermediate extensions L of $K(x_1, \ldots, x_n)$ over K and the subgroups H of G can be assigned a measure: the number field L its degree over K and the subgroup

H its index in G. The relationship between the two is the simplest possible when $H = \mathrm{Gal}_L(x_1, \ldots, x_n)$ as in Theorem 6.1:

Proposition 6.7 *Let $x_1, \ldots, x_n$ be the roots of a normalized separable polynomial over K. Let $G = \mathrm{Gal}_K(x_1, \ldots, x_n)$ be the Galois group of $x_1, \ldots, x_n$ over K. Then the degree of an intermediate extension L of $K(x_1, \ldots, x_n)$ over K is equal to the index of the Galois group $H = \mathrm{Gal}_L(x_1, \ldots, x_n)$ of $x_1, \ldots, x_n$ over L in G, that is*

$$[L : K] = [\mathrm{Gal}_K(x_1, \ldots, x_n) : \mathrm{Gal}_L(x_1, \ldots, x_n)]. \tag{6.2}$$

The Eq. (6.2) also explains the analogous notation for degree and index.

Proof According to Theorem 4.2, we can write $L = K(z)$ for an algebraic number $z \in K(x_1, \ldots, x_n)$. Let $z_1, \ldots, z_k$ be the pairwise different Galois conjugates of z over K where we choose $z_1 = z$. Then k is the degree of z over K, that is, we have to show $[G : H] = k$. If σ is an element of G, i.e., a symmetry over K, then $\sigma \cdot z$ satisfies the same minimal polynomial as z over K, that is, $\sigma \cdot z = z_i$ for some $i \in \{1, \ldots, k\}$. We have thus defined a mapping $\varphi\colon G \to \{1, \ldots, k\}$ with $\sigma \cdot z = z_{\varphi(i)}$.

We claim next that the mapping φ is surjective. To this end, we consider the polynomial

$$g(X) = \prod_{\sigma \in G} (X - \sigma \cdot z).$$

The coefficients of the polynomial are invariant under G. According to Proposition 6.4, $g(X)$ is therefore a polynomial over K. Since $\mathrm{id} \in G$, z is in particular a root of $g(X)$ is, all Galois conjugates of z are also roots of $g(X)$. It follows that every z_i is of the form $\sigma \cdot z$ for a symmetry σ from G. Thus we have shown that φ is surjective.

The mapping φ is certainly not injective in general. If σ and τ are congruent modulo H, i.e., $\tau^{-1} \circ \sigma \in H$, from the invariance of z under the symmetries from H, it then follows that $\tau^{-1} \cdot \sigma \cdot z = z$. Applying τ therefore yields $\sigma \cdot z = \tau \cdot z$, i.e., $\varphi(\sigma) = \varphi(\tau)$. But this is also the only reason why φ cannot be injective:

Suppose $\varphi(\sigma) = \varphi(\tau)$ for two symmetries σ and τ. Then $(\tau^{-1} \circ \sigma) \cdot z = z$. According to Proposition 6.6, $\tau^{-1} \circ \sigma$ is an element of the Galois group of $x_1, \ldots, x_n$ over L, i.e., H. It follows that $\varphi(\sigma) = \varphi(\tau)$ holds if and only if σ and τ are congruent to H modulo H.

Due to the already proven surjectivity of φ, we have thus shown, that φ is a bijection between the congruence classes of elements in G modulo H (of which there are exactly $[G : H]$) and the Galois conjugates of z over K. □

The degrees of the intermediate extensions correspond under the Galois correspondence to the indices of the corresponding subgroups.

Corollary 6.3 *Let $x_1, \ldots, x_n$ be the roots of a normalized separable polynomial over K. Let H be a subgroup of the Galois group G of $x_1, \ldots, x_n$ over K. If L is an intermediate extension of $K(x_1, \ldots, x_n)$ over K that is invariant under all symmetries from H, and the degree of L over K is equal to the index of H in G, then the Galois group of $x_1, \ldots, x_n$ over L is given by H.*

Proof Let L' according to Proposition 6.5 be the intermediate extension of the number field $K(x_1, \ldots, x_n)$ over K such that the Galois group of $x_1, \ldots, x_n$ over L' is given by H. Since L is invariant under H by assumption, $L \subset L'$. According to Proposition 6.7, the degree of L' over K is equal to the index of H in G, that is, L and L' have the same degree over K.

Therefore, the extension $L \subseteq L'$ of number fields must be trivial, that is, we have $L' = L$. Consequently, the Galois group of $x_1, \ldots, x_n$ over L is also H. □

We want to understand the results of this chapter and the fundamental theorem of Galois theory using two longer examples.

Example 6.3 Let

$$f(X) = X^3 - 2.$$

Then $f(X)$ is an irreducible polynomial over the rational numbers. If we set $\omega := \zeta_3$, its roots are given by $x_1 = \sqrt[3]{2}$, $x_2 = \omega\sqrt[3]{2}$ and $x_3 = \omega^2\sqrt[3]{2}$. Since the discriminant $\Delta = -27 \cdot 2^2 = -108$ is not a square in $\mathbf{Q}$, the Galois group G of x_1, x_2 and x_3 is the full symmetric group S_3. Its elements are

$$\mathrm{S}_3 = \{\mathrm{id}, (2, 3), (1, 3), (1, 2), (1, 2, 3), (1, 3, 2)\}.$$

As can be determined, for example by simple checking all cases, the symmetric group has exactly the following six subgroups:

$$\{\mathrm{id}\}, \langle(2, 3)\rangle, \langle(1, 3)\rangle, \langle(1, 2)\rangle, \langle(1, 2, 3)\rangle, \mathrm{S}_3.$$

In Fig. 6.1, these subgroups and their inclusion relationships are shown in the upper diagram: A vertical line means that the subgroup above is contained in the one below. The numbers on the vertical lines are the respective indices of the corresponding group inclusions.

According to the fundamental theorem of Galois theory, Theorem 6.1, there exists for each of these subgroups H an intermediate extension L of $\mathbf{Q}(x_1, x_2, x_3)$ over K such that H is the Galois group of x_1, x_2 and x_3 over L. According to Corollary 6.3, we can characterize L by the fact that L is invariant under all symmetries of H and that the degree of L (over the rational numbers) is exactly the index of H in G.

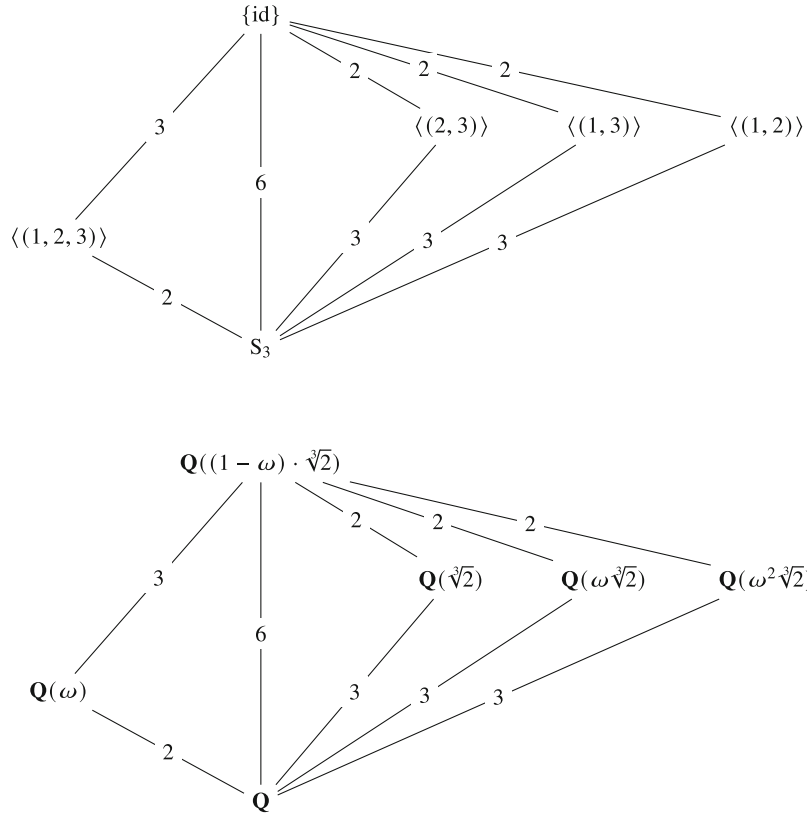


Fig. 6.1 The fundamental theorem of Galois theory illustrated by the cubic equation $X^3 - 2 = 0$

Let us go through the series of subgroups: The subgroup {id} has index 6 in S_3. According to Example 5.5, $t = x_1 - x_2 = (1 - \omega) \cdot \sqrt[3]{2}$ is a primitive element to x_1, x_2 and x_3. Its degree is, according to Proposition 5.3, equal to the order of the Galois group of x_1, x_2 and x_3. Thus, the subgroup {id} is precisely the Galois group of x_1, x_2 and x_3 over t.

As the next subgroup, we consider $\langle(2, 3)\rangle = \{\mathrm{id}, (2, 3)\}$, which has two elements and thus index 3 in S_3. The element $x_1 = \sqrt[3]{2}$ is obviously invariant under $(2, 3)$ and thus also under $\langle(2, 3)\rangle$. Since x_1 with the minimal polynomial $X^3 - 2$ has degree 3 over the rational numbers, $\langle(2, 3)\rangle$ is thus the Galois group of x_1, x_2 and x_3 over $\mathbf{Q}(x_1) = \mathbf{Q}(\sqrt[3]{2})$.

Similarly, we find that $\langle(1, 3)\rangle$ is the Galois group of x_1, x_2 and x_3 over $\mathbf{Q}(x_2) = \mathbf{Q}(\omega\sqrt[3]{2})$ and that $\langle(1, 2)\rangle$ is the Galois group of x_1, x_2 and x_3 over $\mathbf{Q}(x_3) = \mathbf{Q}(\omega^2\sqrt[3]{2})$.

The next subgroup is $\langle(1, 2, 3)\rangle = \{\mathrm{id}, (1, 2, 3), (1, 3, 2)\}$, which has three elements and thus index 2 in S_3. The element $\delta = (x_1 - x_2) \cdot (x_2 - x_3) \cdot (x_3 - x_1) = -6 \cdot (2\omega + 1)$, a square root of the discriminant, is invariant under $(1, 2, 3)$ and thus also under $\langle(1, 2, 3)\rangle$. It follows that ω is also invariant under $\langle(1, 2, 3)\rangle$. Since it also has degree two over the

rational numbers (X^2+X+1 is its minimal polynomial), $\langle(1,2,3)\rangle$ is therefore the Galois group of x_1, x_2, x_3 over $\mathbf{Q}(\omega) = \mathbf{Q}(\delta)$.

The remaining subgroup is S_3, which has index 1. The rational number 0 is invariant under the action of the entire Galois group S_3 and has degree one over the rational numbers. We find that S_3 is the Galois group of x_1, x_2 and x_3 over $\mathbf{Q}$.

We have recorded these findings in the diagram below from Fig. 6.1. The transition from the upper to the lower half of the figure corresponds to the bijection stated in Theorem 6.1.

Example 6.4 We want to illustrate the fundamental theorem of Galois theory using the example of the roots of the polynomial

$$f(X) = X^4 - 2$$

over the rational numbers. First, we count the four different roots of $f(X)$, namely:

$$x_1 = \sqrt[4]{2}, \qquad x_2 = \mathrm{i}\sqrt[4]{2}, \qquad x_3 = -\sqrt[4]{2}, \qquad x_4 = -\mathrm{i}\sqrt[4]{2}.$$

Next, we want to determine the Galois group for $x_1, \ldots, x_4$. We note that $\sqrt[4]{2}$ and i are rational in $x_1, \ldots, x_4$ and conversely $x_1, \ldots, x_4$ are rational in $\sqrt[4]{2}$ and i, that is, we have $\mathbf{Q}(x_1, \ldots, x_4) = \mathbf{Q}(\sqrt[4]{2}, \mathrm{i})$. We claim that the degree of $\mathbf{Q}(\sqrt[4]{2}, \mathrm{i})$ over $\mathbf{Q}$—that is, the degree of a primitive element to $\sqrt[4]{2}$ and i over $\mathbf{Q}$—is equal to 8. To see this, we use the degree formula:

$$[\mathbf{Q}(\sqrt[4]{2}, \mathrm{i}) : \mathbf{Q}] = [\mathbf{Q}(\sqrt[4]{2}, \mathrm{i}) : \mathbf{Q}(\sqrt[4]{2})] \cdot [\mathbf{Q}(\sqrt[4]{2}) : \mathbf{Q}].$$

Since $X^4 - 2$ is the minimal polynomial of $\sqrt[4]{2}$ over $\mathbf{Q}$, the second factor on the right side is $[\mathbf{Q}(\sqrt[4]{2}) : \mathbf{Q}] = 4$. Since $\mathbf{Q}(\sqrt[4]{2})$ is real (that is, every number rational in $\sqrt[4]{2}$ is real), $\mathbf{Q}(\sqrt[4]{2}, \mathrm{i})$ over $\mathbf{Q}(\sqrt[4]{2})$ is a nontrivial extension of number fields, that is, we have $[\mathbf{Q}(\sqrt[4]{2}, \mathrm{i}) : \mathbf{Q}(\sqrt[4]{2})] > 1$. On the other hand, $X^2 + 1$ is a polynomial over $\sqrt[4]{2}$, which has i as a root, that is, we also have $[\mathbf{Q}(\sqrt[4]{2}, \mathrm{i}) : \mathbf{Q}(\sqrt[4]{2})] \leq 2$. Thus, we have $[\mathbf{Q}(\sqrt[4]{2}, \mathrm{i}) : \mathbf{Q}(\sqrt[4]{2})] = 2$ and thus

$$[\mathbf{Q}(\sqrt[4]{2}, \mathrm{i}) : \mathbf{Q}] = 2 \cdot 4 = 8.$$

In particular, the Galois group of $x_1, \ldots, x_4$ contains exactly 8 elements, which we will determine in the following.

Each element of the Galois group σ acts on $\sqrt[4]{2}$ and i. As soon as we know the effect of σ on $\sqrt[4]{2}$ and i, we also know the effect of σ on $x_1, \ldots, x_4$, that is, we know which element of the permutation group S_4 is represented by σ.

Since $\sqrt[4]{2}$ satisfies the algebraic relation $X^4 - 2 = 0$, σ can map the element $\sqrt[4]{2}$ at most to the four roots of $X^4 - 2$, namely $\sqrt[4]{2}$, $\mathrm{i}\sqrt[4]{2}$, $-\sqrt[4]{2}$ and $-\mathrm{i}\sqrt[4]{2}$. Since i also satisfies

the algebraic relation $X^2 + 1$, σ can map the element i at most to the two roots of $X^2 + 1$, namely i and $-$i. Thus, σ can only be one of at most $8 = 4 \cdot 2$ elements. But since we already know that there must be exactly 8 elements, all these possibilities must be realized. We have thus determined the Galois group $\mathrm{Gal}_{\mathbf{Q}}(x_1, \ldots, x_4) = \{\sigma_0, \ldots, \sigma_7\}$ for $x_1, \ldots, x_4$ according to the following table:

$\sigma \in \mathrm{Gal}_{\mathbf{Q}}(x_1, \ldots, x_4)$	$\sigma \cdot \sqrt[4]{2}$	$\sigma \cdot \mathrm{i}$	$\sigma \cdot x_1$	$\sigma \cdot x_2$	$\sigma \cdot x_3$	$\sigma \cdot x_4$
$\mathrm{id} = \sigma_0$	$\sqrt[4]{2}$	i	x_1	x_2	x_3	x_4
$\varrho := \sigma_1$	$\mathrm{i}\sqrt[4]{2}$	i	x_2	x_3	x_4	x_1
$\varrho^2 = \sigma_2$	$-\sqrt[4]{2}$	i	x_3	x_4	x_1	x_2
$\varrho^3 = \sigma_2$	$-\mathrm{i}\sqrt[4]{2}$	i	x_4	x_1	x_2	x_3
$\tau := \sigma_4$	$\sqrt[4]{2}$	$-$i	x_1	x_4	x_3	x_2
$\varrho \circ \tau = \sigma_5$	$\mathrm{i}\sqrt[4]{2}$	$-$i	x_4	x_3	x_2	x_1
$\varrho \circ \tau^2 = \sigma_6$	$-\sqrt[4]{2}$	$-$i	x_3	x_2	x_1	x_4
$\varrho \circ \tau^3 = \sigma_7$	$-\mathrm{i}\sqrt[4]{2}$	$-$i	x_2	x_1	x_4	x_3.

As indicated in the first column, the Galois group is generated by $\varrho = \left(\begin{smallmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{smallmatrix}\right)$ and $\tau = \left(\begin{smallmatrix} 1 & 2 & 3 & 4 \\ 1 & 4 & 3 & 2 \end{smallmatrix}\right)$. The relations $\varrho \circ \tau = \tau \circ \varrho$, $\varrho^4 = \mathrm{id}$ and $\tau^2 = \mathrm{id}$ hold.

We can also visualize this Galois group as the geometric symmetry group of a square. Let's look again at the square in Fig. 5.4, but this time as a square in space. Which movements of space transform the square back into the square?

In the case of the square in the plane, we have the four rotations of 0°, 90°, 180° and 270° of the plane of the square about its center. These are exactly the four permutations id, ϱ, ϱ^2 and ϱ^3. Added to this are the rotary reflections: In space, we can first rotate the square by 180° about an axis running through two opposite corners (that is, effectively mirror it) and then perform a rotation as before by a multiple of 90° about its center in its plane. This corresponds exactly to the four permutations τ, $\tau \circ \varrho$, $\tau \circ \varrho^2$ and $\tau \circ \varrho^3$.

The square, considered as a square in space, has no more symmetries. Its symmetry group is therefore precisely the Galois group of $x_1, \ldots, x_4$. In this context, the Galois group is also called the *Dieder group* D_4 ("Dieder" is the Greek word for dihedral and denotes a regular polygon with front and back.)

As in Example 6.3, we next list all subgroups of D_4. This is essentially a combinatorial task, as each subgroup of D_4 is generated by finitely many elements of the form $\tau^i \circ \sigma^j$ with $i \in \{0, 1\}$ and $j \in \{0, 1, 2, 3\}$ We finally obtain the upper diagram in Fig. 6.2. The vertical lines are between two subgroups, so the upper one has index two in the lower one.

According to the Galois correspondence from Theorem 6.1 and Corollary 6.3, each of these subgroups H corresponds to an intermediate extension L of $K(x_1, \ldots, x_n)$ over K, such that L is invariant under H and such that the degree of L over the rational numbers is precisely the index of H in D_4. Under this correspondence, the upper diagram of Fig. 6.2 yields the lower one. The verification of this fact can be done for each individual subgroup and each individual intermediate extension separately and is elementary.

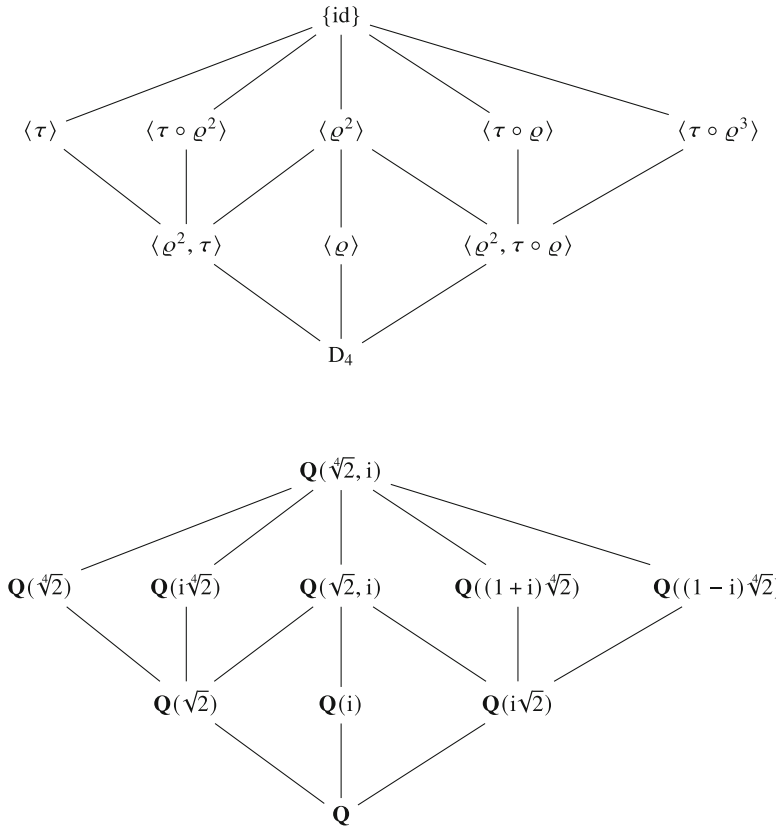


Fig. 6.2 The fundamental theorem of Galois theory illustrated by the quartic equation $X^4 - 2 = 0$

6.3 Algebraically Unique Roots

Let a be an algebraic number. An n-th root of a is an algebraic number x with $x^n = a$. In Sect. 2.4 we saw that for $a \neq 0$ exactly n different ones exist, and we selected one, namely the root which has the smallest argument in the interval from 0 to (exclusively) 2π. We denoted this with $\sqrt[n]{a}$.

From the standpoint of algebra, this definition is unsatisfactory: The definition of the argument depends on the existence of the complex exponential function, which in turn is analytically defined using a series.

If $X^n - a$ is an irreducible polynomial over a, then algebra alone cannot, in principle, distinguish the different n-th roots of a since under the assumption of irreducibility all are Galois conjugate to each other over a. If an n-th root of a thus satisfies a certain polynomial relationship over a, then every n-th root of a that is Galois conjugate to it over a satisfies the same polynomial relationship.

For this reason, from now on we want $\sqrt[n]{a}$ to no longer single out a specific n-th root of a. To say that the symbol $\sqrt[n]{a}$ stands for any n-th root of a is generally not a good idea either. What should, for example,

$$\sqrt[11]{1}$$

mean? To say that it is any of the eleven different eleventh roots of unity would mean to lump all eleven roots of unity together. However, this contradicts the fact that algebra can indeed distinguish certain eleventh roots of unity: For example, 1 is the only eleventh root of unity that satisfies the algebraic relation $X - 1 = 0$. The other ten are primitive eleventh roots of unity, all of which are Galois conjugate to each other and satisfy the algebraic relation $X^{10} + X^9 + \cdots + X + 1 = 0$.

For this reason, we can only give $\sqrt[n]{a}$ a meaning if $X^n - a$ is an irreducible polynomial over a, i.e., if all n-th roots of a are Galois conjugate over a. In this case, $\sqrt[n]{a}$ stands for one of the (algebraically indistinguishable) n-th roots of a. We then say that $\sqrt[n]{a}$ is an *algebraically unique* n-th root of a. If the symbol $\sqrt[n]{a}$ then appears multiple times in a formula or in a context, we agree that $\sqrt[n]{a}$ each time denotes the same n-th root of a. So if $X^n - a$ is irreducible over a, we may for example write

$$\sqrt[n]{a} - \sqrt[n]{a} = 0$$

which in general would not be possible if $\sqrt[n]{a}$ at each point stood for a different n-th root of a. If we fix such an n-th root of a through the symbol $\sqrt[n]{a}$, then the other n-th roots are given by

$$\zeta \sqrt[n]{a}, \quad \zeta^2 \sqrt[n]{a}, \quad \ldots, \zeta^{n-1} \sqrt[n]{a}$$

where ζ is a primitive n-th root of unity.

Let a be an algebraic number which has an algebraically unique n-th root. Let further z be an algebraic number with $z^n = a$. We are tempted to write

$$z = \sqrt[n]{a} \tag{6.3}$$

But what should this equality mean? On the left side stands a specific complex number, on the right side a number, which is only defined up to Galois conjugation. Therefore, we must interpret an equation like (6.3) simply as both sides being Galois conjugate (over a) to each other.

Since the definition of $\sqrt[n]{a}$ depends on the irreducibility of $X^n - a$ over a, we want to investigate this first. To do this, we go one step further and investigate the irreducibility of $X^n - a$ over K, where K is any number field in which a lies. In the case of the irreducibility of $X^n - a$ we say that an algebraically unique n-th root of a exists over K.

We first consider the case where $n = p$ is a prime number. Then $X^p - a = 0$ is obviously not irreducible over K if $X^p - a$ has a root in K, i.e., if the number a is a p-th power of a number in K. But this is also the only condition:

Lemma 6.2 *Let p be a prime number. A p-th root of a is algebraically unique over K if and only if a is not a p-th power over K.*

Proof We have already seen one direction of the claimed equivalence: If $X^p - a$ is irreducible over K, then a is not a p-th power over K.

Since $X^p - a$ over K is either irreducible or not,[1] we can assume for the other direction that $X^p - a$ is not irreducible. We want to deduce from this that a is a p-th power over K:

Let x be a root of $X^p - a = 0$ in the complex numbers. Then, under our assumption of the decomposability of $X^p - a$, the minimal polynomial of x over K is of a degree d with $d < p$. We have $x^p = a$. Let $x_1 = x, x_2, \ldots, x_d$ be the Galois conjugates of x over K. These also satisfy the algebraic relation $X^p = a$, that is $x_i^p = a$ for $i \in \{1, \ldots, d\}$. We conclude

$$(x_1 \cdots x_d)^p = a^d.$$

According to Theorem 3.2, $z := x_1 \cdots x_d$ is up to a sign of $(-1)^d$ the constant coefficient of the minimal polynomial of x over K, thus itself an element in K.

Since d is coprime to p, we can write $1 = rd + sp$ for integers r and s. With this we have

$$a = a^{rd+sp} = (a^d)^r \cdot (a^p)^s = (z^p)^r \cdot (a^s)^p = (z^r a^s)^p,$$

that is, a is a p-th root over K. □

A p-th root of an element that is not a p-th power is algebraically unique.

We can strengthen the lemma if we put an extra assumption on K:

Corollary 6.4 *Let p be a prime number. Assume a primitive p-th root of unity is in K. Then either a p-th root of a number a in K is algebraically unique over K or all solutions of $X^p - a = 0$ are already in K.*

Proof Let ζ be a primitive p-th root of unity that lies in K.

In the case that a p-th root of a over K is algebraically unique, $X^p - a$ is by definition irreducible over K, so the roots of $X^p - a$ certainly do not lie in K.

Otherwise, $X^p - a$ has a root x in K according to Lemma 6.2. All other roots of $X^p - a$ (in the algebraic numbers) are $\zeta x, \zeta^2 x, \ldots, \zeta^{p-1} x$. These are therefore also in K. □

What about the case where n is not a prime number? Then the following still holds:

Lemma 6.3 *Let p and q be two natural numbers. Further, let $n = p \cdot q$. Let a be a number from K. If an n-th root c of a over K is algebraically unique, then there is also an algebraically unique q-th root b of a over K, which we can choose so that c is an algebraically unique p-th root of b over $K(b)$.*

If a thus has a unique n-th root over K, we can write

$$\sqrt[n]{a} = \sqrt[p]{\sqrt[q]{a}}$$

By successively dividing prime divisors from n, we can therefore express each unique n-th root through unique roots whose exponents are prime numbers.

Proof From $X^q - a = f(X) \cdot g(X)$, it follows $X^n - a = f(X^p) \cdot g(X^p)$. Thus, $X^q - a$ must be irreducible over K since $X^n - a$ is irreducible over K by assumption.

Consequently, there exists a unique q-th root b of a over K. We have $(c^p)^q = c^{pq} = c^n = a$, so we can choose $b = c^p$. We note that $X^q - a$ is the minimal polynomial of b over K, so b is of degree q over K.

It remains to show that $X^p - b$ is irreducible over $K(b)$. For this, it suffices to show that $X^p - b$ is the minimal polynomial of c over $K(b)$. Since $X^p - b$ has c as a root, we must therefore show that p is the degree of c over $K(b)$. This is achieved with the degree formula:

$$n = [K(c) : K] = [K(c) : K(b)] \cdot [K(b) : K] = q \cdot [K(c) : K(b)].$$

Dividing the equation by q yields the desired result. □

6.4 Radical Representations of Primitive Roots of Unity

The symbol $\sqrt[11]{1}$ makes no sense in the algebraic context due to the decomposability of $X^{11} - 1$ following the discussion of the last section. We can either speak of the number 1 or of a primitive eleventh unit root ζ, i.e., a solution of

$$X^{10} + X^9 + X^8 + X^7 + X^6 + X^5 + X^4 + X^3 + X^2 + X + 1 = 0. \tag{6.4}$$

However, ζ is not expressed in radicals. Can we represent a solution of Eq. (6.4) using well-defined radical expressions?

We want to give some examples for this. However, for simplicity, we do not start with eleventh, but with the second roots of unity. In this case, there is exactly one primitive second unit root, namely

$$\zeta_2 = -1.$$

In the case of third roots of unity, we are looking for a solution of $X^2 + X + 1 = 0$. Solving this equation algebraically yields:

$$\zeta_3 = \frac{-1 - \sqrt{-3}}{2}.$$

(We are allowed to write $\sqrt{-3}$, since $X^2 + 3$ is irreducible over the rational numbers. The two solutions of $X^2 + 3 = 0$ correspond to the two primitive third roots of unity.)

Let us move on to the case of fourth roots of unity. These are square roots of primitive second roots of unity, so we have

$$\zeta_4 = \sqrt{-1}.$$

The case of fifth roots of unity, i.e., solutions of $X^4 + X^3 + X^2 + X + 1 = 0$, becomes more interesting. As mentioned in the introduction, there is a solution formula for equations of the fourth degree, which we could apply here. However, another approach is more enlightening. Let ζ be a solution. Let θ be a primitive fourth root of unity (so $\theta = \sqrt{-1}$).

We then consider

$$\begin{aligned}
x_1 &:= \zeta + \theta\zeta^2 + \theta^2\zeta^4 + \theta^3\zeta^8 = \zeta + \sqrt{-1}\cdot\zeta^2 - \zeta^4 - \sqrt{-1}\cdot\zeta^3,\\
x_2 &:= \zeta + \theta^2\zeta^2 + \theta^4\zeta^4 + \theta^6\zeta^8 = \zeta - \zeta^2 + \zeta^4 - \zeta^3,\\
x_3 &:= \zeta + \theta^3\zeta^2 + \theta^6\zeta^4 + \theta^9\zeta^8 = \zeta - \sqrt{-1}\cdot\zeta^2 - \zeta^4 + \sqrt{-1}\cdot\zeta^3,\\
x_4 &:= \zeta + \theta^4\zeta^2 + \theta^8\zeta^4 + \theta^{12}\zeta^8 = \zeta + \zeta^2 + \zeta^4 + \zeta^3.
\end{aligned} \tag{6.5}$$

Obviously, $x_4 = -1$, since ζ satisfies the equation $X^4 + X^3 + X^2 + X + 1 = 0$.

To determine the value of x_1, we first square this expression. We get

$$\begin{aligned} x_1^2 &= (\zeta + \sqrt{-1}\cdot\zeta^2 - \zeta^4 - \sqrt{-1}\cdot\zeta^3)^2 \\ &= \zeta^2 - \zeta^4 + \zeta^8 - \zeta^6 \\ &\quad + 2\sqrt{-1}\cdot\zeta^3 - 2 - 2\sqrt{-1}\cdot\zeta^4 - 2\sqrt{-1}\cdot\zeta^6 + 2 + 2\sqrt{-1}\cdot\zeta^7 \\ &= (-1 - 2\sqrt{-1})\, x_2. \end{aligned} \tag{6.6}$$

Then we square x_2 and get

$$\begin{aligned} x_2^2 = (\zeta - \zeta^2 + \zeta^4 - \zeta^3)^2 &= \zeta^2 + \zeta^4 + \zeta^8 + \zeta^6 - 2\zeta^3 + 2\zeta^5 \\ &\quad - 2\zeta^4 - 2\zeta^6 + 2\zeta^5 - 2\zeta^7 \\ &= 4 - \zeta - \zeta^2 - \zeta^3 - \zeta^4 = 5. \end{aligned}$$

Consequently, $x_1^4 = ((-1 - 2\sqrt{-1})x_2)^2 = 20\sqrt{-1} - 15$, so

$$x_1 = \sqrt[4]{20\sqrt{-1} - 15}.$$

(The four choices of the fourth root correspond to the four choices of a primitive fifth root of unity.)

Because of (6.6) we get

$$x_2 = \frac{x_1^2}{-1 - 2\sqrt{-1}} = \frac{(2\sqrt{-1} - 1)\sqrt[4]{20\sqrt{-1} - 15}^2}{5}.$$

To determine x_3, we calculate

$$\begin{aligned} x_1 x_3 &= (\zeta + \sqrt{-1}\cdot\zeta^2 - \zeta^4 - \sqrt{-1}\cdot\zeta^3)\cdot(\zeta - \sqrt{-1}\cdot\zeta^2 - \zeta^4 + \sqrt{-1}\cdot\zeta^3) \\ &= (\zeta - \zeta^4)^2 + (\zeta^2 - \zeta^3)^2 \\ &= -5; \end{aligned}$$

thus

$$x_3 = -\frac{5}{x_1} = -\frac{5x_1^3}{20\sqrt{-1} - 15} = \frac{(4\sqrt{-1} + 3)\sqrt[4]{20\sqrt{-1} - 15}^3}{25}.$$

With this, we know $x_1, \ldots, x_4$, and we can view (6.5) as an inhomogeneous system of equations in ζ, ζ^2, ζ^4 and ζ^3 over the rational numbers.

The sum of all four equations results in

$$x_1 + x_2 + x_3 + x_4 = 4\zeta.$$

Consequently, we obtain the following radical expression for a primitive fifth root of unity:

$$\begin{aligned}\zeta_5 = -\frac{1}{4} + \frac{\sqrt[4]{20\sqrt{-1} - 15}}{4} + \frac{(2\sqrt{-1} - 1)\sqrt[4]{20\sqrt{-1} - 15}^2}{20} \\ + \frac{(4\sqrt{-1} + 3)\sqrt[4]{20\sqrt{-1} - 15}^3}{100}.\end{aligned} \tag{6.7}$$

A primitive sixth root of unity is the negative of a third root of unity, thus

$$\zeta_6 = \frac{1 + \sqrt{-3}}{2}.$$

For primitive seventh roots of unity, a similar calculation as for ζ_5 shows that ζ_7 can be expressed through roots. Thus, we have $\zeta_2, \ldots, \zeta_7$, i.e., primitive solutions of cyclotomic equations, represented only by basic arithmetic operations and forming radicals.

We want to show that arbitrary primitive n-th roots of unity can be represented by radical expressions, i.e., the roots of $\Phi_n(X) = 0$ can be expressed only by basic arithmetic operations and forming radicals. In particular, we will find a radical expression for ζ_{11}, a result that was first found by Vandermonde in 1771 and which was a surprise at the time.

To do this, we first precisely define what "representable by radical expressions" actually means:

Definition 6.1 Let K be a number field. A sequence $x_1, \ldots, x_k$ of algebraic numbers is called *simultaneously expressible by radicals over* K if there is a sequence $z_1, \ldots, z_n$ of algebraic numbers and a sequence $p_1, \ldots, p_n$ of prime numbers such that for all $i \in \{1, \ldots, n\}$ the number z_i is an algebraically unique p_i-th root of a number from $K(z_1, \ldots, z_{i-1})$ and finally $x_1, \ldots, x_k$ are numbers in $K(z_1, \ldots, z_n)$.

(In the situation of the definition, we say that z_i is an algebraically unique root over $K(z_1, \ldots, z_{i-1})$. The occurring p_i are called *radical exponents*.)

With the notations of this definition, there exists for each $i \in \{1, \ldots, n\}$ a polynomial $f_i(Z_1, \ldots, Z_{i-1})$ over K, such that $f_i(z_1, \ldots, z_{i-1})$ is not a p_i-th power over $z_1, \ldots, z_{i-1}$ and such that $z_i = \sqrt[p_i]{f(z_1, \ldots, z_{i-1})}$ holds. Furthermore, there exist polynomials $g_1(Z_1, \ldots, Z_n), \ldots, g_k(Z_1, \ldots, Z_n)$ over K, such that $x_j = g_j(z_1, \ldots, z_n)$ for all $j \in \{1, \ldots, k\}$.

We are going to show in a moment that x_j depends only on the polynomials $f_1, \ldots, f_n$, the prime numbers $p_1, \ldots, p_n$ and the polynomial $g_j(Z_1, \ldots, Z_n)$, except for Galois conjugates.

Once we know this, we can write

$$x_j = g_j\left(\sqrt[p_1]{f_1},\ \sqrt[p_2]{f_2\left(\sqrt[p_1]{f_1}\right)}, \ldots,\ \sqrt[p_n]{f_n\left(\sqrt[p_1]{f_1}, \ldots,\ \sqrt[p_{n-1}]{f_{n-1}(\ldots)}\right)}\right). \tag{6.8}$$

Examples of such radical expressions for algebraic numbers are for example $1 + \sqrt[3]{2}$ or $\sqrt[5]{1+\sqrt{2}}$. Note that the first example stands for three Galois conjugates, the second even for ten Galois conjugates.

To show the independence of x_j of the z_i except for Galois conjugates, the following definition, which generalizes the concept of Galois conjugacy, is useful:

Definition 6.2 Let $z_1, \ldots, z_n$ and $z'_1, \ldots, z'_n$ be two sequences of algebraic numbers. Then $z_1, \ldots, z_n$ are called *(simultaneously) Galois conjugate* over a number field K if for every polynomial $V(Z_1, \ldots, Z_n)$ over K it holds that

$$V(z_1, \ldots, z_n) = 0 \iff V(z'_1, \ldots, z'_n) = 0.$$

Two algebraic numbers z and z' Galois conjugate over K in the old sense if and only if they (as one-element sequences) are simultaneously Galois conjugate in the sense of the new Definition 6.2.

Example 6.5 Let $x_1, \ldots, x_n$ be roots of a normalized separable polynomial over a number field K. An n-letter permutation σ is a symmetry of $x_1, \ldots, x_n$ over K if and only if $x_1, \ldots, x_n$ are Galois conjugate to $x_{\sigma(1)}, \ldots, x_{\sigma(n)}$.

We will also need the following lemma:

Lemma 6.4 *Let the algebraic numbers $z_1, \ldots, z_n$ and $z'_1, \ldots, z'_n$ be simultaneously Galois conjugate over a number field K. Let $f(X, Z_1, \ldots, Z_n)$ be a polynomial such that $f(X, z_1, \ldots, z_n)$ is irreducible over $K(z_1, \ldots, z_n)$. Then $f(X, z'_1, \ldots, z'_n)$ is irreducible over $K(z'_1, \ldots, z'_n)$.*

Proof Let $f(X, z'_1, \ldots, z'_n) = g(X) \cdot h(X)$ be a factorization into normalized polynomials over $K(z'_1, \ldots, z'_n)$. We have to show that $g(X) = 1$ or that $h(X) = 1$. Since the coefficients of $g(X)$ and $h(X)$ over K in $z'_1, \ldots, z'_n$ are rational, we can write

$g(X) = g(X, z'_1, \ldots, z'_n)$ and $h(X) = h(X, z'_1, \ldots, z'_n)$, where $g(X, Z_1, \ldots, Z_n)$ and $h(X, Z_1, \ldots, Z_n)$ are normalized polynomials over K in X. Thus, the following equality holds:

$$f(X, z'_1, \ldots, z'_n) - g(X, z'_1, \ldots, z'_n) \cdot h(X, z'_1, \ldots, z'_n) = 0.$$

Coefficient-wise, this yields algebraic relations of $z'_1, \ldots, z'_n$ over K, so that according to the assumption, after which $z_1, \ldots, z_n$ are Galois conjugate to $z'_1, \ldots, z'_n$ over K, we also have

$$f(X, z_1, \ldots, z_n) - g(X, z_1, \ldots, z_n) \cdot h(X, z_1, \ldots, z_n) = 0.$$

Since $f(X, z_1, \ldots, z_n)$ is irreducible by assumption, we can assume without loss of generality that $g(X, z_1, \ldots, z_n) = 1$. We can again view this coefficient-wise as a series of algebraic relations between $z_1, \ldots, z_n$, that is, we also have $g(X, z'_1, \ldots, z'_n) = 1$, again because $z'_1, \ldots, z'_n$ are Galois conjugate to $z_1, \ldots, z_n$. □

Irreducibility does not distinguish Galois conjugates.

Let's return to the independence of the x_j up to Galois conjugates from any particular choices of the z_i. For this, let another series $z'_1, \ldots, z'_n$ of algebraic numbers be given, so that $z_i'^{p_i} = f_i(z'_1, \ldots, z'_{i-1})$ for all $i \in \{1, \ldots, n\}$. We claim that for all $i \in \{1, \ldots, n\}$ the number z'_i is an algebraically unique p_i-th root over $K(z'_1, \ldots, z'_{i-1})$ and that $z'_1, \ldots, z'_i$ are Galois conjugate to $z_1, \ldots, z_i$ over K.

We want to prove this claim by induction over i. In the case $i = 0$ there is nothing to show. Next, let's assume that the claim has already been proven for an arbitrary i. We want to prove it for $i + 1$. It is $X^p - f_{i+1}(z_1, \ldots, z_i)$ irreducible over $K(z_1, \ldots, z_i)$, where we have abbreviated $p \coloneqq p_{i+1}$. Due to lemma 6.4 and the induction assumption $X^p - f_{i+1}(z'_1, \ldots, z'_i)$ is irreducible over $K(z'_1, \ldots, z'_i)$, that is, z_{i+1} is an algebraically unique p-th root over $K(z'_1, \ldots, z'_i)$.

Next, we consider a polynomial $V(Z_1, \ldots, Z_{i+1})$ over K for which an algebraic relation $V(z_1, \ldots, z_{i+1}) = 0$ holds. We have to show that we then also have $V(z'_1, \ldots, z'_{i+1}) = 0$ (the reverse implication is just as valid for symmetry reasons).

According to the fundamental theorem on elementary symmetric functions there exists a polynomial $H(Z_1, \ldots, Z_n, Y_1, \ldots, Y_p)$ over K with

$$\begin{aligned} &H(Z_1, \ldots, Z_i, e_1(W_1, \ldots, W_p), \ldots, e_p(W_1, \ldots, W_p)) \\ &\quad = V(Z_1, \ldots, Z_i, W_1) \cdots V(Z_1, \ldots, Z_i, W_p). \end{aligned}$$

We then set

$$\tilde{V}(Z_1, \ldots, Z_i) \coloneqq H(Z_1, \ldots, Z_i, \underbrace{0, \ldots, 0}_{p-1}, (-1)^p f_{i+1}(Z_1, \ldots, Z_i)).$$

According to Vieta's theorem, it then applies

$$V(z_1, \ldots, z_i, w_1) \cdots V(z_1, \ldots, z_i, w_p) = \tilde{V}(z_1, \ldots, z_i)$$

where $w_1, \ldots, w_p$ are the p different solutions of $X^p - f_{i+1}(z_1, \ldots, z_i) = 0$ (and among which z_{i+1} is also located). From $V(z_1, \ldots, z_{i+1}) = 0$, it follows $\tilde{V}(z_1, \ldots, z_i) = 0$.

Since $z_1, \ldots, z_i$ are Galois conjugate to $z'_1, \ldots, z'_i$ over K according to the induction hypothesis, it follows that $\tilde{V}(z'_1, \ldots, z'_i) = 0$. If $w'_1, \ldots, w'_p$ are the different solutions of $X^p - f_{i+1}(z'_1, \ldots, z'_i) = 0$, then according to Vieta's theorem, we have that

$$V(z'_1, \ldots, z'_i, w'_1) \cdots V(z'_1, \ldots, z'_i, w'_p) = \tilde{V}(z'_1, \ldots, z'_i) = 0.$$

In particular, a factor $V(z'_1, \ldots, z'_i, w_\ell)$ must vanish for a $\ell \in \{1, \ldots, p\}$. Since the polynomial $X^p - f_{i+1}(z'_1, \ldots, z'_i)$ is irreducible over $K(z'_1, \ldots, z'_i)$, z'_{i+1} is Galois conjugate to w_ℓ over $K(z'_1, \ldots, z'_i)$. Because $V(z'_1, \ldots, z'_i, w_\ell) = 0$, we also have $V(z'_1, \ldots, z'_i, z'_{i+1}) = 0$, which completes the induction step.

Except for simultaneous Galois conjugation, $z_1, \ldots, z_n$ depend only on the polynomials $f_1, \ldots, f_n$ and the prime numbers $p_1, \ldots, p_n$. If $z'_1, \ldots, z'_n$ are Galois conjugate to $z_1, \ldots, z_n$ over K, and we set $x'_j \coloneqq g_j(z'_1, \ldots, z'_n)$ for $j \in \{1, \ldots, k\}$, then $x'_1, \ldots, x'_k$ are Galois conjugate to $x_1, \ldots, x_k$ over K, which means that $x_1, \ldots, x_k$ up to simultaneous Galois conjugation only depend on $f_1, \ldots, f_n$ and $p_1, \ldots, p_n$: If $V(x_1, \ldots, x_k) = 0$ is an algebraic relation over K, then

$$V(g_1(z_1, \ldots, z_n), \ldots, g_k(z_1, \ldots, z_n)) = 0.$$

This is again an algebraic relation of $z_1, \ldots, z_n$ over K; that is, we have $V(g_1(z'_1, \ldots, z'_n), \ldots, g_k(z'_1, \ldots, z'_n)) = 0$, so $V(x'_1, \ldots, x'_k) = 0$.

If we iteratively form algebraically unique roots and form sums and products from them, the numbers obtained from this are unique up to Galois conjugacy.

For later use, here is another lemma:

Lemma 6.5 *Let p be a positive natural number. Let θ be a $(p-1)$-th root of unity. Then*

$$1 + \theta^j + \theta^{2j} + \cdots + \theta^{(p-2)j} = \begin{cases} p-1 & \text{for } j = 0 \\ 0 & \text{for } j \in \{1, \ldots, p-2\}. \end{cases}$$

Proof The case $j = 0$ is clear. Now consider the case $j \in \{1, \ldots, p-2\}$. Then $\omega := \theta^j$ is a primitive root of unity for a divisor k of $p-1$. If $d = \frac{p-1}{k}$, then because $\omega^k = 1$ we have

$$1 + \theta^j + \theta^{2j} + \cdots + \theta^{(p-2)j} = d \cdot (1 + \omega + \omega^2 + \cdots + \omega^{k-1}).$$

Since $\Phi_k(X)$ is a divisor of $X^{k-1} + \cdots + 1$ and $\Phi_k(\omega) = 0$, the result follows. □

After this brief interlude on radical expressions, we can state the main result of this section:

Theorem 6.2 *Let n be a natural number and ζ be an n-th root of unity. Then ζ is expressible by radicals (over the rational numbers).*

Proof The theorem follows from the following more general Lemma 6.6. □

Lemma 6.6 *Let k be a natural number. Then all j-th roots of unity for all $j \in \{1, \ldots, k\}$ can be simultaneously expressed by radicals over K, that is, there exists a sequence $z_1, \ldots, z_m$ of algebraic numbers and a sequence of prime numbers $p_1, \ldots, p_m$, such that for all $i \in \{1, \ldots, m\}$ the number z_i is an algebraically unique p_i-th root over $K(z_1, \ldots, z_{i-1})$ and all j-th roots of unity with $j \in \{1, \ldots, k\}$ are in $K(z_1, \ldots, z_m)$.*

We can even choose the $z_1, \ldots, z_m$ and $p_1, \ldots, p_m$ such that for each $i \in \{1, \ldots, n\}$ a primitive p_i-th root of unity is in $K(z_1, \ldots, z_i)$.

Proof Since all j-th roots of unity are powers of any primitive j-th root of unity, it is sufficient to find the $z_1, \ldots, z_m$ such that for all $j \in \{1, \ldots, k\}$ a primitive j-th root of unity is in $K(z_1, \ldots, z_m)$.

We want to prove this by induction over k. For $k = 0, 1$ there is nothing to show. So let $k > 1$. By induction hypothesis, there exist algebraic numbers $z_1, \ldots, z_\ell$ such that for all $i \in \{1, \ldots, \ell\}$ the number z_i is an algebraically unique root over $K(z_1, \ldots, z_{i-1})$ and all j-th roots of unity with $j < k$ lie in $K(z_1, \ldots, z_\ell)$.

Let ζ be a primitive root of unity. We now reduce to the case that k is a prime number: Otherwise k is a composite number, say $k = p \cdot q$, where p is a prime number and $q > 1$ is a natural number. It is ζ^p a q-th root of unity and because $q < k$ it lies in $K(z_1, \ldots, z_\ell)$. We then consider the polynomial $X^p - \zeta^p$. Since a primitive p-th root of unity because $p < k$ lies in $K(z_1, \ldots, z_\ell)$, according to corollary 6.4 exactly two cases can occur: Either a p-th root of ζ^p over $K(z_1, \ldots, z_\ell)$ is algebraically unique or ζ lies in $K(z_1, \ldots, z_\ell)$. In the first case we set $m := \ell + 1$ and $z_m := \zeta$. In the second case we simply set $m := \ell$.

We can therefore assume that $k = p$ is a prime number. We choose a primitive $(p-1)$-th root of unity θ. (This idea comes from the derivation of the representation (6.7) of a primitive fifth root of unity as a radical expression. The imaginary unit i is indeed a primitive fourth root of unity.) Further we choose a primitive root g modulo p. Then we set

$$\begin{aligned} x_i &:= \zeta + \theta^i \zeta^g + \theta^{2i} \zeta^{g^2} + \cdots + \theta^{(p-2)i} \zeta^{g^{p-2}} \\ &= \sum_{j=0}^{p-2} \theta^{ji} \zeta^{g^j} \end{aligned}$$

for all $i \in \{0, \ldots, p-2\}$. These expressions lie in $\mathbf{Q}(\zeta, \theta)$. We want to show that their $(p-1)$-th powers already lie in $\mathbf{Q}(\theta)$:

It is $\zeta\theta$ a primitive $(p(p-1))$-th root of unity, since p and $p-1$ are coprime. In particular, ζ and θ lie in $\mathbf{Q}(\zeta\theta)$. Also because p and $p-1$ are coprime, $\varphi(p\,(p-1)) = \varphi(p)\,\varphi(p-1) = (p-1)\,\varphi(p-1)$. According to the degree formula, we thus have

$$\begin{aligned} (p-1)\,\varphi(p-1) = \varphi(p(p-1)) &= [\mathbf{Q}(\zeta\theta) : \mathbf{Q}(\theta)] \cdot [\mathbf{Q}(\theta) : \mathbf{Q}] \\ &= \varphi(p-1)[\mathbf{Q}(\zeta, \theta) : \mathbf{Q}(\theta)] \end{aligned}$$

so the degree of ζ over $\mathbf{Q}(\theta)$ is exactly $p-1$.

It follows that $\Phi_p(X) = X^{p-1} + X^{p-2} + \cdots + X + 1$ must also be the minimal polynomial of ζ over $\mathbf{Q}(\theta)$. The considerations about the Galois group of $\zeta, \zeta^2, \ldots, \zeta^{p-1}$ over the rational numbers remain correct also over $\mathbf{Q}(\theta)$: There is an element σ in $G = \mathrm{Gal}_{\mathbf{Q}(\theta)}(\zeta, \zeta^2, \ldots, \zeta^{p-1})$ with $\sigma \cdot \zeta^i = \zeta^{ig}$ for every integer number i, and σ is a generator of G. Let $i \in \{0, \ldots, p-2\}$. Then it holds

$$\sigma \cdot x_i = \sum_{j=0}^{p-2} \theta^{ji}\, \sigma \cdot \zeta^{g^j} = \sum_{j=0}^{p-2} \theta^{ji} \zeta^{g^{j+1}} = \theta^{-i} \sum_{j=1}^{p-1} \theta^{ji} \zeta^{g^j} = \theta^{-i} \sum_{j=0}^{p-2} \theta^{ji} \zeta^{g^j}.$$

(The last equality holds since $\theta^{p-1} = 1 = \theta^0$ and $g^{p-1} \equiv g^0 \equiv 1$ modulo p.) Because $\theta^{p-1} = 1$, it follows that x_i^{p-1} is invariant under σ.

Since σ generates the whole group G, x_i^{p-1} is thus invariant under all elements of the Galois group of $\zeta, \ldots, \zeta^{p-1}$ over $\mathbf{Q}(\theta)$, consequently x_i^{p-1} is in $\mathbf{Q}(\theta)$. Since θ is in $K(z_1, \ldots, z_\ell)$ by the induction hypothesis, x_i is either already in $K(z_1, \ldots, z_\ell)$ or is a uniquely algebraic $(p-1)$-th root over $K(z_1, \ldots, z_\ell)$.

Therefore, we can extend the sequence $z_1, \ldots, z_\ell$ to a sequence $z_1, \ldots, z_m$ so that in particular for $j \in \{1, \ldots, m\}$ the number z_j is a uniquely algebraic root over $K(z_1, \ldots, z_{j-1})$ and so that $x_0, \ldots, x_{p-2}$ are in $K(z_1, \ldots, z_m)$.

Now consider the sum

$$x_0 + x_1 + \cdots + x_{p-2} = \sum_{j=0} \zeta^{g^j} \sum_{i=0}^{p-2} \theta^{ji} = (p-1)\,\zeta^{g^0} = (p-1)\,\zeta,$$

where we have used the previously formulated Lemma 6.5; we recognize that ζ is also in $K(z_1, \ldots, z_m)$. □

If we apply the construction in the proof of Theorem 6.2 to the case of the eleventh roots of unity, we find that the primitive eleventh roots of unity can be expressed through square and fifth roots. With the primitive fifth root of unity

$$\zeta_5 = \frac{1}{100}\left((4\mathrm{i} + 3)\xi^3 + (10\mathrm{i} - 5)\xi^2 + 25\xi - 25\right)$$

Vandermonde's result reads

$$
\begin{aligned}
\zeta_{11} = \frac{1}{2143588810}\Big(&(1965\zeta_5^3 - 8664\zeta_5^2 + 7600\zeta_5 + 6840)\sqrt[10]{21615\zeta_5^3 - 53625\zeta_5^2 - 61985\zeta_5 + 116919}^{\,9} \\
&+ (51821\zeta_5^3 + 28402\zeta_5^2 - 22\zeta_5 + 18348)\sqrt[10]{21615\zeta_5^3 - 53625\zeta_5^2 - 61985\zeta_5 + 116919}^{\,8} \\
&+ (-1219685\zeta_5^3 + 10164\zeta_5^2 + 71995\zeta_5 - 45012)\sqrt[10]{21615\zeta_5^3 - 53625\zeta_5^2 - 61985\zeta_5 + 116919}^{\,7} \\
&+ (620246\zeta_5^3 + 521752\zeta_5^2 + 469843\zeta_5 + 412610)\sqrt[10]{21615\zeta_5^3 - 53625\zeta_5^2 - 61985\zeta_5 + 116919}^{\,6} \\
&+ (658845\zeta_5^3 - 1464100\zeta_5^2 - 439230\zeta_5 + 87846)\sqrt[10]{21615\zeta_5^3 - 53625\zeta_5^2 - 61985\zeta_5 + 116919}^{\,5} \\
&+ (-2254714\zeta_5^3 - 1288408\zeta_5^2 - 6764142\zeta_5 - 1127357)\sqrt[10]{21615\zeta_5^3 - 53625\zeta_5^2 - 61985\zeta_5 + 116919}^{\,4} \\
&+ (7086244\zeta_5^3 + 21258732\zeta_5^2 + 5314683)\sqrt[10]{21615\zeta_5^3 - 53625\zeta_5^2 - 61985\zeta_5 + 116919}^{\,3} \\
&+ (19487171\zeta_5^3 + 38974342\zeta_5^2 - 38974342\zeta_5)\sqrt[10]{21615\zeta_5^3 - 53625\zeta_5^2 - 61985\zeta_5 + 116919}^{\,2} \\
&+ 214358881\zeta_5^3\sqrt[10]{21615\zeta_5^3 - 53625\zeta_5^2 - 61985\zeta_5 + 116919} - 214358881\Big).
\end{aligned}
$$

6.5 Non-solvable Galois Groups

After defining what it means to be "expressible by radicals", we ask ourselves the fundamental question of Galois, namely which polynomial equations are solvable by radicals, that is, the solutions of which polynomial equations are expressible by radicals. Here we want all solutions to be *simultaneously* expressible by radicals:

Definition 6.3 Let K be a number field and $f(X)$ a normalized separable polynomial over K. We say that the polynomial equation $f(X) = 0$ is *solvable* (or, shorter, that $f(X)$ is *solvable*), if the roots $x_1, \ldots, x_n$ over K are simultaneously expressible by radicals.

If $x_1, \ldots, x_n$ are simultaneously expressible by radicals over K, the corresponding radical expressions determine the numbers $x_1, \ldots, x_n$ only up to conjugation. But this must be the case because we cannot algebraically distinguish a particular ordering of the $x_1, \ldots, x_n$.

We have defined that $x_1, \ldots, x_n$ are simultaneously expressible by radicals over K if there exists a sequence $z_1, \ldots, z_m$ of algebraic numbers and a sequence $p_1, \ldots, p_m$ of prime numbers such that for all $i \in \{1, \ldots, m\}$ the number z_i has a p_i-th root in $K(z_1, \ldots, z_{i-1})$ that is algebraically unique and $x_1, \ldots, x_n$ lie in $K(z_1, \ldots, z_m)$.

We can also formulate this a little differently: Namely, we call an extension F of a number field E a *radical extension* (with *exponent* p) if p is a prime number and F is of the form $E(z)$, where z has a p-th root of an element in E that is algebraically unique over K.

With this notation, $x_1, \ldots, x_n$ are exactly then simultaneously expressible by radicals over K when a sequence

$$K =: K_0 \subseteq K_1 \subseteq \cdots \subseteq K_m \tag{6.9}$$

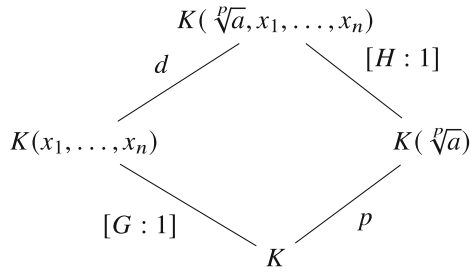
of extensions of K exists, such that for all $i \in \{1, \ldots, m\}$ the extension K_i is a radical extension of K_{i-1} and $x_1, \ldots, x_n$ are elements of K_m. We call the chain $K_0 \subseteq K_1 \subseteq \cdots \subseteq K_m$ in (6.9) a *tower of radical extensions*.

A polynomial equation is solvable if its solutions lie in a tower of radical extensions.

In the last section, we saw that at least the cyclotomic polynomials are solvable. In this section, we want to provide a general necessary criterion for a polynomial equation to be solvable:

For this, we first consider a prime number p and a number field K in which a p-th primitive unit root lies. Let further a be a number from K that is not a p-th power in K. Then there is known to exist a uniquely algebraic p-th root $\sqrt[p]{a}$ over K. In particular, $X^p - a$ is irreducible over K and $[K(\sqrt[p]{a}) : K] = p$.

Let $x_1, \ldots, x_n$ be the roots of a normalized separable polynomial $f(X)$ over K. We want to compare the Galois group H of $x_1, \ldots, x_n$ over $K(\sqrt[p]{a})$ with the Galois group G of $x_1, \ldots, x_n$ over K. For this, we consider the diagram



of number fields, where we have noted the respective degrees of the extensions on the vertical lines. Here, $d := [K(\sqrt[p]{a}, x_1, \ldots, x_n) : K(x_1, \ldots, x_n)]$. According to the degree formula we have

$$d \cdot [G : 1] = [H : 1] \cdot p.$$

Recalling the notion of an index of a subgroup and Lagrange's theorem, we can rearrange this formula to

$$d \cdot [G : H] = p.$$

Since p is a prime number, exactly two cases can occur: Either $[G : H] = 1$ (and thus $d = p$, i.e., a is not a p-th power in $K(x_1, \ldots, x_n)$), or it is $[G : H] = p$ (and thus $d = 1$, i.e., a is a p-th power in $K(x_1, \ldots, x_n)$).

In the first case, $G = H$, i.e., the Galois group of $x_1, \ldots, x_n$ does not reduce when we add a p-th root of a to the number field K.

The more interesting case is the second, in which H is a subgroup of index p in G. In this case, we can even say more. Namely, we claim that the conjugates of all symmetries τ of H lie again in H:

Let σ be a symmetry from G. We have to show, that $\sigma \circ \tau \circ \sigma^{-1}$ lies again in H.

Since in the second case, $\sqrt[p]{a}$ lies in $K(x_1, \ldots, x_n)$, σ acts on $\sqrt[p]{a}$. The element $\sqrt[p]{a}$ satisfies the algebraic relation $X^p - a$ over K, i.e., $\sigma \cdot \sqrt[p]{a}$ must again be a p-th root of a.

Therefore, $\sigma \cdot \sqrt[p]{a} = \zeta \cdot \sqrt[p]{a}$ for a p-th root of unity ζ (depending on σ). Since $\sigma^{-1} \circ \sigma = \mathrm{id}$, it follows that $\sigma^{-1} \cdot \sqrt[p]{a} = \zeta^{-1} \sqrt[p]{a}$. We then calculate

$$(\sigma \circ \tau \circ \sigma^{-1}) \cdot \sqrt[p]{a} = \sigma \cdot \tau \cdot \sigma^{-1} \cdot \sqrt[p]{a} = \sigma \cdot \tau \cdot (\zeta^{-1} \sqrt[p]{a}) = \zeta^{-1} \cdot \sigma \cdot \tau \cdot \sqrt[p]{a}.$$

In the last equality, we have exploited that ζ lies in K and the symmetries thus act trivially on ζ.

Now we use the fact that H is the Galois group of $x_1, \ldots, x_n$ over $K(\sqrt[p]{a})$, so that $\tau \in H$ in particular acts trivially on $\sqrt[p]{a}$. We continue to calculate:

$$\zeta^{-1} \cdot \sigma \cdot \tau \cdot \sqrt[p]{a} = \zeta^{-1} \cdot \sigma \cdot \sqrt[p]{a} = \zeta^{-1} \cdot \zeta \sqrt[p]{a} = \sqrt[p]{a}.$$

We have thus verified that $\sigma \circ \tau \circ \sigma^{-1}$ acts trivially on $\sqrt[p]{a}$. Since σ as a symmetry over K also acts trivially on K, we have

$$K(x_1, \ldots, x_n)^H = K(\sqrt[p]{a}) \subseteq K(x_1, \ldots, x_n)^{\sigma \circ \tau \circ \sigma^{-1}},$$

according to Proposition 6.6, so $\sigma \circ \tau \circ \sigma^{-1} \in H$, which we wanted to show. The conjugates of elements in H are thus again in H.

A subgroup H with such a property gets its own special name:

Definition 6.4 Let H be a subgroup of a group G. Then H is called a *normal subgroup* of G if the conjugates of all elements from H are again in H, i.e.,

$$\sigma \circ \tau \circ \sigma^{-1} \in H$$

for all $\sigma \in G$ and $\tau \in H$.

In general, a subgroup is not closed under conjugation by arbitrary group elements. If this is the case, however, it is called a normal subgroup of the group.

In this section, we have so far proven:

Proposition 6.8 *Let p be a prime number. Let the number field K contain a primitive p-th root of unity. Let a be a number from K that is not a p-th power in K.*

If $x_1, \ldots, x_n$ are the roots of a normalized separable polynomial $f(X)$ over K, G is the Galois group of $x_1, \ldots, x_n$ over K and H is the Galois group of $x_1, \ldots, x_n$ over $K(\sqrt[p]{a})$, then exactly one of the following two cases occurs:

Either H is the full Galois group G, or H is a normal subgroup of index p in G. □

For the following, we also need a lemma:

Lemma 6.7 *Let $x_1, \ldots, x_k$ be simultaneously expressible over K by radicals with radical exponents $p_1, \ldots, p_n$. Let L be an extension of K that contains all p_i-th roots of unity for all $i \in \{1, \cdots, n\}$. Then $x_1, \ldots, x_k$ are also simultaneously expressible over L by radicals with radical exponents from the set $\{p_1, \ldots, p_n\}$.*

Proof By assumption, there exists a tower $K = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_n$ of radical extensions with exponents $p_1, \ldots, p_n$, such that $x_1, \ldots, x_k$ lie in K_n.

If $L = \mathbf{Q}(t_1, \ldots, t_r)$ and $K_i = \mathbf{Q}(u_1, \ldots, u_s)$, we write

$$L \cdot K_i = \mathbf{Q}(t_1, \ldots, t_r, u_1, \ldots, u_s).$$

(This notation does not depend on the choice of the generators of L and K_i. An algebraic number is in $L \cdot K_i$ if and only if it is a rational expression in elements from L and K_i.)

Then

$$L = L \cdot K_0 \subseteq L \cdot K_1 \subseteq \cdots \subseteq L \cdot K_n \tag{6.10}$$

is an extension of number fields. For each $i \in \{1, \ldots, n\}$ it holds that $L \cdot K_i = L \cdot K_{i-1}(z_i)$, where z_i is a p_i-th root of an element a from $L \cdot K_{i-1}$ (in fact, already from K_{i-1}).

We now distinguish two cases. In the first case, z_i is already an element from $L \cdot K_{i-1}$. In this case we have $L \cdot K_i = L \cdot K_{i-1}$. In the other case, not all solutions of $X^{p_i} - a = 0$ are contained in $L \cdot K_{i-1}$. Since $L \cdot K_{i-1}$ contains a primitive p_i-th root of unity, it follows from Corollary 6.4 that $X^{p_i} - a$ is then irreducible over $L \cdot K_{i-1}$, so z_i is a uniquely algebraic p_i-th root of a over $L \cdot K_{i-1}$. That is, in the second case, $L \cdot K_i$ is a radical extension (to the exponent p_i) of $L \cdot K_{i-1}$.

We thus obtain that any two consecutive extensions of the tower (6.10) either coincide or form a radical extension. If we remove the coinciding extensions, we obtain a tower

$$L = L_0 \subseteq L_1 \subseteq \cdots \subseteq L_m = L \cdot K_n$$

of radical extensions with exponents from $\{p_1, \cdots, p_n\}$, and $x_1, \ldots, x_k$ lie in $L \cdot K_n$ (since these numbers already lie in K_n). □

How can we use Proposition 6.8 to study solvable polynomial equations? For this, we consider an equation $f(X) = 0$ with normalized separable polynomial $f(X)$ over a number field K with solutions $x_1, \ldots, x_n$. Let $f(X)$ be solvable over K, where the largest occurring radical exponent is p.

For the moment, we assume that K contains all j-th roots of unity with $j \leq p$. By definition, there exists a tower of radical extensions

$$K = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_n \tag{6.11}$$

with radical exponents at most p such that $x_1, \ldots, x_n$ lie in K_n.

Let G_i be the Galois group of $x_1, \ldots, x_n$ over K_i for all $i \in \{0, \ldots, n\}$. An extension of the number field reduces the Galois group, that is, we have a descending sequence

$$G = G_0 \supseteq G_1 \supseteq \cdots \supseteq G_n = \{\mathrm{id}\}$$

of subgroups, which corresponds to (6.11).

Since the K_i contain enough roots of unity by assumption, it follows from Proposition 6.8 for all $i \in \{1, \ldots, n\}$, that either $G_i = G_{i-1}$ or that G_i is a normal subgroup in G_{i-1} of prime index. By eliminating consecutive coinciding subgroups, it follows that G has a so-called *normal series*, that is, a sequence

$$G = H_0 \supseteq H_1 \supseteq \cdots \supseteq H_m = \{\mathrm{id}\}$$

of subgroups of G exists, such that for all $i \in \{1, \ldots, m\}$ the subgroup H_i is a normal subgroup in H_{i-1} of prime index.

We make this a general definition:

Definition 6.5 A group G is called *solvable*, if it has a normal series, that is, a descending sequence

$$G = G_0 \supseteq G_1 \cdots \supseteq G_n = \{\mathrm{id}\}$$

of subgroups of G exists, such that for each $i \in \{1, \ldots, n\}$ the subgroup G_i is a normal subgroup of prime index in G_{i-1}.

We have thus shown: If $f(X)$ is solvable over K and K contains enough roots of unity, then the Galois group of $f(X)$ over K is solvable. This is also the reason for why solvable groups are called solvable.

We can drop the requirements on K. We will prove the following theorem:

Theorem 6.3 *Let K be a number field. Let $f(X)$ be a normalized separable polynomial over K that is solvable over K. Then the Galois group of the roots $x_1, \ldots, x_n$ of $f(X)$ over K is solvable.*

This is the promised necessary criterion (based on the Galois group) for a polynomial equation to be solvable. And this theorem will be the key to an example of a non-solvable

polynomial equation. For this, we will give a polynomial for which we can prove that its Galois group is not solvable. But first, we should prove Theorem 6.3:

Proof Let p be the largest appearing exponent in the radical representations of $x_1, \ldots, x_n$. We then choose, according to Theorem 6.2, an extension L of K that contains all j-th roots of unity with $j \leq p$. According to Lemma 6.7, $f(X)$ is then also solvable over L, that is, there exists a tower

$$L = L_0 \subseteq L_1 \subseteq \cdots \subseteq L_n$$

of radical extensions, whose exponents are at most p and such that the roots $x_1, \ldots, x_n$ all lie in L_n. After choosing L, there is also a tower

$$K = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_m = L$$

of radical extensions with exponents $p_1, \ldots, p_m$ that we can choose according to the addition in Theorem 6.2 so that for each $j \in \{1, \ldots, m\}$ a primitive p_j-th root of unity lies in K_{j-1}. We can combine the two towers into a tower of radical extensions

$$K = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_m = L_0 \subseteq L_1 \subseteq \cdots \subseteq L_n$$

that has the additional property that for each occurring radical extension F over E of an exponent q in this tower a primitive q-th root of unity already lies in E. With this we can conclude as before and infer that for this tower a descending normal series of the Galois group G of $x_1, \ldots, x_n$ over K exists, that is, G is solvable. □

Galois groups of solvable equations are solvable.

6.6 A Non-solvable Equation of the Fifth Degree

Finally, we can provide an example of a non-solvable polynomial equation of the fifth degree, the central result of Galois' work, after mathematicians before him had been searching for solution formulas (in terms of radicals) for equations of the fifth or higher degree for centuries.

To apply the criterion from Theorem 6.3, we first need a non-solvable group, which we then have to realize as a Galois group of a polynomial equation. To this end, we first define a concept diametrically opposed to solvability:

Definition 6.6 A group G is called *simple* if G has exactly two normal subgroups, namely the subgroups $\{\mathrm{id}\}$ and G.

(It follows that the trivial group $G = \{\mathrm{id}\}$ is not simple by definition.)

Simple groups are obviously solvable if and only if they are of prime order. Examples of simple groups that are not of prime order are provided by the following proposition:

Proposition 6.9 *For every natural number $n \geq 5$ the alternating group A_n is simple.*

Since the order of A_n for $n \geq 2$ is given by $\frac{1}{2}n!$ (every second permutation is even because composition with $(1, 2)$ from the left induces a bijection between the even and the odd permutations in S_n), A_n for $n \geq 5$ is therefore not a solvable group.

Proof Let N be a normal subgroup of A_n. We assume that N consists of more than the element id. We have to show that N then already contains every element from A_n.

Let's first assume that N contains a three-cycle, without loss of generality, we can assume $(1, 2, 3) \in N$. Let $k > 3$. Then $(2, 3, k)$ is an even permutation, thus in A_n. Since N is a normal subgroup, it follows that also

$$(2, 3, k)^{-1} \circ (1, 2, 3) \circ (2, 3, k) = (1, k, 2)$$

is an element in N.

Since N is a subgroup, $(1, k, 2)^2 = (1, 2, k)$ is also in N. Let $a, b \in \{1, \ldots, n\}$ such that a, b, 1 and k are pairwise different. It is $(1, a) \circ (1, b)$ an even permutation and thus in A_n. Consequently,

$$((1, a) \circ (2, b))^{-1} \circ (1, 2, k) \circ (1, a) \circ (2, b) = (a, b, k)$$

is an element of N.

We have thus shown that all three cycles are already in N. We claim that N must already be the entire alternating group A_n: Let σ be an element from A_n. Since every permutation can be written as a composition of permutations swapping two elements, we can write σ in the form

$$\sigma = \tau_1 \circ \cdots \circ \tau_m$$

where the τ_i are transpositions, i.e. permutations of the form (a, b). Since σ lies in the alternating group, σ has an even number of inversions. Thus, $m = 2k$ must be even.

We can therefore write

$$\sigma = (\tau_1 \circ \tau_2) \circ \cdots \circ (\tau_{2k-1} \circ \tau_{2k}).$$

Since the composition of two transpositions according to the following Lemma 6.8 can always be written as a composition of two three cycles, σ can be written entirely as a composition of three cycles (which according to what has already been proven all lie in N), that is, σ is also in N, which proves $N = \mathrm{A}_n$.

It remains to justify our assumption that N contains a three cycle. We have outsourced this fact for the sake of clarity, in Lemma 6.9. □

Lemma 6.8 *Let σ and τ be two transpositions in a permutation group S_n. Then $\sigma \circ \tau$ can be written as a composition (of at most two) three cycles.*

Proof Without loss of generality, the following cases occur: $\sigma = (1, 2)$ and $\tau = (1, 2)$, $\sigma = (1, 2)$ and $\tau = (1, 3)$ and $\sigma = (1, 2)$ and $\tau = (3, 4)$.

In the first case, $\sigma \circ \tau = \mathrm{id}$ and thus the composition of an empty number of three cycles. In the second case, $\sigma \circ \tau = (1, 3, 2)$, so a three cycle. In the last case,

$$\sigma \circ \tau = (1, 2) \circ (3, 4) = (1, 2, 3) \circ (3, 4, 2).$$

□

Lemma 6.9 *Let N be a normal subgroup of A_n with $n \geq 5$, and let N contain a permutation $\tau \neq \mathrm{id}$. Then N also contains a three cycle, i.e. a permutation of the form (i, j, k).*

Before we prove the lemma, we need another notation. Namely, we call two cycles $(p_1, \ldots, p_r)$ and $(q_1, \ldots, q_s)$ from a permutation group S_n *disjoint* if the sets $\{p_1, \ldots, p_r\}$ and $\{q_1, \ldots, q_s\}$ have no element in common.

Two disjoint cycles σ and τ obviously commute, that is $\sigma \circ \tau = \tau \circ \sigma$.

Any permutation σ from S_n is generally not a cycle, but can always be written as a composition of disjoint cycles: If $\sigma = \mathrm{id}$, we can consider σ as an empty composition of disjoint cycles. Otherwise, there exists an $i \in \{1, \ldots, n\}$ with $\sigma(i) \neq i$.

Let k be the natural number such that $i, \sigma(i), \ldots, \sigma^{k-1}(i)$ are pairwise different and $\sigma^k(i) = i$. Then σ can obviously be written in the form $\sigma = (i, \sigma(i), \ldots, \sigma^{k-1}(i)) \circ \varrho$, where ϱ acts trivially on $i, \sigma(i), \ldots, \sigma^{k-1}(i)$.

If ϱ is the identity, we are done. Otherwise, we apply the procedure to a j with $\varrho(j) \neq j$. This procedure must eventually stop. It is easy to see that the resulting decomposition of σ into cycles is unique up to order.

Every permutation is a composition of disjoint cycles. These cycles are unique up to order.

Now let's move on to the proof of Lemma 6.9:

Proof We make a case distinction. In the first case, we assume that in the decomposition of τ into cycles, a k-cycle with $k \geq 4$ occurs. We can then assume without loss of generality that τ is of the form $\tau = (1, 2, 3, 4, \ldots, k) \circ \varrho$, where ϱ is a permutation that only permutes the elements greater than k. We set $\sigma = (1, 3, 2)$. Then

$$\tau' \coloneqq \sigma \circ \tau \circ \sigma^{-1} = \varrho \circ (1, 3, 2) \circ (1, 2, 3, 4, \ldots, k) \circ (1, 2, 3) = \varrho \circ (1, 2, 4, 5, \ldots, k, 3)$$

and τ', because N is a normal subgroup, is an element of N. Thus, $\tau' \circ \tau^{-1} = (1, 3, 4)$ is also an element of N, so N contains a three-cycle.

In the second case, we assume that at least two three-cycles appear in the cycle decomposition of τ. We can then assume without loss of generality that τ is of the form $\tau = (1, 2, 3) \circ (4, 5, 6) \circ \varrho$, where ϱ is a permutation that only permutes the elements greater than 6. Let $\sigma = (2, 4, 3)$. Then N also contains the element

$$\sigma \circ \tau \circ \sigma^{-1} \circ \tau^{-1} = (1, 5, 2, 4, 3),$$

thus a five-cycle. We can assume that τ is already this element, and apply the first case.

In the third case, we assume that in the cycle decomposition of τ there is only one three-cycle and otherwise only transpositions (that is, two-cycles) occur. Then we can assume without loss of generality that τ of the form $\tau = (1, 2, 3) \circ \varrho$, where ϱ is a product of pairwise disjoint two-cycles with $(1, 2, 3)$. Consequently, $\tau^2 = (1, 3, 2)$ is also in N, which also ensures the existence of a three-cycle in N in this case.

What remains is the case where only transpositions occur in the cycle decomposition of τ. Then we can assume without loss of generality that τ is of the form $\tau = (1, 2) \circ (3, 4) \circ \varrho$, where ϱ is a permutation that only permutes elements greater than 4. Let $\sigma = (2, 4, 3)$. Then also

$$\sigma \circ \tau \circ \sigma^{-1} \circ \tau^{-1} = (1, 4) \circ (2, 3)$$

is an element of N.

Let $\upsilon = (1, 5, 4)$. Again, because N is a normal subgroup,

$$\upsilon \circ (1, 4) \circ (2, 3) \circ \upsilon^{-1} = (1, 5) \circ (2, 3)$$

is also an element of N. Thus,

$$(1, 5) \circ (2, 3) \circ (1, 4) \circ (2, 3) = (1, 4, 5)$$

is an element of N, which again proves the existence of a three-cycle in N. □

We have seen that A_n for $n \geq 5$ is not solvable. We want to conclude from this that the full symmetric group S_n for $n \geq 5$ is also not solvable. This follows immediately from:

Proposition 6.10 *If H is a subgroup of a solvable group G, then H is also solvable.*

Subgroups of solvable groups are solvable.

Proof Since G is solvable, G has a normal series of the form

$$G = G_0 \supseteq G_1 \supseteq \cdots \supseteq G_n = \{\mathrm{id}\}.$$

If we intersect this normal series with the elements from H, we obtain a descending sequence of subgroups of the form

$$H = H \cap G_0 \supseteq H \cap G_1 \supseteq \cdots \supseteq H \cap G_n = \{\mathrm{id}\}.$$

It suffices to show that for each $i \in \{1, \ldots, n\}$, the subgroup $H \cap G_i$ is either equal to $H \cap G_{i-1}$ (then we can remove it from the sequence) or is a normal subgroup in $H \cap G_{i-1}$ of index a prime number p.

This follows from the following more general Lemma 6.10. □

Lemma 6.10 *Let G be a group. Let N be a normal subgroup of index a prime number p in G and H a subgroup. Then $H \cap N = H$, or $H \cap N$ is a normal subgroup of index p in H.*

Before we prove the lemma, we make some preliminary considerations: First, we define $H \cdot N$ as the set of group elements that we can write in the form $\sigma \circ \tau$, where σ is in H and τ is in N. It is $\sigma \circ \tau = \sigma \circ \tau \circ \sigma^{-1} \circ \sigma$. Since N is a normal subgroup in G, $\varrho = \sigma \circ \tau \circ \sigma^{-1}$ is again an element in N. Thus, we can write every element of $H \cdot N$ in the form $\varrho \circ \sigma$, where ϱ is in N and σ is in H, and all these elements are in $H \cdot N$.

This implies that $H \cdot N$ is a subgroup of G, the *product* of H and N: Firstly, the identity $\mathrm{id} = \mathrm{id} \circ \mathrm{id}$ is an element of $H \cdot N$. Then we need to show that the inverse of an element of $H \cdot N$, say $\varrho \circ \sigma$, is again in $H \cdot N$. For this, we calculate $(\varrho \circ \sigma)^{-1} = \sigma^{-1} \circ \varrho^{-1}$. Since H and N are each closed under inversion, it follows that $(\varrho \circ \sigma)^{-1}$ is again in $H \cdot N$.

It remains to show that the composition of two elements of $H \cdot N$ is again in $H \cdot N$. For this, we consider an element of the form $\sigma \circ \tau$ and an element of the form $\varrho \circ \sigma'$, where σ and σ' are in H and ϱ and τ are in N. Then we have

$$(\sigma \circ \tau) \circ (\varrho \circ \sigma') = \sigma \circ (\tau \circ \varrho) \circ \sigma' = (\sigma \circ \sigma') \circ (\sigma'^{-1} \circ \tau \circ \varrho \circ \sigma').$$

Again, because N is a normal subgroup in G, the latter parenthesized expression on the right-hand side is in N, so the entire right-hand side is in $H \cdot N$.

We now claim that the index of $H \cap N$ in H is equal to the index of H in $H \cdot N$, because if we choose a representative system $R = \{\varrho_1, \ldots, \varrho_r\}$ of the congruence classes modulo $H \cap N$ in H, this is also a representative system of the congruence classes modulo N in $H \cdot N$.

For this, we need to show two things: Every element in $H \cdot N$ is modulo N congruent to a ϱ_i. Two elements in R are congruent modulo N if and only if they are equal:

For this, we first consider an element $\sigma \circ \tau$ in $H \cdot N$, where σ is in H and τ is in N. By definition of the congruence relation, this is congruent to σ modulo N (we may cancel from the right by elements from N). By definition of R, σ is in turn congruent to a ϱ_i from R modulo $H \cap N$ and thus also modulo N. Every element from $H \cdot N$ is thus congruent to a ϱ_i modulo N.

Finally, we consider two elements ϱ_i and ϱ_j from R that are congruent modulo N, that is, $\varrho_j^{-1} \circ \varrho_i$ is in N. Since ϱ_j and ϱ_i are both also in H, $\varrho_j^{-1} \circ \varrho_i$ is also in H due to the subgroup property of H, so in total it is in $H \cap N$. Thus, ϱ_i and ϱ_j are congruent modulo $H \cap N$, so by definition of R, we have $\varrho_i = \varrho_j$.

In formulas, we have thus shown:

$$[H : H \cap N] = [H \cdot N : N].$$

If H is a subgroup and N is a normal subgroup of a group G, then $H \cdot N$ is again a subgroup of G.

This brings us to the still missing proof of Lemma 6.10, which, after all the preparations, is pleasantly short:

Proof According to Lagrange's theorem, Proposition 5.5, we have

$$[G : N] = [G : H \cdot N] \cdot [H \cdot N : N].$$

According to our preliminary considerations, we can rewrite the latter factor on the right side and obtain

$$[G : N] = [G : H \cdot N] \cdot [H : H \cap N].$$

Since the left side is a prime number p by assumption, the divisor $[H : H \cap N]$ on the right side must be either 1 or again p.

Once we have shown that $H \cap N$ is a normal subgroup in H, we have proven everything. Let σ be in H and τ be in $H \cap N$. We need to show that $\sigma \circ \tau \circ \sigma^{-1}$ is in $H \cap N$. But this

is easy: The expression is in H, since H is a subgroup, and it is in N, since N is a normal subgroup in G. □

After we have shown that the symmetric group S_n for $n \geq 5$ is not solvable, we now want to show that there are polynomial equations over the rational numbers that have one of these groups as Galois groups:

Lemma 6.11 *Let p be a prime number. Let $f(X)$ be a normalized irreducible polynomial of degree p with rational coefficients that has exactly two non-real roots. Then the Galois group of the roots $x_1, \ldots, x_p$ of $f(X)$ over the rational numbers is the full symmetric group S_p.*

We first give an example of a polynomial to which we can apply Lemma 6.11:

Example 6.6 Let

$$f(X) = X^5 - 6X + 3.$$

According to the Eisenstein criterion, Proposition 4.5, $f(X)$ is irreducible over the rational numbers. Looking at the graph of the real values of this polynomial from Fig. 6.3, it appears that $f(X)$ has exactly three real roots. We want to prove this exactly. The image of the graph suggests considering the following values:

x	-2	-1	1	2
$f(x)$	-17	8	-2	$23.$

According to the intermediate value theorem valid for polynomial functions from one-dimensional real analysis, $f(X)$ thus has at least one real root between -2 and -1, one between -1 and 1 and one between 1 and 2. To show that there are no further roots of $f(X)$ exists, we recall Rolle's theorem,[2] which states that between any two real roots of $f(X)$, there must be a real root of the derivative of $f(X)$. To this end, we calculate $f'(X) = 5X^4 - 6$. This derivative has only two real roots, namely $\sqrt[4]{\frac{6}{5}}$ and $-\sqrt[4]{\frac{6}{5}}$ (according to the two local extrema in Fig. 6.3), so $f(X)$ can have no more than three real roots. From Lemma 6.11, it follows that the Galois group of the (complex) roots $x_1, \ldots, x_5$ of $f(X)$ is the entire S_5, a non-solvable group. According to Theorem 6.3, the equation $X^5 - 6X + 3$ is therefore not solvable.

We have thus reproduced the following central conclusion from Galois theory:

Theorem 6.4 *There exist polynomial equations (of fifth or higher degree) over the rational numbers that are not solvable (by radicals).* □

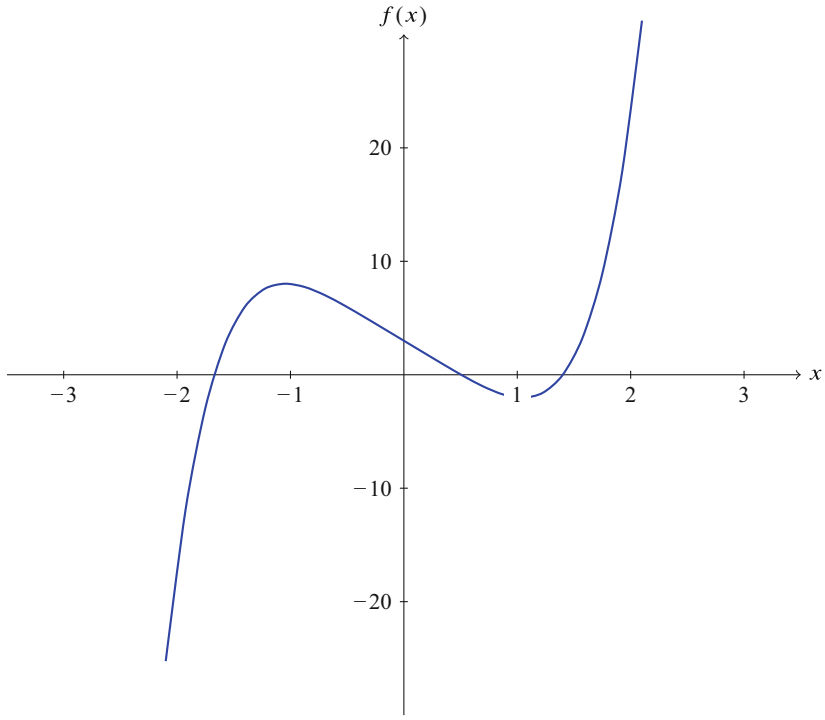


Fig. 6.3 The real values of the polynomial $f(X) = X^5 - 6X + 3$

However, one proof is still due, namely that of Lemma 6.11. For this we need another result from group theory:

Lemma 6.12 *Let p be a prime number. Let G be a transitive subgroup of the symmetric group S_p, that is, for every pair $i,\ j \in \{1, \ldots, p\}$, there exists a permutation σ in G, such that $\sigma(i) = j$. If G contains at least one transposition, then G is already the full symmetric group S_p.*

Proof of Lemma 6.12 For the proof, we want to call two numbers $i,\ j \in \{1, \ldots, p\}$ *equivalent* if the transposition (i, j) is contained in G (here we set $(i, i) := \mathrm{id}$). The so defined relation is obviously reflexive and symmetric. It is also transitive, because if i is equivalent to j and j is equivalent to k, then also

$$(i, k) = (j, k) \circ (i, j) \circ (j, k)$$

is an element of G, that is, i and k are also equivalent to each other.

Let σ be any element of G. Let further $i,\ j \in \{1, \ldots, p\}$ be numbers. We claim, that i is equivalent to j if and only if $\sigma(i)$ is equivalent to $\sigma(j)$:

If i is equivalent to j, then also

$$(\sigma(i), \sigma(j)) = \sigma \circ (i, j) \circ \sigma^{-1}$$

is an element of G, that is, $\sigma(i)$ and $\sigma(j)$ are equivalent.

If we apply the same considerations to σ^{-1} instead of σ, we get that from the equivalence of $\sigma(i)$ and $\sigma(j)$ the equivalence of i and j follows. We can conclude that the number i is equivalent to as many numbers as the number $\sigma(i)$ is equivalent to.

Due to the transitivity of G, it follows from the last paragraph for every pair of numbers $i, i' \in \{1, \ldots, p\}$ that i and i' are each equivalent to as many numbers (we find a σ in G with $\sigma(i) = i'$). Let us call this count m. Since the elements of the set $\{1, \ldots, p\}$ fall into equivalence classes and these equivalence classes all have the same size m, it follows that m is a divisor of p.

Finally, we take advantage of the fact that G contains a transposition $\tau = (i, j)$ by assumption. (Since τ is a transposition, we have $i \neq j$.) Consequently, i and j are equivalent, meaning that the equivalence class of i contains at least two elements, so we have $m \geq 2$. As a divisor of the prime number p, m must therefore be $m = p$. There is therefore only one equivalence class, meaning every number $i \in \{1, \ldots, p\}$ is equivalent to every number $j \in \{1, \ldots, p\}$. Translated into properties of G, this means that G contains all transpositions. Since every permutation from S_p can be written as a product of transpositions, G therefore contains all permutations, so it is the full symmetric group. □

If p is a prime number and f is an irreducible polynomial of degree p whose Galois group contains a transposition, then the Galois group is already the full symmetric group S_p.

Proof of Lemma 6.11 Let $x_3, \ldots, x_p$ be the $p - 2$ real roots of $f(X)$ and x_1 and x_2 the two non-real ones. Let G be the Galois group of $x_1, \ldots, x_p$ over the rational numbers. We want to show that G is the full symmetric group S_p. Since $f(X)$ is irreducible, the Galois group G operates transitively on the roots of $f(X)$ according to Corollary 5.1; in the language of lemma 6.12, G is therefore a transitive subgroup of the symmetric group S_p. It remains to show that G contains a transposition.

We claim that $\tau := (1, 2)$ is in G: Since $f(X)$ is a polynomial with rational coefficients, $0 = \overline{f(x_1)} = f(\overline{x_1})$, that is, the complex conjugate of x_1 is again a root of $f(X)$. Since x_1 is non-real, $\overline{x_1}$ is unequal to x_1 and again non-real, so $\overline{x_1} = x_2$ must hold. For the

remaining roots, we have $\overline{x_i} = x_i$, $i \geq 3$ since they are all real. If then $H(x_1, \ldots, x_n) = 0$ is an algebraic relation between $x_1, \ldots, x_n$ over the rational numbers, we see that

$$0 = \overline{H(x_1, \ldots, x_n)} = H(\overline{x_1}, \ldots, \overline{x_n}) = H(x_2, x_1, x_3, \ldots, x_p) = \tau \cdot H(x_1, \ldots, x_p).$$

That is, τ preserves every algebraic relation between $x_1, \ldots, x_p$. □

Complex conjugation induces an element of the Galois groups of polynomials over purely real number fields.

6.7 About Solvable Equations

After we have learned about an equation that is not solvable, the question about which equations are solvable on the other hand arises naturally. Interestingly, the formal reverse of Theorem 6.3 applies, that is, we have overall:

Theorem 6.5 *Let K be a number field. Let $f(X)$ be a normalized separable polynomial over K. Then $f(X)$ is solvable if and only if the Galois group of the roots $x_1, \ldots, x_n$ of $f(X)$ over K is solvable.*

Whether a Galois group is solvable can be determined, at least in principle, in finite time. Thus, we can decide for each normalized separable polynomial over K using Theorem 6.5 whether it is solvable or not. The aim of this section is the proof of Theorem 6.5, one direction of which we have already seen in the proof of Theorem 6.3.

For the other direction, we first need the following important statement, which we can consider as a reversal of Proposition 6.8. Here, K is a number field.

Lemma 6.13 *Let $f(X)$ be a normalized separable polynomial over K with roots $x_1, \ldots, x_n$. Let G be the Galois group of $x_1, \ldots, x_n$ over K. If G contains a normal subgroup H of index p where p is a prime number and if a primitive p-th root of unity is contained in K, then there exists an element a in K such that the Galois group of $x_1, \ldots, x_n$ over $K(\sqrt[p]{a})$ is given by H.*

Proof Let σ be an element from G that is not contained in H, that is, an element from G that is not congruent modulo H to the identity. (Such an element exists, since the index of H is greater than one.) Let $\tilde{G}$ be the set of those group elements which can be written in the form $\sigma^k \circ \tau$, where τ is from H and k is an integer. Since H is a normal subgroup, it can be shown easily that $\tilde{G}$ is a subgroup of G (it is indeed $\tilde{G} = \langle \sigma \rangle \cdot H$).

Because of

$$[G : H] = [G : \tilde{G}] \cdot [\tilde{G} : H],$$

$[G : \tilde{G}]$ is a divisor of p, the index of H in G. But since σ is not in H, $\tilde{G}$ is strictly larger than H, so $[\tilde{G} : H]$ is strictly larger than 1.

Thus $[G : \tilde{G}]$ is a proper divisor of p, so $[G : \tilde{G}] = 1$. From this it follows that there is no element in G that is not congruent modulo $\tilde{G}$ to the identity, that is $G = \tilde{G}$.

So far, we have shown that every element of G can be written in the form $\sigma^k \circ \tau$.

Two elements $\sigma^k \circ \tau$ and $\sigma^\ell \circ \tau'$ are congruent modulo H if and only if σ^k and σ^ℓ are congruent modulo H. In particular, every element from G is congruent modulo H to an element of the form σ^k.

Since there are only p different congruence classes modulo H, there exist $i, j \in \{0, \ldots, p\}$ with $i < j$, so that σ^i and σ^j are congruent modulo H, that is, $\sigma^{-i} \circ \sigma^j$ is from H. Setting $k := j - i$, we obtain a $k \in \{1, \ldots, p\}$ such that σ^k is congruent to the identity modulo H (that is, it lies in H itself). It follows that for every integer i, the element σ^i is congruent to σ^{i+k} modulo H. Consequently, the sequence id, $\sigma, \ldots, \sigma^{k-1}$ runs through all congruence classes of G modulo H. Since there are a total of p of them, we have $k = p$, so we have that σ^p is an element of H.

After these considerations, according to Theorem 6.1, let L be the intermediate extension of $K(x_1, \ldots, x_n)$ over K so that the Galois group of $x_1, \ldots, x_n$ over L is exactly H. According to the theorem about the primitive element, there exists a number z from L with $L = K(z)$. Since z is invariant under H, it follows in particular $\sigma^p \cdot z = 1$. Since H has index p in G, the degree of z over K is exactly p.

We now follow the proof idea of Theorem 6.2:

Choose a primitive p-th root of unity ζ in K and then set

$$\begin{aligned}
y_1 &:= z + \zeta\, \sigma \cdot z + \zeta^2 \sigma^2 \cdot z + \cdots + \zeta^{p-1} \sigma^{p-1} \cdot z, \\
y_2 &:= z + \zeta^2 \sigma \cdot z + \zeta^4 \sigma^2 \cdot z + \cdots + \zeta^{2(p-1)} \sigma^{p-1} \cdot z, \\
&\vdots \\
y_p &:= z + \zeta^p \sigma \cdot z + \zeta^{2p} \sigma^2 \cdot z + \cdots + \zeta^{(p-1)p} \sigma^{p-1} \cdot z,
\end{aligned}$$

that is $y_i = \sum_{j=0}^{p-1} \zeta^{ij} \sigma^j \cdot z$ for $i \in \{1, \ldots, p\}$. A priori, each y_i is a number rational in $x_1, \ldots, x_n$ over K.

In fact, $y_i \in K(z) = K(x_1, \ldots, x_n)^H$: To show this, it is sufficient to show that y_i is invariant under every symmetry τ from H: It is $\tau \circ \sigma^j = \sigma^j \circ \varrho_j$, where $\varrho_j := \sigma^{-j} \circ \tau \circ \sigma^j$ is again an element from the normal subgroup H. In particular, $\varrho_j \cdot z = z$. It follows that

$$\tau \cdot y_i = \sum_{j=0}^{p-1} \zeta^{ij}\, \tau \cdot \sigma^j \cdot z = \sum_{j=0}^{p-1} \zeta^{ij}\, \sigma^j \cdot \varrho_j \cdot z = \sum_{j=0}^{p-1} \zeta^{ij}\, \sigma^j \cdot z = y_i.$$

Because of $\sigma^p \cdot z = z$ and $\zeta^{ip} = 1$, we can write the sum y_i also as $y_i = \sum_{j=1}^{p} \zeta^{ij}\, \sigma^j \cdot z$. We have thus

$$\sigma \cdot y_i = \sum_{j=0}^{p-1} \zeta^{ij}\, \sigma^{j+1} \cdot z = \sum_{j=1}^{p} \zeta^{i\,(j-1)}\, \sigma^j \cdot z = \zeta^{-i} \sum_{j=1}^{p} \zeta^{ij}\, \sigma^j \cdot z = \zeta^{-i} y_i.$$

For all $i \in \{1, \ldots, p\}$ it follows that

$$\sigma \cdot y_i^p = (\sigma \cdot y_i)^p = \zeta^{-p} y_i^p = y_i^p,$$

that is, $a_i := y_i^p$ is invariant under the action of σ. Since a_i as a polynomial in y_i is invariant under the action of H and every element from G is of the form $\sigma^k \circ \tau$ with τ from H, consequently a_i is invariant under the action of the entire Galois group and according to Proposition 6.4 is thus an element from K.

Due to Lemma 6.5 (applied to $p+1$ instead of p), we have

$$y_1 + \cdots + y_p = \sum_{j=0}^{p-1} \left(\sigma^j \cdot z \cdot \sum_{i=1}^{p} \zeta^{ij} \right) = p \cdot z.$$

Since z cannot be zero (otherwise $L = K$, which cannot be the case because $H \subsetneq G$), there exists at least one $i \in \{1, \ldots, p\}$, such that $y := y_i$ does not vanish. In particular, $\sigma \cdot y = \zeta^{-1} y$ is not equal to y, that is, y is not invariant under G, which means the second factor on the right side of the degree formula

$$p = [K(z) : K] = [K(z) : K(y)] \cdot [K(y) : K]$$

is nontrivial.

Since p is a prime number, we thus have $[K(z) : K(y)] = 1$, so $L = K(z) = K(y)$. Since $y^p = a := a_i$, y is a p-th root of a number a from K. Thus we have proven the existence of an a from K, such that the Galois group of $x_1, \ldots, x_n$ over $K(\sqrt[p]{a})$ is exactly H.

(Different choices of a p-th root of a give the same $K(\sqrt[p]{a})$, as they only differ by a power of a p-th root of unity, and this lies in K.) □

Proof of Theorem 6.5 We need to show that a tower

$$K = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_n$$

of radical extensions exists, so that $x_1, \ldots, x_n$ all lie in K_n. First, according to the addendum in Theorem 6.2, we choose a tower

$$K = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_m$$

of radical extensions such that K_m contains all primitive p-th roots of unity, where p runs through all prime divisors of the order g of the Galois group G of $x_1, \ldots, x_n$.

Next, we consider the Galois group G_m of $x_1, \ldots, x_n$ over K_m. Since every algebraic relation over K is also one over K_m, G_m is certainly a subgroup of G. According to Lagrange's theorem, Proposition 5.5, its order is a divisor of g, the order of G. Furthermore, G_m is solvable according to Proposition 6.10. If G_m is the trivial group {id}, then $x_1, \ldots, x_n$ must already lie in K_m, in which case we have found a tower of radical extensions in which the roots of $f(X)$ lie.

Otherwise, G_m is nontrivial. Then G_m contains, by definition of solvability, a normal subgroup G_{m+1} of the index of a prime number p in G_m. The prime p is a divisor of g, so we can apply Lemma 6.13 and thus obtain radical extension K_{m+1} of K_m so that the Galois group of $x_1, \ldots, x_n$ over K_{m+1} is given by G_{m+1}.

At this point, we continue with G_{m+1} as we argued with G_m: If G_{m+1} is trivial, then $x_1, \ldots, x_n$ are already contained in K_{m+1}, and we are done. Otherwise, G_{m+1} has a normal subgroup G_{m+2} of the index of a prime number and so on. This process must eventually come to a halt, as the orders of G_m, G_{m+1}, G_{m+2} and so on are strictly monotonically decreasing. □

An equation is solvable if and only if its Galois group is solvable.

6.8 Solving the Cubic and the Quartic Equations

According to Theorem 6.5, a polynomial equation is solvable if and only if its Galois group is solvable. We have already shown that the symmetric group S_n, where n is greater or equal to 5, is not solvable and then found an example of a fifth-degree polynomial whose roots are not given by radical expressions. To find examples or example classes for

solvable polynomials, we need to look for solvable Galois groups. To this end, we first show:

Proposition 6.11 *The symmetric group* S_4 *in four letters is solvable. More precisely,* S_4 *has the normal series*

$$S_4 \supseteq A_4 \supseteq V_4 \supseteq \tilde{C}_2 \supseteq \{\mathrm{id}\}.$$

Here,

$$V_4 := \{\mathrm{id}, (1,2)\circ(3,4), (1,3)\circ(2,4), (1,4)\circ(2,3)\}$$

is the so-called Klein[3] four-group*, and* $\tilde{C}_2 := \{\mathrm{id}, (1,2)\circ(3,4)\}$ *is a cyclic group with two elements.*

The Klein four-group is indeed a subgroup of S_4, as composition and inversion obviously do not lead out of V_4 and the identity is in V_4.

Since we need to prove that certain subgroups are normal subgroups, here is a lemma:

Lemma 6.14 *Let* H *be a subgroup of index* 2 *in a group* G*. Then* H *is a normal subgroup of* G*.*

Proof of Theorem 6.14 We make a preliminary consideration: We claim that for any permutation σ from G, σ^2 is in H. If σ is in H, this is obviously correct. Otherwise, both σ and σ^{-1} are not in H. But since there are only two congruence classes modulo H, σ and σ^{-1} must be congruent to each other modulo H. This is equivalent to $(\sigma^{-1})^{-1}\circ\sigma = \sigma^2$ being in H.

Let's now prove the lemma: Let σ be from G and τ be from H. We need to show that $\sigma\circ\tau\circ\sigma^{-1}$ is also from H. For this, we write

$$\sigma\circ\tau\circ\sigma^{-1} = \sigma\circ\tau\circ\sigma\circ\tau\circ\tau^{-1}\circ\sigma^{-2} = (\sigma\circ\tau)^2\circ\tau^{-1}\circ\sigma^{-2}.$$

The individual terms are from H according to our preliminary consideration, thus also the composition of all. □

Subgroups of index 2 are always normal subgroups.

Example 6.7 As an example, we find that the alternating group A_n for any natural number $n \geq 2$ is a normal subgroup in S_n, because A_n has index 2 in S_n: The transposition $(1, 2)$

is not in A_n as it is an odd permutation. Thus, there are at least two congruence classes modulo A_n in S_n.

If σ is any permutation, then σ modulo A_n is congruent to the identical permutation id if σ is even. If σ is otherwise an odd permutation, then $(1, 2) \circ \sigma$ is an even permutation, thus in A_n; that means, $\sigma = (1, 2) \circ ((1, 2) \circ \sigma)$ is congruent to $(1, 2)$ modulo A_n. Thus, there are exactly two congruence classes modulo A_n; that means, the index of A_n in S_n is two.

Proof of Proposition 6.11 The proof of the theorem is divided into showing that A_4 is a normal subgroup of index of a prime number in S_4, that V_4 is a normal subgroup of index of a prime number in A_4, that $\tilde{C}_2$ is a normal subgroup of index of a prime number in V_4 and that $\{\mathrm{id}\}$ is a normal subgroup of index of a prime number in $\tilde{C}_2$. For A_4 we have already seen in Example 6.7 that this subgroup is a normal subgroup of index 2 in S_4. In particular, A_4 has twelve elements. Furthermore, V_4 has four, $\tilde{C}_2$ two and $\{\mathrm{id}\}$ one element. According to the Lagrange's theorem, Proposition 5.5, it follows for the indices that

$$[A_4 : V_4] = 3, \qquad [V_4 : \tilde{C}_2] = 2, \qquad [\tilde{C}_2 : \{\mathrm{id}\}] = 2.$$

In view of Lemma 6.14 we only need to show that V_4 in A_4 is a normal subgroup. This follows from the following more general Lemma 6.15, because if a subgroup of A_4 is a normal subgroup in the larger group S_4, then also in A_4. □

The symmetric group S_4 (and thus all S_n and A_n with $n \leq 4$) is solvable.

Lemma 6.15 *The Klein four-group* V_4 *is a normal subgroup in the symmetric group* S_4 *on four letters.*

Proof The Klein four-group contains exactly all 4-letter permutations of the form $(i, j) \circ (k, \ell)$, where $i, j, k, l \in \{1, 2, 3, 4\}$ are pairwise different. If σ is any permutation in S_4, it follows that

$$\sigma \circ (i, j) \circ (k, \ell) \circ \sigma^{-1} = (\sigma(i), \sigma(j)) \circ (\sigma(k), \sigma(\ell))$$

is again an element of the Klein four-group. □

We can consider the symmetric group S_3 on three letters as a subgroup of the symmetric group S_4 on four letters by considering a three-letter permutation as a four-letter permutation that holds the last letter 4 fixed. Since S_4 is solvable according to Proposition 6.11 and

subgroups of solvable groups are solvable according to Proposition 6.10, the solvability of S_3 follows.

But we can also directly specify a normal series, namely

$$S_3 \supseteq A_3 \supseteq \{\mathrm{id}\}.$$

Since the Galois group of the roots of a separable polynomial of degree less than or equal to 4 over the rational numbers is a subgroup of a symmetric group on at most 4 letters and is therefore solvable, polynomial equations of degree at most four are solvable by radicals. This is the reason for the existence of the formulas mentioned in the introduction for the solutions of equations of the third and fourth degree.

There are solution formulas for equations of degree 3 and 4.

At the end of this chapter and thus at the end of this introductory book into the Galois theory, we want to derive these formulas.

For this, we first consider a general normalized polynomial equation

$$X^n + a_{n-1}X^{n-1} + \cdots + a_1 X + a_0 = 0 \tag{6.12}$$

with coefficients $a_0, \ldots, a_{n-1}$ in a number field K.

The left side of the equation can be transformed as follows according to the binomial theorem:

$$\begin{aligned} &X^n + a_{n-1}X^{n-1} + a_{n-2}X^{n-2} + \cdots + a_1 X + a_0 \\ &\quad = \left(X + \frac{a_{n-1}}{n}\right)^n + \left(a_{n-2} - \binom{n}{2}\left(\frac{a_{n-1}}{n}\right)^2\right) X^{n-2} + \cdots \\ &\qquad + \left(a_1 - \binom{n}{n-1}\left(\frac{a_{n-1}}{n}\right)^{n-1}\right) X + \cdots + \left(a_0 - \binom{n}{n}\left(\frac{a_{n-1}}{n}\right)^n\right). \end{aligned}$$

So if we set $Y := X + \frac{a_{n-1}}{n}$ in (6.12), we get an equation of the form

$$Y^n + b_{n-2}Y^{n-2} + \cdots + b_1 Y + b_0 = 0, \tag{6.13}$$

where the $b_0, \ldots, b_{n-2}$ are numbers from K. A number x is a solution of (6.12) if and only if $y = x + \frac{a_{n-1}}{n}$ is a solution of (6.13).

To solve the Eq. (6.12), it is therefore sufficient, to provide a solution method for Eq. (6.13). We can therefore assume in the following derivation of solution formulas that

our polynomial equations are in *depressed form*, i.e., the $(n-1)$-th coefficient vanishes when n is the degree of the polynomial equation.

The reason why we make this reduction is that the following solution formulas are significantly simplified when they are formulated only in terms of the coefficients of the depressed equation.

In the case $n = 1$, only the equation $X = 0$ remains, which has $x = 0$ as its only solution. In the case $n = 2$, we have equations of the form $X^2 + a = 0$, where a is from K. A number x is a solution of this equation if and only if x is a root of a, that is

$$x = \sqrt{-a},$$

where both choices of the square root are allowed. After the preliminary considerations to bring a polynomial equation to a depressed form, the case of quadratic equations has become quite simple and is fully understood. The interesting cases are therefore $n = 3$ and $n = 4$.

Let's start with a general depressed cubic equation, i.e., an equation of the form

$$X^3 + pX + q = 0, \tag{6.14}$$

where p and q are two numbers from K.

For the moment, we assume that the polynomial $X^3 + pX + q$ is separable, i.e., that (6.14) has three different roots x_1, x_2, and x_3 in the algebraic numbers and that we can particularly apply the Galois theory. We further assume for the moment that the Galois group of x_1, x_2, and x_3 over K is the full symmetric group S_3.

If we go through the proof of Theorem 6.5, we see that we should first extend K by a primitive second and a third unit root (since 2 and 3 are the prime divisors of $6 = [S_3 : 1]$). A primitive second unit root is -1, which is already contained in K. A primitive third unit root is the solution of $X^2 + X + 1 = 0$, so

$$\omega = -\frac{1}{2} - \frac{1}{2}\sqrt{-3}.$$

From now on, we continue to calculate in $K(\omega)$.

In principle, the Galois group of x_1, x_2, and x_3 over $K(\omega)$ could have become smaller, but we want to assume that it is still the full symmetric group S_3.

The group A_3 is a normal subgroup in S_3 of index 2. In the proof of Theorem 6.5, we use Lemma 6.13 at this point to find an element a such that the Galois group of $x_1, \ldots, x_3$ over $K(\omega, \sqrt{a})$ is reduced to A_3. If we look at the proof of Lemma 6.13, we first need to find an element δ in $K(\omega, x_1, x_2, x_3)$ such that the Galois group of x_1, x_2, and x_3 over $K(\omega, \delta)$ is exactly A_3. According to the considerations we have already made about the Galois group of a general irreducible polynomial of third degree, this is given by

$$\delta = (x_1 - x_2) \cdot (x_2 - x_3) \cdot (x_3 - x_1).$$

It is $\delta^2 = \Delta = -4p^3 - 27q^2$ the discriminant of $X^3 + pX + q$ and thus a number from K. Therefore, we obviously do not need to further apply Lemma 6.13 and find that the Galois group of x_1, x_2 and x_3 over $K(\omega, \sqrt{\Delta})$ reduces to A_3.

The set {id} is a normal subgroup of index 3 in A_3. Since the only element of A_3 that fixes x_1 is the identity, we find that the Galois group of x_1, x_2 and x_3 over $K(\omega, \sqrt{\Delta}, x_1)$ is given by the trivial group {id}. We set $\sigma := (1, 2, 3)$. This is a symmetry, which is contained in A_3, but not in {id}.

According to the proof of Lemma 6.13, we next consider the following numbers:

$$\begin{aligned} y_1 &:= x_1 + \omega\sigma \cdot x_1 + \omega^2\sigma^2 \cdot x_1 = x_1 + \omega x_2 + \omega^2 x_3, \\ y_2 &:= x_1 + \omega^2\sigma \cdot x_1 + \omega^4\sigma^2 \cdot x_1 = x_1 + \omega^2 x_2 + \omega x_3, \\ y_3 &:= x_1 + \omega^3\sigma \cdot x_1 + \omega^6\sigma^2 \cdot x_1 = x_1 + x_2 + x_3 = 0, \end{aligned}$$

where $x_1+x_2+x_3 = 0$ is due to the fact that the equation $X^3+pX+q = 0$ has no quadratic term (whose coefficient according to Vieta's theorem is exactly $-x_1 - x_2 - x_3$). Since $y_1+y_2 = y_1+y_2+y_3 = 3x_1$, either y_1 or y_2 must be non-zero (after all, $x_1 \neq 0$, as it is non-rational since we have assumed the Galois group of $X^3 + pX + q$ to be the full symmetric group). Without loss of generality, we assume that $u := \frac{y_1}{3}$ is non-zero. According to the proof of Lemma 6.13, u^3 is invariant under the Galois group A_3, and indeed a longer calculation (which we can verify by expanding both sides using $x_1 + x_2 + x_3 = 0$), yields

$$y_1^3 = (x_1 + \omega x_2 + \omega^2 x_3)^3 = -\frac{27}{2}q - 3\left(\omega + \frac{1}{2}\right)\delta,$$

so

$$u^3 = -\frac{q}{2} + \frac{1}{2}\sqrt{-\frac{\Delta}{27}}.$$

In particular, we obtain

$$u = \sqrt[3]{-\frac{q}{2} + \frac{1}{2}\sqrt{-\frac{\Delta}{27}}},$$

thus a radical expression over the rational numbers.

According to the proof method of Theorem 6.5, the Galois group of x_1, x_2 and x_3 over $K(\omega, \delta, u)$ is the trivial group, in particular, the roots x_1, x_2 and x_3 must therefore be rational in ω, δ and u. And indeed,

$$y_1 \cdot y_2 = x_1^2 + x_2^2 + x_3^2 - x_1x_2 - x_2x_3 - x_1x_3 = (x_1+x_2+x_3)^2 - 3(x_1x_2 + x_1x_3 + x_2x_3) = -3p.$$

If we set $v := \frac{y_2}{3}$, it follows that $v = -\frac{p}{y_1} = -\frac{p}{3u}$, thus

$$x_1 = \frac{1}{3}(y_1 + y_2) = u + v = \sqrt[3]{-\frac{q}{2} + \frac{1}{2}\sqrt{-\frac{\Delta}{27}}} - \frac{p}{3\sqrt[3]{-\frac{q}{2} + \frac{1}{2}\sqrt{-\frac{\Delta}{27}}}},$$

and this is obviously a radical expression in the coefficients p and q of the original equation.

In the considerations above, we have chosen a primitive unit root $-\frac{1}{2} - \frac{1}{2}\sqrt{-3}$. If we had chosen the Galois conjugate, $-\frac{1}{2} + \frac{1}{2}\sqrt{-3}$, i.e., ω^2 instead of ω, then u and v would swap in our equations above, that is, we have

$$v^3 = -\frac{q}{2} - \frac{1}{2}\sqrt{-\frac{\Delta}{27}}.$$

Because $uv = -\frac{p}{3}$, in $K(\omega, \delta, u)$ only one of the up to three roots of

$$\sqrt[3]{-\frac{q}{2} - \frac{1}{2}\sqrt{-\frac{\Delta}{27}}}$$

stands for v.

In summary, we can finally give the following solution formula for a third degree equation:

Theorem 6.6 *Let*

$$X^3 + pX + q = 0 \tag{6.15}$$

be a (depressed) cubic equation over a number field K. Let $\Delta := -4p^3 - 27q^2$ be its discriminant. Further let

$$u = \sqrt[3]{-\frac{q}{2} + \frac{1}{2}\sqrt{-\frac{\Delta}{27}}}, \qquad v = \sqrt[3]{-\frac{q}{2} - \frac{1}{2}\sqrt{-\frac{\Delta}{27}}},$$

where we choose one of the two third roots so that $u \cdot v = -\frac{p}{3}$. Then the three solutions (with multiplicities) of (6.15) are given by

$$x = u + v.$$

(The possible choices of the third root for u or v provide the three solutions.)

Proof We have already proven the theorem in our considerations for the case that $X^3 + pX + q$ is separable and that the Galois group of this polynomial over $K(\omega)$, where ω is a primitive third unit root, is the full symmetric group. If we go through the above statements about u and v without considering the Galois groups (that is the formulas that connect u and v with the solutions x_1, x_2 and x_3) once again, we see that they remain correct for any choices of p and q, which proves the theorem also in the general case. □

Finally, we want to provide the solution formula for the general depressed equation of fourth degree

$$X^4 + pX^2 + qX + r = 0, \tag{6.16}$$

where we again assume that p, q and r are numbers from a number field K. We could again proceed completely analogous to the proof of Theorem 6.5. Then, however, the derivation of the solution formula would become quite lengthy, so it is worth abbreviating in between and including our knowledge about the solutions of a cubic equation:

We want to assume, analogous to the above procedure, that the solutions $x_1, \ldots, x_4$ of (6.16) are pairwise different and that their Galois group over K is given by the full symmetric group S_4.

Let

$$\delta := (x_1 - x_2) \cdot (x_1 - x_3) \cdot (x_1 - x_4) \cdot (x_2 - x_3) \cdot (x_2 - x_4) \cdot (x_3 - x_4).$$

be the product over all six differences $(x_i - x_j)$ with $i < j$. Then the Galois group of $x_1, \ldots, x_4$ over $K(\delta)$ is the alternating group A_4 (because δ is invariant under the A_4, but not under the full S_4). Furthermore,

$$\delta^2 = \Delta = 256r^3 - 128p^2r^2 - 27q^4 + 4p^3q^2 - 16p^4r - 144pq^2r$$

is the discriminant (whose explicit form given here can be derived as in the case of the depressed cubic), which we will actually not need in the following, as we will see).

In the next step, we want to reduce the Galois group of $x_1, \ldots, x_4$ to the Klein four-group V_4 within the alternating group. The structure of the elements within the Klein four-group suggests considering the following numbers:

$$y_1 = (x_1 + x_2) \cdot (x_3 + x_4), \quad y_2 = (x_1 + x_4) \cdot (x_2 + x_3), \quad y_3 = (x_1 + x_3) \cdot (x_2 + x_4).$$

All these elements are invariant under the symmetries of the V_4. On the other hand, they are not invariant under the full alternating group A_4: Let $\sigma := (1, 2, 3)$, a symmetry, which is in the alternating group, but not in the Klein four-group.

Then $\sigma \cdot y_1 = y_2$, $\sigma \cdot y_2 = y_3$ and $\sigma \cdot y_3 = y_1$. For the difference of y_1 and $\sigma \cdot y_1$ we have

$$y_1 - \sigma \cdot y_1 = y_1 - y_2 = (x_2 - x_4) \cdot (x_3 - x_1).$$

The number on the right side is not zero, since we have assumed the four roots $x_1, \ldots, x_4$ to be different under our separability assumption. Thus, y_1 is not invariant under σ, and in particular, y_1 is different from zero.

Since there is no proper subgroup of the alternating group A_4 in which the Klein four-group V_4 is properly contained, it follows that the Galois group of $x_1, \ldots, x_4$ over $K(\sqrt{\Delta}, y_1)$ is the Klein four-group V_4. By the same reasoning, it follows that y_2 and y_3 are also non-zero and not invariant under all symmetries of the alternating group.

To represent y_1, y_2 and y_3, we note that the polynomial

$$g(X) \coloneqq (X - y_1) \cdot (X - y_2) \cdot (X - y_3)$$

is invariant under the permutation σ, so overall under all symmetries from A_4 (since A_4 is generated by σ and V_4). If we apply the symmetry $\tau \coloneqq (1, 2)$ to $g(X)$, we get $\tau \cdot g(X) = g(X)$ because $\tau \cdot y_1 = y_1$, $\tau \cdot y_2 = y_3$ and $\tau \cdot y_3 = y_2$.

Consequently, the polynomial $g(X)$ (its coefficients) are invariant under all symmetries from the symmetric group S_4 since an odd transposition generates the entire symmetric group from the alternating group A_4 (after all, we have already seen that every permutation not in A_4 can be written as a composition of a fixed odd with an even permutation).

According to the general theory, the coefficients of $g(X)$ must all be in K. And indeed, a brief calculation yields

$$g(X) = X^3 - 2pX^2 + (p^2 - 4r)X + q^2.$$

This polynomial is classically called the *cubic resolvent* of the originally given Eq. (6.16). The roots of $g(X)$ are precisely the y_i. We can therefore represent y_1, y_2 and y_3 by radical expressions over K according to Theorem 6.6.

In the next step, we need to adjoin an element to $K(\sqrt{\Delta}, y_1)$ so that the Galois group of V_4 is reduced to $\tilde{C}_2$.

A sensible choice is

$$t_1 \coloneqq x_1 + x_2$$

because $(x_1 + x_2) + (x_3 + x_4) = 0$, $t^2 = (x_1 + x_2)^2 = -y_1$, which means $t_1 = \sqrt{-y_1}$.

In particular, t is not zero. The number t_1 is invariant under the symmetries from $\tilde{C}_2$, but not under all symmetries from V_4, for example $((1, 3) \circ (2, 4)) \cdot t_1 = x_3 + x_4 = -t_1$. Thus, we find that the Galois group of $x_1, \ldots, x_4$ over $K(\sqrt{\Delta}, y_1, \sqrt{-y_1}) = K(\sqrt{\Delta}, \sqrt{-y_1})$ is only $\tilde{C}_2$.

In the last step, we still need to find a number which is not invariant under the nontrivial symmetry $(1,2)\circ(3,4)$ in $\tilde{C}_2$. An obvious choice is x_1.

However, we reach our goal faster if we choose as a number

$$t_2 \coloneqq x_1 + x_4$$

It holds that $t_2^2 = (x_1+x_4)^2 = -y_2$, which means, $t_2 = \sqrt{-y_2}$, and t_2 is not zero. Also, $((1,2)\circ(3,4))\cdot t_2 = x_2 + x_3 = -t_2$. So t_2 is indeed not invariant under $\tilde{C}_2$. Thus, we find that the Galois group of $x_1, \ldots, x_4$ over $K(\sqrt{\Delta}, \sqrt{-y_1}, \sqrt{-y_2})$ is the trivial group $\{\mathrm{id}\}$. Consequently, our sought solution x_1 should be expressible rationally in $\delta = \sqrt{\Delta}$ and the square roots $t_1 = \sqrt{-y_1}$ and $t_2 = \sqrt{-y_2}$:

To arrive at the corresponding formula, we first define

$$t_3 \coloneqq x_1 + x_3.$$

Because of $x_1 + x_2 + x_3 + x_4 = 0$, it is then obviously $x_1 = \frac{1}{2}(t_1 + t_2 + t_3)$. Further, a short calculation (again considering $x_1 + x_2 + x_3 + x_4 = 0$) shows that

$$t_1 \cdot t_2 \cdot t_3 = (x_1 + x_2)\cdot(x_1 + x_4)\cdot(x_1 + x_3) = x_1x_2x_3 + x_1x_2x_4 + x_1x_3x_4 + x_2x_3x_4 = -q,$$

where we have taken the last equality from Vieta's theorem.

Consequently, $t_3 = -\frac{q}{t_1\cdot t_2}$. Thus, we have derived the formula

$$x_1 = \frac{1}{2}\left(\sqrt{-y_1} + \sqrt{-y_2} - \frac{q}{\sqrt{-y_1}\cdot\sqrt{-y_2}}\right).$$

This is a radical expression in the coefficients p, q and r of the original equation. Because $t_3^2 = (x_1 + x_3)^2 = -y_3$, we can also write the last summand as $\sqrt{-y_3}$, where we have to choose the sign of $\sqrt{-y_3}$ in the number field $K(\sqrt{\Delta}, \sqrt{-y_1}, \sqrt{-y_2})$ so that $\sqrt{-y_1}\cdot\sqrt{-y_2}\cdot\sqrt{-y_3} = -q$ because of $t_1t_2t_3 = -q$.

From our considerations, we obtain the solution formula for an equation of fourth degree, which was originally found by Ludovico Ferrari:

Theorem 6.7 *Let*

$$X^4 + p\,X^2 + q\,X + r = 0 \tag{6.17}$$

be a (depressed) quartic equation over a number field K. Let y_1, y_2 and y_3 be the three solutions of its cubic resolvent equation

$$Y^3 - 2p\,Y^2 + (p^2 - 4r)\,Y + q^2 = 0.$$

Further, let

$$t_1 = \sqrt{-y_1}, \qquad t_2 = \sqrt{-y_2}, \qquad t_3 = \sqrt{-y_3}$$

where we choose the sign of one of the three roots so that $t_1 \cdot t_2 \cdot t_3 = -q$. *Then the four solutions with multiplicities of* (6.17) *are given by*

$$x = \frac{1}{2}(t_1 + t_2 + t_3).$$

(The possible choices of the square roots of $-y_1$, $-y_2$ *and* $-y_3$ *provide the four solutions.)*

(The possible six permutations of y_1, y_2 and y_3 do not provide any further solutions.)

Proof We have already proven the theorem for the case that $X^4 + pX^2 + qX + r$ is separable and that the Galois group of this polynomial over K is the full symmetric group. If we go through the above derivation again, we see that the claimed relationships (without considering the explicit form of the Galois groups) remain correct in the general case. □

Summary

- We can investigate Galois groups of roots of polynomials also for **extensions** E of number fields. The larger the extension, the smaller the associated group.
- Conversely, each subgroup of a Galois group corresponds to an extension of the number field that consists precisely of the numbers that are **invariant** under the action of the elements of the subgroup.
- The **fundamental theorem of Galois theory** describes an inclusion-reversing bijection between the subgroups of a Galois group and the (intermediate) extensions.
- An algebraic number is said to be **representable by radicals** if it can be represented by sums, products, and algebraically unique roots, up to Galois conjugacy.
- The **primitive roots of unity**, i.e., the roots of the cyclotomic polynomials, have an explicit representation by radicals.
- A group is called **solvable** if it is given as an ascending sequence of subgroups, each subgroup being of prime index in the next.

(continued)

- If a polynomial is **solvable**, that is the roots of a polynomial can be simultaneously represented by radicals, then the associated Galois group is solvable. The proof uses the fact that the primitive roots of unity can be represented by radicals.
- The symmetric groups S_n with $n \geq 5$ are not solvable. There is a polynomial of degree 5 over the rational numbers, whose Galois group is S_5. Thus, there are **unsolvable** polynomial equations.
- If a Galois group of the roots $x_1, \ldots, x_n$ of a polynomial is solvable, radical expressions for $x_1, \ldots, x_n$ can be constructed from the associated sequence of subgroups. Thus, a polynomial is **solvable if and only if** its Galois group is solvable.
- The Galois groups of polynomials of degree 4 or less are solvable. Thus, there exist **solution formulas** for polynomial equations of degree 4 or less.

Exercises

Relative Galois Groups

6.1 Show that $\sqrt{2} + \sqrt{7}$ and $-\sqrt{2} - \sqrt{7}$ are Galois conjugate over the rational numbers, but not over $\sqrt{7}$.

6.2 Let K be a number field. Show that the relation of being Galois conjugate for sequences of algebraic numbers $y_1, \ldots, y_n$ and $z_1, \ldots, z_m$ over K is indeed an equivalence relation.

6.3 Let $t \in \mathbf{Q}(\sqrt{2})$. Show that t is either a rational number or that $\mathbf{Q}(t) = \mathbf{Q}(\sqrt{2})$. Conclude that $\mathbf{Q}(\sqrt{2})$ has no proper intermediate extension over the rational numbers.

6.4 Let K and L be two number fields, such that K is an intermediate extension of L. Let x be an algebraic number. Show that a Galois conjugate of x over L is also a Galois conjugate of x over K.

6.5 Let $f(X)$ be a normalized separable polynomial over a number field K, and let $x_1, \ldots, x_n$ be its roots. Let y be a number from $K(x_1, \ldots, x_n)$. Show that the Galois group H of $x_1, \ldots, x_n$ over $K(y)$ consists of those permutations σ of the Galois group G of $x_1, \ldots, x_n$ over K under which y remains invariant, that is, for which $\sigma \cdot y = y$ holds.

6.6 Let $f(X)$ and $g(X)$ be two different normalized irreducible polynomials over a number field K. Let y be a root of $g(X)$. Let x_1 and x_2 be two roots of $f(X)$. Show

that x_1 and x_2 are Galois conjugate over $K(y)$ if and only if a permutation σ of the Galois group of the roots of $f(X) \cdot g(X)$ over K exists such that $\sigma \cdot x_1 = x_2$ and $\sigma \cdot y = y$.

6.7 Let $f(X)$ and $g(X)$ be two different normalized irreducible polynomials over a number field K. Let x be a root of $f(X)$ and y be a root of $g(X)$. Let further σ be a permutation of the Galois group G of the roots of $f(X) \cdot g(X)$ over K. Let $\tilde{f}(X)$ be the minimal polynomial of $\sigma \cdot x$ over $K(y)$, and let $\tilde{g}(X)$ be the minimal polynomial of $\sigma^{-1} \cdot y$ over $K(x)$. Let N be the subset of those permutations τ from G such that $\tau \cdot x$ is a root of $\tilde{f}(X)$ and M be the subset of those permutations τ from G such that $\tau \cdot y$ is a root of $\tilde{g}(X)$.

Show that

$$\frac{\deg \tilde{f}}{\deg f} = N = M = \frac{\deg \tilde{g}}{\deg g}.$$

6.8 Let $f(X)$ and $g(X)$ be two different normalized irreducible polynomials over a number field K. Let x be a root of $f(X)$ and y be a root of $g(X)$. Let $f(X) = f_1(X) \cdots f_n(X)$ be the decomposition of $f(X)$ into irreducible factors over $K(y)$, and let $g(X) = g_1(X) \cdots g_m(X)$ be the decomposition of $g(X)$ into irreducible factors over $K(x)$. Show that then $m = n$ holds and that we can arrange the factors such that the ratio $\frac{\deg f_k(X)}{\deg g_k(X)}$ for all $k \in \{1, \ldots, n\}$ is constant.

(This statement is the so-called Dedekind reciprocity theorem of Galois theory.)

The Fundamental Theorem of Galois Theory

6.9 Let $x_1, \ldots, x_n$ be the roots of a separable polynomial with rational coefficients and G their Galois group. Let σ and τ be two symmetries in G. Let further t be a number rational in $x_1, \ldots, x_n$ that is invariant under σ and τ. Show that t is also invariant under id, $\tau \circ \sigma$ and σ^{-1}.

6.10 In the following, we abbreviate a set of variables of the form $X_1, \ldots, X_n$ with $\underline{X}$. Accordingly, for example, $\underline{Y}$ stands for $Y_1, \ldots, Y_n$ or $\underline{X}_i$ for $X_{1i}, \ldots, X_{ni}$. We call a polynomial

$$f(\underline{X}_1, \ldots, \underline{X}_m)$$

with integer coefficients *symmetric* in $\underline{X}_1, \ldots, \underline{X}_m$ if for every m-letter permutation σ it holds that

$$\sigma \cdot f(\underline{X}_1, \ldots, \underline{X}_m) \coloneqq f(\underline{X}_{\sigma(1)}, \ldots, \underline{X}_{\sigma(m)}) = f(\underline{X}_1, \ldots, \underline{X}_m).$$

Let the polynomials

$$e_\lambda(\underline{X}_1, \ldots, \underline{X}_m) \in \mathbf{Z}[\underline{X}_1, \ldots, \underline{X}_m]$$

be those polynomials, such that the equality

$$\sum_{k_1+\cdots+k_n \le m} e_{(k_1,\ldots,k_n)}(\underline{X}_1, \ldots, \underline{X}_m) \cdot T_1^{k_1} \cdots T_n^{k_n} = \prod_{j=1}^{m} (1 + X_{1j} T_1 + \cdots + X_{nj} T_n)$$

holds in $\mathbf{Z}[\underline{X}_1, \ldots, \underline{X}_m, \underline{T}]$, where the sum is over all tuples $\lambda = (k_1, \ldots, k_n)$ of natural numbers with $k_1 + \ldots + k_n \le m$. Show that the $e_\lambda(\underline{X}_1, \ldots, \underline{X}_m)$ are each symmetric in $\underline{X}_1$, $\ldots$, $\underline{X}_m$. These polynomials are called *Mac Mahon's*[4] *generalized symmetric functions*.

6.11 Let $g(\underline{X}_1, \ldots, \underline{X}_m)$ be a polynomial symmetric in $\underline{X}_1, \ldots, \underline{X}_m$ with integer (or rational or algebraic) coefficients. Show analogously to the fundamental theorem about the elementary symmetric functions that $g(\underline{X}_1, \ldots, \underline{X}_m)$ can then be written as a polynomial in Mac Mahon's generalized symmetric functions $e_\lambda(\underline{X}_1, \ldots, \underline{X}_m)$ with integer (or rational or algebraic) coefficients.

6.12 Let K be a number field. Let $x_1, \ldots, x_n$ be the roots of a separable polynomial $f(X)$ over K. Let $H = \{\sigma_1, \ldots, \sigma_m\}$ be a subgroup of the Galois group G of $x_1, \ldots, x_n$ over K. Show that

$$K(x_1, \ldots, x_n)^H = K(e_\lambda(\sigma_1 \cdot \underline{x}, \ldots, \sigma_m \cdot \underline{x})),$$

where λ runs over all tuples $(k_1, \ldots, k_n)$ of natural numbers with $k_1 + \cdots + k_n \le m$.

6.13 Determine the Galois group of the roots $x_1, \ldots, x_4$ of the polynomial $X^4 + 1$ over the rational numbers, their subgroups and the intermediate extensions of $\mathbf{Q}(x_1, \ldots, x_4)$ over $\mathbf{Q}$ corresponding to these subgroups, according to theorem 6.1.

6.14 Determine the Galois group over the rational numbers of the roots $x_1, \ldots, x_6$ of the polynomial $X^6 - 2X^3 - 1$ over the rational numbers, their subgroups and the intermediate extensions of $\mathbf{Q}(x_1, \ldots, x_6)$ over $\mathbf{Q}$ corresponding to these subgroups, according to theorem 6.1.

6.15 Determine explicitly all subgroups of the Dieder group D_4 and illustrate them using their action on the square in space.

6.16 Determine explicitly the symmetry group of a regular n-gon in space, the so-called *Dieder group* D_n. Show that this group can be generated by two elements and has a total of $2n$ elements.

Algebraically Unique Roots

6.17 Give a number field in which 7 does not have an algebraically unique 5-th root and in which $X^5 - 7$ does not decompose into linear factors.

6.18 Specify a number field in which 7 does not have an algebraically unique 5-th root and in which $X^5 - 7$ decomposes into linear factors.

6.19 Let p be a prime number. Let a be a number from a number field K. Show that $X^p - a$ is either irreducible over K or that at least one linear factor appears in the prime factor decomposition of $X^p - a$ over K.

6.20 Give an example that Lemma 6.2 becomes false if we do not assume that p is a prime number.

6.21 Let p be a prime number. Let b be a number from a number field K. Show that a primitive p-th root of unity exists in K if and only if the polynomial $X^p - b^p$ decomposes into linear factors over K.

Radical Representations of Primitive Roots of Unity

6.22 Let $x_1, \ldots, x_4$ be as in the derivation of formula (6.7). Show that $x_2^2 = 5$ and $x_3^4 = -20\sqrt{-1} - 15$. Why is it problematic to specify a fifth primitive root of unity by the radical expression

$$\frac{1}{4}\left(-1 + \sqrt[4]{20\sqrt{-1} - 15} + \sqrt{5} + \sqrt[4]{-20\sqrt{-1} - 15}\right)?$$

6.23 Let the sequence of algebraic numbers $z_1, \ldots, z_n$ satisfy the equations

$$z_1^5 = 2, \qquad z_2^3 = z_1^2 - 2z_1 + 4, \qquad z_3^2 = z_2^2 - z_1^2 z_2 + 7.$$

Give a radical expression in the form of (6.8) for $x = z_2 + z_3$.

6.24 Let K be a number field. Let $z_1, \ldots, z_n$ be algebraic numbers and $f_1(X)$, $f_2(X, Z_1)$, $f_3(X, Z_1, Z_2)$, $\ldots$, $f_n(X, Z_1, \ldots, Z_{n-1})$ be polynomials over K such that for all $i \in$

$\{1, \ldots, n\}$ the polynomial $f_i(X, z_1, \ldots, z_{i-1})$ is the minimal polynomial of z_i over $K(z_1, \ldots, z_{i-1})$. Finally, let $x := g(z_1, \ldots, z_n)$, where $g(Z_1, \ldots, Z_n)$ is a polynomial over K. Show that x, up to Galois conjugation over K, depends only on the polynomials $f_1, \ldots, f_n$ and $g(Z_1, \ldots, Z_n)$.

6.25 Let x be expressible by radicals. Let x' be Galois conjugate to x. Show that x' is also expressible by radicals, and indeed by the same radical expression as x.

6.26 Let x be an algebraic number whose Galois conjugates are given by $x_1 = x, x_2, \ldots, x_n$. Show that x is constructible if and only if the degree of a primitive element to $x_1, \ldots, x_n$ over the rational numbers is a power of two.

6.27 Let n be a positive natural number. Show that a primitive n-th root of unity can be expressed by radicals whose exponents are at most the maximum of 2 and $\frac{n-1}{2}$.

6.28 Let p be a prime number. Let ζ be a primitive p-th root of unity, and let θ be a primitive $(p-1)$-th root of unity. Show that $\zeta\theta$ is a primitive $(p\,(p-1))$-th root of unity.

6.29 Let p be a prime number. Let ζ be a primitive p-th root of unity, and let θ be a primitive $(p-1)$-th root of unity. Let g be a primitive root modulo p. Then we set

$$x_i := \zeta + \theta^i \zeta^g + \cdots + \theta^{(p-2)i} \zeta^{g^{p-2}}$$

for all $i \in \{0, \ldots, p-2\}$. Let $d \in \{1, \ldots, p-2\}$. Show that

$$\zeta^d = \frac{1}{p-1} \left(\sum_{i=0}^{p-2} \theta^{-ji} x_i \right),$$

when j is an integer with $g^j \equiv d$ modulo p.

6.30 Give a radical expression for a primitive seventh root of unity.

Non-solvable Galois Groups

6.31 Let $f(X)$ be a normalized irreducible polynomial with rational coefficients. Let one of the roots x of $f(X)$ be expressible by radicals. Show that $f(X)$ is then already solvable.

6.32 Let K be a number field of degree 2 over the rational numbers. Show that K is a radical extension with exponent 2.

6.33 Give an example of a normalized separable polynomial $f(X)$ over the rational numbers such that the Galois group of the roots of $f(X)$ over $\mathbf{Q}(\sqrt[3]{2})$ is equal to the Galois group of the roots of $f(X)$ over the rational numbers.

6.34 Give an example of a normalized separable polynomial $f(X)$ over the rational numbers such that the Galois group of the roots of $f(X)$ over $\mathbf{Q}(\sqrt[3]{2})$ is a subgroup of index 3 in the Galois group of the roots of $f(X)$ over the rational numbers. (Is this subgroup a normal subgroup?)

6.35 Show that every nontrivial group has at least two normal subgroups.

6.36 Show that the center of a group G is a normal subgroup in itself.

6.37 Show that the dihedral group D_4 is a subgroup, but not a normal subgroup of index 3 in the symmetric group S_4.

6.38 Let N be a normal subgroup of order two in a group G. Show that N lies in the center of G.

6.39 Let N be a normal subgroup in a group G. Let H be another subgroup of G. Let the order of H be coprime to the index of N in G. Show that then H is contained in N.

6.40 Let $x_1, \ldots, x_n$ be the roots of a normalized separable polynomial over a number field K. Let $g(X)$ be another normalized separable polynomial over K all of whose roots $y_1, \ldots, y_m$ already lie in $K(x_1, \ldots, x_n)$. Show that the Galois group N of $x_1, \ldots, x_n$ over $K(y_1, \ldots, y_m)$ is a normal subgroup in the Galois group G of $x_1, \ldots, x_n$ over K.

6.41 Let p be a prime number. Let $f(X)$ be a normalized irreducible polynomial over a number field K that contains a primitive p-th root of unity. Let L over K be a radical extension of exponent p. Let the Galois group of the roots of $f(X)$ over K have order p. Show that either $f(X)$ is irreducible over L or that $f(X)$ over L splits into linear factors.

A Non-solvable Equation of the Fifth Degree

6.42 Where did we use in the proof of Lemma 6.9 that $n \geq 5$?

6.43 Is the symmetric group S_5 a simple group?

6.44 Show that the equation $X^5 - 23X + 1 = 0$ is not solvable over the rational numbers.

6.45 Let p be a prime number. Show that for all positive natural numbers k, the equation $X^5 - kpX + p = 0$ is not solvable.

6.46 Show that for every natural number n with $n \geq 5$, there is a normalized separable polynomial of degree n over the rational numbers that is not solvable.

6.47 Give an example of a normalized irreducible polynomial $f(X)$ of degree five over the rational numbers such that the equation $f(X) = 0$ is solvable.

6.48 Give an example of why Lemma 6.12 becomes false if p is not assumed to be a prime number.

6.49 Let p be a prime number. Let G be a subgroup of the symmetric group S_p on p elements. Suppose G contains a p-cycle and a transposition. Show that G is already the full symmetric group S_p.

6.50 Let $f(X)$ be a normalized separable polynomial with rational coefficients that has at least one non-real root. Show that the Galois group of the roots $x_1, \ldots, x_n$ of $f(X)$ contains at least one element of order 2.

6.51 Let $f(X)$ be a normalized irreducible polynomial of degree a prime number p over a number field K. If $f(X)$ is solvable over K, show that the roots $x_1, \ldots, x_n$ of $f(X)$ can be arranged in such a way that for each element σ of the Galois group of $x_1, \ldots, x_n$ over K there exist integers r and s such that $\sigma(i) \equiv r \cdot i + s$ modulo p for all $i \in \{1, \ldots, p\}$.

On Solvable Equations

6.52 Let G be a cyclic group. Let H be a subgroup of G. Show that H is a normal subgroup of G.

6.53 Show that every cyclic group is solvable.

6.54 Let the Galois group of the roots $x_1, \ldots, x_n$ of a polynomial $f(X)$ over a number field K be a cyclic group. Show that the equation $f(X) = 0$ is solvable.

6.55 Let $f(X)$ be a normalized irreducible polynomial of degree a prime number p over a number field K. Let all roots of $f(X)$ be expressible rationally in one of the roots of $f(X)$ over K. Show that the Galois group of the roots of $f(X)$ is cyclic, and infer that the equation $f(X) = 0$ is thus solvable.

Solving the Cubic and the Quartic Equations

6.56 Show that all elements of the Klein four-group have order 1 or 2.

6.57 Is the alternating group A_4 solvable?

6.58 Let H be a subgroup of a group G. Let N be a normal subgroup of H. Is N also a normal subgroup of G?

6.59 Why does the order of the three solutions y_1, y_2 and y_3 of the cubic resolvent not matter in the solution formula from Theorem 6.7?

6.60 Let $f(X)$ be a normalized depressed (i.e., not having a cubic term) fourth degree polynomial over a number field K. Let $g(X)$ be its cubic resolvent. Show that the discriminant of $f(X)$ is equal to the discriminant of $g(X)$.

6.61 Let $f(X) = X^4 + pX^2 + qX + r$ be a normalized (depressed) fourth degree polynomial over a number field K. Show that its discriminant is given by

$$\Delta = 256r^3 - 128p^2r^2 - 27q^4 - 4p^3q^2 + 16p^4r + 144pq^2r.$$

6.62 Let $x_1, \ldots, x_4$ be the solutions of a depressed quartic equation over a number field K. Let G be the Galois group of $x_1, \ldots, x_4$ over K. Let H be the Galois group of the solutions y_1, y_2, y_3 of the cubic resolvent equation of the quartic equation. Show that $[H : 1] = [G : G \cap V_4]$.

6.63 Give a radical expression for the solutions of the cubic equation

$$X^3 + 2X^2 - 10X - 20 = 0.$$

Endnotes

[1] This means that given the polynomial $X^p - a$ we can (constructively) decide whether it is irreducible or not.

[2] Michel Rolle, 1652–1719, French mathematician.

[3] Felix Christian Klein, 1849–1925, German mathematician.

[4] Percy Alexander Mac Mahon, 1854–1929, British mathematician.

Constructive Mathematics

A

The existence of an object is only justified by its construction.

Abstract

In constructive mathematics, the existence of an object can only be proven by providing a construction of the same. In contrast, in classical logic one is allowed to prove the existence of an object by assuming its non-existence and leading this to a contradiction.

A constructive proof of existence contains an algorithm to compute the object whose existence is claimed. On the other hand, if only a classical proof of a statement of the form "The polynomial $f(X)$ has a real root x." is given, we may still do not know how to find such a root x.

Every constructive proof is in particular a classical proof, but not vice versa. So if we manage to prove a theorem constructively, it is not only classically correct, but we can also be sure that it also has an algorithmic content.

Classical proofs are done within the framework of classical logic. We cannot use this unrestrictedly for constructive proofs, because the classical law of excluded middle is not constructive. But that is, in some sense, the only restriction. Without the law of excluded middle, classical logic becomes intuitionistic, with the help of which we can do proofs in constructive mathematics.

The classical fundamental theorem of algebra states that every non-constant polynomial over the complex numbers has a complex root. In this book, we have shown this statement constructively for such polynomials whose coefficients are algebraic numbers. And indeed, the classical statement is constructively false. We outline in this appendix that there can be no constructive proof for the existence of roots of arbitrary non-constant polynomials with complex coefficients.

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A.1 The Law of Excluded Middle

In classical logic, the law of excluded middle, states that for every statement φ the statement $\varphi \vee \neg\varphi$, i.e., the statement "φ or not φ", is provable.

However, the content of this law is empty in a constructive sense. Consider, for example, a face-down deck of cards . Let φ be the statement that the top card is an ace. We can then ask whether the top card is an ace or not, i.e., $\varphi \vee \neg\varphi$. According to the law of excluded middle, this statement is true. But from a practical point of view, this doesn't help much, because we can neither infer that the top card is an ace nor that the top card is not an ace.

The law of excluded middle is also the basis of the proof by contradiction: To prove a statement φ, we assume its opposite $\neg\varphi$. Then we try to lead this to a contradiction. If we succeed, $\neg\varphi$ cannot be true. Due to $\neg\varphi \vee \varphi$, φ must therefore be true.

The difference between classical and constructive mathematics can be particularly well illustrated by the following example. We consider the following statement:

> There exist two positive irrational numbers a and b such that a^b is rational.

Classically, there is a short proof using the law of excluded middle. We consider the number $\sqrt{2}^{\sqrt{2}}$. If this number is rational, we have proven the above statement because then we can set $a = b = \sqrt{2}$, as $\sqrt{2}$ is known to be irrational.

Otherwise (according to the law of excluded middle!), the number $\sqrt{2}^{\sqrt{2}}$ is irrational; we can then set $a = \sqrt{2}^{\sqrt{2}}$ and $b = \sqrt{2}$ because then,

$$a^b = (\sqrt{2}^{\sqrt{2}})^{\sqrt{2}} = \sqrt{2}^{\sqrt{2}\cdot\sqrt{2}} = \sqrt{2}^2 = 2$$

is a rational number.

From the standpoint of constructive mathematics, however, this is not a proof; after all, this proof does not allow us to specify a pair a, b of positive irrational numbers such that a^b is irrational.

(According to a proof by Rodion Kuzmin,[1] the *Gelfond*[2]*–Schneider*[3] *constant* $2^{\sqrt{2}}$ is transcendental, and therefore irrational, which is why $a = 2^{\sqrt{2}}$ and $b = \sqrt{2}$ is a correct choice.)

A.2 Intuitionistic Logic

The logical operators, i.e., the logical connectives, allow us to combine propositional formulas into further propositional formulas. Let φ and ψ be propositional formulas. We can combine these into a *conjunction* $\varphi \wedge \psi$, into a *disjunction* $\varphi \vee \psi$ and into an *implication* $\varphi \implies \psi$.

If x is a (term) variable that appears free in the propositional formula $\varphi(x)$, we can further form the *universal quantification* $\forall x : \varphi(x)$ and the *existential quantification* $\exists x : \varphi(x)$.

Finally, there is the *falsum* $\bot$.

We can obtain further propositional formulas by multiple application of these operators. So the *negation* $\neg\varphi$ is defined as $\varphi \implies \bot$, the *verum* $\top$ as $\bot \implies \bot$ and the *equivalence* $\varphi \Leftrightarrow \psi$ as $(\varphi \implies \psi) \wedge (\psi \implies \varphi)$.

The rules of logic state what a proof of propositional formulas formed by operators is. Let's start with the *intuitionistic logic*:

A proof of a conjunction $\varphi \wedge \psi$ is the specification of a proof of φ and a proof of ψ.

A proof of a disjunction $\varphi \vee \psi$ is the specification of a proof of φ or the specification of a proof of ψ and the specification of which case occurs.

A proof of an implication $\varphi \implies \psi$ is the specification of a proof of ψ, whenever a proof of φ is given.

A proof of a universal quantification $\forall x : \varphi(x)$ is the specification of a proof of $\varphi(t)$ for every term t.

A proof of an existential quantification $\exists x : \varphi(x)$ is the specification of a term t and a proof for the propositional formula $\varphi(t)$.

The falsum has no proof.

Classical logic also has all these proof rules; only one more proof rule is added:

For every propositional formula φ, there is a proof of $\varphi \vee \neg\varphi$.

A.3 The Fundamental Theorem of Algebra in Constructive Mathematics

Classically, the fundamental theorem of algebra states the following: If $f(X)$ is a non-constant polynomial with complex coefficients, then there is a complex number w with $f(w) = 0$. In the first chapter of this book, we have given a constructive proof for polynomials $f(X)$ with algebraic coefficients; the root w is then automatically an algebraic number.

In this section, we want to sketch an argument for why the more general fundamental theorem for arbitrary non-constant polynomials over the complex numbers is constructively unprovable; in other words, why there can be no algorithm that computes, for every non-constant complex polynomial $f(X)$, a complex number w with $f(w) = 0$.

We need a few terms from analysis for this argument: Let Y be a topological space (for our argument we can assume $Y = \mathbf{C}$). We want to develop an interpretation of the language of logic *over* Y. To avoid confusion with the usual interpretation, we use the adjective *local* when we talk about the interpretation over Y.

Let's start with the concept of a term or a local term. Let U be an open subset of Y. Then a *(local) term* t over U is simply a (term-valued) function with domain U. If $V \subseteq U$ is an open subset, then we denote by $t|V$ the restriction of t to V.

Next, we consider *local propositions*. These are also defined over open sets U of Y and each make statements about local terms over U. If $V \subseteq U$ is an open subset of U, then we can restrict each local proposition φ over U to a local proposition $\varphi|V$ over V.

We further make the following locality assumption: If $(U_i)_{i\in I}$ is an open cover of U, then a local proposition φ over U is true if and only if for all $i \in I$ the local proposition $\varphi|U_i$ over U_i is true.

We then have to interpret the logical connectives and quantifiers as follows: Let φ and ψ be local propositions over the open set U. Then $\varphi \wedge \psi$ is the proposition that both φ and ψ are true over U. Furthermore, $\varphi \vee \psi$ is the proposition that there exists an open cover $(U_i)_{i\in I}$ of U such that for all $i \in I$ the proposition $\varphi|U_i$ or the proposition $\psi|U_i$ over $i \in I$ is true. The proposition $\varphi \implies \psi$ is true if for every open subset $V \subseteq U$ of U, for which $\varphi|V$ is true, $\psi|V$ over V is also true.

Furthermore, $\forall x\colon \varphi(x)$ is the proposition that for all open subsets $V \subseteq U$ and all terms t over V the proposition $(\varphi|V)(t)$ holds. The proposition $\exists x\colon \varphi(x)$ is true if there exists an open subset $(U_i)_{i\in I}$ of U and for each $i \in I$ a term t_i over U_i such that $(\varphi|U_i)(t_i)$ is true.

Finally, $\bot$ is only true over the empty open set $\emptyset$.

It now turns out that intuitionistic logic has an interpretation in the local propositions defined over Y as described. If there is a derivation of a local proposition φ over U according to the rules of intuitionistic logic, then φ over U is true.

If we replace intuitionistic logic with classical logic, this is no longer the case: Let $Y = \mathbf{R}$, and let φ be the local proposition that the (identical) function x has no zero. According to classical logic, $\varphi \vee \neg\varphi$ would hold, that is, there is an open cover of $\mathbf{R}$ consisting of U and V such that $\varphi|U$ and $\neg\varphi|V$ hold. The proposition $\neg\varphi|V$ is equivalent to $\varphi|V \implies \bot$ and thus means that for every open subset $W \subseteq V$, for which $\varphi|W$ holds, falsehood also holds, that is, φ may only hold on $\emptyset$, that is, $\neg\varphi|V$ is true when the zeros of x in V are dense. However, this is only possible for $V = \emptyset$. Thus $U = Y = \mathbf{R}$ must hold. But then φ holds on $\mathbf{R}$, that is, x has no zero on $\mathbf{R}$, a contradiction.

We define a *local natural number* over an open set U as a locally constant function $f\colon U \to \mathbf{N}_0$, we define its *successor* as the function $f + 1\colon U \to \mathbf{N}_0$ and as *local zero* the constant zero function $0\colon U \to \mathbf{N}_0$, so the properties of the usual natural numbers are transferred mutatis mutandis to the local natural numbers over open sets of Y.

Mutatis mutandis, the constructions of the integers and then the rational numbers from the natural numbers are also transferred. We can therefore speak of *local integers* and *local rational numbers*. It follows that a local integer or rational number is a locally constant function over an open set to $\mathbf{Z}$ or $\mathbf{Q}$, respectively.

A *local real number* x over U is then an object that can be approximated arbitrarily closely by local rational numbers, that is, for each $\varepsilon > 0$ there exists a rational number r, such that x deviates from r by less than ε. (Here, the existence of r is again to be interpreted locally, that is, there exist an open cover $(U_i)_{i\in I}$ and rational numbers r_i over U_i, such that for all $i \in I$ each $x|U_i$ deviates by less than ε from r_i.

A local complex number is again an expression of the form $a + b \cdot \mathrm{i}$, where a and b are local real numbers.

After defining continuity, it turns out that every continuous real-valued function $U \to \mathbf{R}$ defines a local real number over U and that conversely every local real number over U defines a continuous function $U \to \mathbf{R}$. Similarly, we can can equate local complex numbers with continuous functions $U \to \mathbf{C}$.

Now suppose we had a constructive proof for the fundamental theorem of algebra for arbitrary non-constant polynomials with complex coefficients. Since every constructive proof can also be interpreted locally over Y, we would thus have a proof of the following statement:

> Let $f(X)$ be a non-constant polynomial, whose coefficients are continuous complex-valued functions over a topological space Y. Then there exists a continuous complex-valued function w, so that $f(w)$ is the zero function.

That this statement cannot be true, we see from the following example: We choose as the topological space the complex plane $Y = \mathbf{C}$. With $z\colon \mathbf{C} \to \mathbf{C}$ we denote the identical function. Then $X^2 - z$ is a non-constant polynomial, so, according to the above statement, there would be a continuous function $w\colon \mathbf{C} \to \mathbf{C}$ with $w^2 = z$, i.e., a globally defined continuous square root on $\mathbf{C}$. From complex analysis, however, we know that this cannot be the case. (A short argument for this is as follows: According to the theorem on implicitly defined functions, w would be holomorphic outside the origin, and according to the Riemann[4] removable singularity theorem then also on the whole $\mathbf{C}$, since w is continuous at the origin. Differentiating $w^2 = z$ with respect to z yields $2\,w\,w' = 1$. At the origin $z = 0$ we then obtain the contradiction $0 = 1$ due to $w(0) = 0$ (since 0 is the only square root of 0)).

Thus, the fundamental theorem of algebra for non-constant polynomials whose coefficients are continuous complex-valued functions is incorrect. Therefore, there can be no constructive proof for the fundamental theorem for non-constant polynomials with complex-valued coefficients.

The general motto we have exploited here is: If a statement is constructively true, it remains true when the objects continuously vary in families. This property is essentially the reason why constructive reasoning is also of great interest in classical mathematics.

Linear Algebra

B

There is little mathematics that one can do without linear algebra.

Abstract

Galois theory does not completely do without linear algebra, another important subfield of algebra. In the main text of this book, this has become clear because we have used, for example, matrix notation in proofs and talked about determinants.

To make this book fully accessible even to readers who have no prior knowledge of linear algebra beyond solving linear equation systems, we have compiled the necessary background material in this appendix.

In addition, we discuss the theorems from linear algebra that we used in the main part. These are Cramer's rule for the solutions of a linear equation system and the formula for the Vandermonde determinant.

B.1 Matrices

Definition B.1 Let n and m be two natural numbers. An $(n \times m)$-*matrix* A is a rectangular scheme with n rows and m columns, in which a total of $n \cdot m$ numbers a_{ij}, $1 \leq i \leq n$, $1 \leq j \leq m$ are arranged:

$$A = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \dots & a_{nn} \end{pmatrix}.$$

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Depending on the ring the numbers a_{ij} come from, we speak of *integer*, *rational*, *real* or *complex* matrices. We can also form matrices with even more general entries, such as matrices whose entries are polynomials.

Example B.1 The $(n \times n)$-*identity matrix* is the matrix that has 1 on the diagonal and 0 everywhere else:

$$I = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \dots & 0 & 1 \end{pmatrix}$$

Example B.2 Let $A = (a_{ij})$ be an $(n \times n)$-matrix. Then the $(n \times n)$-matrix

$$X \cdot I - A = \begin{pmatrix} X - a_{11} & -a_{12} & \dots & \dots & -a_{1n} \\ -a_{21} & X - a_{22} & \ddots & & \vdots \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ \vdots & & \ddots & X - a_{n-1,n-1} & -a_{n-1,n} \\ -a_{n1} & \dots & \dots & -a_{n,n-1} & X - a_{nn} \end{pmatrix}$$

is the *characteristic matrix* of A. The characteristic matrix is a matrix of polynomials in the variable X.

B.2 Determinants

Definition B.2 Let $A = (a_{ij})$ be an $(n \times n)$-matrix. Then the number

$$0 = \sum_{\sigma \in \mathrm{S}_n} \operatorname{sgn} \sigma \cdot \prod_{i=1}^{n} a_{i\sigma(i)} = a_{11}a_{22}a_{33} \cdots a_{nn} - a_{12}a_{21}a_{33} \cdots a_{nn} + \cdots$$

is called the *determinant* of A. Here, the sum runs over all $n!$ permutations σ of $\{1, \dots, n\}$.

The determinant of A is thus a sum of $n!$ terms. Each term is, except for its sign, an n-fold product of one matrix entry from each row and column.

The signs are chosen exactly so that the determinant changes its sign when two rows or when two columns are swapped. Moreover, the determinant vanishes when two rows or two columns are identical.

Furthermore, the determinant is linear in both its rows and its columns. In particular, the determinant of a matrix that has a row or a column of zeros vanishes. Also, the determinant does not change when a multiple of a row or column is added to another row or column.

So if a row or column of a matrix is a linear combination (that is, a sum of multiples) of the other rows or columns, then the determinant of this matrix vanishes.

Proposition B.1 *Let n be a natural number. Let $x_1, x_2, \ldots, x_n$ be further arbitrary numbers. Then the* Vandermonde determinant *is given by*

$$\det \begin{pmatrix} 1 & \ldots & 1 \\ x_1 & \ldots & x_n \\ x_1^2 & \ldots & x_n^2 \\ \vdots & & \vdots \\ x_1^{n-1} & \ldots & x_n^{n-1} \end{pmatrix} = \prod_{1 \le i < j \le n} (x_j - x_i).$$

Proof It suffices to show the following polynomial equality in the variables $X_1, \ldots, X_n$:

$$\det \begin{pmatrix} 1 & \ldots & 1 \\ X_1 & \ldots & X_n \\ X_1^2 & \ldots & X_n^2 \\ \vdots & & \vdots \\ X_1^{n-1} & \ldots & X_n^{n-1} \end{pmatrix} = \prod_{1 \le i < j \le n} (X_j - X_i).$$

We do this by induction over n. In the case of $n = 0$, the left side is the determinant of the (0×0)-matrix, thus 1. Furthermore, in this case, the right side is the empty product, also 1.

For the induction conclusion on n, we assume that we have already shown the assertion for $n - 1$.

According to the definition of determinants, we can consider the left side of the asserted equality for n as a polynomial of degree $n - 1$ in X_n. The right side is also a polynomial of degree $n - 1$ in X_n. The coefficients of these polynomials in X_n can again be interpreted as rational functions in the variables $X_1, \ldots, X_{n-1}$.

We need to show that the two sides match as polynomials in X_n. To do this, it is sufficient to show that they have the same $n - 1$ roots and the same leading term.

The left side vanishes when we substitute $X_1, \ldots, X_{n-1}$ for X_n because the determinant of a matrix with two identical columns vanishes. The right side obviously also vanishes when $X_1, \ldots, X_{n-1}$ is substituted for X_n.

Thus, we have found the respective $n-1$ roots of both sides, and they match. It remains to calculate the leading coefficient in X_n on both sides. On the left side, this is by definition

of the determinant just

$$\det \begin{pmatrix} 1 & \dots & 1 \\ X_1 & \dots & X_{n-1} \\ X_1^2 & \dots & X_{n-1}^2 \\ \vdots & & \vdots \\ X_1^{n-2} & \dots & X_{n-1}^{n-2} \end{pmatrix},$$

and on the right side

$$\prod_{1 \le i < j \le n-1} (X_j - X_i).$$

By induction hypothesis, these match. □

B.3 Cramer's Rule

The following statement is known as *Cramer's rule*.

Proposition B.2 *Let the linear system of equations*

$$\begin{aligned} b_1 &= a_{11}\, x_1 + a_{12}\, x_2 + a_{13}\, x_3 + \cdots + a_{1n}\, x_n, \\ b_2 &= a_{21}\, x_1 + a_{22}\, x_2 + a_{23}\, x_3 + \cdots + a_{2n}\, x_n, \\ &\vdots \\ b_n &= a_{n1}\, x_1 + a_{n2}\, x_2 + a_{n3}\, x_3 + \cdots + a_{nn}\, x_n \end{aligned} \tag{B.1}$$

be given in $x_1, \dots, x_n$.

We define the matrix

$$A := \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix},$$

and for all $1 \leq j \leq n$ *the matrix*

$$A_j := \begin{pmatrix} a_{11} & \dots & a_{1,j-1} & b_1 & a_{1,j+1} & \dots & a_{1n} \\ a_{21} & \dots & a_{2,j-1} & b_2 & a_{2,j+1} & \dots & a_{2n} \\ \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & \dots & a_{n,j-1} & b_n & a_{n,j+1} & \dots & a_{nn} \end{pmatrix}.$$

If $\det A \neq 0$*, then Eq.* (B.1) *has exactly one solution, and it holds*

$$x_j = \frac{\det A_j}{\det A}$$

for all $1 \leq j \leq n$.

Proof We fix a $1 \leq j \leq n$. According to the definition of the determinant of A, there exist numbers $c_1, \dots, c_n$ such that

$$\det A = c_1\, a_{1j} + c_2\, a_{2j} + \cdots + c_n\, a_{nj},$$

and these numbers depend only on the entries of A that are not in the j-th column.

We next combine the equations from Eq. (B.1), by adding the c_1-multiple of the first, the c_2-multiple of the second, etc., up to the c_n-multiple of the n-th equation. We obtain

$$c_1\, b_1 + c_2\, b_2 + \cdots + c_n\, b_n = d_1\, x_1 + d_2\, x_2 + \cdots + d_n\, x_n, \tag{B.2}$$

where the coefficients d_i are given by:

$$d_i = c_1\, a_{1i} + c_2\, a_{2i} + \cdots + c_n\, a_{ni}.$$

It follows that $d_j = \det A$. Furthermore, d_i for $i \neq j$ is the determinant of the matrix that results from A by replacing the j-column with the i-th column. However, the determinant of such a matrix vanishes, i.e., $d_i = 0$ for $i \neq j$.

With the same consideration, the left side of Eq. (B.2) is the determinant of the matrix that results from A by replacing the entries of the j-th column with $b_1, b_2, \dots, b_n$, so A_j.

Thus, from Eq. (B.2), it follows that

$$\det A_j = \det A \cdot x_j.$$

Since by assumption $\det A \neq 0$, the x_j are thus, as claimed, uniquely determined in the case of existence.

It remains to show the existence of a solution to the linear system of equations. To do this, we first consider that adding multiples of rows to other rows does not change the

determinant of the coefficient matrix, so it does not vanish in particular. When swapping rows, only the sign changes, at most.

Since the determinant of the coefficient matrix does not vanish, not all coefficients in the first column can vanish, without loss of generality the coefficient a_{11}. By adding suitable multiples of the first row to the others, we obtain an equivalent linear system of equations with $a_{21} = \cdots = a_{n1} = 0$, so that the determinant still does not vanish.

But then not all $a_{22}, \ldots, a_{n2}$ can vanish either, so we can again assume without loss of generality, that $a_{22} \neq 0$. Again by adding suitable multiples, this time of the second row to the others, we obtain an equivalent system of equations, for which also $a_{32} = \cdots = a_{n2} = 0$.

If we continue this procedure, we finally obtain an equivalent system of equations, for which the diagonal entries $a_{11}, \ldots, a_{nn}$ do not vanish and for which the entries below the diagonal a_{ij}, $i > j$ vanish. Such a linear system of equations *in triangular form* has an obvious solution. □

Analysis

C

The complex numbers are an object of analysis and not of algebra. The fundamental theorem of algebra does not really deserve its name.

Abstract

Galois theory deals with expressions of the form $\sqrt[3]{1-\sqrt{2}}$ and, more generally, with solutions of polynomial equations. Before properties of these terms or, more generally, these solutions can be examined, the first step must be to ensure that there are numbers that can play the role of $\sqrt[3]{1-\sqrt{2}}$ and that there are always solutions to (nontrivial) polynomial equations.

In this book, we have solved this problem by extending the ring of of rational numbers to the ring of complex numbers and then showing through the fundamental theorem of algebra that every nontrivial polynomial equation (with rational or, more generally, algebraic coefficients) has a solution in the complex numbers.

The complex numbers are an object of analysis. They arise in an algebraic way from the real numbers, as described in the main text of this book, but the real numbers are defined by limit processes. It is clear that the proof of the fundamental theorem of algebra requires some facts from analysis, which we have compiled in this chapter from the perspective of a constructive mathematician.

From the standpoint of pure algebra, it may be a little unsatisfactory that we have used analysis to build our Galois theory. Indeed, for the pure Galois theory, the introduction of complex numbers would not have been necessary. As we will show in the second volume, there is a purely algebraic construction of the field $\overline{\mathbf{Q}}$ of algebraic numbers in which every nontrivial polynomial equation has a solution, and which is therefore algebraically closed.

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There are two reasons why we did not take this path in this book. On the one hand, the algebraic construction is comparatively abstract, while the construction of complex numbers from the already known real numbers is relatively simple. On the other hand, some applications of Galois theory, such as the numerical calculation of roots or the various construction problems with compass and straightedge, lead directly to the complex numbers.

C.1 Real Numbers

The length of the diagonal of a unit square, i.e., $\sqrt{2}$, can be approximated as closely as desired by rational numbers, but is not itself a rational number (as we saw in the first chapter). The real numbers $\mathbf{R}$ fill such gaps between the rational numbers. And a real number should be an object that can be approximated as closely as desired by rational numbers. What exactly is meant by this will be explained in the following:

If x is a real number and q and $\varepsilon > 0$ are rational numbers, then firstly

$$|x - q| < \varepsilon$$

should be a well-formed statement. If this statement is true, we say that x is approximated by q with an error less than ε. If x is approximated by q with an error less than ε, then also with an error less than ε' for all $\varepsilon' \geq \varepsilon$.

Two further properties apply to this relation: Firstly, every real number x can be approximated as closely as desired, i.e., for every rational number $\varepsilon > 0$ there also exists a rational number q through which x is approximated with an error less than ε, i.e.,

$$\forall x \in \mathbf{R}\, \forall \varepsilon \in \mathbf{Q}_+\, \exists q \in \mathbf{Q}\colon |x - q| < \varepsilon.$$

Furthermore, two approximations for x are close to each other. More precisely: If q_1 and q_2 are two rational numbers that approximate x with an error less than ε_1 and ε_2 respectively, then the distance between q_1 and q_2 is at most $\varepsilon_1 + \varepsilon_2$, so

$$\forall x \in \mathbf{R}\, \forall \varepsilon_1, \varepsilon_2 \in \mathbf{Q}_+\, \forall q_1, q_2 \in \mathbf{Q}\colon$$
$$(|x - q_1| < \varepsilon_1 \wedge |x - q_2| < \varepsilon_2 \implies |q_1 - q_2| < \varepsilon_1 + \varepsilon_2)\,.$$

With this, we have explained the approximation relation for real numbers. What remains is to provide a construction rule for real numbers. Specifically:

If a rule is given that allows us to construct a rational number q_ε for every rational number $\varepsilon > 0$ such that $|q_{\varepsilon_1} - q_{\varepsilon_2}| < \varepsilon_1 + \varepsilon_2$ for every pair $\varepsilon_1, \varepsilon_2 > 0$, then a real number x exists with $|x - q_\varepsilon| < \varepsilon$ for all $\varepsilon > 0$.

We have deliberately spoken of a *construction rule* and not of a mapping $\varepsilon \mapsto q_\varepsilon$. The difference between the two is that in a construction rule, choices can be made at each step. Only with the (countable) axiom of choice, which is not compatible with our constructive view, is a mapping determined by a construction rule.

In Appendix A, we gave an interpretation of intuitionistic logic locally over a topological space Y. Here, the difference between a construction rule and a mapping becomes even clearer: If a mapping $\varepsilon \mapsto q_\varepsilon$ is given over Y, then a rational number q_ε over Y is assigned to each rational number $\varepsilon > 0$ over Y. In contrast, with a construction rule, for each $\varepsilon > 0$, we would choose an open cover $(U_i)_{i\in I}$ of Y and locally over each U_i a local rational number $q_{\varepsilon,i}$ over U_i, without these numbers being restrictions $q_{\varepsilon,i} = q_\varepsilon|U_i$ of a rational number q_ε defined over Y.

But back to our definition of real numbers. We now know the elementary property of real numbers, namely that they can be approximated arbitrarily accurately by rational numbers, and that each real number is given by such approximation rule.

We can immediately give examples of real numbers. Namely, we can regard each rational number q as a real number, namely as the real number which is approximated by q with an error less than ε for all $\varepsilon > 0$. We write this real number again as q, without there being any confusion.

We call two real numbers x_1 and x_2 *different*, written $x_1 \neq x_2$, if rational approximations q_1 and q_2 exist with $|x_1 - q_1| < \varepsilon_1$ and $|x_2 - q_2| < \varepsilon_2$ but $|q_1 - q_2| \geq \varepsilon_1 + \varepsilon_2$.

Two real numbers x_1 and x_2 are called *equal*, written $x_1 = x_2$, if they are not different, so we have

$$x_1 = x_2 \iff \neg(x_1 \neq x_2).$$

The reason we must not define inequality as the negation of equality lies again in our constructive viewpoint. This can be most easily explained again with our local interpretation of intuitionistic logic over a topological space Y:

For this, we choose $Y = \mathbf{R}$. We then remember that a local real number over $Y = \mathbf{R}$ can be understood as a continuous function over $\mathbf{R}$. Let x_1 be the identical function on $\mathbf{R}$, thus understood as a local real number, and x_2 be the zero function. Then the (local) statement $x_1 \neq x_2$ is true on the open subset $\mathbf{R} \setminus \{0\}$, but not on all of $\mathbf{R}$. By definition of negation, the local statement $\neg(x_1 \neq x_2)$, i.e. $x_1 = x_2$, is true on the largest open subset in the complement of $\mathbf{R} \setminus \{0\}$ in $\mathbf{R}$, i.e. nowhere. It follows that $\neg(x_1 = x_2)$ is true everywhere; in particular, $\neg(x_1 = x_2)$ is not equivalent to $x_1 \neq x_2$ (the latter statement implies the former).

C.2 The Real Numbers as a Ring

If x_1 and x_2 are two real numbers, their *sum* $x_1 + x_2$ is the real number that is approximated by $q_1 + q_2$ up to $\varepsilon_1 + \varepsilon_2$ when x_1 is approximated by q_1 up to ε_1 and x_2 is approximated by q_2 up to ε_2.

Furthermore, the *product* $x_1 \cdot x_2$ of x_1 and x_2 is the real number that is approximated by $q_1 \cdot q_2$ up to $q_2\,\varepsilon_1 + q_1\,\varepsilon_2 + \varepsilon_1\,\varepsilon_2$ when x_1 is approximated by q_1 up to ε_1 and x_2 is approximated by q_2 up to ε_2.

To find a rational approximation of the product $x_1 \cdot x_2$ to a given $\varepsilon > 0$, we can proceed as follows. First, we find rational approximations $\tilde{q}_1$ and $\tilde{q}_2$ of x_1 and x_2 up to 1. Then we choose a rational number $0 < \tilde{\varepsilon} < 1$ such that $(4 + |\tilde{q}_1| + |\tilde{q}_2|)\,\tilde{\varepsilon} + \tilde{\varepsilon}^2 \leq \varepsilon$. If q_1 and q_2 are approximations of x_1 and x_2 up to $\tilde{\varepsilon}$, then $q_1 \cdot q_2$ is an approximation of $x_1 \cdot x_2$ up to ε. We leave the short calculation for this to the reader.

Together with the real numbers 0 and 1 (which can be approximated arbitrarily accurately by the rational numbers of the same name), the addition and multiplication defined in this way satisfy the usual arithmetic laws as known from the integers or rational numbers. In particular, for every real number x, there is a real number $-x$ such that $x + (-x) = 0$.

We can also assign the *absolute value* $|x|$ to every real number x: If x is approximated by q up to ε, then $|x|$ is approximated by $|q|$ up to ε.

Restricted to the rational numbers, all the operations introduced here agree with the usual ones. This also applies to the following comparison rule:

If x is a real number, it is called *positive*, written $x > 0$, if a rational $c > 0$ exists such that x can be approximated arbitrarily closely by rational numbers $q \geq c$, i.e.

$$x > 0 \iff \exists c \in \mathbf{Q}_+\, \forall \varepsilon \in \mathbf{Q}_+\, \exists q \in \mathbf{Q}\colon q \geq c \wedge |x - q| < \varepsilon.$$

If x_1 and x_2 are two real numbers, x_1 is said to be *smaller* than x_2, written $x_1 < x_2$, if $x_2 - x_1$ is positive.

C.3 Cauchy Processes and Cauchy Sequences

A *Cauchy process* is a rule according to which we can construct a real number x_n for each natural number n in such a way that $|x_n - x_{n'}|$ becomes arbitrarily small for sufficiently large n, n', i.e.

$$\forall \varepsilon > 0\, \exists m\, \forall n, n' \geq m\colon |x_n - x_{n'}| < \varepsilon.$$

A *Cauchy sequence* is a Cauchy process where $n \mapsto x_n$ is a mapping, i.e. is well-defined.

Every real number defines a Cauchy process: If x is a real number, we choose for each natural number n a rational approximation q_n of x with an error smaller than $\frac{1}{n}$. Because $|q_n - q_{n'}| < \frac{1}{n} + \frac{1}{n'} \to 0$ for $n, n' \to \infty$, it follows that q_n is a Cauchy process.

Conversely, if a Cauchy process x_n is given, it defines a real number x that is approximated by rational numbers as follows: Let $\varepsilon > 0$ be a rational number. Then we choose a natural number n with $\frac{1}{n} < \frac{\varepsilon}{3}$. If q_n is an approximation of x_n with an error of less than $\frac{1}{n}$, then q_n is an approximation of x with an error of less than ε.

We call the thus defined real number x the limit of the Cauchy process x_n, and write $x = \lim_{n\to\infty} x_n$.

Every real number is therefore, in particular, a limit of the form $\lim_{n\to\infty} q_n$ where the q_n form a Cauchy process of rational numbers.

Sum and product are compatible with formation of limits, that is, we have

$$\lim_{n\to\infty} (x_n + y_n) = \lim_{n\to\infty} x_n + \lim_{n\to\infty} y_n$$

and

$$\lim_{n\to\infty} (x_n \cdot y_n) = \lim_{n\to\infty} x_n \cdot \lim_{n\to\infty} y_n$$

for any two Cauchy processes x_n and y_n.

In general, we call a function f of real numbers *continuous* if

$$f\left(\lim_{n\to\infty} x_n\right) = \lim_{n\to\infty} f\,(x_n)$$

for Cauchy processes x_n.

Sum and product are thus (in both arguments) continuous. More generally, it follows that every polynomial function is continuous. Furthermore, the absolute value function is continuous.

C.4 Roots of Nonnegative Real Numbers

We can use the theory of Cauchy processes to show that every nonnegative real number has (unique) k-th roots for all $k = 1, 2, 3, \ldots$.

First, we note that k-th roots of rational numbers $q \geq 0$ can be approximated arbitrarily accurately by rational numbers, i.e., for every natural number n there exists a rational number $r_n \geq 0$ with $|q - r_n^k| < \frac{1}{n}$. The existence of r_n follows constructively; there are algorithms such as the shifting nth root algorithm or the Newtonian method,[5] which allow to determine such an r_n from q.

From $|q - r_n^k| < \frac{1}{n}$ for all n, it follows after a small calculation with inequalities that the r_n form a Cauchy sequence, that is, there exists a real number $y \geq 0$ with $y = \lim_{n\to\infty} r_n$,

in particular $y^k = q$. We thus obtain that every rational number $q \geq 0$ has a (unique) k-th root $\sqrt[k]{q} \geq 0$ in the real numbers.

Finally, let an arbitrary real number $x \geq 0$ be given. Then x is the limit of a Cauchy sequence q_n, where the q_n are all nonnegative rational numbers. A brief calculation with inequalities shows that then also the $\sqrt[k]{q_n}$ form a Cauchy sequence, thus possess a real number $y \geq 0$ as limit. It follows $y^k = x$. Thus, for every nonnegative real number x, there is a (unique) non-negative k-th root $\sqrt[k]{x}$ in the real numbers.

C.5 The Exponential Series

If x is an arbitrary real number, then the sums

$$s_n := \sum_{k=0}^{n-1} \frac{x^k}{k!}$$

form a Cauchy sequence because for $0 \ll m \leq n$, we have

$$\begin{aligned} |s_n - s_m| &= \left| \sum_{k=m}^{n-1} \frac{x^k}{k!} \right| \\ &\leq \sum_{k=m}^{n-1} \frac{|x|^k}{k!} \\ &\leq \frac{|x|^m}{m!} \cdot \sum_{\ell=0}^{n-m-1} \left(\frac{|x|}{m} \right)^\ell \\ &< \frac{|x|^m}{m!} \cdot \frac{1}{1 - \frac{|x|}{m}}, \end{aligned}$$

and the right side becomes arbitrarily small for sufficiently large m.

We obtain that the limit

$$\mathrm{e}^x := \sum_{k=0}^{\infty} \frac{x^k}{k!} := \lim_{n \to \infty} \sum_{k=0}^{n-1} \frac{x^k}{k!}$$

exists. We call the thus obtained real function $x \mapsto \mathrm{e}^x$ *(real) exponential function.*

In a very similar way, it turns out that the *cosine*

$$\cos x := \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$$

and the *sine*

$$\sin x := \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!}$$

are real numbers.

Proposition C.1 *If x_1 and x_2 are two real numbers, the following addition theorem holds:*

$$\mathrm{e}^{x_1+x_2} = \mathrm{e}^{x_1} \cdot \mathrm{e}^{x_2}.$$

Proof According to the binomial theorem:

$$\begin{aligned}
\mathrm{e}^{x_1+x_2} &= \sum_{n=0}^{\infty} \frac{(x_1+x_2)^n}{n!} \\
&= \sum_{n=0}^{\infty} \sum_{k=0}^{n} \binom{n}{k} \frac{x_1^k \, x_2^{n-k}}{n!} \\
&= \sum_{n=0}^{\infty} \sum_{k=0}^{n} \frac{x_1^k}{k!} \frac{x_2^{n-k}}{(n-k)!} \\
&= \sum_{n_1,n_2=0}^{\infty} \frac{x_1^{n_1}}{n_1!} \frac{x_2^{n_2}}{n_2!} \\
&= \left(\sum_{n_1=0}^{\infty} \frac{x_1^{n_1}}{n_1!} \right) \cdot \left(\sum_{n_2=0}^{\infty} \frac{x_2^{n_2}}{n_2!} \right) \\
&= \mathrm{e}^{x_1} \cdot \mathrm{e}^{x_2}.
\end{aligned}$$

□

Corollary C.1 *If x is a real number, then for every integer, we have*

$$\mathrm{e}^{n \cdot x} = \left(\mathrm{e}^{x}\right)^{n}.$$

Glossary

Algebraic Independence A system of numbers is called algebraically independent if there is no non-trivial polynomial relation between them.

Algebraic Integer Complex number that is the zero of a normalized polynomial with integer coefficients.

Algebraic Number Complex number that is the zero of a non-trivial polynomial with rational coefficients.

Algebraic Relation Polynomial relation between algebraic numbers, often the zeros of a separable polynomial.

Algebraically Unique Root Zero of an irreducible polynomial of the form $X^n - a$.

Alternating Group The subgroup formed by the even permutations in the symmetric group.

Angle Trisection Problem to divide a given angle into three equal parts using only compass and straightedge.

Center The subset of all group elements that commute with all other group elements.

Centralizer of an Element The centralizer of a group element σ is the subset of group elements that commute with σ.

Complex Numbers The complex numbers form the smallest extension of the real numbers numbers in which a square root i of -1 exists.

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Congruence Modulo a Subgroup Two elements τ and τ' of a group G are called congruent modulo H if there is a $\varrho \in H$ with $\tau = \tau' \circ \varrho$.

Constructible Number Point in the complex number plane that can be constructed with compass and straightedge from the points 0 and 1.

Cube Duplication Problem of constructing a cube with double volume for a given cube using only compass and straightedge.

Cycle Permutation of the form $(i_1, i_2, i_3, \ldots, i_k)$, i.e. a permutation that maps i_1 to i_2, i_2 to i_3 etc., and i_k to i_1 and the remaining elements to themselves.

Cyclic Group Group generated by a single element.

Cyclotomic Equation Equation of the form $X^n - 1 = 0$.

Cyclotomic Polynomial Minimal polynomial of a primitive root of unity.

Degree of an Algebraic Number Degree of the minimal polynomial of the algebraic number.

Discriminant Polynomial in the coefficients of an equation that vanishes if and only if the equation has a multiple solution (in an algebraic closure).

Elementary Symmetric Function Certain symmetric polynomials in n indeterminates.

Equivalence Relation Reflexive, symmetric, and transitive relation.

Even Permutation Permutation whose signum is positive.

Galois Conjugate Two numbers are Galois conjugate if they have the same minimal polynomial.

Galois Group Group of all symmetries of the zeros of a separable polynomial.

Index of a Subgroup Number of congruence classes modulo the subgroup.

Inversion Pair of a Permutation A pairs of letters whose relative order is swapped by the permutation.

Minimal Polynomial of an Algebraic Number Normalized polynomial of smallest degree that has a given algebraic number as a root.

Normal Series Descending sequence of subgroups of a group so that the individual factors are each formed by inclusions of normal subgroups.

Normal Subgroup Subgroup of a group that is invariant under conjugation with any group element.

Number Field Extension of $\mathbf{Q}$ of the form $\mathbf{Q}(t_1, \ldots, t_n)$ with algebraic numbers $t_1, \ldots, t_n$.

Order of a Group Number of elements of a group.

Order of an Element Smallest positive exponent so that the power of the element with this exponent is 1.

p-Group Group whose order is a prime power.

Permutation Group Subgroup of a symmetric group.

Polynomial Equation Equation of the form $a_n X^n + a_{n-1} X^{n-1} + \cdots + a_1 X + a_0 = 0$, where the coefficients $a_0, \ldots, a_n$ are elements of an arithmetic domain.

Product of Subsets of a Group Subset of the form $H \cdot N = \{h \cdot n \mid h \in H, n \in N\}$ for subsets H and N of a group.

Radical Extension Extension of the form $K(a)$ over K, where a is an algebraically unique square root over K.

Resolvable Group Group that has a normal series whose factors are of prime order.

Root of Unity Complex zero of $X^n - 1$.

Separable Polynomial Polynomial that has no multiple zeros (in an algebraic closure).

Simple Group Group that has exactly two normal subgroups (the trivial subgroup and itself).

Signum of a Permutation The signum of a permutation is $+1$ if the permutation has an even number of inversion pairs, otherwise -1.

Squaring of the Circle Problem of constructing a square with the same area for a given circle using only compass and straightedge.

Subgroup Subset of a group that is closed under composition and formation of inverses and contains the identity.

Symmetric Group Full permutation group.

Transitive Operation Operation of a group on a set such that for each pair x, y of elements of the set there is a group element g with $g \cdot x = y$.

Transcendental Number Complex number that is non-algebraic.

Vandermonde Determinant Polynomial in n indeterminates that is the determinant of a certain $(n \times n)$ matrix.

Endnotes

[1]Rodion Osievich Kuzmin, 1891–1949, Soviet mathematician.
[2]Alexander Osipovich Gelfond, 1906–1968, Soviet mathematician.
[3]Theodor Schneider, 1911–1988, German mathematician.
[4]Georg Friedrich Bernhard Riemann, 1826–1866, German mathematician.
[5]Sir Isaac Newton, 1642–1727, English mathematician, physicist, astronomer, and theologian.

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